A Hipótese do Consenso Perceptivo Distribuído na Anestesia Geral

Um modelo de rede com restrição temporal para a inconsciência induzida por anestésicos

> **Versão em português brasileiro.** Em caso de divergência, prevalece o original em inglês. Os apêndices a partir do B e as referências permanecem em inglês, e os títulos das obras citadas, no idioma original, como é praxe acadêmica.

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Resumo

Propomos a **hipótese do Consenso Perceptivo Distribuído (CPD):** a experiência exige que os sinais entre áreas cerebrais cheguem dentro de uma janela curta de um ritmo compartilhado, e que as conexões que o fazem abranjam um quórum, uma fração mínima da rede. Anestésicos quimicamente diversos eliminariam essas conexões por falha de transmissão, por variabilidade temporal ou por ambas. Estendendo o modelo de percolação de Zhou et al. (2015), a CPD prevê que o componente gigante das conexões cuja latência evocada entre áreas fica dentro da janela colapsa na perda de responsividade, e que a variabilidade dessa latência de um ensaio para outro, e não sua média, aumenta primeiro, uma cláusula também esperada pela comunicação por coerência. Simulações em três conectomas mostram apenas que o mecanismo pode operar, quando a janela temporal não é muito mais estreita que o *jitter* temporal (ambos de 3 ms, não medidos entre áreas). Num modelo de osciladores de fase, a falha de transmissão abole o quórum, como os máximos escolhidos garantem, mas em geral só além dos 62% de falha medidos em cultura, exceto no conectoma humano quando os sinais nos dois sentidos de uma conexão precisam chegar a tempo. Nele, a variabilidade temporal sozinha só aboliu o quórum sob regras de contagem adotadas depois que a regra especificada de antemão falhou, com um erro temporal máximo de 40 ms, um parâmetro livre, e apenas porque um sinal com um ciclo inteiro de atraso ou mais conta como perdido, e não como chegado a tempo num ciclo posterior. Os testes especificados de antemão em dados abertos foram inconclusivos (eletrocorticografia em macacos) ou não puderam ser executados (ressonância magnética funcional); em caráter exploratório, por um critério descritivo, sua rede de acoplamento permaneceu intacta durante a perda de responsividade, mas tinha pouca margem para cair. Seis mecanismos propostos falharam.

1. Introdução e escopo

1.1 O problema

180 anos após a primeira demonstração pública da anestesia com éter, não há consenso sobre como os anestésicos produzem a inconsciência (Franks, 2008; Alkire et al., 2008; Hemmings et al., 2019). Os agentes não têm relação estrutural entre si (do gás monoatômico xenônio ao alquilfenol propofol), atuam sobre muitos alvos moleculares e produzem assinaturas eletroencefalográficas (EEG) distintas, mas convergem para um desfecho comportamental semelhante. Uma explicação satisfatória precisa mostrar como *causas moleculares heterogêneas* produzem um *efeito comum no nível do sistema*. A busca por um único mecanismo molecular não teve êxito, porque os fatos moleculares são genuinamente diversos (Seção 2.1). Por isso, a CPD busca a unificação um nível acima: muitas vias moleculares convergindo para uma única forma de falha de rede.

1.2 O que este artigo afirma e o que não afirma

Esta é uma **hipótese**, não um teorema: seu valor está em saber se suas previsões resistem ao teste experimental. Ela trata das condições em que a experiência e o conteúdo relatável estão presentes, e não de qual é esse conteúdo nem de por que ele é acompanhado de experiência. As afirmações são marcadas com **[E]** (estabelecida, com citação) ou **[H]** (afirmação nova ou controversa

deste artigo, formulada de modo a poder ser falseada); as duas analogias mantidas da computação são marcadas com **[A]** e não carregam nenhuma afirmação.

Uma condição limita todos os resultados das simulações. Os resultados valem apenas quando a janela temporal entre áreas W não é muito mais estreita que o *jitter* temporal σ_b (referência $W = \sigma_b$ = 3 ms; com W = 1 ms ou σ_b = 10 ms, o modelo na vigília não tem quórum), e nenhum dos dois foi medido entre áreas (Seção 4.7).

Os postulados. O núcleo da hipótese é formado por P1 a P5, P4c e P7 (Seção 3.2). P6, P8 e P9 são propostas auxiliares das quais o núcleo não depende (Seção 3.3), e a afirmação de que atrasos constantes são inofensivos é uma questão em aberto (Seção 3.4).

Postulado	Em uma linha	Situação nos modelos (Seção 6)
P1 Ciclos de integração	A experiência é gerada em ciclos repetidos nos quais populações distribuídas convergem para um estado compartilhado	Suposto, não testado
P2 Condição de atraso limitado	Uma população só contribui para um ciclo se seus sinais chegarem dentro de uma janela W em torno de um ritmo compartilhado; um sinal fora dela é perdido para esse ciclo, e não contado num ciclo posterior (uma regra afirmada, não sustentada)	Implementado no modelo de percolação e, no modelo de osciladores, por uma leitura acrescentada depois
P3 Quórum	Qualquer experiência exige um componente gigante de conexões temporalmente coerentes que contenha pelo menos uma fração q^* das populações; a consciência conectada (relatável, responsiva) exige, além disso, que as populações sensoriais pertençam a ele. Ainda não está especificado quando uma conexão entre duas populações conta	Como em P2
P4 Dois modos de falha	Os anestésicos eliminam conexões temporalmente coerentes ao fazer a transmissão falhar ou ao tornar variável, de um ensaio para outro, a temporização dos sinais entre áreas em relação ao ritmo compartilhado, e não a temporização de neurônios isolados	A falha de transmissão abole o quórum por construção; a variabilidade temporal o faz nos dois conectomas de invertebrados pela regra especificada de antemão, e no conectoma humano apenas sob regras adotadas depois de uma falha
P4c Inconsciência anestésica	Onde um agente abole a experiência, e não só a responsividade, o componente gigante fica abaixo de q^*; onde a experiência persiste sem responsividade, ele fica acima de q^*	Suposto: os modelos leem a perda do quórum como o desfecho anestésico; não testado
P5 Retroalimentação da referência	O ritmo compartilhado é sustentado pelo próprio grupo coerente, de modo que um grupo menor o enfraquece	Produz inércia no modelo de percolação depois de calibrado; ausente do modelo de osciladores
P7 Alerta	Os sistemas de alerta promotores da vigília sustentam parte do ritmo compartilhado independentemente da coerência cortical	Presente nas equações; a recuperação causada pelo alerta em concentração fixa é reproduzida no modelo de percolação, uma verificação de consistência
P6 Dinâmica padrão (auxiliar)	Quando o quórum se perde, o córtex passa à sua própria dinâmica local, específica de cada agente	Não testado
P8 Degradação gradual (auxiliar)	À medida que a folga temporal diminui, o ritmo desacelera e a atividade passa a ocorrer em rajadas	Sua previsão distintiva falhou
P9 Controle glial (auxiliar)	Alças gliais mais lentas mantêm as condições para a coerência temporal	Apenas verificações de consistência; seu único teste não poderia ter passado

O que é novo e o que é aproveitado. A comunicação por coerência (Fries, 2005, 2015), o processamento recorrente (Lamme e Roelfsema, 2000; Lamme, 2006), a desconexão funcional como via final comum da anestesia (Hudetz e Mashour, 2016), os modelos de percolação da anestesia e da percepção (Zhou et al., 2015; Arese Lucini et al., 2019) e a histerese produzida por retroalimentação (Kim et al., 2018) precedem este artigo (Seção 2.3). P3 e P7 se sobrepõem ao espaço de trabalho neuronal global (GNW, do inglês *global neuronal workspace*): um estado global de disponibilidade, do tipo tudo ou nada, cujo limiar é fixado por uma entrada de alerta (Dehaene e Changeux, 2005; Mashour et al., 2020). O modelo mais próximo é o de Zhou et al. (2015), no qual a anestesia reduz uma única probabilidade de transmissão numa rede em camadas, a perda da consciência aparece como uma transição de percolação e as assinaturas de EEG da anestesia são reproduzidas, o que este artigo não tenta fazer. As contribuições do artigo são:

1. A variabilidade temporal entre áreas como uma segunda via para a perda de conexões utilizáveis, combinada com a falha de transmissão em uma única probabilidade por conexão; as duas vias só podem ser distinguidas medindo cada uma.
2. A proposta [H] de que o que os anestésicos degradam é a *variabilidade* da temporização entre áreas em relação a um ritmo compartilhado, e não sua média nem a coerência dos ritmos em curso (previsões 3 e 8), e de que, por meio de P3, essa variabilidade fixa um limiar na rede (previsão 1, com sua regra temporal para as arestas). A cláusula da variabilidade é compartilhada com a comunicação por coerência; o limiar, não.
3. Uma aplicação da estimativa padrão por funções geradoras (Callaway et al., 2000) a um limiar de quórum, e não ao surgimento do componente gigante, para a distribuição de graus de cada conectoma; a estimativa é sabidamente precisa em redes agrupadas cujo comprimento médio do caminho mais curto é pequeno (Melnik et al., 2011) e é comparada com a medição na conectividade de um nematoide, de um inseto e de um ser humano.
4. A proposta [H] de que o tônus de alerta restaura o quórum em concentração fixa ao fortalecer o ritmo compartilhado (P7). Sua previsão, a de que, entre as concentrações de recuperação e de perda, a restauração dura mais que o estímulo, é com-

partilhada por qualquer explicação com dois estados estáveis (Seção 8).
5. Uma leitura de um modelo de osciladores de fase nos três conectomas que aplica a janela e o quórum da CPD, estendendo o componente gigante de ligações sincronizadas (Gómez-Gardeñes et al., 2007) ao erro temporal de cada conexão.
6. Quinze previsões falseáveis, um protocolo de análise, testes especificados de antemão em registros abertos e um simulador aberto e testado em que cada parâmetro é identificado como medido, calibrado, suposto ou livre.

1.3 Escopo e definições

A anestesia geral compreende inconsciência, amnésia, analgesia e imobilidade. Brown et al. (2010) a descrevem como um coma reversível induzido por fármacos, e não como sono: a surto-supressão, vista na anestesia profunda e no coma, não ocorre no sono. **Esta hipótese trata apenas da inconsciência.** A imobilidade, base do padrão de concentração alveolar mínima (CAM; Eger et al., 1965), não depende do prosencéfalo (Rampil et al., 1993) e é mediada em grande parte pela medula espinal (Antognini e Schwartz, 1993; Rampil, 1994). A memória e a responsividade podem ser prejudicadas separadamente (Veselis et al., 1997).

Seguimos Sanders et al. (2012). *Responsividade* é uma resposta comportamental a um comando ou estímulo; sua perda é a perda de responsividade (LOR, do inglês *loss of responsiveness*) em humanos e a perda do reflexo de endireitamento (LORR) em roedores. *Consciência conectada* é a experiência que inclui a percepção do ambiente; *consciência desconectada* é a experiência sem ela, como no sonho; *inconsciência* é a ausência de qualquer experiência. P3 corresponde diretamente a essas distinções: um componente gigante de conexões temporalmente coerentes abaixo do quórum significa inconsciência; acima do quórum, mas sem as populações sensoriais, consciência desconectada; acima do quórum e incluindo-as, consciência conectada. Portanto, irresponsividade não equivale a inconsciência.

1.4 Terminologia

Cada termo é usado em um único sentido definido (Apêndice A). O argumento se apoia em seis deles: conexões *temporalmente coerentes*, a *janela de integração*, a *folga temporal*, o *componente gigante*, o *quórum* e a *inércia neural*. Uma conexão temporalmente coerente também é chamada de *ocupada*, o termo da percolação; o ritmo compartilhado também é chamado de *referência de fase*. Neste artigo, "transição de percolação" significa a perda do quórum, a concentração em que o componente gigante cai abaixo de q^*, e não o ponto crítico da percolação (Seção 4.2). Usa-se perda de responsividade (LOR) sempre que a medida é comportamental, e perda da consciência apenas quando o estudo citado usou esse termo ou tinha evidência além do comportamento.

Palavras emprestadas. *Consenso* e *quórum* são emprestados da computação distribuída em seu sentido cotidiano: uma fração mínima das partes que precisam participar para que um resultado conte. Nenhuma propriedade da teoria do consenso é afirmada ou transferida: as populações não concordam sobre nenhum valor, não há condição de validade nem interseção de quóruns, e nenhum resultado de impossibilidade ou limite de tolerância (Fischer et al., 1985; Dwork et al., 1988; Lamport et al., 1982) se aplica ao cérebro. Esses resultados motivaram P2 e P4 e não restringem nada. Na CPD, concordância significa apenas que as populações estão ligadas em um único grupo por conexões cujos sinais chegam a tempo; não há votação nem votante. *Perceptivo*, no nome da hipótese, é histórico: a hipótese trata de qualquer experiência (P3), e não apenas da percepção.

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2. Fundamentação

2.1 Mecanismos moleculares [E]

A correlação de Meyer-Overton entre potência e solubilidade em lipídios (Meyer, 1899; Overton, 1901) motivou um século de teorias lipídicas, mas a solubilidade em lipídios, por si só, é insuficiente: os anestésicos inibem com potências coincidentes uma enzima sem lipídios, a luciferase de vaga-lume (Franks e Lieb, 1984); isômeros ópticos do isoflurano diferem em seus efeitos sobre canais iônicos (Franks e Lieb, 1991); alguns compostos lipofílicos, os não imobilizadores, violam a correlação (Koblin et al., 1994); uma mutação pontual (N265M) na subunidade β3 do receptor do ácido γ-aminobutírico do tipo A (GABA-A) reduz fortemente os efeitos do propofol e do etomidato em camundongos (Jurd et al., 2003); e o xenônio quase não afeta os receptores GABA-A, mas inibe os receptores de N-metil-D-aspartato (NMDA) (Franks et al., 1998). Vias mediadas pela membrana continuam plausíveis como contribuintes (Cantor, 1997; Pavel et al., 2020). Os anestésicos atuam sobre múltiplos alvos, principalmente receptores GABA-A, receptores NMDA e canais de potássio com dois domínios de poro (K2P), com perfis específicos para cada agente (Rudolph e Antkowiak, 2004; Hemmings et al., 2019). P4 atribui a cada via um tipo de falha [H] (Apêndice F.1, com os agentes, os fluxos iônicos e as fontes): a potencialização do receptor GABA-A (Mihic et al., 1997; Jurd et al., 2003), a ativação de canais K2P (Patel et al., 1999; Gruss et al., 2004; Heurteaux et al., 2004), a inibição de canais de Na^+ pré-sinápticos (Herold e Hemmings, 2012; Baumgart et al., 2015) e a desorganização de domínios lipídicos (Pavel et al., 2020) causariam falha de transmissão; o bloqueio do receptor NMDA (Orser et al., 1997; Franks et al., 1998; Jevtović-Todorović et al., 1998; Dickinson et al., 2007), variabilidade temporal e falha de transmissão; a inibição de canais ativados por hiperpolarização e regulados por nucleotídeos cíclicos (HCN) (Chen et al., 2005; Ying et al., 2006; Chen et al., 2009), alteração do ritmo e da temporização; o agonismo dos receptores α2-adrenérgicos pela dexmedetomidina (Nelson et al., 2003), supressão do alerta (P7); e o complexo I mitocondrial é um fator de sensibilidade (Kayser et al., 1999). Duas propostas não confirmadas (microtúbulos e spin nuclear) também estão no Apêndice F.1. **Implicação:** a CPD exige apenas que, no nível dos circuitos, os agentes aumentem a falha de transmissão, a variabilidade temporal ou ambas.

2.2 Achados no nível dos sistemas [E]

- **Envolvimento talamocortical.** Agentes voláteis suprimem a atividade do tálamo e da formação reticular mesencefálica em humanos (Alkire et al., 2000); nicotina no tálamo medial central reverte a perda do reflexo de endireitamento induzida por sevoflurano em ratos (Alkire et al., 2007); a estimulação talâmica restaura o alerta em macacos anestesiados (Redinbaugh et al., 2020). Se o tálamo ou o córtex muda primeiro é uma questão em aberto (Velly et al., 2007; Fan et al., 2026).
- **Colapso da comunicação cortical.** Propofol, sevoflurano e cetamina perturbam a conectividade de retroalimentação frontoparietal (U. Lee et al., 2013). Na inconsciência induzida por propofol, os disparos ficam confinados a breves janelas de fase de uma oscilação lenta, assíncronas ao longo do córtex (Lewis et al., 2012; lido como uma temporização fragmentada por Kordas, 2026). Sob anestesia, a conectividade funcional em macacos passa a espelhar a anatomia (Barttfeld et al., 2015; Uhrig et al., 2018). Revisões concluem que a irresponsividade acompanha a desconexão frontoparietal (Hudetz e Mashour, 2016), a redução da complexidade e do repertório (Mashour e Hudetz, 2018) e a redução da eficiência da rede (Lee e Mashour, 2018).
- **Assinaturas de EEG específicas de cada agente.** O propofol produz potência beta na sedação e, depois, alfa frontal coerente (8 a 12 Hz) com oscilações lentas e delta (0,1 a 4 Hz) na LOR (Purdon et al., 2013, 2015; Ching et al., 2010); a cetamina produz potência gama e nenhum alfa frontal (Akeju et al., 2016). Os anestésicos também desacoplam os dendritos apicais dos neurônios piramidais da camada 5 de camundongos de seus corpos celulares (Suzuki e Larkum, 2020).
- **Perda de coordenação, e não de atividade, em várias espécies.** O isoflurano mantém ativos os neurônios da cabeça de *Caenorhabditis elegans*, mas faz colapsar suas correlações (Awal et al., 2020); em *Drosophila*, reduz a coerência sincronizada ao estímulo em regiões centrais do cérebro (Cohen et al., 2016), prejudica a retroalimentação de baixa frequência (Cohen et al., 2018) e fragmenta a atividade integrada (Leung et al., 2021). A conectividade humana sob anestesia torna-se menos individual (Luppi et al., 2025; um único estudo).
- **Complexidade.** O índice de complexidade perturbacional (PCI) é alto na vigília, baixo sob propofol e xenônio e alto sob cetamina, em consonância com os relatos de sonhos ao despertar (Sarasso et al., 2015; Casali et al., 2013).
- **Inércia neural.** A consciência retorna em uma concentração menor do que aquela em que é perdida, o que a farmacocinética não explica totalmente, em camundongos e em *Drosophila* (Friedman et al., 2010; Joiner et al., 2013); a razão entre as concentrações de indução e de despertar é de cerca de 1,08 para o isoflurano e 2,85 para o halotano em camundongos (Friedman et al., 2010). As razões humanas são de cerca de 1,07 e 1,11 para concentrações de propofol no sítio efetor previstas (infusão alvo-controlada, TCI), cujos autores não encontraram histerese consistente ou descrevem os valores como semelhantes (Warnaby et al., 2017; Iwakiri et al., 2005), e de cerca de 1,63 para o propofol sérico medido, cujos autores advertem que um equilíbrio incompleto poderia explicá-la (Sepúlveda et al., 2018). Um estudo prospectivo desenhado para testar a inércia neural em humanos, com 36 voluntários, não a encontrou para o propofol, mas a encontrou para o sevoflurano na maioria dos desfechos, no caso da perda de responsividade apenas com remifentanil (Kuizenga et al., 2018). Se ela existe em humanos é, portanto, controverso, e a explicação de P5 depende disso. A recuperação passa por estados metaestáveis discretos (Hudson et al., 2014).
- **Células gliais.** Os anestésicos suprimem os sinais de cálcio dos astrócitos (Thrane et al., 2012), manipular astrócitos ou micróglia altera a indução e o despertar de formas que dependem da região e estão em parte sujeitas a fatores de confusão, e a mielina determina a temporização da condução (Ford et al., 2015; Kato et al., 2020); os detalhes estão no Apêndice F.2.

2.3 Modelos anteriores e a posição da CPD

Referencial	Relação com a CPD
Comunicação por coerência (Fries, 2005, 2015): grupos se comunicam quando suas janelas para enviar e para receber coincidem, e sua influência mútua depende da relação de fase entre seus ritmos (Womelsdorf et al., 2007)	P2 é essa condição. A CPD acrescenta que os anestésicos tornam variável a temporização entre áreas, e pergunta quantas conexões precisam cumpri-la para que haja integração em toda a rede
Ciclos perceptivos (VanRullen e Koch, 2003; VanRullen, 2016); processamento recorrente (Lamme e Roelfsema, 2000; Lamme, 2006)	P1 adota ciclos recorrentes; a CPD acrescenta uma condição temporal e um quórum no nível da rede
Integração em larga escala por sincronização de fase entre bandas de frequência (Varela et al., 2001); ligação talamocortical por coincidência temporal (Llinás et al., 1998)	P1, P2 e P5 reformulam essas propostas; a CPD acrescenta um limiar de contagem (P3) e atribui aos anestésicos a falha de transmissão e a variabilidade temporal
Desvinculação cognitiva (*cognitive unbinding*; Mashour, 2013)	A CPD propõe um motivo específico para a falha da síntese: a violação de uma condição de atraso limitado
Teoria da informação integrada (Tononi et al., 2016)	O parâmetro de ordem da CPD não é Φ nem o aproxima
Espaço de trabalho neuronal global (GNW; Dehaene e Changeux, 2011; Mashour et al., 2020)	O quórum da CPD é uma operacionalização, por percolação, da disponibilidade global. Os modelos do GNW já contêm atrasos sinápticos e ignição do tipo tudo ou nada (Dehaene et al., 2003; Dehaene e Changeux, 2005) e uma entrada de vigilância que fixa o limiar da ignição (Dehaene e Changeux, 2005), e o GNW já foi aplicado à anestesia (Mashour et al., 2020). A CPD difere ao tomar como variável controlada a variabilidade da temporização entre áreas, e não a excitabilidade, e ao derivar o limiar da distribuição de graus

Referencial	Relação com a CPD
Biestabilidade e períodos OFF (Sanchez-Vives et al., 2017; Pigorini et al., 2015; D'Andola et al., 2018; Rosanova et al., 2018)	Depois de uma entrada, o córtex biestável cai em períodos OFF silenciosos que interrompem as interações causais e reduzem a complexidade perturbacional, no sono sem movimentos oculares rápidos (NREM), *in vitro* e na síndrome de vigília sem resposta; as respostas à estimulação magnética transcraniana permanecem locais no sono NREM e sob midazolam (Massimini et al., 2005; Ferrarelli et al., 2010). Os estados DOWN e as falhas por omissão da CPD são um caso particular disso; a CPD acrescenta a via temporal e um limiar de rede
Interruptor talamocortical (Alkire et al., 2000); transições de fase em campo médio (Steyn-Ross et al., 1999; Bojak e Liley, 2005); integração dendrítica (Aru et al., 2020)	A CPD atribui ao tálamo uma referência compartilhada, e não um interruptor; trata a perda como uma transição em uma rede de conexões discretas, sem modelar o EEG; o desacoplamento dendrítico é uma via celular candidata para as falhas por omissão

Modelos quantitativos já contêm elementos da CPD e devem ser creditados (detalhes no Apêndice F.2): a percolação do componente gigante quando as sinapses são enfraquecidas em culturas (Breskin et al., 2006); a percolação de *k-core* da percepção consciente e subliminar (Arese Lucini et al., 2019); uma única probabilidade de transmissão numa rede talamocortical que reproduz características do EEG na anestesia (Zhou et al., 2015), que já colocava todos os agentes em um único parâmetro de transmissão; a histerese da surto-supressão por percolação em redes autossimilares (Su et al., 2020); a percolação da conectividade funcional no conectoma da mosca (Tian e Sun, 2022); uma rede recorrente ajustada à eletrocorticografia de macacos sob anestesia (Du e Huang, 2025); a sincronização explosiva após a inconsciência induzida por anestésicos (Kim et al., 2016); a histerese por retroalimentação adaptativa em osciladores sobre conectomas, relacionada à potência (Kim et al., 2018), da qual P5 é muito próximo; as relações de fase entre *hubs* e periferia e a reconfiguração dos *hubs* sob anestesia (Moon et al., 2015; H. Lee et al., 2013); a retroalimentação pela qual as ligações de uma unidade dependem do estado das outras, que torna a percolação abrupta em redes interdependentes (Buldyrev et al., 2010) e a sincronização explosiva quando as ligações são controladas por um parâmetro de ordem local (Zhang et al., 2015), a classe à qual P5 pertence; e os estados metaestáveis de recuperação com inércia impulsionada por ruído (Hudson et al., 2014; Proekt e Hudson, 2018). O que a CPD afirma como novo é, portanto, mais restrito do que um modelo de percolação ou de histerese em si (Seção 1.2).

3. A hipótese

3.1 Motivação

Na computação distribuída, a concordância determinística é impossível em um sistema totalmente assíncrono se um processo puder falhar por parada (*crash*) (Fischer et al., 1985) e se torna alcançável sob *sincronia parcial*, quando os atrasos são limitados (Dwork et al., 1988); uma mensagem que nunca chega é uma falha por *omissão*, e uma que chega fora do tempo combinado, uma falha de *temporização* (Cristian, 1991). A CPD toma emprestado o modelo síncrono de rodadas como sua única analogia estrutural [A]: em cada rodada, as mensagens que perdem a rodada são descartadas. A sincronia parcial motivou a pergunta, mas não é o modelo, já que seus limites são desconhecidos ou só valem a partir de algum momento (Apêndice F.3). O cérebro chega repetidamente a um único estado coerente a partir de sinais que alcançam áreas diferentes em momentos diferentes, e P2 propõe que ele só consegue fazê-lo enquanto os atrasos entre áreas permanecem dentro de uma janela fixa. Nada mais é transferido (Seção 1.4); os sistemas de engenharia por trás de cada postulado estão tabulados no Apêndice F.3.

O problema temporal é real na escala cortical: nas vias corticocorticais humanas, o atraso axonal mediano é de cerca de 10 ms (Lemaréchal et al., 2022), e a 40 Hz um ciclo gama dura 25 ms, de modo que um atraso típico corresponde a cerca de 40% de um ciclo. Nesse regime, a CPD propõe que a temporização seja uma restrição de primeira ordem para a comunicação [H]. Um atraso constante pode ser absorvido por um deslocamento de fase correspondente (Fries, 2015); a proposta diz respeito à sua variabilidade.

3.2 Postulados do núcleo [H]

Os postulados mantêm a numeração das versões anteriores, para que todos os relatórios e apêndices continuem a se referir a eles corretamente: o núcleo é formado por P1 a P5, P4c e P7; P6, P8 e P9 são propostas auxiliares (Seção 3.3).

A afirmação central. A experiência exige *folga temporal positiva*: os sinais entre populações precisam chegar dentro da janela em que podem ser combinados, avaliados em relação a um ritmo compartilhado. Os anestésicos corroem essa folga ao fazer a transmissão falhar ou os instantes de chegada variarem, e, quando ela se torna negativa em conexões suficientes, as populações deixam de estar ligadas em um único grupo. A afirmação diz respeito a margens temporais, e não a uma frequência: a hierarquia dos ritmos cerebrais é amplamente preservada entre mamíferos cujos volumes cerebrais diferem vários milhares de vezes, possivelmente graças a axônios de longo alcance mais rápidos nos cérebros maiores (Buzsáki et al., 2013), o que preservaria as margens temporais [H].

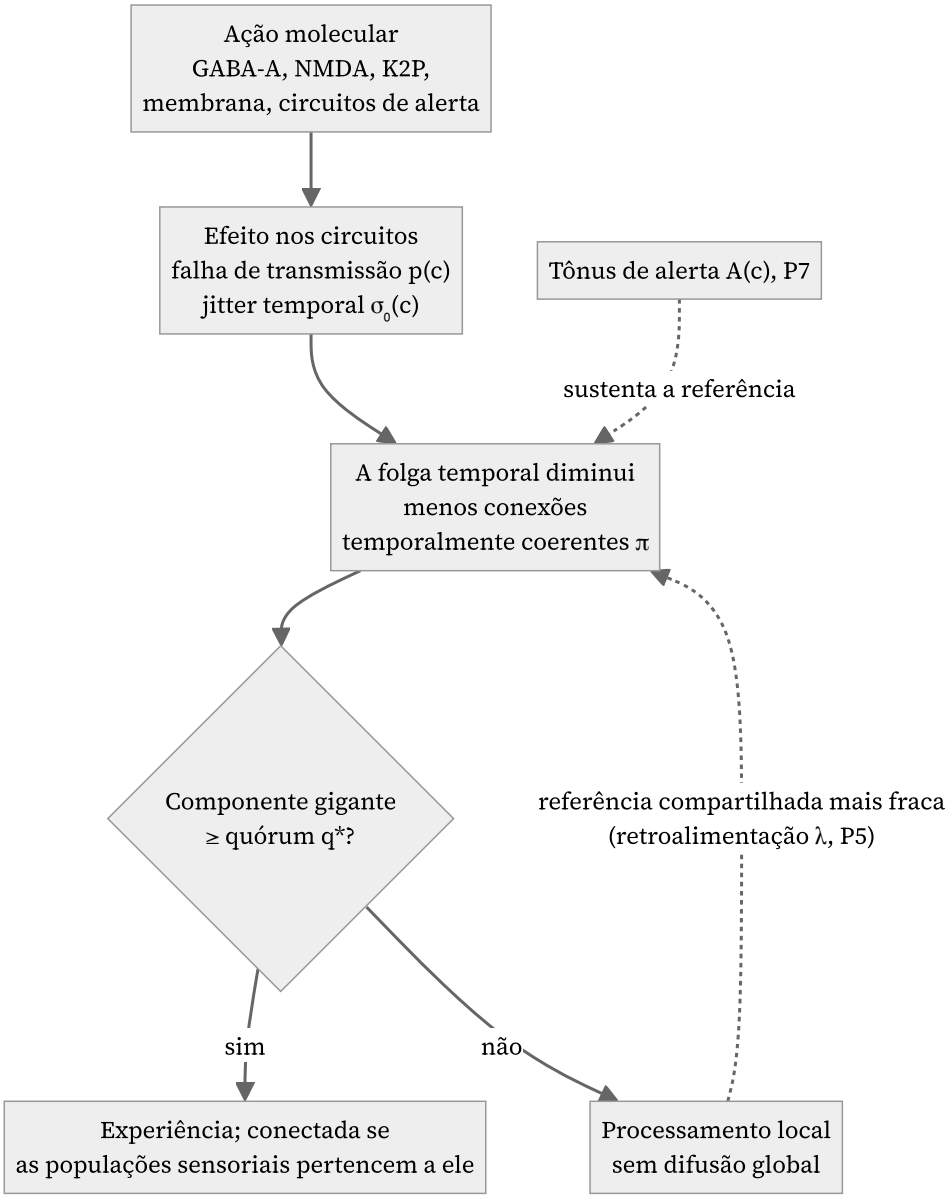


Figura A. Cadeia causal do núcleo da hipótese. As setas tracejadas indicam influências de retroalimentação e de modulação; as propostas auxiliares P8 e P9 (Seção 3.3) atuariam sobre o mesmo nó C. Os símbolos são definidos na Seção 4 e no Apêndice A.2.

- **P1 (Ciclos de integração).** A experiência é gerada em ciclos de integração repetidos nos quais populações distribuídas convergem para um estado compartilhado. O postulado adota as épocas perceptivas discretas (VanRullen e Koch, 2003) e o processamento recorrente entre áreas (Lamme e Roelfsema, 2000; Lamme, 2006); abordagens relacionadas estão listadas no Apêndice F.4. A percepção é amostrada por vários ritmos, e não por um só (VanRullen, 2016), de modo que uma referência única (P5) é uma simplificação. Em vários estudos, a anestesia perturba o processamento recorrente mais do que o *feedforward*, embora não de modo uniforme (Lee et al., 2009; Imas et al., 2005).
- **P2 (Condição de atraso limitado).** Uma população só contribui para um ciclo se seus sinais chegarem dentro de uma janela de integração W em relação à referência de fase compartilhada. Um sinal fora da janela é perdido para esse ciclo, e um sinal com um ciclo inteiro de atraso ou mais conta como perdido, e não como chegado a tempo num ciclo posterior, porque o conteúdo que carrega pertence ao ciclo em que foi enviado [H, não sustentada: não se oferece nenhuma evidência de que o conteúdo seja específico de um ciclo, e a via das falhas de temporização depende inteiramente dessa regra (Seção 4.7)]. Essa é a condição da comunicação por coerência (Fries, 2005, 2015), formulada como uma regra do tipo tudo ou nada; a janela comportamental de integração de eventos multissensoriais, de cerca de 100 ms ou mais (Wallace e Stevenson, 2014), é uma contrapartida psicofísica muito mais larga.
- **P3 (Quórum).** Qualquer experiência, conectada ou desconectada, exige que as conexões temporalmente coerentes liguem um componente gigante que contenha pelo menos uma fração q^* das populações. A consciência conectada (relatável, responsiva) exige, além disso, que as populações sensoriais pertençam a esse componente. P3 é uma condição necessária, e não suficiente (Seção 7.9). A CPD ainda não especifica quando uma conexão entre duas populações conta: quando um sinal em qualquer um dos sentidos chega a tempo, com uma única temporização por conexão, ou apenas quando os sinais nos dois sentidos chegam. O veredito humano sobre a variabilidade temporal depende dessa escolha (Apêndices C.12 a C.14), e

ela seria decidida medindo quantas conexões corticais humanas são recíprocas e se a temporização de seus dois sentidos varia de forma independente. Nenhuma regra adotada depois da falha do Apêndice C.12 faz parte da hipótese.

- **P4 (Dois modos de falha).** Os anestésicos reduzem a fração de conexões temporalmente coerentes de uma ou de ambas as formas a seguir: (a) aumentando a probabilidade de que a transmissão falhe (falhas por omissão, por exemplo o aumento da inibição mediada por GABA-A e a redução da liberação de glutamato; Baumgart et al., 2015); (b) aumentando a variabilidade, de um ensaio para outro, da temporização dos sinais entre áreas em relação ao ritmo compartilhado (falhas de temporização). O que se torna variável é a temporização dos sinais *entre* áreas, que perde precisão à medida que o ritmo compartilhado enfraquece (P5), e não a temporização dos disparos de neurônios isolados, que as poucas medições locais encontram preservada (neurônios auditivos do tronco encefálico sob cetamina-xilazina, Jing et al., 2021; córtex insular, Koyanagi et al., 2021, e Kato et al., 2016). Propõe-se que a via molecular afete a rede apenas por meio dessas duas grandezas e da parcela do ritmo sustentada pelo alerta (P7). Atribuir a cetamina às falhas de temporização é uma suposição, e não um achado: nenhum estudo mostra que a cetamina aumenta a variabilidade temporal entre áreas, o xenônio, também antagonista NMDA, reduz a complexidade cortical como o propofol (Sarasso et al., 2015), e o efeito de circuito estabelecido da inibição dos receptores NMDA em ratos acordados é a desinibição (Homayoun e Moghaddam, 2007). O *jitter* acrescentado pelo agente, $\sigma_0(c)$ (Seção 4.1), é um marcador provisório: não se propõe nenhum mecanismo de circuito para ele.
- **P4c (Inconsciência anestésica).** Em qualquer concentração na qual um agente abole a experiência, e não apenas a responsividade, o componente gigante fica abaixo de q^*. Onde a responsividade é perdida mas a experiência persiste, o componente fica acima de q^*: com as populações sensoriais fora dele na experiência desconectada, como nos sonhos relatados depois da cetamina (Sarasso et al., 2015) e sob dexmedetomidina (Radek et al., 2018), e com elas dentro dele quando comandos ainda são seguidos de forma encoberta, como por um participante sob propofol (Huang et al., 2018a).
- **P5 (Retroalimentação da referência).** A referência de fase compartilhada é sustentada pelo próprio núcleo talamocortical coerente, de modo que um componente gigante menor produz uma referência mais fraca e um erro temporal efetivo maior em todas as conexões restantes.
- **P7 (Alerta).** Os sistemas de alerta promotores da vigília sustentam parte da referência compartilhada independentemente da coerência cortical, e os anestésicos os suprimem: o isoflurano e o sevoflurano reduzem a expressão de c-Fos em neurônios orexinérgicos (Kelz et al., 2008), e o isoflurano despolariza neurônios promotores do sono (Moore et al., 2012). No sono natural, o tônus de alerta é retirado, e durante os estados DOWN da oscilação lenta populações locais inteiras silenciam (Vyazovskiy et al., 2009).

3.3 Propostas auxiliares (fora do núcleo)

O núcleo não depende destas propostas, e suas falhas não o afetam.

- **P6 (Dinâmica padrão).** Quando o quórum é perdido, o córtex não silencia, mas passa à sua dinâmica local intrínseca, como a oscilação lenta (Steriade et al., 1993a), o padrão de atividade padrão do córtex (Sanchez-Vives et al., 2017), cuja forma é específica de cada agente. Não testado nos modelos (limitação 30).
- **P8 (Degradação gradual).** À medida que a folga temporal diminui, o ritmo de referência desacelera, de modo que o mesmo erro temporal custa um erro de fase menor ($2\pi f\sigma$), ao preço de menos ciclos por segundo, e a atividade contínua dá lugar a rajadas sincronizadas. A sequência do propofol, do beta ao alfa frontal com oscilações lentas (Purdon et al., 2013, 2015), e a surto-supressão, cujos períodos de supressão se alongam com a dose (Brown et al., 2010), são propostas como seus estágios; as leituras dominantes atribuem o alfa frontal a mecanismos talâmicos que impedem a comunicação (Purdon et al., 2015; Ching et al., 2010; Vijayan et al., 2013) e a surto-supressão à redução do metabolismo (Ching et al., 2012). Sua previsão distintiva falhou no modelo (Seção 6).
- **P9 (Controle glial).** Alças gliais mais lentas mantêm as condições para a coerência temporal: os oligodendrócitos podem ajustar os atrasos de condução para que sinais distantes cheguem juntos (Pajevic et al., 2014, 2023; Noori et al., 2020), os astrócitos regulam o estado cortical ao longo de segundos a horas (Poskanzer e Yuste, 2016; Ingiosi et al., 2020), e a micróglia participa de uma alça noradrenérgica (Liu et al., 2019; Stowell et al., 2019; Ma et al., 2024), proposta como uma fonte biológica da retroalimentação de P5. O que P9 acrescenta à regulação lenta já estabelecida (Turrigiano, 2008) é uma ação sobre a coerência temporal (Apêndice F.4).

3.4 Uma questão em aberto: atrasos constantes

Versões anteriores deste artigo propunham (como "P4b") que um atraso constante é inofensivo porque o cérebro integra as entradas em relação a expectativas aprendidas sobre quando elas chegam. Os modelos não têm essa antecipação, e neles erros temporais constantes não são inofensivos; uma expectativa aprendida da fase de chegada, especificada de antemão e testada, falhou (Seção 6). Um receptor que vê apenas a fase de chegada não consegue distinguir um atraso constante da relação de fase que o acoplamento mantém, de modo que continua em aberto como os atrasos constantes se tornariam inofensivos. A adaptação comportamental a atrasos (Fujisaki et al., 2004; Yamamoto e Kawabata, 2011; Beech et al., 2025, 2026) diz respeito a atrasos muito mais longos que a janela. A afirmação sobre a variabilidade, testada pelas previsões 3 e 8, faz parte de P4 e não depende desta questão.

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4. Modelo formal [H]

4.1 Rede e coerência

Seja o cérebro um grafo $G = (V, E)$ de N populações neurais com distribuição de graus $P(k)$. Na concentração c no sítio efetor, expressa em relação à concentração do agente que produz metade de seu efeito máximo (CE50), aqui o efeito máximo sobre os *circuitos*, cada conexão transmite com probabilidade $1 - p(c)$. Cada agente atua por uma curva dose-resposta de Hill com coeficiente n:

$$h(c) = \frac{c^n}{1 + c^n}, \qquad p(c) = p_{\max}\, h(c), \qquad \sigma_0(c) = \sigma_{\max}\, h(c),$$

em que $p_{\max}$ e $\sigma_{\max}$ são a falha de transmissão máxima e o *jitter* acrescentado máximo do agente (Apêndice C.1). O erro no instante de chegada δ_e de um sinal transmitido, em relação à referência de fase, é gaussiano, com média zero e desvio-padrão σ_{eff} (Seção 4.3). Uma conexão é **temporalmente coerente** se transmite e $|\delta_e| \le W/2$, com probabilidade

$$\pi(c) = \big(1 - p(c)\big) \cdot \operatorname{erf}\!\left(\frac{W}{2\sqrt{2}\,\sigma_{\text{eff}}}\right).$$

No modelo de percolação, cada ligação não dirigida do grafo tem um sorteio de falha e um sorteio de temporização. A leitura da CPD do modelo de osciladores deu inicialmente dois sorteios a cada ligação humana (um por sentido, bastando qualquer um deles), e foi assim que surgiu seu resultado humano especificado de antemão (Seção 6).

4.2 Limiares de percolação e de quórum

Tratando a coerência como percolação de ligações em um grafo aleatório com a distribuição de graus dada, um componente gigante surge pela primeira vez em (Callaway et al., 2000)

$$\pi_c = \frac{\langle k \rangle}{\langle k^2 \rangle - \langle k \rangle}.$$

O quórum exige mais do que a existência. Com as funções geradoras $G_0(x) = \sum_k P(k) x^k$ e $G_1(x) = G_0'(x)/G_0'(1)$, o componente gigante na ocupação π é $S = 1 - G_0(u)$, em que $u = 1 - \pi + \pi\, G_1(u)$ (Callaway et al., 2000), e o limiar de quórum π_q resolve $S(\pi_q) = q^*$. Nos três conectomas, π_q/π_c é 1,7 (*C. elegans*), 7,9 (*Drosophila*) e 1,5 (humano) (Apêndice B.5), de modo que, com q^* = 0,5, a perda do quórum não é o ponto crítico da percolação. Sem retroalimentação da referência, ela é o cruzamento de um nível por uma curva $S(\pi)$ suave, em que não se esperam expoentes críticos; com retroalimentação forte, o quórum é perdido na dobra da solução autoconsistente (Seção 4.3), um ponto de sela-nó perto do qual se esperaria desaceleração, mas não expoentes de percolação. Redes dominadas por *hubs* têm $\langle k^2 \rangle$ grande e π_c baixo, são robustas a falhas aleatórias e vulneráveis nos *hubs* (Callaway et al., 2000; Albert et al., 2000). Dos três conectomas, apenas o de *Drosophila* tem uma cauda fortemente pesada: o coeficiente de variação do grau não dirigido, que decorre do Apêndice B.5 como $\sqrt{(1/\pi_c + 1)/\langle k \rangle - 1}$, é de cerca de 2,1 nele, contra 0,61 no de *C. elegans* e 0,44 no conectoma humano.

4.3 Retroalimentação da referência e histerese

P5 é modelado fazendo o *jitter* efetivo crescer à medida que o componente gigante encolhe:

$$\sigma_{\text{eff}} = \sqrt{\sigma_b^2 + \sigma_0(c)^2 + \sigma_{\text{ref}}^2}, \qquad \sigma_{\text{ref}} = \lambda\,\big(1 - S\big)\big(1 - A(c)\big), \qquad A(c) = A_0\,\big(1 - a\,h(c)\big),$$

em que σ_b é o *jitter* basal independente do agente, os três erros são independentes, $\lambda \ge 0$ é a intensidade da retroalimentação, $A(c)$ a parcela da referência sustentada pelo alerta (P7), A_0 seu valor na vigília e a a fração do efeito do agente que atua sobre os circuitos de alerta. No sono natural, A_0 = 0, e os estados DOWN silenciam a cada momento uma fração d de populações inteiras (ou, numa explicação alternativa, fazem cada conexão falhar com probabilidade d). O estado estacionário satisfaz $S = F(\pi(c, S))$, sendo F a função de componente gigante do grafo. Para λ = 0, a perda é contínua e não tem histerese. Para λ grande o suficiente, a equação tem duas soluções estáveis numa faixa de concentrações, de modo que o quórum é perdido em c_{off} e só é recuperado numa concentração menor, c_{on}: **a inércia neural surge quando λ é grande o suficiente**. Como λ é escolhido para reproduzir uma razão de inércia medida, reproduzir essa razão é uma calibração, e não um teste.

4.4 Profundidade graduada

O quórum é um limiar sobre S, mas o próprio S é contínuo, em consonância com a sedação graduada. Os registros divergem quanto à abruptude da transição: a conectividade funcional medida por ressonância magnética funcional (RMf) caiu gradualmente durante a indução lenta com propofol, enquanto a conectividade subcortical-cortical se recuperou de forma abrupta (Huang et al., 2021b), e várias medidas de acoplamento persistem ou aumentam após a LOR (Liang et al., 2025; Dong et al., 2023; Lv et al., 2016; Zhang et al., 2023; Apêndice F.5). A magnitude do acoplamento pode aumentar com a hipersincronia (Seção 7.8), de

modo que não testa a CPD, que prevê uma mudança abrupta apenas no componente gigante das conexões temporalmente coerentes (previsão 1).

4.5 Modelo de osciladores de fase

Na implementação com osciladores de fase (Apêndice C), a fase θ_i da população i evolui como

$$\dot{\theta}_i = \omega_i + \frac{K}{s_i}\sum_j w_{ji}\, m_{ji}\, \sin\!\big(\theta_j - \theta_i - \alpha_{ji}\big) + \lambda_{\text{osc}}\big[R + A(c)(1-R)\big]\sin(\psi - \theta_i) + \xi_i,$$

com frequência natural ω_i, acoplamento K, intensidade total das entradas s_i, peso w_{ji}, m_{ji} = 0 para uma conexão que falhou e 1 caso contrário, erro de fase α_{ji} (atraso mais *jitter*), magnitude R e fase ψ do ritmo médio, intensidade adimensional da referência λ_{osc} e ruído ξ_i. Um nó é uma região no conectoma humano e um neurônio nos de invertebrados, nos quais o modelo testa a conectividade, e não as células (limitação 17). Redes de osciladores de fase sobre conectomas são uma descrição fenomenológica estabelecida da sincronia em larga escala (Kuramoto, 1984; Brown et al., 2004; Cabral et al., 2011; Breakspear, 2017); ficam de fora os disparos, a cinética sináptica, a geração dos ritmos e a neuromodulação.

Escala de tempo. O tempo do modelo é adimensional, com a escala fixada por K; apenas os erros temporais e os atrasos de condução, dados em milissegundos, são convertidos em fase, como $2\pi f\sigma$ na frequência de referência f = 40 Hz, dentro da banda gama (Fries, 2015). Os resultados temporais do modelo são, portanto, livres de escala: seus tempos de estabilização, de flutuação e de correlação não podem ser comparados com milissegundos, e só a razão entre um erro temporal e o ciclo tem significado físico. Um erro de fase gaussiano de desvio-padrão s reduz em média o acoplamento de uma conexão pelo fator $e^{-s^2/2}$, a média de $\cos\delta$ sobre o erro. A referência que desacelera segundo P8, a convergência numérica e as configurações das execuções estão nos Apêndices F.5, C.1 e C.4.

4.6 Regulação glial (P9, auxiliar)

Cada alça glial é um parâmetro mantido fixo durante uma varredura. A mielina danificada eleva o *jitter* basal, $\sigma_b(\mu) = \sigma_b(1 + \gamma(1-\mu))$; a alça da micróglia enfraquece o alerta à medida que o componente gigante colapsa, $A(c,S) = A(c)(1 - g(1-S))$, com g = 0 para a depleção da micróglia; silenciar os astrócitos remove sua parcela dos estados DOWN, $d \to d(1-s_a)$. Nenhuma medição fixa γ, g ou s_a; eles são testados com γ = 1, $\mu \in$ {1; 0,75; 0,5}, $g \in$ {0; 0,25; 0,5} e s_a = 0,5, e λ foi calibrado sem a alça da micróglia, de modo que os testes dizem respeito a direções, e não a magnitudes (Apêndice B.7).

4.7 Situação do modelo e de seus parâmetros

Cada parâmetro é *medido* (uma medição publicada da própria grandeza), *calibrado* (ajustado a uma medição e depois mantido fixo), *suposto* (um valor publicado de uma grandeza relacionada, aplicado onde a própria grandeza não foi medida) ou *livre* (não medido). A janela W = 3 ms e o *jitter* basal σ_b = 3 ms são **supostos**: W está dentro das janelas medidas para entradas sinápticas coincidentes (de 1 a 10 ms no córtex de barris de roedores, Gabernet et al., 2005; menos de 2 ms no hipocampo, Pouille e Scanziani, 2001; vários milissegundos nos dendritos da camada 5, Larkum et al., 1999), e σ_b é a precisão de disparo, de cerca de 3 ms, na área temporal média (MT) de macacos (Buracas et al., 1998; menos de 1 ms em neurônios isolados *in vitro*, Mainen e Sejnowski, 1995); nenhum estudo mede a variabilidade, de um ensaio para outro, da latência entre áreas corticais. Nos valores de referência, cerca de 38% das conexões são temporalmente coerentes na vigília ($\operatorname{erf}(W/(2\sqrt{2}\,\sigma_b)) \approx 0{,}38$). Os dois valores variam de 1 a 10 ms, e os resultados dependem deles: com W = 1 ms, σ_b = 10 ms ou grau médio 4, a rede na vigília não tem quórum em nenhuma semente, de modo que todas as afirmações ficam indefinidas, e não refutadas (Apêndice B.6). λ é calibrado, a velocidade de condução é medida (Lemaréchal et al., 2022), e q^*, os efeitos máximos dos agentes, o coeficiente de Hill, o nível de alerta e as constantes dos osciladores são livres. Entre os parâmetros que a análise de sensibilidade não varia (Apêndice C.1) estão os efeitos máximos do agente semelhante aos antagonistas de NMDA (40 ms de erro temporal acrescentado, 20% de falha de transmissão), a fração a e as frequências de referência. Um terceiro modelo, de populações excitatórias e inibitórias (Wilson e Cowan, 1972), testa o congestionamento (Seção 5.1).

Como o modelo de osciladores lê P2 e P3. Em sua construção original, ele não implementava nenhum dos dois: o erro temporal enfraquecia o acoplamento de forma contínua, e o quórum era lido como uma coerência global de pelo menos metade de seu valor na vigília, de modo que seus resultados anteriores (Apêndices C.3 a C.11) dizem respeito à sincronia entre osciladores. Uma leitura acrescentada depois, com seus testes fixados antes de serem executados (Apêndice C.12), conta uma conexão num ciclo amostrado quando ela transmite e seu erro temporal em relação ao ritmo compartilhado fica dentro de $W/2$, e pergunta se as conexões que contam formam um componente gigante de pelo menos q^*. Se a variabilidade temporal consegue abolir o quórum passa então a depender da regra de P2 segundo a qual um sinal atrasado é perdido. Se a temporização for avaliada no receptor, com todo erro reduzido a um único ciclo, como na leitura da comunicação por coerência, um sinal que chega com um ciclo inteiro de atraso conta como chegado a tempo, de modo que pelo menos $Wf \approx 12\%$ das conexões que transmitem contam por acaso, por maior que seja o erro, e a variabilidade temporal não abole o quórum em nenhum conectoma. A CPD conta um sinal com um ciclo inteiro de atraso ou mais como perdido, e não como chegado a tempo num ciclo posterior, porque o conteúdo que ele carrega pertence ao ciclo em que foi enviado (P2); não se oferece nenhuma evidência para essa regra, e a via das falhas de temporização depende inteiramente dela.

5. Correspondência com observações clínicas e fisiológicas

5.1 Indução: excitação paradoxal

Concentrações baixas de agentes GABAérgicos podem causar desinibição e agitação, com aumento das potências beta e gama [E] (Brown et al., 2010; Purdon et al., 2013), o que um modelo de rede atribui a uma passagem dos interneurônios da sincronia à antissincronia sob propofol em dose baixa (McCarthy et al., 2008). A CPD propôs primeiro uma contribuição complementar [H], a concentração da carga nas vias sobreviventes à medida que π cai, que prevê que a excitação aumenta com a carga sensorial e recorrente (previsão 4). Num modelo de populações excitatórias e inibitórias do conectoma humano, isso não apareceu (Seção 6). O que os modelos mostraram perto do limiar foram flutuações coletivas da coerência e, em um de três acoplamentos, um pico de sincronia logo antes do colapso, enquanto a atividade dos *hubs* cai. A explicação revista da CPD [H] é que a excitação da indução reflete um remanescente instável e hipersincronizado da rede (Seção 7.8). Ela não explica o aumento da atividade, o pico não foi robusto entre acoplamentos, e flutuações crescentes são genéricas perto de limiares; a previsão 2 a testa.

5.2 Perda da consciência

Abaixo do quórum, o componente gigante já não abrange a rede. Propõe-se que o núcleo reticular talâmico (NRT), que controla a transmissão talamocortical, amplifique essa perda [H]; modelos talamocorticais mostram como o propofol produz alfa frontal por meio de circuitos talâmicos (Ching et al., 2010). A bicoerência das ondas lentas frontais mudou em um degrau discreto na perda da consciência induzida por propofol, enquanto o acoplamento alfa posterior declinou de forma contínua (Tong et al., 2026), como prevê um limiar, embora não necessariamente um limiar de percolação. A perda não é puramente passiva: o isoflurano despolariza neurônios promotores do sono do núcleo pré-óptico ventrolateral (VLPO; Moore et al., 2012), a sedação passa pela via do sono (Nelson et al., 2002, 2003), e uma população hipotalâmica ativada por anestésicos diversos aprofunda a anestesia (Jiang-Xie et al., 2019); ainda assim, os neurônios do VLPO não são necessários (Eikermann et al., 2011). P7 formula isso de forma testável: a ação sobre os circuitos de alerta acelera a perda do quórum sem ser necessária para ela. A CPD não afirma que esse desligamento seja protetor.

5.3 Anestesia leve e processamento local isolado

Sob propofol, as respostas do lobo temporal à fala persistem, enquanto as respostas a frases ambíguas se perdem (Davis et al., 2007); após a LOR, o córtex auditivo primário continua a responder, enquanto os córtices de associação não (Krom et al., 2020); e neurônios do hipocampo de pacientes anestesiados ainda codificam características da fala e mostram plasticidade (Katlowitz et al., 2026). Segundo P3 [H], as populações fora do componente gigante, ou de um componente abaixo do quórum, ainda computam e aprendem, mas seu conteúdo não faz parte de nenhuma experiência e não está disponível para relato. Os pacientes que movem a mão sob comando na técnica do antebraço isolado (Tunstall, 1977; Sanders et al., 2012), 12 de 260 (4,6%) após a intubação em um estudo multicêntrico (Sanders et al., 2017), são responsivos e, portanto, segundo P3, mantêm acima do quórum um componente gigante que inclui populações sensoriais. A memória implícita sem recordação explícita durante a anestesia (Lubke et al., 1999; Merikle e Daneman, 1996; Deeprose et al., 2004) se encaixa na mesma leitura: o aprendizado local continua, mas um episódio só pode ser armazenado para um estado que alcançou o quórum. A memória e a responsividade são afetadas separadamente por agentes diferentes (Veselis et al., 1997), de modo que não se afirma que a perda do quórum seja a única causa da amnésia (Apêndice F.6).

5.4 Despertar e inércia neural

No modelo de percolação, o ramo de S baixo persiste até c_{on} (Seção 4.3), reproduzindo a histerese medida depois que λ é ajustado: uma calibração, e não uma explicação. O despertar também é ativo: o metilfenidato (Solt et al., 2011) e a estimulação dopaminérgica (Taylor et al., 2016) e colinérgica pré-frontal (Pal et al., 2018) restauram um comportamento semelhante ao da vigília durante a administração contínua de anestésico em roedores, e a estimulação talâmica central restaura o alerta em macacos anestesiados (Redinbaugh et al., 2020; Bastos et al., 2021), de modo liga-desliga e com assinaturas de acesso consciente (Tasserie et al., 2022). P7 explica isso [H]: elevar o alerta reduz $\sigma_{\text{ref}} = \lambda(1 - S)(1 - A)$, de modo que S pode subir acima de q^* com $p(c)$ e $\sigma_0(c)$ inalterados. No modelo de percolação, um pulso do tônus de alerta restaura o quórum em concentração fixa em todas as sementes, em todos os tamanhos de rede, uma verificação de consistência da equação, e não um teste (Apêndice C.14). Entre c_{on} e c_{off}, onde o modelo de percolação tem dois estados estáveis, um breve estímulo de despertar deveria restaurar um quórum que dura mais que ele; acima de c_{off}, o quórum deveria voltar apenas enquanto o estímulo dura (previsão 15). O modelo de fase não mostra inércia robusta, de modo que isso se apoia apenas no modelo de percolação. A CPD propõe [H] que os estados metaestáveis da recuperação (Hudson et al., 2014), em que a ativação subcortical e límbica precede a recuperação frontoparietal (Långsjö et al., 2012), sejam componentes gigantes de tamanho intermediário, e que esse componente, ao contrário da correlação média entre regiões, que permanece abaixo do nível pré-anestesia por pelo menos 1,5 hora após o despertar em camundongos (Zhang et al., 2026), volte junto com a responsividade. A retroalimentação forte também pode prender redes pequenas do modelo na inconsciência sem nenhum agente presente, uma pista notada depois dos resultados (Apêndice F.6). A micróglia é uma contribuinte candidata (P9): a perda genética de seu receptor P2Y12 acelera a recuperação (Cao et al., 2023).

5.5 Delírio do despertar [H]

O delírio do despertar, classificado separadamente do delírio pós-operatório (Evered et al., 2018) e descrito sobretudo em crianças (Vlajkovic e Sindjelic, 2007), pode refletir em parte uma recuperação assíncrona: componentes límbicos e subcorticais que recuperam a coerência antes dos frontoparietais (Långsjö et al., 2012). Tal como formulada, essa proposta não o distingue da agitação causada pela dor, e não foi testada.

5.6 Sono e sonho [H]

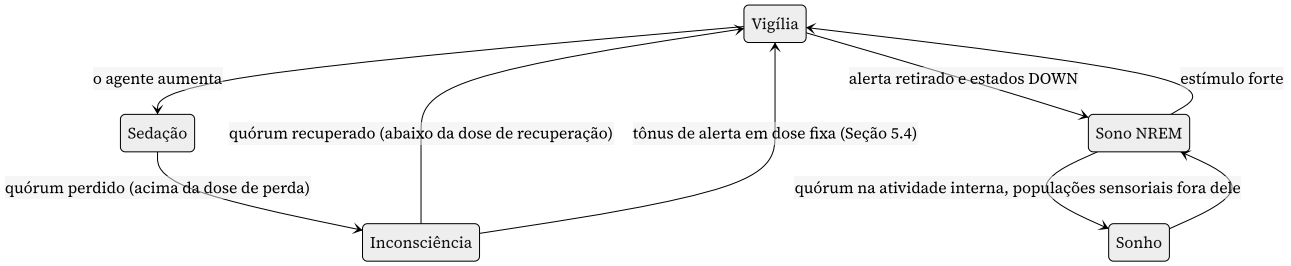


Figura B. Estados distinguidos pela hipótese. No modelo de percolação, a recuperação da anestesia ocorre numa dose menor do que a perda; acima da dose de perda, apenas o tônus de alerta restaura o quórum, e só enquanto dura (previsão 15).

O sono natural abole a consciência sem fármaco e difere neurofisiologicamente da anestesia (Akeju e Brown, 2017). Segundo P7, o sono NREM combina a retirada do alerta com estados DOWN, nos quais populações locais inteiras silenciam (Vyazovskiy et al., 2009). O modelo pergunta qual fração das populações precisa estar silenciosa ao mesmo tempo para que o quórum se perca e a compara com a fração de tempo medida em estados DOWN: 0,095 no sono NREM de ratos (Vyazovskiy et al., 2009) e uma estimativa superior de cerca de 0,25 em humanos (Cash et al., 2009; Nir et al., 2011); a derivação e a explicação alternativa, na qual os estados DOWN fazem falhar conexões isoladas, estão no Apêndice F.6, e os resultados, na Seção 6. A comparação trata uma fração de tempo como uma fração de populações silenciosas ao mesmo tempo. Por que o sono pausaria a integração, para liberar recursos para a manutenção, é uma hipótese adicional [H] sustentada apenas indiretamente (Maquet et al., 1990; Tononi e Cirelli, 2014; Apêndice F.6); ela prevê, com a mesma profundidade de ondas lentas, um retorno da responsividade mais rápido e com menos histerese a partir do sono do que a partir da anestesia (previsão 14).

O sonho é consciência desconectada: segundo P3, ele exige um componente gigante acima do quórum sobre a atividade gerada internamente, com a entrada sensorial bloqueada no tálamo (McCormick e Bal, 1994; Steriade et al., 1993b), de modo que as populações sensoriais ficam fora dele (Nir e Tononi, 2010). Relatam-se sonhos no sono com movimentos oculares rápidos (REM) e no sono NREM, associados a reduções locais da atividade de baixa frequência no córtex posterior (Siclari et al., 2017). A mesma explicação se aplica à cetamina, depois da qual voluntários relatam sonhos vívidos desligados do ambiente, enquanto a complexidade cortical permanece semelhante à da vigília (Sarasso et al., 2015): as vias sensoriais estariam desligadas de um quórum sobrevivente. Uma segunda proposta situava as falhas de temporização da cetamina nas conexões de retroalimentação, já que os receptores NMDA carregam sinais dependentes de retroalimentação no córtex visual primário de macacos (Self et al., 2012) e a cetamina reduz a conectividade do córtex frontal para o parietal, embora o propofol e o sevoflurano também o façam (U. Lee et al., 2013). No modelo, a explicação pela via sensorial se manteve em *C. elegans*, com seu controle; a explicação pela retroalimentação falhou, e nenhuma das duas explica a complexidade preservada (Seção 6).

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6. Resultados

As simulações perguntam se os modelos se comportam como a hipótese diz, e não se os cérebros o fazem. O modelo de percolação é executado em redes aleatórias de 302 a 1.000.000 de nós e nos grafos dos conectomas; o modelo de osciladores de fase, nos conectomas de *C. elegans*, *Drosophila* e humano; um modelo de populações excitatórias e inibitórias, no conectoma humano. Uma semente fixa os sorteios aleatórios (grafo, frequências, ruído, falhas); nos conectomas, a conectividade é fixa, de modo que as contagens descrevem a variabilidade própria do modelo, e não indivíduos. Cada entrada é de um de quatro tipos, usados em todas as tabelas dos apêndices: um **teste** poderia ter falhado; uma **verificação de construção** não poderia ter falhado (ou passado) nas configurações escolhidas; uma **verificação de consistência** decorre das equações ou da teoria de percolação estabelecida para os grafos usados, embora ainda possa falhar numa única rede pequena; uma **análise *post hoc*** foi definida depois dos resultados que lê. As regras de aprovação, e quando cada uma foi fixada, estão no Apêndice A.6; os identificadores dos testes estão indexados no Apêndice A.5. A tabela mantém os resultados em que o argumento se apoia; todos os demais, com suas contagens, estão no Apêndice F.7. As linhas que relatam o modelo de percolação em redes aleatórias (inércia neural, previsão 15 e sono) põem suas contagens na primeira coluna de contagens, que nessas linhas não se refere a *C. elegans*.

Afirmação	Tipo	Especificada de antemão?	Poderia falhar?	*C. elegans*	*Drosophila*	Humano	Apêndice
A falha de transmissão abole o quórum (C1; T2a; T2a-D; T2a-1)	Verificação de construção	Fixada antes da execução relatada, depois de execuções anteriores (Apêndice D); T2a-D e T2a-1, antes de suas próprias execuções	Não, com os máximos escolhidos (mas 0 de 5 sementes com dispersão de frequências de 0,25)	20/20	6/6	20/20	B.4, B.6, C.3, C.12, C.13
A falha necessária excede a inibição de 62% da liberação por disparo único medida em cultura (Baumgart et al., 2015)	Comparação *post hoc*	Não	Sim	Não dirigida, 85 a 90% (nos três); dirigida, 72%; modelo de fase, 74 a 81% (um sorteio), 75 a 87% (regra de C.12), 43 a 75% (os dois sentidos), nos três	Dirigida, 61%	Os dois sentidos, 43%	B.5, C.12, C.13, F.9
A variabilidade temporal abole o quórum, critério de coerência (T2b)	Verificação de construção	Sim	Não: o erro excede um ciclo	20/20	6/6	20/20	C.3
O mesmo, leitura da CPD, bastando um dos sentidos de uma conexão chegar a tempo (T2b-D)	Teste	Sim	Sim	20/20, a 1,21 CE50	6/6, a 1,10	**0/20, falha** em todas as condições (componente gigante de 0,645 a 3 CE50)	C.12, C.13
O mesmo, um sorteio de temporização por conexão (T2b-1); os dois sentidos (sem veredicto)	Teste	Sim, depois da falha de T2b-D e de um diagnóstico *post hoc*	Sim, embora em grande parte previsto	20/20, a 0,903; os dois sentidos, 0,708	6/6, a 1,000; os dois, 1,000	20/20, a 1,15; os dois sentidos, 0,60	C.13
O resultado humano com qualquer um dos sentidos, em função da reciprocidade e do erro temporal máximo	Estimativa estática *post hoc*, sem veredicto	Sim, depois de C.13	Não se aplica			Quórum mantido com os 40 ms supostos apenas acima de uma fração recíproca de cerca de 0,62 e, mesmo com todas as conexões nos dois sentidos, apenas abaixo de um erro temporal máximo de cerca de 49,5 ms	C.14
O mesmo, com a temporização avaliada no receptor e os erros reduzidos a um ciclo	Leitura secundária	Sim	Sim	Nunca abolido, 0/20	0/6	0/20	C.12
Os osciladores antecipam a perda em relação à percolação estática, para a falha de transmissão (T0a-D) e para a variabilidade temporal (T0b-D)	Teste	Sim	Sim	T0a-D 20/20 (1,52 contra 2,06 CE50); T0b-D 9/20, falha	6/6 (1,40 contra 1,90); 0/6, falha	20/20 (1,82 contra nenhuma perda); 0/20, falha	C.12
O limiar de quórum decorre da distribuição de graus: não dirigido, medido contra a estimativa (B.5); dirigido, estimativa contra o medido (C.14)	Verificação de consistência; estimativa dirigida sem veredicto	Não; a estimativa dirigida, sim	Sim	Não dirigido, 2,0% acima da estimativa (10 sorteios), dentro do erro amostral; estimativa dirigida de 0,91 do limiar medido	5,5% acima, além dele; 0,96	3,4% acima, dentro dele; 0,96	B.5, C.14

Afirmação	Tipo	Especificada de antemão?	Poderia falhar?	*C. elegans*	*Drosophila*	Humano	Apêndice
As flutuações excedem as de fases independentes (T1d) e tornam-se mais lentas no limiar (T1e); o pico no colapso fixado primeiro (T1) falhou	Teste	Sim (T1: fixada antes da execução relatada, depois de execuções anteriores (Apêndice D))	Sim, mas os critérios são pouco exigentes (limitação 33)	T1d 17/20; T1e 20/20; T1 7/20	6/6; 6/6; T1 0/6	16/20; 20/20; T1 5/20	C.3, C.12
Inércia neural, modelo de percolação (C2, C3, calibração)	Verificações de consistência; calibração	Fixada antes da execução relatada, depois de execuções anteriores (Apêndice D); repetida com sementes novas	Não	Razões de 1,07 a 1,63 reproduzidas nas redes de calibração; a de 2,85 do halotano não (teto de 2,04 a 2,40); $c_{\text{off}} \approx 0{,}88$ e $c_{\text{on}} \approx 0{,}83$ CE50 com 1.000.000 de nós. Com sementes novas, a razão calibrada para 1,08 é 1,071 com 10.000 nós e 1,066 com 1.000.000, dentro da faixa publicada em 9 de 20 sementes nesse tamanho			B.3, B.4, C.14
O alerta restaura o quórum em concentração fixa, e a restauração dura mais que o pulso dentro da janela, mas não acima dela (P15a, P15b; previsão 15)	Verificações de consistência	Sim	Não	20/20 em todos os tamanhos; janela de 0,057 CE50 de largura em torno de 0,854 com 1.000.000 de nós			C.14
O sono perde o quórum com a fração humana de estados DOWN (O4, O5), e não com a de rato (O3)	Teste	Fixada antes da execução relatada, depois de execuções anteriores (Apêndice D)	Sim	Fração limiar de 0,152; O3 0/20, O4, O5 20/20 em todos os tamanhos; entre as duas frações em 6 das 11 configurações fora da referência com quórum na vigília; com sementes novas, O4 19/20 com 302 nós			B.4, B.6, C.14
Falhas do tipo da cetamina nas vias sensoriais mantêm o quórum, mas interrompem a difusão global (K1; K1c; K1d; controle K1e)	Teste	K1c a K1e	Sim	K1 12/20; K1c 20/20; K1d 20/20; K1e 18/20	K1 2/6	K1 10/20; K1c a K1e não avaliáveis	C.3, C.11, C.12
Mecanismos propostos anteriormente: tempos de chegada aprendidos tornam inofensivos os atrasos constantes (L1 a L4); o congestionamento eleva a atividade dos *hubs* antes do colapso (W1 a W4); falhas nas conexões de retroalimentação abolem o quórum sob cetamina (K2); ilhas locais de coerência sobrevivem à coerência global (S1, S2); uma referência mais lenta protege mais contra falhas de temporização (F3); inércia no modelo de fase (S3; I1, I2)	Teste	Sim (S3: fixada antes da execução relatada, depois de execuções anteriores (Apêndice D))	Sim	Todos falharam: L1, L4 0/20 (L2 20/20 só porque o quórum foi perdido antes); K2 1/20; S1, S2 0/20; F3 0/20; S3 2/20, I1 0/10, I2 0/20	K2 0/6; S1, S2 0/6; F3 0/6; S3 0/6	L1, L3, L4 0/20 (L2 20/20); W1 a W4 no máximo 1 de 10 sementes por configuração, nas duas execuções; S1, S2 0/20; F3 0/20; S3 5/20, I1 0/10, I2 0/20	C.2, C.3, C.5 a C.7, C.10, C.12

Dados abertos. As previsões 1, 2, 3 e 8 foram testadas, como especificado de antemão no Apêndice E, em três conjuntos de dados públicos: RMf de 26 voluntários durante infusão de propofol em degraus (o conjunto de dados de Michigan; Huang et al., 2025, nas citações de dados), eletrocorticografia (ECoG) subdural de quatro macacos sob cetamina-medetomidina, propofol, cetamina ou medetomidina (Neurotycho; Yanagawa et al., 2013) e respostas de ECoG à estimulação elétrica por pulsos isolados em pacientes acordados e sob propofol (PRIOS; van Blooijs et al., 2024). Os testes são identificados aqui como OD1 a OD5 e, nos Apêndices E.1 a E.12, como H1 a H5a:

Teste	Dados	Confirmatório (código registrado)	Exploratório (desvios documentados)
OD1, o componente gigante colapsa num limiar (previsão 1)	RMf de Michigan	Não avaliável: o código rejeitou 19 dos 26 participantes	Pelo ponto de corte descritivo fixado em E.5, o componente gigante permaneceu intacto (25 participantes analisáveis, 24 com uma janela depois da LOR: mediana de 0,95 do valor de vigília; pelo menos 0,9 em 21 de 24), com a rede de vigília quase totalmente conectada, de modo que a medida tinha pouca margem para cair; teste de inclinação inconclusivo e instável sob nova semeadura (p não ajustado 0,10 → 0,048; sub-24 0,11 → 0,94); o resultado se reproduziu com um gerador por participante (Dev8, Apêndice E.15). No estudo que originou esse conjunto de dados, um participante seguiu comandos de forma encoberta depois da LOR (Huang et al., 2018a), o que a CPD lê como quórum mantido sem responsividade, de modo que OD1 só diz respeito à previsão 1 onde a inconsciência está estabelecida
OD2a, OD2b, desaceleração antes da LOR (previsão 2)	RMf de Michigan; ECoG do Neurotycho	OD2a não avaliável; OD2b inconclusivo (mediana de τ de 0,06, p 0,45)	Ambos inconclusivos. *Post hoc*, R2: o intervalo de 95% de OD2b entre macacos vai de −0,51 a 0,52 no conjunto confirmatório e de −0,47 a 0,39 no exploratório, um limite inferior da incerteza (Apêndice E.15)
OD3, a variabilidade do atraso acompanha a LOR entre agentes (previsão 3, forma com a atividade em curso)	ECoG do Neurotycho	Inconclusivo (intervalo de 95% de −0,23 a 0,20)	Inconclusivo (intervalo de 95% de −0,21 a 0,23). A variabilidade do atraso é aqui uma transformação monótona do valor de travamento de fase (PLV) entre janelas, de modo que isto testa o acoplamento em curso, e não a temporização evocada. Área sob a curva característica de operação do receptor (AUC) de 0,92 sob cetamina, 0,64 sob cetamina-medetomidina, 0,47 sob propofol e 0,44 sob medetomidina (a medetomidina, um agonista α2, age por supressão do alerta, P7; não está estabelecido se a CPD prevê a perda do quórum sob ela, e esta leitura foi feita depois dos resultados); para o propofol, prevê-se a abolição do quórum, e sua AUC próxima do acaso conta contra a forma com a atividade em curso para os agentes GABAérgicos. *Post hoc*, R2: entre macacos, de −0,25 a 0,22 (conjunto confirmatório) e de −0,26 a 0,24 (exploratório), um limite inferior da incerteza: dois macacos para três dos quatro agentes (Apêndice E.15)
OD4, histerese entre indução e despertar	RMf de Michigan	Não avaliável	Inconclusivo; não interpretável (Apêndice E.14)
OD5, a variabilidade da latência aumenta mais do que a latência (previsão 8, secundária)	ECoG do PRIOS	Relatado sem veredicto	Relatado sem veredicto (mediana da razão logarítmica de 1,28: a variabilidade da latência aumentou cerca de 3,6 vezes em relação à latência; no conjunto confirmatório, aumentou mais que a latência nos cinco pacientes analisáveis, com mediana do grupo de cerca de 3,8 vezes). A relação sinal-ruído (SNR) não foi menor sob propofol (mediana do grupo de 3,54, contra 3,10 em vigília), o que contraria um artefato de amplitude, mas, em simulação, com a SNR dos dados (cerca de 3) e 10 ensaios, o estimador de latência mediu um *jitter* real de 2 ms como no máximo 0,14 ms (ruído branco), e o pequeno aumento da SNR sozinho produz razões logarítmicas de 0,15 a 1,43 com 2 a 4 ms de *jitter*, incluindo os 1,28 observados. A comparação com SNR igualada foi não informativa pela regra fixada antes da reanálise, depois de conhecidos os resultados de E.14 (Apêndice E.15). É a única observação cujo sentido concorda com a previsão 8, sem veredicto

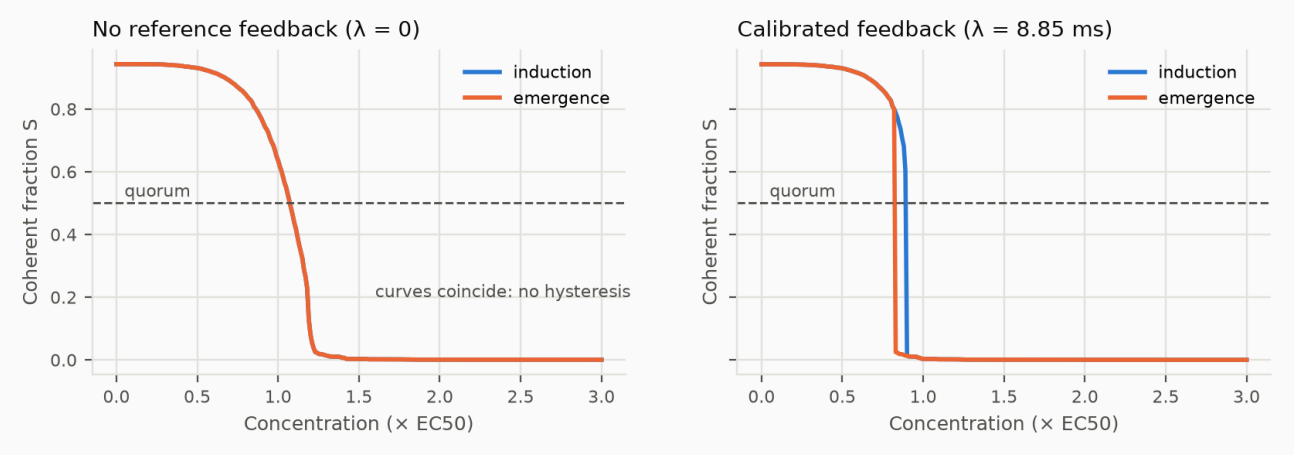


Figura 1. Fração coerente durante a indução e o despertar no modelo de percolação, sem e com retroalimentação da referência calibrada.

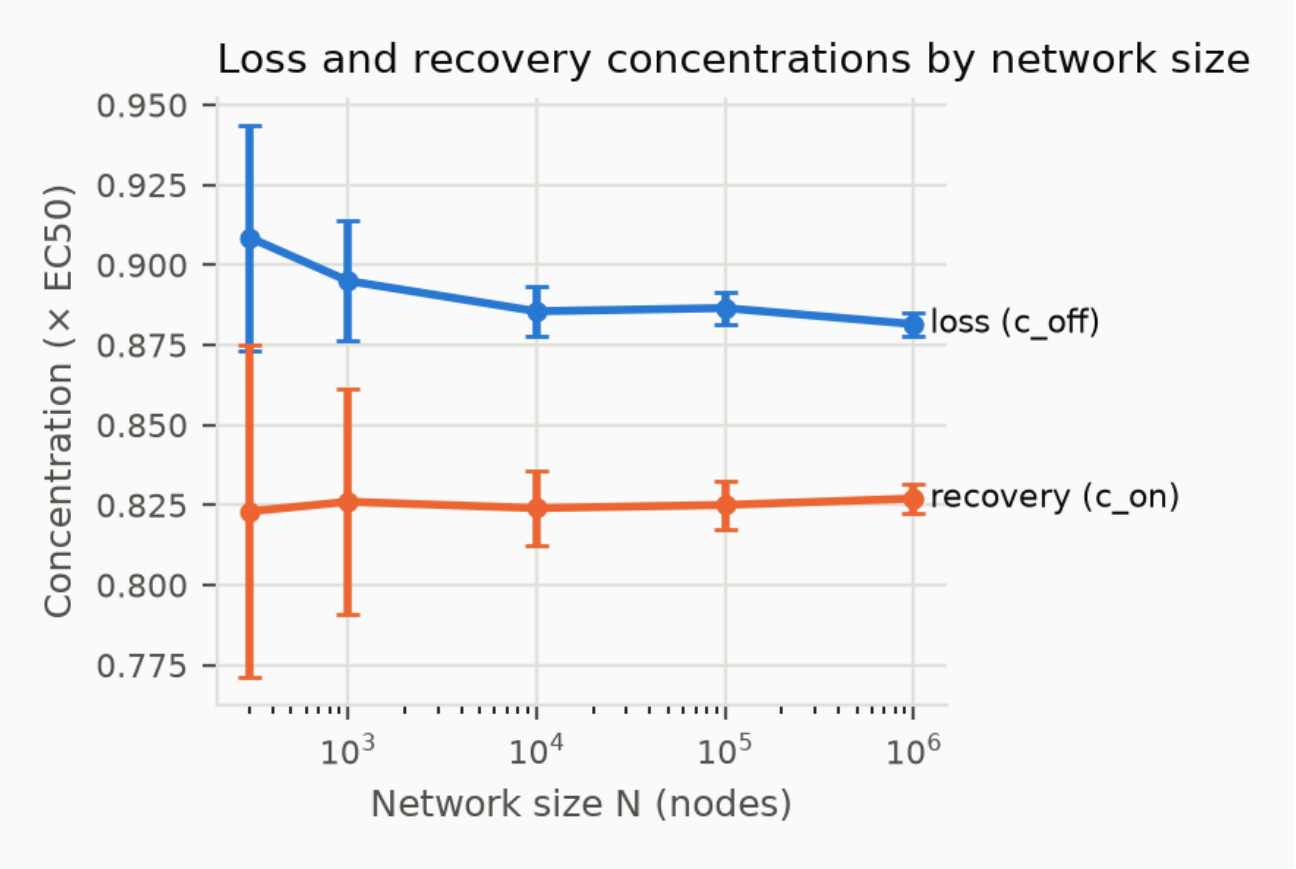


Figura 2. Concentrações de perda e de recuperação em função do tamanho da rede (média ± desvio-padrão, DP, entre sementes).

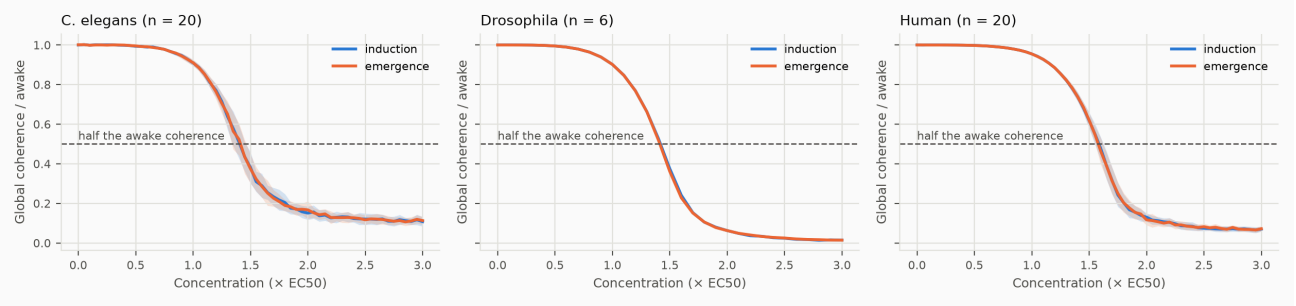


Figura 3. Coerência global em relação ao valor de vigília durante a indução e o despertar em três conectomas (média ± DP entre sementes). Linha tracejada: metade da coerência de vigília, o critério de perda dos estudos com osciladores, que não é o quórum da CPD (Seção 4.7).

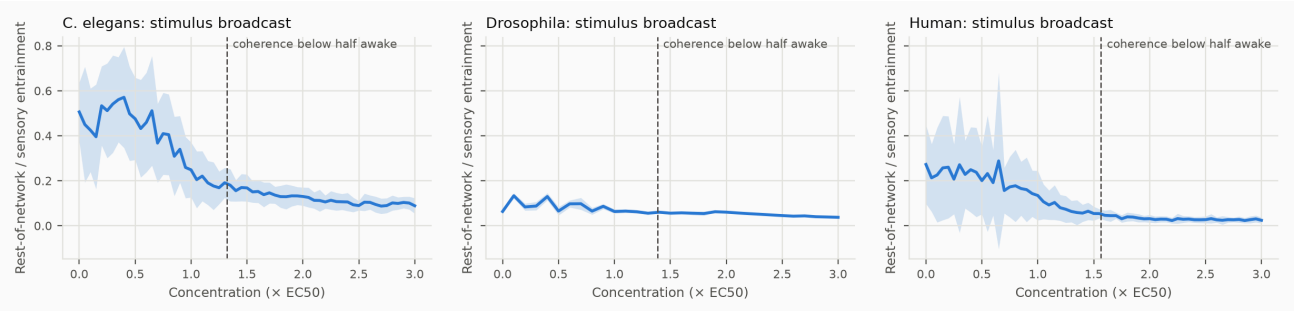


Figura 4. Propagação de um estímulo sensorial além das populações estimuladas durante a indução (média ± DP entre sementes). Linha tracejada: concentração média em que a coerência global cai abaixo da metade do seu valor de vigília.

Como ler a tabela. Três resultados têm mais peso. O resultado humano para a variabilidade temporal depende de como se conta uma conexão inferida da ressonância magnética de difusão (RM de difusão), que não registra o sentido das conexões, e, portanto, da reciprocidade não medida das conexões corticais humanas (conexões num único sentido são comuns no córtex de macaco; Markov et al., 2014), do erro temporal máximo de 40 ms, um parâmetro livre, e da regra de P2 segundo a qual um sinal atrasado é perdido; numa estimativa estática, o resultado com qualquer um dos sentidos só se mantém, com os 40 ms supostos, se mais de cerca de 62% das conexões humanas forem recíprocas e, mesmo com todas recíprocas, apenas abaixo de um erro temporal máximo de cerca de 49,5 ms (Apêndice C.14). O resultado das flutuações sobrevive a uma comparação com fases independentes,

mas com um critério pouco exigente e sobre a coerência global, e não sobre S. A inércia é uma propriedade apenas do modelo de percolação, e, em redes não usadas na calibração, sua razão fica um pouco abaixo do alvo da calibração (Apêndice C.14). A cetamina continua sendo uma lacuna dos modelos da CPD. A estimulação evoca uma onda lenta e uma resposta de baixa complexidade sob propofol e xenônio, e uma resposta complexa, semelhante à da vigília, sob cetamina (Sarasso et al., 2015); a explicação pelos períodos OFF (Seção 2.3) pode ler isso como córtex biestável sob os dois primeiros agentes e não sob a cetamina, e a CPD não tem nenhuma previsão que a separe dessa explicação. Os relatos detalhados estão no Apêndice F.7.

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7. Principais objeções e respostas

7.1 Inibição na origem

Os anestésicos potencializam sobretudo a inibição; por que haveria uma fase de aumento da atividade de alta frequência? A antissincronia dos interneurônios (McCarthy et al., 2008) explica boa parte dela e é compatível com a CPD, cuja própria explicação falhou no modelo (Seção 5.1).

7.2 Heterogeneidade molecular

Etomidato, xenônio e cetamina atuam sobre alvos diferentes, com efeitos diferentes; como um único referencial pode abarcá-los? A CPD é agnóstica quanto ao mecanismo no nível molecular (P4): propõe [H] que os agentes GABAérgicos atuem sobretudo por falha de transmissão e a cetamina sobretudo por variabilidade temporal, e ambos reduzem $\pi(c)$. A cetamina altera a temporização de intervalos dentro de uma área (Waters et al., 2026, um *preprint*; Apêndice F.8), o que não testa a temporização entre áreas. O PCI preservado e os relatos de sonhos sob cetamina indicam consciência desconectada (Sarasso et al., 2015), que os modelos explicam apenas em parte (Seção 6).

7.3 Estereosseletividade e especificidade de receptores

Os efeitos específicos de enantiômeros e a resistência de animais *knock-in* excluem a perturbação lipídica como mecanismo único (Seção 2.1); a CPD não precisa de nenhum mecanismo molecular em particular.

7.4 Consciência conectada e processamento local preservado

Se a integração colapsa, por que o processamento auditivo complexo persiste? Segundo P3, as populações fora do componente gigante podem continuar processando sua entrada; o que falha é sua inclusão no componente e, com ela, o relato. A CPD prevê que uma área sensorial estimulada continue acompanhando sua entrada enquanto a propagação para o restante da rede falha, o que corresponde a registros em que o córtex auditivo primário continua a responder após a LOR, enquanto o córtex de associação não (Krom et al., 2020); a afirmação diz respeito à difusão global, e não à integridade dos módulos (limitação 10), e a mudança mais nítida é prevista na passagem de respostas locais para respostas em toda a rede, e não ao longo de um gradiente contínuo. A CPD seria enfraquecida por conteúdo relatável enquanto o componente gigante estivesse comprovadamente abaixo do quórum, ou por responsividade enquanto as populações sensoriais estivessem comprovadamente fora dele.

7.5 Organismos sem córtex

C. elegans (302 neurônios; White et al., 1986) e *Drosophila* são anestesiados pelos mesmos agentes. Os efeitos moleculares são conservados (Zalucki et al., 2015; Kayser et al., 1999), e nos dois o efeito é uma perda de coordenação, e não de atividade (Seção 2.2). A CPD afirma apenas que, onde o comportamento coordenado depende de comunicação temporalmente coerente, os anestésicos o abolem num limiar fixado pela conectividade; ela não tira nenhuma conclusão sobre a consciência de invertebrados.

7.6 Imobilidade

A imobilidade anestésica é espinal e está fora do escopo (Seção 1.3).

7.7 Plantas e células isoladas

Os anestésicos interrompem o movimento de órgãos e bloqueiam potenciais de ação em plantas (Yokawa et al., 2018) e afetam organismos unicelulares (Kelz e Mashour, 2019). São efeitos de membrana conservados, compatíveis com P4; não dizem respeito aos postulados de rede.

7.8 Hipersincronia sob anestesia

O propofol produz um alfa frontal forte e espacialmente coordenado (Cimenser et al., 2011; Purdon et al., 2013), e a oscilação lenta é compartilhada por todo o córtex; uma explicação baseada na perda de coerência parece prever o oposto. A CPD distingue o ritmo compartilhado da temporização do conteúdo transportado em relação a ele [A: na recuperação de relógio a partir dos dados, o receptor extrai o relógio do próprio fluxo de dados, mas um relógio recuperado forte não salva símbolos isolados que chegam fora da janela de amostragem]. Quando o propofol abole a consciência, o alfa frontal torna-se hipersincrônico enquanto os componentes tardios e

propagados das respostas somatossensoriais desaparecem, mais ainda nos ensaios com alfa mais forte (Supp et al., 2011), e os disparos ficam confinados a janelas da oscilação lenta não alinhadas entre locais distantes (Lewis et al., 2012); outros registros desse tipo estão no Apêndice F.8 (Bardon et al., 2025; Bharioke et al., 2022). A medida que a CPD exige é, portanto, a temporização entre populações sincronizada ao estímulo, e não a coerência dos ritmos em curso (previsão 1); os modelos não geram os próprios ritmos (limitação 18).

7.9 Integração sem diferenciação

As ondas lentas globais do sono NREM e as crises de ausência são estados amplamente sincrônicos em que a consciência é perdida ou prejudicada, num grau que varia entre pacientes e episódios (Blumenfeld, 2005); a CPD prevê consciência sempre que $S \geq q^*$? Não: P3 formula uma condição necessária, e não suficiente. Mas a CPD não tem um requisito de diferenciação, de modo que não consegue, por si só, dizer por que esses estados são inconscientes. Se um componente gigante sincronizado ao estímulo e acima do quórum fosse encontrado durante crises de ausência, a condição se mostraria insuficiente e seria necessário um termo de diferenciação; as crises de ausência testam, portanto, se a CPD é completa, e não P3, que um quórum durante a inconsciência não consegue refutar. Tampouco testam P4c, que trata da inconsciência produzida por um anestésico: P4c é refutado por um componente gigante acima do quórum numa concentração anestésica em que a inconsciência esteja estabelecida por mais do que a ausência de resposta (previsão 1). Uma medida candidata do compartilhamento diferenciado de informação é a informação mútua simbólica ponderada (wSMI), que aumentou com o estado de consciência em 181 registros de pacientes em estado vegetativo, em estado minimamente consciente e conscientes, numa comparação transversal (King et al., 2013); a troca de informação, que ela mede, esteve entre as assinaturas de EEG mais confiáveis do estado consciente numa comparação sistemática (Sitt et al., 2014).

7.10 A CPD redescreve explicações existentes

A CPD é a comunicação por coerência somada a um modelo de percolação ao estilo de Zhou e à desconexão funcional. No nível da rede, em grande parte, sim: as previsões 1, 2 e 6 são compartilhadas com essas explicações e sustentariam a CPD apenas fracamente. O que não é compartilhado é que a variabilidade, de um ensaio para outro, da latência evocada entre áreas fixa, por meio de P3, um limiar de rede, o que a previsão 1, com a regra temporal para as arestas, testa em conjunto com as previsões 3 e 8.

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8. Previsões falseáveis

#	Previsão	Medição	Resultado que falsearia a CPD
1	A LOR coincide com um colapso abrupto do componente gigante da rede temporal evocada, mais abrupto que a mudança na potência local; a coerência do alfa em curso pode aumentar ao mesmo tempo (Seção 7.8)	Estimulação magnética transcraniana com EEG (EMT-EEG; Massimini et al., 2005; Ferrarelli et al., 2010) ou estimulação elétrica por pulsos isolados (Zelmann et al., 2023) durante rampas lentas. Para cada par de estimulação e registro, $\hat{\pi}$ é a fração de ensaios com resposta detectada cuja latência no ensaio fica dentro de $\pm W/2$ da latência mediana do par na vigília (secundariamente, da fase do ritmo de referência no alvo). Existe uma aresta quando $\hat{\pi}$ excede um ponto de corte escolhido, separadamente para cada participante, de modo que a rede de vigília fique perto de seu limiar de percolação. W, fixado de antemão em milissegundos, entra apenas por meio de $\hat{\pi}$. Uma rede exige muitos locais de estimulação, e não uma estrela em torno de um único local. A previsão 1 acrescenta a forma do colapso em função da concentração	Declínio gradual do componente gigante sem limiar na LOR, ou um componente gigante acima de q^* em concentrações nas quais a inconsciência esteja estabelecida por mais do que a ausência de resposta (nenhuma experiência relatada em despertares seriados, testes negativos do antebraço isolado e de imaginação)
2	A variância e a autocorrelação de S aumentam antes da LOR (desaceleração crítica, um sinal antecipatório genérico, ou *early-warning signal*; Scheffer et al., 2009)	$S(t)$ durante indução lenta	Nenhum aumento sob indução lenta e cuidadosamente controlada. Os dados existentes tendem a contrariá-la: a dinâmica da ECoG tornou-se mais estável à medida que a consciência se perdia sob propofol em humanos (Alonso et al., 2014) e sob cetamina-medetomidina e propofol em macacos (Solovey et al., 2015); apenas registros talamocorticais em camundongos sob cetamina-xilazina mostraram flutuações divergentes e um ciclo de histerese (Hwang et al., 2012)

#	Previsão	Medição	Resultado que falsearia a CPD
3	A *variabilidade* do atraso entre áreas, e não o atraso médio, prediz a LOR entre agentes; nos agentes GABAérgicos, ela surge por meio de P5, de modo que se prevê que as falhas de resposta aumentem antes da variabilidade da latência ou junto com ela, ao passo que, na cetamina, se prevê que a variabilidade venha primeiro	Distribuição, entre ensaios, do atraso das respostas frontoparietais evocadas	O atraso médio ou a potência local predizem a LOR tão bem quanto, ou melhor que, a variabilidade do atraso com propofol, sevoflurano e cetamina
4	A excitação da indução aumenta com a carga sensorial e recorrente	Picos de atividade na indução com e sem privação sensorial, com concentração e alerta equiparados e atividade muscular controlada	Nenhuma diferença entre cargas; a versão do modelo falhou
5	A inércia neural se correlaciona com o acoplamento talamocortical da referência (λ)	Travamento de fase talamocortical com manipulação do impulso talâmico em animais	Os limiares mudam sem mudança na histerese
6	Perturbar conexões dos *hubs* reduz a concentração para a LOR mais do que perturbar o mesmo número de conexões aleatórias	Inativação focal de regiões *hub* e não *hub*	Deslocamentos iguais
7	A cetamina reduz a coerência com estímulos externos mais do que a coerência entre atividades geradas internamente	Travamento de fase sincronizado ao estímulo contra espontâneo, sob cetamina	Perda igual ou maior da coerência interna enquanto o PCI permanece alto
8	A variabilidade, de um ensaio para outro, da latência das respostas entre áreas aumenta logo abaixo da LOR, antes que a latência média mude, enquanto a temporização de neurônios isolados permanece precisa	Registros intracranianos em concentrações no sítio efetor aumentadas em degraus, em pacientes com eletrodos implantados por razões clínicas (Krom et al., 2020), com microeletrodos para a temporização de unidades isoladas: a ECoG clínica sozinha não pode testar a parte sobre neurônios isolados	Variabilidade da latência inalterada até a LOR, ou aumentando apenas depois da latência média. A variabilidade evocada depende da profundidade (Kisley e Gerstein, 1999), e sob propofol as respostas a pulsos isolados tornam-se mais variáveis (Zelmann et al., 2023) e mais lentas (van Blooijs et al., 2024), mas não foram medidas ao longo de concentrações graduadas
9	No início do alfa induzido por propofol, o travamento de fase de longo alcance é mais preservado na banda alfa do que na banda larga, e o início do alfa segue um aumento da variabilidade temporal (P8)	Registros durante indução lenta	Nenhuma preservação, ou início sem relação com a variabilidade temporal
10	O estímulo necessário para acordar quem dorme aumenta com a fração e a extensão medidas dos estados DOWN, além do que a potência de ondas lentas prevê (P7)	EEG intracraniano ou de alta densidade com estímulos de despertar graduados nos estágios N1 a N3 do sono NREM (de leve a profundo)	Nenhuma relação depois de considerada a potência de ondas lentas; a relação com a potência de ondas lentas já é conhecida (Neckelmann e Ursin, 1993; Bastuji et al., 2021)
11	A mielina danificada reduz a concentração para a LOR (P9)	Curva dose-resposta da LORR em camundongos desmielinizados; integridade da substância branca comparada à concentração de propofol na LOR na esclerose múltipla	Nenhuma diferença, ou sensibilidade reduzida; os resultados existentes são conflitantes (Wang et al., 2024; Tot et al., 2024)
12	A depleção da micróglia estreita a inércia neural (P9, P5)	Rampas lentas em camundongos com depleção e em controles	Inércia inalterada (a depleção genética não antecipou o despertar; Cao et al., 2026)
13	Suprimir a sinalização dos astrócitos corticais reduz o limiar de despertar em proporção à queda dos estados DOWN (P9, P7), ampliando Bojarskaite et al. (2020)	Supressão do cálcio dos astrócitos com estímulos de despertar graduados	Menos estados DOWN, limiar inalterado
14	Com a mesma profundidade de ondas lentas, a responsividade volta mais rápido e com menos histerese ao acordar do sono NREM do que no despertar da anestesia (P7)	Estímulos de despertar graduados no sono N3 comparados com a redução do propofol em degraus nos mesmos voluntários	Retorno igual ou mais lento a partir do sono; a parte distintiva é a histerese menor

#	Previsão	Medição	Resultado que falsearia a CPD
15	Entre as concentrações de recuperação e de perda, um breve estímulo de despertar restaura uma responsividade que dura mais que ele; acima da concentração de perda, a responsividade volta apenas enquanto o estímulo dura (P5, P7)	Estimulação talâmica central ou colinérgica em animais, em concentrações de equilíbrio, dentro e fora da janela entre indução e despertar no mesmo animal, comparando por quanto tempo a responsividade persiste depois do estímulo	Responsividade que nunca dura mais que o estímulo dentro da janela, ou que dura mais que ele igualmente fora dela

Quais previsões discriminam. O comportamento de limiar de uma rede funcional (previsão 1) também é esperado pelo modelo de Zhou et al. (2015) e pela ignição do GNW; o aumento da variância antes de uma transição (previsão 2) é genérico (Scheffer et al., 2009); a vulnerabilidade dos *hubs* (previsão 6) decorre da percolação em qualquer rede com graus heterogêneos e só é forte nas de cauda pesada (Albert et al., 2000); C5 foi executada em grafos de ligação preferencial, ao passo que o conectoma humano é apenas levemente heterogêneo (Seção 4.2), de modo que o efeito previsto em humanos é menor. A previsão 15 também não é específica da CPD: qualquer explicação com dois estados estáveis numa faixa de concentrações a prevê, como a alternância impulsionada por ruído numa paisagem de múltiplos poços (Proekt e Hudson, 2018) ou a retroalimentação adaptativa em redes de osciladores (Kim et al., 2018); ela separa explicações biestáveis de monoestáveis. Sua janela decorre da razão de inércia: mirando o ponto médio, a concentração precisa ficar dentro de $(\rho-1)/(\rho+1)$ do alvo, em que $\rho = c_{\text{off}}/c_{\text{on}}$. No modelo de percolação calibrado para ρ = 1,08, cuja própria razão é de cerca de 1,07 com 1.000.000 de nós e de 1,12 com 302 nós, isso dá cerca de 3% e cerca de 6% (janela de 0,057 CE50 de largura em torno de 0,854 CE50 com 1.000.000 de nós; Apêndice C.14), uma reformulação da calibração, e não um resultado independente. Um experimento deve, portanto, usar um agente e uma espécie com inércia medida grande (halotano em camundongos, $\rho \approx 2{,}85$, cerca de ±48%) e fixar a concentração a partir de c_{on} e c_{off} medidos no próprio animal. A estimulação talâmica em concentração fixa já restaura o alerta durante a estimulação (Redinbaugh et al., 2020; Bastos et al., 2021), de modo liga-desliga (Tasserie et al., 2022), o que é coerente com a segunda cláusula, mas não se sabe onde essas concentrações ficavam em relação às de recuperação e de perda. Duas previsões são específicas da CPD, e apenas em parte. Que a variabilidade, de um ensaio para outro, da latência evocada entre áreas, e não a latência média, a amplitude da resposta nem a coerência dos ritmos em curso, acompanhe a perda de responsividade (previsões 3 e 8) decorre de P2 tal como a comunicação por coerência o aplicaria (Fries, 2015; Womelsdorf et al., 2007); o acréscimo da CPD é que essa variabilidade, por meio de P3, fixa um limiar de rede.

Um resultado nulo falseador (previsões 1, 2, 4, 6, 10, 11, 12 e 14) só será declarado por meio de um teste de equivalência contra um menor tamanho de efeito de interesse fixado antes da coleta dos dados (dois testes unilaterais; Lakens et al., 2018). Para a previsão 1, comportamento de limiar significa que uma sigmoide se ajusta ao componente gigante em função da concentração melhor do que uma reta, pelo critério de informação bayesiano; como aplicar um limiar a uma conectividade que declina de forma contínua também produz uma sigmoide, e a análise de percolação de redes funcionais com limiar é um método estabelecido (Gallos et al., 2012), o componente gigante é calculado em vários limiares, junto com a conectividade sem limiar, e o comportamento de limiar só é declarado se o colapso for mais abrupto do que a aplicação do limiar produz por si só. Do mesmo modo, um componente gigante intacto só será declarado por dois testes unilaterais que mostrem que a razão entre S depois da LOR e na vigília fica acima de um limite fixado de antemão. A densidade de arestas precisa ser fixada de modo que a rede de vigília fique perto de seu próprio limiar de percolação; uma rede muito acima dele tem pouca margem para cair, como no teste de RMf da Seção 6.

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9. Protocolo de análise proposto

9.1 Índice de complexidade perturbacional

O PCI (Casali et al., 2013), a complexidade de Lempel-Ziv normalizada (Lempel e Ziv, 1976) da resposta cortical à estimulação magnética transcraniana (fórmula no Apêndice F.8), é estimado em platôs estáveis, separadamente das rampas que medem S. A CPD prevê que o PCI caia junto com S de um platô para outro para os agentes GABAérgicos e permaneça relativamente preservado sob cetamina; é um dado de apoio, e não um teste específico.

9.2 Medidas específicas da CPD

1. Estimar a sincronia de fase em bandas definidas de antemão. O índice ponderado de atraso de fase (wPLI), robusto à condução de volume, descarta o acoplamento com atraso próximo de zero que P2 conta, de modo que é combinado com uma medida sensível ao atraso em dados intracranianos ou reconstruídos no espaço das fontes (mudanças dirigidas na LOR por propofol: Chen et al., 2022), e seu componente gigante é um limite inferior.
2. Construir uma rede por janela de tempo, com limiares baseados em dados substitutos (*surrogates*); calcular $S(t)$, sua variância e sua autocorrelação de defasagem 1, e a distribuição, entre ensaios, da variabilidade do atraso entre áreas.
3. Ajustar o modelo da Seção 4 a S em função da concentração no sítio efetor, separadamente para a indução e o despertar, relatando a incerteza conjunta de π_c, q^* e λ, que podem não ser identificáveis separadamente.
4. Comparar agentes e o sono natural (Akeju e Brown, 2017). A irresponsividade sob dexmedetomidina em geral é reversível por estímulos e, na maioria das vezes, acompanhada de sonhos (Radek et al., 2018), de modo que a CPD prevê [H] ali um

componente gigante no quórum ou perto dele, com a difusão global sensorial reduzida.
5. Controlar os opioides, que mudaram se as concentrações de indução e de recuperação diferiam (Kuizenga et al., 2018), e a estimulação cirúrgica.
6. Especificar de antemão as hipóteses, as bandas e os limiares antes da análise, idealmente num registro público. O Apêndice E faz isso, no repositório dos autores, para o que os registros abertos permitem testar.

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10. Limitações

Os itens mantêm a numeração usada nos apêndices; o texto completo de cada um, com seus valores, está no Apêndice F.9.

1. **Palavras emprestadas.** Consenso e quórum são usados apenas em seu sentido cotidiano (Seção 1.4); as previsões são formuladas em termos neurofisiológicos.
2. **Parâmetros.** W e σ_b são supostos, e não medidos entre áreas; λ é calibrado para uma única razão (1,08), adotada depois de vistos os resultados com um alvo anterior (1,11); as razões humanas são alvos fracos; o modelo não alcança os 2,85 do halotano; em redes não usadas na calibração, o λ calibrado dá 1,071, e não 1,08, com 10.000 nós, e 1,066 com 1.000.000, logo abaixo da faixa publicada (Apêndice C.14).
3. **Topologia simplificada.** A percolação em grafos aleatórios ignora a disposição espacial, a modularidade, o sentido das conexões e a dinâmica sináptica, e trata um feixe de fibras como uma única conexão do tipo tudo ou nada.
4. **Fatores de confusão na medição.** A sincronia de fase do EEG é afetada pela condução de volume, pela escolha da referência e pela relação sinal-ruído.
5. **Cetamina.** A explicação pela via sensorial se mantém em um conectoma; a explicação pela retroalimentação falhou; nenhuma das duas explica a complexidade preservada; o veredicto humano para as falhas de temporização depende da regra de contagem e, sob a regra de qualquer um dos sentidos, da reciprocidade não medida das conexões humanas e do erro temporal máximo (Apêndice C.14).
6. **Criticalidade.** Se o córtex opera perto da criticalidade é uma questão em aberto (Beggs e Plenz, 2003; Tagliazucchi et al., 2016; Toker et al., 2022); dois estudos situam o ponto crítico entre estados sincronizados e dessincronizados, e não entre quiescência e atividade (di Santo et al., 2018; Fontenele et al., 2019), e a inconsciência anestésica é caracterizada pelo afastamento da criticalidade de avalanches e da borda do caos no EEG em repouso (Maschke et al., 2024). A CPD exige apenas um limiar no subgrafo temporalmente coerente.
7. **Experiência.** A CPD trata das condições para a experiência e o relato, e não de por que a integração é acompanhada de experiência.
8. **O alerta como um único número.** Sem alerta, o modelo pode permanecer inconsciente em concentração zero; com ele, algumas redes pequenas ainda permanecem; o nível de alerta é livre.
9. **A inércia depende da classe do modelo.** Ela aparece no modelo de percolação e não no modelo de fase.
10. **O processamento local é mais restrito que a integridade dos módulos.** O que persistiu nos modelos foi o arrastamento da área estimulada pelo estímulo, e não a coerência espontânea dos módulos.
11. **Excitação paradoxal.** O congestionamento falhou; a explicação revista se apoia em flutuações genéricas e num pico visto em um único acoplamento.
12. **Sono natural.** Os estados DOWN são modelados de duas formas, com resultados diferentes; a fração de populações silenciosas ao mesmo tempo não foi medida; o sono REM não é modelado.
13. **Restringir cada agente.** A CPD só é falseável se $p(c)$ e $\sigma_0(c)$ forem medidos de forma independente para cada agente.
14. **Atrasos fixos.** Os atrasos constantes não são inofensivos nos modelos, e continua em aberto como o cérebro os tornaria inofensivos (Seção 3.4).
15. **Uma mosca.** Os resultados de *Drosophila* se apoiam em um único cérebro de fêmea (Dorkenwald et al., 2024), 6 sementes e uma grade mais grosseira; já existe um conectoma de macho (Berg et al., 2026).
16. **Glia como parâmetros.** As intensidades gliais são livres, as evidências vêm de camundongos, e achados centrais estão sujeitos a fatores de confusão.
17. **Temporização de mamíferos em todas as espécies.** Todos os conectomas usam uma referência de 40 Hz e janela e *jitter* de 3 ms; os resultados em invertebrados testam a conectividade, e não a integração na escala de milissegundos (Goodman et al., 1998; Liu et al., 2018).
18. **Os modelos não geram seus ritmos.** Nenhum dos dois produz alfa, oscilações lentas ou surto-supressão, e a coerência do modelo de fase mistura ritmo e conteúdo.
19. **Uma única referência.** A percepção é amostrada por vários ritmos (VanRullen, 2016).
20. **O conectoma humano não tem tálamo** (Hagmann et al., 2008).
21. **Muitos testes, alguns acrescentados depois,** sem correção para comparações múltiplas.
22. **Verificações de consistência e de construção não constituem evidência a favor.**
23. **A concentração é relativa.** Nos modelos, a CE50 é a concentração de metade do efeito máximo sobre os circuitos, e não a da LOR; o quórum é perdido entre 0,88 e 1,8 CE50, conforme a rede, de modo que não decorre nenhuma correspondência única com concentrações clínicas. As concentrações de propofol previstas no sítio efetor na LOR, no conjunto de dados de Michigan, variaram de 0,8 a 3,0 μg/ml (Apêndice E.2); os valores publicados diferem conforme o modelo farmacocinético e a ke0.
24. **Nenhuma comparação quantitativa com explicações rivais.**

25. **A perda do quórum é garantida pelos máximos escolhidos** para a falha de transmissão em todas as leituras e para a temporização na leitura por coerência.
26. **Mais falha de transmissão do que a medida.** O modelo precisa, na maioria dos casos, de mais falha do que a única medição comparável, e essa comparação não é entre grandezas equivalentes.
27. **No modelo, os ratos deveriam permanecer conscientes no sono NREM.**
28. **As sementes são realizações de um único diagrama de conexões,** e não cérebros independentes.
29. **Os resultados perto dos limites da calibração são frágeis.** A validação com 10.000 nós compartilhava sementes com a calibração; repetida com sementes disjuntas, C5 falhou em 1 de 20 sementes com 302 nós, e a razão de inércia ficou abaixo do alvo da calibração (Apêndice C.14).
30. **P6 não é testado.**
31. **A medida de difusão global tem um piso** de travamento ao acaso.
32. **O modelo de osciladores não implementava P2 e P3 tal como formulados** até a leitura do Apêndice C.12.
33. **O resultado das flutuações diz respeito à coerência global, e não a S,** e seus critérios são pouco exigentes.
34. **O sono NREM e o sonho entram em conflito no modelo,** que silencia populações ao acaso e não consegue manter um quórum posterior enquanto silencia populações em outras regiões.
35. **Outras causas de inconsciência.** Uma direção futura: a síncope, na qual a parada circulatória abrupta pode abolir a consciência em cerca de 7 s (Wieling et al., 2009), seria, segundo a CPD, uma via para falhas por omissão, com a previsão do mesmo tipo de limiar.

11. Conclusão

O que se afirma. A experiência exige que os sinais entre áreas cerebrais cheguem dentro de uma janela curta de um ritmo compartilhado, e que as conexões que o fazem liguem um componente gigante que contenha pelo menos um quórum das populações; a consciência conectada e responsiva exige, além disso, que as populações sensoriais pertençam a ele. Os anestésicos, qualquer que seja seu alvo molecular, eliminariam essas conexões por falha de transmissão, por variabilidade temporal entre áreas de um ensaio para outro, ou por ambas, e também suprimindo os sistemas de alerta que sustentam parte do ritmo. O ritmo compartilhado é sustentado pelo próprio grupo coerente. Onde um anestésico abole a experiência, e não só a responsividade, o quórum se perde; onde a experiência persiste sem responsividade, o quórum se mantém (P4c). Um sinal com um ciclo inteiro de atraso ou mais conta como perdido, uma regra afirmada, e não sustentada, e ainda não está especificado quando uma conexão entre duas populações conta. As palavras consenso e quórum carregam apenas seu sentido cotidiano. Três outras propostas (dinâmica padrão, degradação gradual e controle glial) são auxiliares.

O que os modelos mostraram. As simulações mostram que o mecanismo pode operar, e não que os cérebros o utilizam, e apenas quando a janela temporal entre áreas não é muito mais estreita que o *jitter* temporal (ambos de 3 ms), nenhum dos dois medido entre áreas. Com a leitura pela janela e pelo quórum da própria CPD, a falha de transmissão abole o quórum em três conectomas reais, o que, com as doses escolhidas, não poderia deixar de fazer, e na maioria dos casos apenas com mais falha de transmissão do que a medida. A variabilidade temporal sozinha abole o quórum nos dois conectomas de invertebrados; no conectoma humano, não o aboliu sob a regra de contagem fixada de antemão e o aboliu sob duas regras adotadas depois dessa falha, com o erro temporal máximo escolhido de 40 ms, de modo que o veredicto humano depende da reciprocidade não medida das conexões corticais humanas (numa estimativa estática, o quórum só sobrevive se mais de cerca de 62% delas forem recíprocas). E só chega a fazê-lo porque a CPD conta como perdido, e não como chegado a tempo num ciclo posterior, um sinal com um ciclo inteiro de atraso ou mais, como P2 afirma sem evidência; avaliada no receptor, com todo erro reduzido a um ciclo, nunca o faz. O limiar de quórum de cada conectoma é aproximado por sua distribuição de graus, com diferença de 2,0 a 5,5%, como se espera de redes com caminhos curtos. As flutuações no limiar excedem as de fases independentes, um critério pouco exigente. A inércia neural aparece apenas no modelo de percolação, depois de calibrado e abaixo de um teto; em redes não usadas na calibração, sua razão fica um pouco abaixo do alvo calibrado, perto da menor razão medida. Seis mecanismos propostos não sobreviveram: o congestionamento como fonte da excitação da indução, a explicação da cetamina pelas conexões de retroalimentação, as ilhas locais de coerência, o mecanismo específico da degradação gradual, a inércia no modelo de fase e os tempos de chegada aprendidos como forma de tornar inofensivos os atrasos constantes. A cetamina continua sendo uma lacuna dos modelos da CPD; a explicação pelos períodos OFF pode ler a complexidade preservada sob cetamina como ausência de biestabilidade (Sarasso et al., 2015; Seção 6), e a CPD não tem nenhuma previsão que a separe dessa explicação.

O que os dados mostraram. Os testes confirmatórios especificados de antemão em registros abertos foram inconclusivos (dois testes de eletrocorticografia em macacos) ou não puderam ser executados, porque o código registrado rejeitou os arquivos de RMf. Por um critério descritivo fixado de antemão, numa análise exploratória cuja medida de rede tinha pouca margem para cair (a rede de vigília ficava muito acima de seu limiar de percolação), a forma da previsão 1 baseada no acoplamento em curso não foi sustentada: o componente gigante da RMf manteve uma mediana de 0,95 de seu tamanho de vigília durante a perda de responsividade. No estudo que originou esse conjunto de dados, um participante seguiu comandos de forma encoberta depois da LOR (Huang et al., 2018a), o que a CPD lê como quórum mantido sem responsividade, de modo que o resultado só diz respeito à previsão 1 onde a inconsciência está estabelecida. A RMf mede o acoplamento em curso, e não a rede sincronizada ao estímulo de que trata a previsão 1, e nenhum conjunto de dados aberto permitiu testar essa rede nem a variabilidade temporal evocada das previsões 3 e 8. Nas respostas intracranianas a pulsos isolados, a variabilidade da latência aumentou mais que a latência sob propofol nos cinco pacientes analisáveis do conjunto confirmatório (mediana do grupo de cerca de 3,8 vezes), a única observação cujo sen-

tido concorda com a previsão 8, relatada sem veredicto: com a relação sinal-ruído dos dados, o estimador de latência não conseguia ver um *jitter* de 2 ms em simulação (Apêndice E.15).

O que refutaria o núcleo. Se o componente gigante da rede temporal evocada declinar gradualmente durante a perda de responsividade, ou permanecer acima do quórum onde a inconsciência estiver estabelecida por mais do que a ausência de resposta (P4c), numa indução lenta com a densidade de arestas fixada de modo que a rede de vigília fique perto de seu limiar (previsão 1), ou se a variabilidade temporal entre áreas, de um ensaio para outro, não aumentar antes da perda de responsividade e antes que a latência média mude (previsões 3 e 8), a CPD está errada, quaisquer que sejam as revisões que seus mecanismos tenham recebido. Medir essa rede com EMT-EEG ou com estimulação por pulsos isolados ao longo de uma indução lenta é o teste mais urgente.

Apêndice A. Termos, símbolos, siglas, índice de testes e regras de aprovação

A.1 Termos

Termo	Definição neste artigo
População neural	Um grupo de neurônios tratado como uma unidade: uma região cerebral no conectoma humano, um único neurônio nos conectomas de *C. elegans* e de *Drosophila*. É um nó da rede. A rede de *C. elegans* contém apenas neurônios: 300 dos 302 do diagrama completo de conexões do hermafrodita (Cook et al., 2019), porque dois (CANL e CANR) não têm sinapses na lista de arestas do OpenWorm; músculos e outras células ficam de fora.
Conexão funcional	Uma via pela qual uma população influencia outra: uma sinapse ou um conjunto de sinapses entre neurônios, ou um feixe de fibras de substância branca entre regiões. É uma aresta da rede. Um feixe de fibras não é uma única sinapse: a falha sináptica enfraquece o feixe em vez de removê-lo (Seção 10).
Falha de transmissão	Um sinal enviado por uma conexão que não produz seu efeito no alvo, por exemplo porque o neurotransmissor não é liberado. Sua probabilidade é p.
Jitter temporal	A variabilidade, de um ensaio para outro, do instante de chegada de um sinal, medida como desvio-padrão σ em milissegundos. Um atraso constante não é *jitter*.
Referência de fase	Um ritmo compartilhado entre populações, em relação ao qual os instantes de chegada são avaliados; P5 o atribui ao sistema talamocortical.
Janela de integração	O intervalo W em torno da referência de fase dentro do qual um sinal que chega pode contribuir para o ciclo de integração em curso; um sinal fora dele é perdido para esse ciclo (P2).
Ciclo de integração	Uma repetição do processo pelo qual as populações convergem para um estado compartilhado (P1), ritmada pelo ritmo de referência.
Conexão ocupada	O termo da percolação para uma conexão que permanece na rede. Aqui uma conexão está ocupada quando transmite e seu sinal chega dentro da janela de integração, o que o artigo também chama de *temporalmente coerente*. A probabilidade disso é a probabilidade de ocupação π. Esse sentido temporal de "coerente" é diferente da *coerência espectral*, que o artigo sempre nomeia por extenso.
Folga temporal	A margem que resta na janela de integração depois do erro temporal: positiva quando os sinais chegam dentro da janela, negativa quando caem fora dela.
Componente gigante	O maior conjunto de populações ligadas entre si por conexões temporalmente coerentes. Sua fração em relação a todas as populações é S.
Quórum	A fração mínima q^* de populações que o componente gigante precisa conter para que haja qualquer experiência; a consciência conectada exige, além disso, que as populações sensoriais pertençam a ele (P3). Entende-se apenas o sentido cotidiano da palavra (Seção 1.4).
Concentração no sítio efetor	A concentração de um agente em seu local de ação; nos modelos, é expressa em relação à concentração do agente que produz metade de seu efeito máximo sobre os circuitos (CE50), de modo que 1 significa metade do efeito máximo sobre os circuitos, e não a concentração na perda de responsividade.
Perda e recuperação da responsividade	Os desfechos comportamentais da indução e do despertar; nos modelos, a perda e a recuperação do quórum, nas concentrações c_{off} e c_{on}.
Inércia neural	Recuperação em concentração menor do que a da perda, $c_{\text{on}} < c_{\text{off}}$, não explicada pela distribuição do fármaco no organismo.
Despertar	Neste texto, a recuperação da consciência ao fim da anestesia (*emergence*); para o sono, usa-se *acordar*. Em *estímulo de despertar*, *limiar de despertar* e *microdespertar*, o termo se refere ao alerta (*arousal*).
Tônus de alerta	A atividade de sistemas cerebrais promotores da vigília que sustenta a referência de fase independentemente da coerência cortical (P7).
Estado DOWN (*down-state*)	Um período de quase silêncio em uma população cortical durante a oscilação lenta do sono NREM.
Difusão global (*broadcast*)	A propagação da atividade das populações que recebem uma entrada para o restante da rede.
Arrastamento (*entrainment*)	O travamento de fase de uma população a um ritmo externo, como um estímulo periódico.

Termo	Definição neste artigo
Controle em cascata	O termo da engenharia de controle para alças externas mais lentas que ajustam as condições em que uma alça interna mais rápida opera (P9).
Parâmetro de ordem	Um único número que resume o estado coletivo de um sistema; aqui, S.
Histerese	Dependência de um estado em relação ao sentido da mudança: aqui, o quórum é perdido em uma concentração maior do que aquela em que retorna.
Biestabilidade	A existência de dois estados estáveis nas mesmas condições.
Desaceleração crítica	À medida que um sistema se aproxima de uma transição, ele se recupera mais lentamente de pequenas perturbações, o que aparece como aumento da variância e da autocorrelação de suas flutuações.
Travamento de fase (*phase locking*)	Uma diferença constante, ao longo do tempo, entre as fases de dois ritmos; o atraso de fase é essa diferença.
Surto-supressão	Um padrão de EEG da anestesia profunda em que surtos de atividade se alternam com períodos de quase silêncio.
Complexo K	Uma onda lenta grande e isolada do sono NREM, que se mostrou um estado DOWN cortical isolado (Cash et al., 2009).
Limiar de despertar; microdespertar	O estímulo mais fraco que acorda quem dorme; um microdespertar é um despertar breve, de segundos.
Classes de parâmetros	*Medido*: uma medição publicada da própria grandeza, usada tal como está. *Calibrado*: ajustado a uma medição publicada e depois mantido fixo. *Suposto*: um valor publicado de uma grandeza relacionada, aplicado onde a própria grandeza não foi medida. *Livre*: sem valor medido; variado na análise de sensibilidade, quando indicado (Apêndice C.1).
Tipos de resultado	Um *teste* poderia ter falhado. Uma *verificação de construção* não poderia ter falhado (ou, no caso do A1, não poderia ter passado) nas configurações escolhidas. Uma *verificação de consistência* decorre das equações do modelo ou da teoria de percolação estabelecida para os grafos usados, e só confirma que o código as implementa; ainda pode falhar numa única rede pequena (C4 e C5 com 302 nós). Uma *análise post hoc* foi definida depois dos resultados que lê.
Semente; intervalo de confiança	Uma semente fixa os sorteios aleatórios (grafo, frequências, ruído, falhas); nos conectomas, a conectividade é fixa. As tabelas dos apêndices dão cada contagem com seu intervalo de Wilson de 95%.
Modelo nulo	Uma cópia aleatorizada dos dados que mantém o número de conexões de cada nó, usada para testar se um resultado depende da conectividade real.
Componente fortemente conexo	Em uma rede cujas conexões têm direção, um conjunto de nós em que cada um alcança todos os outros pelas conexões dirigidas.

A.2 Símbolos

Símbolo	Significado	Unidade
c	Concentração no sítio efetor em relação à CE50	adimensional
$p(c)$	Probabilidade de falha de transmissão	adimensional
$\sigma_0(c)$, σ_{eff}	*Jitter* temporal induzido pelo agente e *jitter* efetivo	ms
W	Janela de integração	ms
π, π_c, π_q	Probabilidade de ocupação; seus limiares de surgimento e de quórum	adimensional
G_0, G_1, u	Funções geradoras das distribuições de graus e de graus excedentes; probabilidade de que uma ligação não leve ao componente gigante (Seção 4.2)	adimensional
S	Fração de populações no componente gigante	adimensional
q^*	Fração de quórum	adimensional
λ	Intensidade da retroalimentação da referência no modelo de percolação	ms
λ_{osc}	Intensidade da retroalimentação da referência no modelo de osciladores de fase	adimensional
σ_b	*Jitter* temporal basal, independente do agente	ms
$h(c)$	Efeito fracionário do agente, uma função de Hill da concentração	adimensional
a	Fração do efeito do agente que atua sobre os circuitos de alerta	adimensional
$A_0, A(c)$	Parcela da referência sustentada pelo alerta, na vigília e em função da concentração	adimensional
d	Fração das populações (ou, na explicação alternativa, das conexões) silenciadas por estados DOWN a cada momento	adimensional
f	Frequência do ritmo de referência	Hz
μ	Integridade da mielina (1 = intacta)	adimensional
γ	Aumento do *jitter* basal quando a mielina é totalmente perdida	adimensional
g	Intensidade da alça da micróglia (0 após a depleção da micróglia)	adimensional
s_a	Fração dos estados DOWN que depende da sinalização dos astrócitos	adimensional
$\langle k \rangle$, $\langle k^2 \rangle$	Grau médio e média do quadrado do grau da rede	conexões

Símbolo	Significado	Unidade
χ, χ_{ind}	Suscetibilidade da coerência global e seu valor para fases independentes (Apêndice C.12)	adimensional

A.3 Siglas

Sigla	Significado
AMPA	Ácido α-amino-3-hidroxi-5-metil-4-isoxazolpropiônico, um tipo de receptor de glutamato
AUC; ROC	Área sob a curva característica de operação do receptor; característica de operação do receptor (do inglês *receiver operating characteristic*)
BIC	Critério de informação bayesiano (do inglês *Bayesian information criterion*)
BOLD	Sinal dependente do nível de oxigênio no sangue (do inglês *blood-oxygen-level-dependent*), a grandeza que a RMf registra
CEP; CONEP	Comitê de Ética em Pesquisa; Comissão Nacional de Ética em Pesquisa, que supervisiona os CEPs no Brasil
CNS	Conselho Nacional de Saúde
DANDI	*Distributed Archives for Neurophysiology Data Integration*, o arquivo que contém os conjuntos de dados 000458 e 001285
CPD	Consenso Perceptivo Distribuído
CE50	Concentração que produz metade do efeito máximo; nos modelos, metade do efeito máximo sobre os circuitos
EEG, MEG, ECoG	Eletroencefalografia, magnetoencefalografia, eletrocorticografia
erfcx	Função erro complementar escalonada, $\operatorname{erfcx}(x) = e^{x^2}\operatorname{erfc}(x)$
RMf; RM	Ressonância magnética funcional; ressonância magnética
GABA-A, NMDA	Receptor do ácido γ-aminobutírico do tipo A; receptor de N-metil-D-aspartato
GNW	Espaço de trabalho neuronal global (do inglês *global neuronal workspace*)
HCN; TREK-1, TASK; K2P	Canais ativados por hiperpolarização e regulados por nucleotídeos cíclicos; canais de potássio com dois domínios de poro (K2P)
ke0	Constante de velocidade de equilíbrio do sítio efetor de um modelo farmacocinético, por minuto
LOC	Perda da consciência (do inglês *loss of consciousness*); no Apêndice E.6, o momento marcado pelos experimentadores do Neurotycho, uma perda comportamental de responsividade
LOR, LORR; ROR	Perda de responsividade; perda do reflexo de endireitamento (Seção 1.3); retorno da responsividade
CAM	Concentração alveolar mínima: a concentração alveolar, em estado de equilíbrio e a 1 atm, em que metade dos indivíduos não se move em resposta a um estímulo nocivo supramáximo, como a incisão cirúrgica
MNI	Coordenadas cerebrais padrão do Instituto Neurológico de Montreal (*Montreal Neurological Institute*)
MT; V1	Área visual temporal média; córtex visual primário
N1 a N3	Estágios do sono NREM, do mais leve (N1) ao mais profundo (N3); apenas no Apêndice E.15, N1 é o primeiro pico negativo de uma resposta evocada
NREM, REM	Sono sem movimentos oculares rápidos e sono com movimentos oculares rápidos
NWB	*Neurodata Without Borders*, um formato de arquivo para dados de neurofisiologia
OD1 a OD5; Dev1 a Dev8	Identificadores usados fora dos Apêndices E.1 a E.12 para os testes em dados abertos H1 a H5a e os desvios D1 a D8 (Apêndice A.5)
PCI	Índice de complexidade perturbacional (Seção 9.1)
PLD2	Fosfolipase D2
PLV	Valor de travamento de fase (do inglês *phase-locking value*)
SESOI	Menor tamanho de efeito de interesse (do inglês *smallest effect size of interest*)
SNR	Relação sinal-ruído (do inglês *signal-to-noise ratio*)
TCI	Infusão alvo-controlada (do inglês *target-controlled infusion*), cujas concentrações no sítio efetor e no plasma são previsões de um modelo
EMT, EMT-EEG	Estimulação magnética transcraniana; EMT com registro de EEG
TR	Tempo de repetição de uma aquisição de RMf (o intervalo de amostragem)
NRT; VLPO	Núcleo reticular talâmico; núcleo pré-óptico ventrolateral
wPLI	Índice ponderado de atraso de fase (do inglês *weighted phase lag index*)
wSMI	Informação mútua simbólica ponderada (do inglês *weighted symbolic mutual information*)
XCP-D	Um *pipeline* de pós-processamento de RMf cujas saídas parceladas são fornecidas pelo conjunto de dados de Michigan

A.4 Analogias

Duas analogias são mantidas no texto principal, ambas marcadas com [A]: o modelo síncrono de rodadas (Seção 3.1) e a recuperação de relógio a partir dos dados (Seção 7.8). As demais que motivaram os postulados estão tabuladas, com o ponto em que cada uma falha, no Apêndice F.3. Os termos da classificação de falhas são usados como termos definidos, e não como analogias.

Termo da computação	O que significa na computação	Termo definido usado na afirmação
Rodadas síncronas	Os processos agem em rodadas sincronizadas, e uma mensagem que perde sua rodada é descartada. A sincronia parcial, na qual os atrasos são limitados mas os limites são desconhecidos ou só valem a partir de algum momento (Dwork et al., 1988), motivou a pergunta, mas não é o modelo	Condição de atraso limitado (P2)
Recuperação de relógio a partir dos dados	O receptor extrai o relógio do próprio fluxo de dados; um relógio recuperado forte não salva símbolos que chegam fora da janela de amostragem	Um ritmo compartilhado coerente, gerado pelo próprio grupo coerente (P5), enquanto a temporização do conteúdo entre populações falha (Seção 7.8)
Quórum; consenso	O número mínimo de participantes cuja concordância é necessária para que uma decisão valha; a concordância sobre um único valor	Fração mínima q^* de populações no componente gigante (P3); apenas no sentido cotidiano (Seção 1.4)
Falha por omissão; falha de temporização	Uma mensagem que nunca é entregue; uma mensagem entregue fora de sua janela de tempo combinada (Cristian, 1991)	Falha de transmissão, probabilidade p; *jitter* temporal, $\sigma_0(c)$
Folga temporal (*timing slack*)	No projeto de circuitos, a margem com que um sinal chega antes do momento em que precisa ser amostrado	Margem que resta na janela de integração depois do erro temporal (Apêndice A.1)

A.5 Índice dos identificadores de testes

Os identificadores de testes aparecem apenas na tabela da Seção 6 e nos apêndices. Os tipos são os do Apêndice A.1; "Especificado de antemão em" indica onde o teste foi fixado antes da execução que o avalia.

Identificadores	Tipo	Especificado de antemão em	Relatado em
C1	Verificação de construção	Estudo de validação, executado de novo depois de resultados anteriores (Apêndice D)	B.4, B.6
C2, C3, C4, C5, C6, C7	Verificações de consistência	Como C1	B.4, B.6
O1 a O5	Testes, relatados sem veredicto	Como C1	B.4
E1 a E3	Testes, fora do veredicto	Como C1	B.4
M1, G1 a G3	Verificações de consistência	B.7, antes da execução com várias sementes	B.7
A1	Verificação de construção (não poderia passar); especificado de novo	B.7	B.7
T1, T3, S1 a S4, H0, H1, D1, SG1, SG2, K1, F1 a F3, N1, N2	Testes (H0 a S4, N1 e N2 secundários)	Desenho do estudo entre espécies, executado de novo depois de correções no código	C.3
T2a, T2b	Verificações de construção	Como T1	C.3, B.6
T1b, T3b	Análises *post hoc*	Nenhum	C.3
W1 a W4, B1, B2	Testes	Desenho do estudo de congestionamento	C.5, C.6
X1, X2	Análises *post hoc*, seguidas de uma execução de replicação sem regra de aprovação	Nenhum	C.5, C.6
K2a a K2c, T1c	Testes	C.7	C.7
L1 a L4	Testes (L2 passou só porque o quórum foi perdido antes)	C.10	C.10
K1c	Teste	C.11	C.11
T2a-D, T2a-DN, W2a, T2a-M	Verificações de construção	C.12	C.12
T2b-D, T2b-DN, W2b, T2b-M, T0a-D, T0b-D, SP2, T1d, T1e, T1f, WNa, WNb, GNa, GNb, K1d, K1e, I1, I2, PI1, PI2, PI4	Testes	C.12	C.12
SP1	Verificação de consistência (reclassificada depois da execução)	C.12, como teste	C.12
PI3	Verificação de consistência	C.12	C.12

Identificadores	Tipo	Especificado de antemão em	Relatado em
T2a-1	Verificação de construção	C.13	C.13
T2b-1	Teste	C.13	C.13
SP1-1	Verificação de consistência (reclassificada depois da execução)	C.13, como teste	C.13
OD1 (H1), OD2a (H2a), OD2b (H2b), OD3 (H3), OD4 (H4)	Testes (confirmatórios com o código registrado; exploratórios com desvios)	E.4 a E.8	E.14
OD5 (H5a)	Análise secundária, sem veredicto	E.9	E.14
Dev1 a Dev8 (D1 a D7 do Apêndice E.13, D8 do Apêndice E.15)	Desvios do plano registrado para os dados abertos	Nenhum (D8: E.15)	E.13, E.15
P15a, P15b	Verificações de consistência	C.14	C.14
C1 a C7, O1 a O5, E1 a E3 em sementes disjuntas da calibração	Tipos como em B.4	C.14	C.14
Estimativa do limiar dirigido; estimativas de reciprocidade e de erro temporal máximo	Estimativas, sem veredicto (a segunda *post hoc* na origem)	C.14	C.14
R1 (OD5 com controle da relação sinal-ruído), R2 (*bootstrap* entre macacos de OD2b e OD3)	Análises exploratórias, sem veredicto	E.15	E.15
R3 (Michigan reanalisado com um gerador por sujeito, Dev8)	Análise exploratória	E.15	E.15

Os identificadores de dados abertos H1 a H5a e D1 a D7 coincidem com os testes H1 e D1 de C.3; os Apêndices E.1 a E.12 são texto registrado e os mantêm, e o restante do artigo usa OD1 a OD5 e Dev1 a Dev7. A pré-especificação registrada do Apêndice E.15 chama seu novo desvio de D8; nas demais partes, ele é Dev8. R1 a R3 são as reanálises de E.15; o relatório com sementes novas de C.14 também identifica como R1 sua linha de razões entre perda e recuperação, que o artigo chama de fração de sementes dentro da faixa publicada. Nos Apêndices E.1 e E.2, que são texto registrado, "Huang et al. (2021)" aparece sem letra: é Huang et al. (2021b) em E.1 e Huang et al. (2021a, 2021b) em E.2 (Apêndice E.13).

A.6 Regras de aprovação por estudo

Estudo	Regra de aprovação	Quando foi fixada
Validação, estudo glial (B.4, B.7)	Uma verificação só se mantém se valer em todas as sementes, em todos os tamanhos	Junto com as afirmações, antes das execuções relatadas, que repetem execuções anteriores (Apêndice D)
Sensibilidade (B.6), entre espécies (C.3), congestionamento (C.5)	Contagens relatadas sem regra de aprovação	Não se aplica
Convergência (C.4)	As médias variam menos que um passo da grade ou um desvio-padrão	00:56 UTC, 19 de outubro de 2026, antes da primeira execução
Execução de replicação do pico de sincronia (C.6)	Nenhuma: apenas contagens	Não se aplica
K2, T1c (C.7); L1 a L4 (C.10); K1c (C.11)	Vale em pelo menos 80% das sementes (das sementes aplicáveis, no caso de K1c)	Antes de cada execução
Leitura da CPD (C.12); regra dos pares (C.13)	Limite inferior do intervalo de Wilson de 95% acima de metade	Antes de cada execução
Verificações do pulso P15a, P15b (C.14)	Limite inferior do intervalo de Wilson de 95% acima de metade	Antes da execução, *commit* `74f9cb0`
Validação com sementes novas (C.14)	Como em B.4: vale apenas se valer em todas as sementes, em todos os tamanhos	Antes da execução, *commit* `74f9cb0`
Estimativas (C.14); reanálises de dados abertos (E.15)	Nenhuma: relatadas sem veredicto; R1 com uma regra de leitura fixada de antemão	Antes de cada execução, *commit* `74f9cb0`
Dados abertos (Apêndice E)	Testes unilaterais com correção de Holm, com equivalência para resultados nulos, mais pontos de corte descritivos	E.4 a E.9, *commit* adc63f6

A passagem da regra de 80% para a regra de Wilson foi feita para C.12, depois de vistos os resultados anteriores. Nenhum veredicto relatado neste artigo se inverte sob qualquer das duas regras: todas as contagens em C.7 e em C.10 a C.13 estão em 16 de 20 (ou 6 de 6) ou acima, ou em 13 de 20 (ou 0 de 6) ou abaixo, e a contagem humana de K1c (2 de 2 sementes aplicáveis) não tem veredicto sob nenhuma das regras, porque menos da metade das sementes era aplicável.

Declarações

- **Informações sobre os autores:** G.S.O. é engenheiro de software (nível *staff*), bacharel em Ciência da Computação pela Universidade Federal Rural do Rio de Janeiro (UFRRJ), Brasil (2015–2021). C.G.F. é estudante de graduação em Psicologia na Universidade Veiga de Almeida (UVA), Brasil, desde fevereiro de 2025. O trabalho foi realizado de forma independente, fora de qualquer instituição e sem endosso institucional; a UVA é citada apenas para identificar C.G.F. Nenhum dos autores tem formação em anestesiologia clínica ou em neurociência experimental; por isso, o artigo se apoia em trabalhos experimentais publicados para cada afirmação empírica, e a revisão crítica por especialistas dessas áreas é bem-vinda.
- **Contribuições dos autores** (CRediT): G.S.O.: conceituação, metodologia, software, análise formal, investigação, curadoria de dados, redação (rascunho original, revisão e edição), visualização. C.G.F.: conceituação (desenvolvimento da hipótese por meio de discussão), redação (revisão e edição). Ambos os autores leram e aprovaram a versão final, incluindo o Apêndice E e a lista de autores, e concordam em responder por todos os aspectos do trabalho.
- **Ética:** nenhum experimento com seres humanos ou animais foi realizado neste trabalho, e nenhum dado novo foi coletado. O Apêndice E é uma análise secundária de três conjuntos de dados públicos e anonimizados: OpenNeuro ds006623 (Huang et al., 2025; com Huang et al., 2018a, 2021a, 2021b), OpenNeuro ds004370 (van Blooijs et al., 2023; com van Blooijs et al., 2024, e Demuru et al., 2022) e os registros de anestesia e sono do Neurotycho (RIKEN Brain Science Institute; Yanagawa et al., 2013). Cada um foi coletado sob a aprovação ética de sua instituição de origem. Os arquivos foram usados tal como publicados, sem nenhuma tentativa de reidentificação e sem vinculação a outros dados, e apenas valores derivados são publicados, sob os identificadores dos próprios conjuntos de dados. A Resolução nº 510/2016 do Conselho Nacional de Saúde (CNS), art. 1º, parágrafo único, estabelece que não será registrada nem avaliada pelo sistema brasileiro de ética em pesquisa, formado pelos Comitês de Ética em Pesquisa (CEP) sob a Comissão Nacional de Ética em Pesquisa (CONEP), a "pesquisa que utilize informações de domínio público" (inciso III) (Conselho Nacional de Saúde, 2016). Esta análise secundária se enquadra no inciso III porque usa apenas conjuntos de dados que quem os disponibilizou anonimizou e liberou abertamente para reutilização (dois sob uma dedicação ao domínio público CC0 e um sob renúncia de direitos autorais), sem contato com participantes e sem coleta de dados novos. A resolução trata de pesquisas em Ciências Humanas e Sociais, e essa dispensa é a disposição mais próxima deste caso; não se buscou avaliação ética. O conectoma humano é uma rede estrutural com média de grupo derivada de dados de imagem já publicados e anonimizados (Hagmann et al., 2008).
- **Uso de ferramentas de IA:** A hipótese, seus postulados e todas as decisões deste artigo são dos próprios autores. G.S.O. concebeu o estudo, escolheu e aprovou cada análise, conjunto de dados, regra de aprovação e mudança na teoria, fixou cada pré-especificação registrando-a em um *commit* antes da respectiva execução e revisou cada resultado. G.S.O. usou duas ferramentas de IA generativa como assistentes. O Gemini na Pesquisa Google (Modo IA, Google; a versão do modelo não é informada) foi usado em 17 de outubro de 2026 para explorar a literatura. O Claude Opus 5.5 (Anthropic; identificador do modelo claude-opus-5-5), acessado pelo Claude Code, foi utilizado sob a direção de G.S.O. para redigir e editar textos, escrever e testar código (os simuladores, as análises dos dados abertos, o material explicativo interativo e as figuras) e conferir referências. Todas as simulações e análises foram executadas no computador do primeiro autor (Apple M2 Pro). A ferramenta, que redigiu as pré-especificações, também viu os valores listados no Apêndice E.1, de modo que o relato, em E.1, do que foi visto antes de os testes serem fixados abrange também a ferramenta, e não apenas os autores. Cada referência indexada foi conferida no Crossref ou no PubMed (os identificadores de conjuntos de dados, no DataCite), e cada atribuição, no texto completo da fonte quando disponível em acesso aberto e, caso contrário, em seu resumo; G.S.O. revisou essas conferências, e as referências que não puderam ser verificadas, ou cujos dados estavam incorretos, foram removidas ou corrigidas. O código foi verificado por testes automatizados, por comparação com resultados exatos quando existem (os limiares de percolação de grafos aleatórios e o limiar de percolação de ligações da rede quadrada) e por uma segunda implementação do modelo de percolação em JavaScript, cujos valores coincidem com os do modelo em Python com diferença de até 10^{-6}. Todo "parecerista", "painel" e "rodada de revisão" mencionado neste artigo, inclusive os pedidos que levaram aos Apêndices C.12 a C.14 e E.15, foi uma revisão conduzida por G.S.O. com o Claude nos papéis de pareceristas da área; não houve revisão por pares humana. Uma dessas revisões leu resultados intermediários de um estudo especificado de antemão (Apêndice C.7), e outra encontrou erros no código do modelo de osciladores de fase, corrigidos antes das execuções finais (Apêndice D). Uma auditoria com erros plantados, em 20 de outubro de 2026, conferiu cada número e veredito então presentes no texto contra os relatórios e encontrou os dez números alterados de propósito para testá-la. O texto foi conferido pela última vez em 22 de outubro de 2026: cada número contra os relatórios, por um script de rastreamento; os resultados acrescentados depois de 20 de outubro (Apêndices C.12 a C.14, E.14 e E.15), também por código de recálculo independente; cada referência no Crossref, no PubMed ou no DataCite; e a versão em português contra o texto em inglês. Os autores assumem total responsabilidade pelo conteúdo deste artigo. Nenhuma das ferramentas de IA é autora.
- **Conflitos de interesse:** nenhum declarado.
- **Financiamento:** nenhum; sem apoio em espécie.
- **Disponibilidade dos dados e do código:** o simulador (Apache License 2.0), todos os relatórios de estudo com as medições por semente e os metadados de execução (`simulator/results/`, incluindo as execuções substituídas em `simulator/results/archive/`), o código de análise dos dados abertos (`analysis/open_data/`), com suas pré-especificações (`analysis/`) e os resultados por participante e de grupo (`analysis/open_data/results/`), e o material explicativo interativo estão no repositório público https://github.com/DrGabriel/neuroscience, com o histórico completo de versões, e cada versão publicada dele é arquivada no Zenodo (https://doi.org/10.5281/zenodo.23670829, todas as versões). O material explicativo interativo está disponível em https://drgabriel.github.io/neuroscience/explainer/?lang=pt-BR. Os caminhos de arquivos citados neste artigo, como `simulator/results/` ou `NOTICE.md`, são relativos à raiz desse repositório. Cada relatório registra o

commit do Git e o hash SHA-256 de cada módulo que o produziu, e o resumo SHA-256 e o tamanho de cada arquivo de entrada do Apêndice E (os arquivos de Michigan e do PRIOS e os 28 arquivos compactados do Neurotycho) estão listados em `analysis/open_data/results/2026-10-21/input_manifest.tsv`. Os conjuntos de dados abertos não são redistribuídos; cada um é citado nas citações de dados com sua licença (ds006623 e ds004370, CC0; Neurotycho, com renúncia de direitos autorais no site do provedor; DANDI 000458 e 001285 e ds005620, CC BY 4.0) e com os artigos cuja citação seus autores solicitam. Os três conectomas também são citados ali, com suas licenças (FlyWire v783, CC BY 4.0; OpenWorm c302, MIT; conectividade de 998 regiões do The Virtual Brain, GPL-3.0); formas derivadas dos dois primeiros, e do conectoma de consenso de Shafiei et al. (2019; CC BY 4.0), usado apenas no material explicativo, são redistribuídas no material explicativo interativo, conforme `NOTICE.md`. Os dados e softwares de terceiros estão listados com suas licenças em `NOTICE.md`.

- **Pré-especificação:** nenhum estudo foi registrado em um registro público. Os testes identificados como especificados de antemão, incluindo os do Apêndice E (*commit* `adc63f6`, 22:45:02 UTC, 20 de outubro de 2026), foram registrados no repositório Git dos autores antes das execuções que os avaliam. Os horários dos *commits* são definidos pela máquina dos autores e não são certificados de forma independente; o histórico completo é público (https://github.com/DrGabriel/neuroscience), e o GitHub registra quando cada *commit* foi enviado. As análises *post hoc* estão identificadas como tais.
- **Aviso:** este artigo apresenta uma hipótese teórica e simulações computacionais. Não constitui aconselhamento médico e não deve ser usado para orientar a prática clínica.

Citações de dados

Os identificadores dos conjuntos de dados são DOIs do DataCite, resolvidos no DataCite, e não no Crossref; os registros do Neurotycho e os dois arquivos do GitHub não têm DOI, e os arquivos do GitHub são identificados por seus resumos SHA-256.

Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J. F., & Juel, B. E. (2024). A repeated awakening study exploring the capacity of complexity measures to capture dreaming during propofol sedation [Data set]. OpenNeuro, ds005620 v1.0.0. https://doi.org/10.18112/openneuro.ds005620.v1.0.0

Claar, L. D., Rembado, I., Kuyat, J. R., Russo, S., Marks, L. C., Olsen, S. R., & Koch, C. (2023b). Simultaneous electroencephalography, extracellular electrophysiology, and cortical electrical stimulation in head-fixed mice (Version 0.230317.0039) [Data set]. DANDI Archive, 000458. https://doi.org/10.48324/dandi.000458/0.230317.0039

FlyWire Consortium (2024). FlyWire Whole-brain Connectome Connectivity Data (Version 783.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.10676866. File `proofread_connections_783.feather`, SHA-256 `24f960ae3e7d4f8cd30db3b62e99fb5179cc3d1e76d8c155bfb441e9737d3faf`. Cited with Dorkenwald et al. (2024), Buhmann et al. (2021), Heinrich et al. (2018) and Eckstein et al. (2024).

Huang, Z., et al. (2025). Michigan Human Anesthesia fMRI Dataset-1 [Data set]. OpenNeuro, ds006623 v1.0.0. https://doi.org/10.18112/openneuro.ds006623.v1.0.0. The 452 files downloaded came from the dataset's unversioned path and match snapshot 1.0.0: every one is listed there with the same size, the 343 annexed files have the SHA-256 digest of their snapshot key and the other 109 their snapshot's Git object hash (checked on 22 October 2026; digests in `analysis/open_data/results/2026-10-21/input_manifest.tsv`).

OpenWorm. c302, `c302/data/herm_full_edgelist.csv` [Data set]. GitHub, https://github.com/openworm/c302 (master branch; file SHA-256 `142693f17556148d7f962835b18ac6dd5af18b7467eef61815ebc1dd5474c0ca`). Cited with Cook et al. (2019).

RIKEN Brain Science Institute. (n.d.). Neurotycho Anesthesia and Sleep task [Data set]. https://neurotycho.org/anesthesia-and-sleep-task (accessed 20 October 2026). Copyright waived on the provider's site. The 28 archives read are listed with their SHA-256 digests in `analysis/open_data/results/2026-10-21/input_manifest.tsv`. Cited with Yanagawa et al. (2013).

Shafiei, G., Markello, R., Talpalaru, T., Makowski, C., Kirschner, M., Devenyi, G. A., et al. (2019). Consensus Structural and functional connectome from 70 young healthy adults [Data set]. Zenodo. https://doi.org/10.5281/zenodo.3067849. Used only in the interactive explainer.

The Virtual Brain. tvb-data, `tvb_data/connectivity/connectivity_998.zip` [Data set]. GitHub, https://github.com/the-virtual-brain/tvb-data (master branch; file SHA-256 `31c65d4b6bf1760d06e0c4ab95c71c8dd499e48814e47bf2284e62a2b9301afd`). Cited with Hagmann et al. (2008) and Sanz Leon et al. (2013).

van Blooijs, D., Blok, S., Huiskamp, G. J. M., & Leijten, F. S. S. (2023). PRIOS [Data set]. OpenNeuro, ds004370 v1.0.2. https://doi.org/10.18112/openneuro.ds004370.v1.0.2. The subject files were listed from snapshot 1.0.2 and match it (all 96 sidecar files byte-identical, all 15 recordings of identical size); the top-level `README.txt` and `dataset_description.json` read here are the v1.0.0 copies served at the dataset's unversioned path. The 117 files downloaded are listed with their SHA-256 digests in `analysis/open_data/results/2026-10-21/input_manifest.tsv`. Cited with van Blooijs et al. (2024), as the v1.0.2 description asks, and Demuru et al. (2022), as the v1.0.0 description asks.

Yousefi, A. (2025). EEG Anesthesia Dataset (Version 0.250110.2125) [Data set]. DANDI Archive, 001285. https://doi.org/10.48324/dandi.001285/0.250110.2125

Referências

Abeysuriya, R. G., Hadida, J., Sotiropoulos, S. N., et al. (2018). A biophysical model of dynamic balancing of excitation and inhibition in fast oscillatory large-scale networks. *PLOS Computational Biology*, 14(2), e1006007. https://doi.org/10.1371/journal.pcbi.1006007

Agulhon, C., Fiacco, T. A., & McCarthy, K. D. (2010). Hippocampal short- and long-term plasticity are not modulated by astrocyte Ca2+ signaling. *Science*, 327(5970), 1250–1254. https://doi.org/10.1126/science.1184821

Akeju, O., & Brown, E. N. (2017). Neural oscillations demonstrate that general anesthesia and sedative states are neurophysiologically distinct from sleep. *Current Opinion in Neurobiology*, 44, 178–185. https://doi.org/10.1016/j.conb.2017.04.011

Akeju, O., Song, A. H., Hamilos, A. E., et al. (2016). Electroencephalogram signatures of ketamine anesthesia-induced unconsciousness. *Clinical Neurophysiology*, 127(6), 2414–2422. https://doi.org/10.1016/j.clinph.2016.03.005

Albert, R., Jeong, H., & Barabási, A.-L. (2000). Error and attack tolerance of complex networks. *Nature*, 406(6794), 378–382. https://doi.org/10.1038/35019019

Alkire, M. T., Haier, R. J., & Fallon, J. H. (2000). Toward a unified theory of narcosis: brain imaging evidence for a thalamocortical switch as the neurophysiologic basis of anesthetic-induced unconsciousness. *Consciousness and Cognition*, 9(3), 370–386. https://doi.org/10.1006/ccog.1999.0423

Alkire, M. T., Hudetz, A. G., & Tononi, G. (2008). Consciousness and anesthesia. *Science*, 322(5903), 876–880. https://doi.org/10.1126/science.1149213

Alkire, M. T., McReynolds, J. R., Hahn, E. L., & Trivedi, A. N. (2007). Thalamic microinjection of nicotine reverses sevoflurane-induced loss of righting reflex in the rat. *Anesthesiology*, 107(2), 264–272. https://doi.org/10.1097/01.anes.0000270741.33766.24

Alonso, L. M., Proekt, A., Schwartz, T. H., et al. (2014). Dynamical criticality during induction of anesthesia in human ECoG recordings. *Frontiers in Neural Circuits*, 8, 20. https://doi.org/10.3389/fncir.2014.00020

Antognini, J. F., & Schwartz, K. (1993). Exaggerated anesthetic requirements in the preferentially anesthetized brain. *Anesthesiology*, 79(6), 1244–1249. https://doi.org/10.1097/00000542-199312000-00015

Arese Lucini, F., Del Ferraro, G., Sigman, M., et al. (2019). How the Brain Transitions from Conscious to Subliminal Perception. *Neuroscience*, 411, 280–290. https://doi.org/10.1016/j.neuroscience.2019.03.047

Aru, J., Suzuki, M., & Larkum, M. E. (2020). Cellular mechanisms of conscious processing. *Trends in Cognitive Sciences*, 24(10), 814–825. https://doi.org/10.1016/j.tics.2020.07.006

Awal, M. R., Wirak, G. S., Gabel, C. V., & Connor, C. W. (2020). Collapse of global neuronal states in *Caenorhabditis elegans* under isoflurane anesthesia. *Anesthesiology*, 133(1), 133–144. https://doi.org/10.1097/ALN.0000000000003304

Barabási, A.-L., & Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439), 509–512. https://doi.org/10.1126/science.286.5439.509

Bardon, A. G., Ballesteros, J. J., Brincat, S. L., et al. (2025). Convergent effects of different anesthetics on changes in phase alignment of cortical oscillations. *Cell Reports*, 44(5), 115685. https://doi.org/10.1016/j.celrep.2025.115685

Barttfeld, P., Uhrig, L., Sitt, J. D., et al. (2015). Signature of consciousness in the dynamics of resting-state brain activity. *Proceedings of the National Academy of Sciences*, 112(3), 887–892. https://doi.org/10.1073/pnas.1418031112

Bastos, A. M., Donoghue, J. A., Brincat, S. L., et al. (2021). Neural effects of propofol-induced unconsciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824. https://doi.org/10.7554/eLife.60824

Bastuji, H., Cadic-Melchior, A., Magnin, M., et al. (2021). Intracortical Functional Connectivity Predicts Arousal to Noxious Stimuli during Sleep in Humans. *Journal of Neuroscience*, 41(23), 5115–5123. https://doi.org/10.1523/jneurosci.2935-20.2021

Baumgart, J. P., Zhou, Z.-Y., Hara, M., et al. (2015). Isoflurane inhibits synaptic vesicle exocytosis through reduced Ca2+ influx, not Ca2+-exocytosis coupling. *Proceedings of the National Academy of Sciences*, 112(38), 11959–11964. https://doi.org/10.1073/pnas.1500525112

Beech, S., Stanton Fraser, D., & Gilchrist, I. D. (2025). Visuomotor adaptation to constant and varying delays in a target acquisition task. *Journal of Vision*, 25(6), 8. https://doi.org/10.1167/jov.25.6.8

Beech, S., Stanton Fraser, D., & Gilchrist, I. D. (2026). Visuomotor adaptation and savings to constant and varying visual feedback delays in a driving simulator. *Journal of Vision*, 26(4), 7. https://doi.org/10.1167/jov.26.4.7

Beggs, J. M., & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35), 11167–11177. https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003

Ben-Or, M. (1983). Another advantage of free choice (extended abstract): Completely asynchronous agreement protocols. In *Proceedings of the Second Annual ACM Symposium on Principles of Distributed Computing* (pp. 27–30). https://doi.org/10.1145/800221.806707

Berg, S., Beckett, I. R., Costa, M., et al. (2026). Sexual dimorphism in the complete *Drosophila* male central nervous system connectome. *Cell*, 189(18), 5504–5526.e15. https://doi.org/10.1016/j.cell.2026.08.015

Bharioke, A., Munz, M., Brignall, A., et al. (2022). General anesthesia globally synchronizes activity selectively in layer 5 cortical pyramidal neurons. *Neuron*, 110(12), 2024–2040.e10. https://doi.org/10.1016/j.neuron.2022.03.032

Blumenfeld, H. (2005). Consciousness and epilepsy: why are patients with absence seizures absent? *Progress in Brain Research*, 150, 271–286. https://doi.org/10.1016/S0079-6123(05)50020-7

Boguñá, M., & Serrano, M. Á. (2005). Generalized percolation in random directed networks. *Physical Review E*, 72(1), 016106. https://doi.org/10.1103/PhysRevE.72.016106

Bojak, I., & Liley, D. T. J. (2005). Modeling the effects of anesthesia on the electroencephalogram. *Physical Review E*, 71(4), 041902. https://doi.org/10.1103/PhysRevE.71.041902

Bojarskaite, L., Bjørnstad, D. M., Pettersen, K. H., et al. (2020). Astrocytic Ca2+ signaling is reduced during sleep and is involved in the regulation of slow wave sleep. *Nature Communications*, 11(1), 3240. https://doi.org/10.1038/s41467-020-17062-2

Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature Neuroscience*, 20(3), 340–352. https://doi.org/10.1038/nn.4497

Breskin, I., Soriano, J., Moses, E., & Tlusty, T. (2006). Percolation in living neural networks. *Physical Review Letters*, 97(18), 188102. https://doi.org/10.1103/PhysRevLett.97.188102

Briere, D., & Traverse, P. (1993). AIRBUS A320/A330/A340 electrical flight controls: a family of fault-tolerant systems. In *Proceedings of the 23rd International Symposium on Fault-Tolerant Computing (FTCS-23)* (pp. 616–623). IEEE. https://doi.org/10.1109/FTCS.1993.627364

Brown, E. N., Lydic, R., & Schiff, N. D. (2010). General anesthesia, sleep, and coma. *New England Journal of Medicine*, 363(27), 2638–2650. https://doi.org/10.1056/NEJMra0808281

Brown, E., Moehlis, J., & Holmes, P. (2004). On the phase reduction and response dynamics of neural oscillator populations. *Neural Computation*, 16(4), 673–715. https://doi.org/10.1162/089976604322860668

Buhmann, J., Sheridan, A., Malin-Mayor, C., et al. (2021). Automatic detection of synaptic partners in a whole-brain Drosophila electron microscopy data set. *Nature Methods*, 18(7), 771–774. https://doi.org/10.1038/s41592-021-01183-7

Buldyrev, S. V., Parshani, R., Paul, G., et al. (2010). Catastrophic cascade of failures in interdependent networks. *Nature*, 464(7291), 1025–1028. https://doi.org/10.1038/nature08932

Buracas, G. T., Zador, A. M., DeWeese, M. R., & Albright, T. D. (1998). Efficient discrimination of temporal patterns by motion-sensitive neurons in primate visual cortex. *Neuron*, 20(5), 959–969. https://doi.org/10.1016/s0896-6273(00)80477-8

Buzsáki, G., Logothetis, N., & Singer, W. (2013). Scaling Brain Size, Keeping Timing: Evolutionary Preservation of Brain Rhythms. *Neuron*, 80(3), 751–764. https://doi.org/10.1016/j.neuron.2013.10.002

Cabral, J., Hugues, E., Sporns, O., & Deco, G. (2011). Role of local network oscillations in resting-state functional connectivity. *NeuroImage*, 57(1), 130–139. https://doi.org/10.1016/j.neuroimage.2011.04.010

Callaway, D. S., Newman, M. E. J., Strogatz, S. H., & Watts, D. J. (2000). Network robustness and fragility: percolation on random graphs. *Physical Review Letters*, 85(25), 5468–5471. https://doi.org/10.1103/PhysRevLett.85.5468

Canolty, R. T., & Knight, R. T. (2010). The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11), 506–515. https://doi.org/10.1016/j.tics.2010.09.001

Cantor, R. S. (1997). The lateral pressure profile in membranes: a physical mechanism of general anesthesia. *Biochemistry*, 36(9), 2339–2344. https://doi.org/10.1021/bi9627323

Cao, K., Cheng, W., Qiu, L., et al. (2026). More than microglial depletion: PLX5622 activates the hepatic constitutive androstane receptor to alter anesthesia and addiction. *Neuron*, 114(8), 1419–1432.e7. https://doi.org/10.1016/j.neuron.2025.12.044

Cao, K., Qiu, L., Lu, X., et al. (2023). Microglia modulate general anesthesia through P2Y12 receptor. *Current Biology*, 33(11), 2187–2200.e6. https://doi.org/10.1016/j.cub.2023.04.047

Casali, A. G., Gosseries, O., Rosanova, M., et al. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198), 198ra105. https://doi.org/10.1126/scitranslmed.3006294

Cash, S. S., Halgren, E., Dehghani, N., et al. (2009). The Human K-Complex Represents an Isolated Cortical Down-State. *Science*, 324(5930), 1084–1087. https://doi.org/10.1126/science.1169626

Celesia, G. G. (1976). Organization of auditory cortical areas in man. *Brain*, 99(3), 403–414. https://doi.org/10.1093/brain/99.3.403

Chandra, T. D., & Toueg, S. (1996). Unreliable failure detectors for reliable distributed systems. *Journal of the ACM*, 43(2), 225–267. https://doi.org/10.1145/226643.226647

Chen, X., Sirois, J. E., Lei, Q., et al. (2005). HCN subunit-specific and cAMP-modulated effects of anesthetics on neuronal pacemaker currents. *Journal of Neuroscience*, 25(24), 5803–5814. https://doi.org/10.1523/JNEUROSCI.1153-05.2005

Chen, X., Shu, S., & Bayliss, D. A. (2009). HCN1 channel subunits are a molecular substrate for hypnotic actions of ketamine. *Journal of Neuroscience*, 29(3), 600–609. https://doi.org/10.1523/JNEUROSCI.3481-08.2009

Chen, Y., Li, S., Wu, F., et al. (2022). Altered functional and directed connectivity in propofol-induced loss of consciousness: a source-space resting-state EEG study. *Clinical Neurophysiology*, 142, 209–219. https://doi.org/10.1016/j.clinph.2022.08.003

Chennu, S., O'Connor, S., Adapa, R., et al. (2016). Brain connectivity dissociates responsiveness from drug exposure during propofol-induced transitions of consciousness. *PLOS Computational Biology*, 12(1), e1004669. https://doi.org/10.1371/journal.pcbi.1004669

Ching, S., Cimenser, A., Purdon, P. L., Brown, E. N., & Kopell, N. J. (2010). Thalamocortical model for a propofol-induced α-rhythm associated with loss of consciousness. *Proceedings of the National Academy of Sciences*, 107(52), 22665–22670. https://doi.org/10.1073/pnas.1017069108

Ching, S., Purdon, P. L., Vijayan, S., et al. (2012). A neurophysiological–metabolic model for burst suppression. *Proceedings of the National Academy of Sciences*, 109(8), 3095–3100. https://doi.org/10.1073/pnas.1121461109

Cimenser, A., Purdon, P. L., Pierce, E. T., et al. (2011). Tracking brain states under general anesthesia by using global coherence analysis. *Proceedings of the National Academy of Sciences*, 108(21), 8832–8837. https://doi.org/10.1073/pnas.1017041108

Claar, L. D., Rembado, I., Kuyat, J. R., et al. (2023a). Cortico-thalamo-cortical interactions modulate electrically evoked EEG responses in mice. *eLife*, 12, RP84630. https://doi.org/10.7554/eLife.84630

Cohen, D., van Swinderen, B., & Tsuchiya, N. (2018). Isoflurane impairs low-frequency feedback but leaves high-frequency feedforward connectivity intact in the fly brain. *eNeuro*, 5(1), ENEURO.0329-17.2018. https://doi.org/10.1523/ENEURO.0329-17.2018

Cohen, D., Zalucki, O. H., van Swinderen, B., & Tsuchiya, N. (2016). Local versus global effects of isoflurane anesthesia on visual processing in the fly brain. *eNeuro*, 3(4), ENEURO.0116-16.2016. https://doi.org/10.1523/ENEURO.0116-16.2016

Conselho Nacional de Saúde (2016). Resolução nº 510, de 7 de abril de 2016 [Resolution 510 of 7 April 2016, on research in the human and social sciences]. *Diário Oficial da União*, 98 (24 May 2016), Seção 1, 44–46.

Cook, S. J., Jarrell, T. A., Brittin, C. A., et al. (2019). Whole-animal connectomes of both *Caenorhabditis elegans* sexes. *Nature*, 571, 63–71. https://doi.org/10.1038/s41586-019-1352-7

Cristian, F. (1991). Understanding fault-tolerant distributed systems. *Communications of the ACM*, 34(2), 56–78. https://doi.org/10.1145/102792.102801

Dakos, V., Carpenter, S. R., Brock, W. A., et al. (2012). Methods for detecting early warnings of critical transitions in time series illustrated using simulated ecological data. *PLoS ONE*, 7(7), e41010. https://doi.org/10.1371/journal.pone.0041010

D'Andola, M., Rebollo, B., Casali, A. G., et al. (2018). Bistability, causality, and complexity in cortical networks: an in vitro perturbational study. *Cerebral Cortex*, 28(7), 2233–2242. https://doi.org/10.1093/cercor/bhx122

Davis, M. H., Coleman, M. R., Absalom, A. R., et al. (2007). Dissociating speech perception and comprehension at reduced levels of awareness. *Proceedings of the National Academy of Sciences*, 104(41), 16032–16037. https://doi.org/10.1073/pnas.0701309104

Deeprose, C., Andrade, J., Varma, S., et al. (2004). Unconscious learning during surgery with propofol anaesthesia. *British Journal of Anaesthesia*, 92(2), 171–177. https://doi.org/10.1093/bja/aeh054

Dehaene, S., & Changeux, J.-P. (2005). Ongoing spontaneous activity controls access to consciousness: a neuronal model for inattentional blindness. *PLoS Biology*, 3(5), e141. https://doi.org/10.1371/journal.pbio.0030141

Dehaene, S., & Changeux, J.-P. (2011). Experimental and theoretical approaches to conscious processing. *Neuron*, 70(2), 200–227. https://doi.org/10.1016/j.neuron.2011.03.018

Dehaene, S., Sergent, C., & Changeux, J.-P. (2003). A neuronal network model linking subjective reports and objective physiological data during conscious perception. *Proceedings of the National Academy of Sciences*, 100(14), 8520–8525.

https://doi.org/10.1073/pnas.1332574100

Demuru, M., van Blooijs, D., Zweiphenning, W., et al. (2022). A practical workflow for organizing clinical intraoperative and long-term iEEG data in BIDS. *Neuroinformatics*, 20(3), 727–736. https://doi.org/10.1007/s12021-022-09567-6

di Santo, S., Villegas, P., Burioni, R., & Muñoz, M. A. (2018). Landau–Ginzburg theory of cortex dynamics: scale-free avalanches emerge at the edge of synchronization. *Proceedings of the National Academy of Sciences*, 115(7), E1356–E1365. https://doi.org/10.1073/pnas.1712989115

Dickinson, R., Peterson, B. K., Banks, P., et al. (2007). Competitive inhibition at the glycine site of the *N*-methyl-D-aspartate receptor by the anesthetics xenon and isoflurane: evidence from molecular modeling and electrophysiology. *Anesthesiology*, 107(5), 756–767. https://doi.org/10.1097/01.anes.0000287061.77674.71

Diekelmann, S., & Born, J. (2010). The memory function of sleep. *Nature Reviews Neuroscience*, 11(2), 114–126. https://doi.org/10.1038/nrn2762

Dong, K., Wei, Q., Zhang, D., et al. (2023). Recovery of brain network integration and segregation during the loss and recovery of consciousness induced by sevoflurane. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 31, 304–315. https://doi.org/10.1109/TNSRE.2022.3221965

Dorkenwald, S., Matsliah, A., Sterling, A. R., et al. (2024). Neuronal wiring diagram of an adult brain. *Nature*, 634, 124–138. https://doi.org/10.1038/s41586-024-07558-y

Dorogovtsev, S. N., Mendes, J. F. F., & Samukhin, A. N. (2001). Giant strongly connected component of directed networks. *Physical Review E*, 64(2), 025101. https://doi.org/10.1103/PhysRevE.64.025101

Du, W., & Huang, H. (2025). Response function as a quantitative measure of consciousness in brain dynamics. *Physical Review Research*, 7(4), 043249. https://doi.org/10.1103/6lx7-qsv7

Dwork, C., Lynch, N., & Stockmeyer, L. (1988). Consensus in the presence of partial synchrony. *Journal of the ACM*, 35(2), 288–323. https://doi.org/10.1145/42282.42283

Eckstein, N., Bates, A. S., Champion, A., et al. (2024). Neurotransmitter classification from electron microscopy images at synaptic sites in *Drosophila melanogaster*. *Cell*, 187(10), 2574–2594.e23. https://doi.org/10.1016/j.cell.2024.03.016

Edelman, G. M., & Gally, J. A. (2013). Reentry: a key mechanism for integration of brain function. *Frontiers in Integrative Neuroscience*, 7, 63. https://doi.org/10.3389/fnint.2013.00063

Eger, E. I., Saidman, L. J., & Brandstater, B. (1965). Minimum alveolar anesthetic concentration: a standard of anesthetic potency. *Anesthesiology*, 26(6), 756–763. https://doi.org/10.1097/00000542-196511000-00010

Eikermann, M., Vetrivelan, R., Grosse-Sundrup, M., et al. (2011). The ventrolateral preoptic nucleus is not required for isoflurane general anesthesia. *Brain Research*, 1426, 30–37. https://doi.org/10.1016/j.brainres.2011.10.018

Emerson, D. J., Weiser, B. P., Psonis, J., et al. (2013). Direct Modulation of Microtubule Stability Contributes to Anthracene General Anesthesia. *Journal of the American Chemical Society*, 135(14), 5389–5398. https://doi.org/10.1021/ja311171u

Erdős, P., & Rényi, A. (1959). On random graphs I. *Publicationes Mathematicae Debrecen*, 6, 290–297. https://doi.org/10.5486/PMD.1959.6.3-4.12

Evered, L., Silbert, B., Knopman, D. S., et al. (2018). Recommendations for the nomenclature of cognitive change associated with anaesthesia and surgery, 2018. *British Journal of Anaesthesia*, 121(5), 1005–1012. https://doi.org/10.1016/j.bja.2017.11.087

Fan, L., Wen, X., Lu, F., et al. (2026). Local and interregional thalamocortical dynamics show temporally dissociable changes during propofol induced loss of consciousness. *NeuroImage*, 340, 122235. https://doi.org/10.1016/j.neuroimage.2026.122235

Ferrarelli, F., Massimini, M., Sarasso, S., et al. (2010). Breakdown in cortical effective connectivity during midazolam-induced loss of consciousness. *Proceedings of the National Academy of Sciences*, 107(6), 2681–2686. https://doi.org/10.1073/pnas.0913008107

Filatrella, G., Nielsen, A. H., & Pedersen, N. F. (2008). Analysis of a power grid using a Kuramoto-like model. *European Physical Journal B*, 61(4), 485–491. https://doi.org/10.1140/epjb/e2008-00098-8

Fischer, M. J., Lynch, N. A., & Paterson, M. S. (1985). Impossibility of distributed consensus with one faulty process. *Journal of the ACM*, 32(2), 374–382. https://doi.org/10.1145/3149.214121

FitzHugh, R. (1961). Impulses and physiological states in theoretical models of nerve membrane. *Biophysical Journal*, 1(6), 445–466. https://doi.org/10.1016/S0006-3495(61)86902-6

Fontenele, A. J., de Vasconcelos, N. A. P., Feliciano, T., et al. (2019). Criticality between cortical states. *Physical Review Letters*, 122(20), 208101. https://doi.org/10.1103/PhysRevLett.122.208101

Ford, M. C., Alexandrova, O., Cossell, L., et al. (2015). Tuning of Ranvier node and internode properties in myelinated axons to adjust action potential timing. *Nature Communications*, 6(1), 8073. https://doi.org/10.1038/ncomms9073

Foxe, J. J., & Simpson, G. V. (2002). Flow of activation from V1 to frontal cortex in humans: a framework for defining "early" visual processing. *Experimental Brain Research*, 142(1), 139–150. https://doi.org/10.1007/s00221-001-0906-7

Franks, N. P. (2008). General anaesthesia: from molecular targets to neuronal pathways of sleep and arousal. *Nature Reviews Neuroscience*, 9(5), 370–386. https://doi.org/10.1038/nrn2372

Franks, N. P., & Lieb, W. R. (1984). Do general anaesthetics act by competitive binding to specific receptors? *Nature*, 310(5978), 599–601. https://doi.org/10.1038/310599a0

Franks, N. P., & Lieb, W. R. (1991). Stereospecific effects of inhalational general anesthetic optical isomers on nerve ion channels. *Science*, 254(5030), 427–430. https://doi.org/10.1126/science.1925602

Franks, N. P., Dickinson, R., de Sousa, S. L. M., Hall, A. C., & Lieb, W. R. (1998). How does xenon produce anaesthesia? *Nature*, 396(6709), 324. https://doi.org/10.1038/24525

Friedman, E. B., Sun, Y., Moore, J. T., et al. (2010). A conserved behavioral state barrier impedes transitions between anesthetic-induced unconsciousness and wakefulness: evidence for neural inertia. *PLoS ONE*, 5(7), e11903. https://doi.org/10.1371/journal.pone.0011903

Fries, P. (2005). A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends in Cognitive Sciences*, 9(10), 474–480. https://doi.org/10.1016/j.tics.2005.08.011

Fries, P. (2015). Rhythms for cognition: communication through coherence. *Neuron*, 88(1), 220–235. https://doi.org/10.1016/j.neuron.2015.09.034

Fujisaki, W., Shimojo, S., Kashino, M., et al. (2004). Recalibration of audiovisual simultaneity. *Nature Neuroscience*, 7(7), 773–778. https://doi.org/10.1038/nn1268

Fujita, T., Chen, M. J., Li, B., et al. (2014). Neuronal Transgene Expression in Dominant-Negative SNARE Mice. *Journal of Neuroscience*, 34(50), 16594–16604. https://doi.org/10.1523/jneurosci.2585-14.2014

Gabernet, L., Jadhav, S. P., Feldman, D. E., Carandini, M., & Scanziani, M. (2005). Somatosensory integration controlled by dynamic thalamocortical feed-forward inhibition. *Neuron*, 48(2), 315–327. https://doi.org/10.1016/j.neuron.2005.09.022

Gallos, L. K., Makse, H. A., & Sigman, M. (2012). A small world of weak ties provides optimal global integration of self-similar modules in functional brain networks. *Proceedings of the National Academy of Sciences*, 109(8), 2825–2830. https://doi.org/10.1073/pnas.1106612109

Golomb, D., & Rinzel, J. (1993). Dynamics of globally coupled inhibitory neurons with heterogeneity. *Physical Review E*, 48(6), 4810–4814. https://doi.org/10.1103/physreve.48.4810

Gómez-Gardeñes, J., Moreno, Y., & Arenas, A. (2007). Paths to synchronization on complex networks. *Physical Review Letters*, 98(3), 034101. https://doi.org/10.1103/PhysRevLett.98.034101

Goodman, M. B., Hall, D. H., Avery, L., & Lockery, S. R. (1998). Active currents regulate sensitivity and dynamic range in *C. elegans* neurons. *Neuron*, 20(4), 763–772. https://doi.org/10.1016/s0896-6273(00)81014-4

Gruss, M., Bushell, T. J., Bright, D. P., et al. (2004). Two-Pore-Domain K+ Channels Are a Novel Target for the Anesthetic Gases Xenon, Nitrous Oxide, and Cyclopropane. *Molecular Pharmacology*, 65(2), 443–452. https://doi.org/10.1124/mol.65.2.443

Hagmann, P., Cammoun, L., Gigandet, X., et al. (2008). Mapping the structural core of human cerebral cortex. *PLoS Biology*, 6(7), e159. https://doi.org/10.1371/journal.pbio.0060159

Haruwaka, K., Ying, Y., Liang, Y., et al. (2024). Microglia enhance post-anesthesia neuronal activity by shielding inhibitory synapses. *Nature Neuroscience*, 27(3), 449–461. https://doi.org/10.1038/s41593-023-01537-8

He, Y., Liu, T., He, Q., et al. (2023). Microglia facilitate and stabilize the response to general anesthesia via modulating the neuronal network in a brain region-specific manner. *eLife*, 12, RP92252. https://doi.org/10.7554/elife.92252

Heinrich, L., Funke, J., Pape, C., Nunez-Iglesias, J., & Saalfeld, S. (2018). Synaptic cleft segmentation in non-isotropic volume electron microscopy of the complete *Drosophila* brain. In *Medical Image Computing and Computer Assisted Intervention (MICCAI 2018)*, Lecture Notes in Computer Science, vol. 11071, 317–325. https://doi.org/10.1007/978-3-030-00934-2_36

Hemmings, H. C., Riegelhaupt, P. M., Kelz, M. B., et al. (2019). Towards a comprehensive understanding of anesthetic mechanisms of action: a decade of discovery. *Trends in Pharmacological Sciences*, 40(7), 464–481. https://doi.org/10.1016/j.tips.2019.05.001

Herold, K. F., & Hemmings Jr., H. C. (2012). Sodium Channels as Targets for Volatile Anesthetics. *Frontiers in Pharmacology*, 3, 50. https://doi.org/10.3389/fphar.2012.00050

Heurteaux, C., Guy, N., Laigle, C., et al. (2004). TREK-1, a K+ channel involved in neuroprotection and general anesthesia. *The EMBO Journal*, 23(13), 2684–2695. https://doi.org/10.1038/sj.emboj.7600234

Hill, A. V. (1910). The possible effects of the aggregation of the molecules of haemoglobin on its dissociation curves. In Proceedings of the Physiological Society, January 22, 1910. *Journal of Physiology*, 40(Suppl.), iv–vii. https://doi.org/10.1113/jphysiol.1910.sp001386

Homayoun, H., & Moghaddam, B. (2007). NMDA receptor hypofunction produces opposite effects on prefrontal cortex interneurons and pyramidal neurons. *Journal of Neuroscience*, 27(43), 11496–11500. https://doi.org/10.1523/JNEUROSCI.2213-07.2007

Huang, Y., Qiu, Z., Yu, X., et al. (2026). Brain-penetrant microtubule-stabilizer epothilone B delays isoflurane-induced unconsciousness in mice. *Neuropharmacology*, 287, 110834. https://doi.org/10.1016/j.neuropharm.2026.110834

Huang, Z., Tarnal, V., Vlisides, P. E., et al. (2021a). Anterior insula regulates brain network transitions that gate conscious access. *Cell Reports*, 35(5), 109081. https://doi.org/10.1016/j.celrep.2021.109081

Huang, Z., Tarnal, V., Vlisides, P. E., et al. (2021b). Asymmetric neural dynamics characterize loss and recovery of consciousness. *NeuroImage*, 236, 118042. https://doi.org/10.1016/j.neuroimage.2021.118042

Huang, Z., Vlisides, P. E., Tarnal, V. C., et al. (2018a). Brain imaging reveals covert consciousness during behavioral unresponsiveness induced by propofol. *Scientific Reports*, 8, 13195. https://doi.org/10.1038/s41598-018-31436-z

Huang, Z., Zhang, J., Wu, J., et al. (2018b). Disrupted neural variability during propofol-induced sedation and unconsciousness. *Human Brain Mapping*, 39(11), 4533–4544. https://doi.org/10.1002/hbm.24304

Hudetz, A. G., & Mashour, G. A. (2016). Disconnecting consciousness: is there a common anesthetic end point? *Anesthesia & Analgesia*, 123(5), 1228–1240. https://doi.org/10.1213/ANE.0000000000001353

Hudson, A. E., Calderon, D. P., Pfaff, D. W., & Proekt, A. (2014). Recovery of consciousness is mediated by a network of discrete metastable activity states. *Proceedings of the National Academy of Sciences*, 111(25), 9283–9288. https://doi.org/10.1073/pnas.1408296111

Hwang, E., Kim, S., Han, K., et al. (2012). Characterization of Phase Transition in the Thalamocortical System during Anesthesia-Induced Loss of Consciousness. *PLoS ONE*, 7(12), e50580. https://doi.org/10.1371/journal.pone.0050580

IEEE. (2020). *IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems* (IEEE Std 1588-2019). https://doi.org/10.1109/IEEESTD.2020.9120376

Imas, O. A., Ropella, K. M., Ward, B. D., et al. (2005). Volatile anesthetics disrupt frontal-posterior recurrent information transfer at gamma frequencies in rat. *Neuroscience Letters*, 387(3), 145–150. https://doi.org/10.1016/j.neulet.2005.06.018

Ingiosi, A. M., Hayworth, C. R., Harvey, D. O., et al. (2020). A Role for Astroglial Calcium in Mammalian Sleep and Sleep Regulation. *Current Biology*, 30(22), 4373–4383.e7. https://doi.org/10.1016/j.cub.2020.08.052

ISO. (2024). *Road vehicles – Controller area network (CAN) – Part 1: Data link layer and physical coding sublayer* (ISO 11898-1:2024). International Organization for Standardization.

Iwakiri, H., Nishihara, N., Nagata, O., Matsukawa, T., Ozaki, M., & Sessler, D. I. (2005). Individual effect-site concentrations of propofol are similar at loss of consciousness and at awakening. *Anesthesia & Analgesia*, 100(1), 107–110. https://doi.org/10.1213/01.ANE.0000139358.15909.EA

Jevtović-Todorović, V., Todorović, S. M., Mennerick, S., et al. (1998). Nitrous oxide (laughing gas) is an NMDA antagonist, neuroprotectant and neurotoxin. *Nature Medicine*, 4(4), 460–463. https://doi.org/10.1038/nm0498-460

Jiang-Xie, L., Yin, L., Zhao, S., et al. (2019). A Common Neuroendocrine Substrate for Diverse General Anesthetics and Sleep. *Neuron*, 102(5), 1053-1065.e4. https://doi.org/10.1016/j.neuron.2019.03.033

Jing, Z., Pecka, M., & Grothe, B. (2021). Ketamine-xylazine anesthesia depth affects auditory neuronal responses in the lateral superior olive complex of the gerbil. *Journal of Neurophysiology*, 126(5), 1660–1669. https://doi.org/10.1152/jn.00217.2021

Joiner, W. J., Friedman, E. B., Hung, H.-T., et al. (2013). Genetic and anatomical basis of the barrier separating wakefulness and anesthetic-induced unresponsiveness. *PLoS Genetics*, 9(9), e1003605. https://doi.org/10.1371/journal.pgen.1003605

Jurd, R., Arras, M., Lambert, S., et al. (2003). General anesthetic actions in vivo strongly attenuated by a point mutation in the GABA_A receptor β3 subunit. *FASEB Journal*, 17(2), 250–252. https://doi.org/10.1096/fj.02-0611fje

Kandeepan, S., Rudas, J., Gomez, F., et al. (2020). Modeling an auditory stimulated brain under altered states of consciousness using the generalized Ising model. *NeuroImage*, 223, 117367. https://doi.org/10.1016/j.neuroimage.2020.117367

Katlowitz, K. A., Cole, E. R., Mickiewicz, E. A., et al. (2026). Plasticity and language in the anaesthetized human hippocampus. *Nature*, 654(8119), 714–723. https://doi.org/10.1038/s41586-026-10448-0 (see also the Author Corrections, *Nature* 655, E1, and 657, E15–E16).

Kato, D., Wake, H., Lee, P. R., et al. (2020). Motor learning requires myelination to reduce asynchrony and spontaneity in neural activity. *Glia*, 68(1), 193–210. https://doi.org/10.1002/glia.23713

Kato, R., Yamanaka, M., Yokota, E., et al. (2016). Spike Timing Rigidity Is Maintained in Bursting Neurons under Pentobarbital-Induced Anesthetic Conditions. *Frontiers in Neural Circuits*, 10, 86. https://doi.org/10.3389/fncir.2016.00086

Kayser, E. B., Morgan, P. G., & Sedensky, M. M. (1999). GAS-1: a mitochondrial protein controls sensitivity to volatile anesthetics in the nematode *Caenorhabditis elegans*. *Anesthesiology*, 90(2), 545–554. https://doi.org/10.1097/00000542-199902000-00031

Kelz, M. B., & Mashour, G. A. (2019). The biology of general anesthesia from paramecium to primate. *Current Biology*, 29(22), R1199–R1210. https://doi.org/10.1016/j.cub.2019.09.071

Kelz, M. B., Sun, Y., Chen, J., et al. (2008). An essential role for orexins in emergence from general anesthesia. *Proceedings of the National Academy of Sciences*, 105(4), 1309–1314. https://doi.org/10.1073/pnas.0707146105

Kesten, H. (1980). The critical probability of bond percolation on the square lattice equals 1/2. *Communications in Mathematical Physics*, 74(1), 41–59. https://doi.org/10.1007/BF01197577

Khan, S., Huang, Y., Timuçin, D., et al. (2024). Microtubule-Stabilizer Epothilone B Delays Anesthetic-Induced Unconsciousness in Rats. *eNeuro*, 11(8), ENEURO.0291-24.2024. https://doi.org/10.1523/eneuro.0291-24.2024

Kim, H., Moon, J.-Y., Mashour, G. A., & Lee, U. (2018). Mechanisms of hysteresis in human brain networks during transitions of consciousness and unconsciousness: theoretical principles and empirical evidence. *PLoS Computational Biology*, 14(8), e1006424. https://doi.org/10.1371/journal.pcbi.1006424

Kim, M., Mashour, G. A., Moraes, S.-B., et al. (2016). Functional and topological conditions for explosive synchronization develop in human brain networks with the onset of anesthetic-induced unconsciousness. *Frontiers in Computational Neuroscience*, 10, 1. https://doi.org/10.3389/fncom.2016.00001

King, J.-R., Sitt, J. D., Faugeras, F., et al. (2013). Information sharing in the brain indexes consciousness in noncommunicative patients. *Current Biology*, 23(19), 1914–1919. https://doi.org/10.1016/j.cub.2013.07.075

Kisley, M. A., & Gerstein, G. L. (1999). Trial-to-Trial Variability and State-Dependent Modulation of Auditory-Evoked Responses in Cortex. *Journal of Neuroscience*, 19(23), 10451–10460. https://doi.org/10.1523/jneurosci.19-23-10451.1999

Koblin, D. D., Chortkoff, B. S., Laster, M. J., et al. (1994). Polyhalogenated and perfluorinated compounds that disobey the Meyer-Overton hypothesis. *Anesthesia & Analgesia*, 79(6), 1043–1048. https://doi.org/10.1213/00000539-199412000-00004

Kopetz, H., & Bauer, G. (2003). The time-triggered architecture. *Proceedings of the IEEE*, 91(1), 112–126. https://doi.org/10.1109/JPROC.2002.805821

Kordas, B. (2026). Cellular and microcircuit mechanisms of anesthetic disruption of conscious-state organization. *Cells*, 15(16), 1486. https://doi.org/10.3390/cells15161486

Koyanagi, Y., Oi, Y., & Kobayashi, M. (2021). Fast-spiking Interneurons Contribute to Propofol-induced Facilitation of Firing Synchrony in Pyramidal Neurons of the Rat Insular Cortex. *Anesthesiology*, 134(2), 219–233. https://doi.org/10.1097/aln.0000000000003653

Krom, A. J., Marmelshtein, A., Gelbard-Sagiv, H., et al. (2020). Anesthesia-induced loss of consciousness disrupts auditory responses beyond primary cortex. *Proceedings of the National Academy of Sciences*, 117(21), 11770–11780. https://doi.org/10.1073/pnas.1917251117

Kuizenga, M. H., Colin, P. J., Reyntjens, K. M. E. M., et al. (2018). Test of neural inertia in humans during general anaesthesia. *British Journal of Anaesthesia*, 120(3), 525–536. https://doi.org/10.1016/j.bja.2017.11.072

Kuramoto, Y. (1984). *Chemical Oscillations, Waves, and Turbulence*. Springer Series in Synergetics, vol. 19. Berlin: Springer. https://doi.org/10.1007/978-3-642-69689-3

Lakatos, P., Shah, A. S., Knuth, K. H., et al. (2005). An Oscillatory Hierarchy Controlling Neuronal Excitability and Stimulus Processing in the Auditory Cortex. *Journal of Neurophysiology*, 94(3), 1904–1911. https://doi.org/10.1152/jn.00263.2005

Lakens, D., Scheel, A. M., & Isager, P. M. (2018). Equivalence testing for psychological research: A tutorial. *Advances in Methods and Practices in Psychological Science*, 1(2), 259–269. https://doi.org/10.1177/2515245918770963

Lamme, V. A. F. (2006). Towards a true neural stance on consciousness. *Trends in Cognitive Sciences*, 10(11), 494–501. https://doi.org/10.1016/j.tics.2006.09.001

Lamme, V. A. F., & Roelfsema, P. R. (2000). The distinct modes of vision offered by feedforward and recurrent processing. *Trends in Neurosciences*, 23(11), 571–579. https://doi.org/10.1016/s0166-2236(00)01657-x

Lamport, L. (1978). Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM*, 21(7), 558–565. https://doi.org/10.1145/359545.359563

Lamport, L., Shostak, R., & Pease, M. (1982). The Byzantine generals problem. *ACM Transactions on Programming Languages and Systems*, 4(3), 382–401. https://doi.org/10.1145/357172.357176

Långsjö, J. W., Alkire, M. T., Kaskinoro, K., et al. (2012). Returning from oblivion: imaging the neural core of consciousness. *Journal of Neuroscience*, 32(14), 4935–4943. https://doi.org/10.1523/JNEUROSCI.4962-11.2012

Larkum, M. E., Zhu, J. J., & Sakmann, B. (1999). A new cellular mechanism for coupling inputs arriving at different cortical layers. *Nature*, 398(6725), 338–341. https://doi.org/10.1038/18686

Lee, H. S., Ghetti, A., Pinto-Duarte, A., et al. (2014). Astrocytes contribute to gamma oscillations and recognition memory. *Proceedings of the National Academy of Sciences*, 111(32), E3343–E3352. https://doi.org/10.1073/pnas.1410893111

Lee, H., Mashour, G. A., Noh, G.-J., et al. (2013). Reconfiguration of Network Hub Structure after Propofol-induced Unconsciousness. *Anesthesiology*, 119(6), 1347–1359. https://doi.org/10.1097/aln.0b013e3182a8ec8c

Lee, H.-C., Park, Y., Yoon, S. B., et al. (2022). VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients. *Scientific Data*, 9, 279. https://doi.org/10.1038/s41597-022-01411-5

Lee, U., & Mashour, G. A. (2018). Role of network science in the study of anesthetic state transitions. *Anesthesiology*, 129(5), 1029–1044. https://doi.org/10.1097/ALN.0000000000002228

Lee, U., Kim, S., Noh, G., et al. (2009). The directionality and functional organization of frontoparietal connectivity during consciousness and anesthesia in humans. *Consciousness and Cognition*, 18(4), 1069–1078. https://doi.org/10.1016/j.concog.2009.04.004

Lee, U., Ku, S., Noh, G., et al. (2013). Disruption of frontal–parietal communication by ketamine, propofol, and sevoflurane. *Anesthesiology*, 118(6), 1264–1275. https://doi.org/10.1097/ALN.0b013e31829103f5

Lemaréchal, J.-D., Jedynak, M., Trebaul, L., et al. (2022). A brain atlas of axonal and synaptic delays based on modelling of cortico-cortical evoked potentials. *Brain*, 145(5), 1653–1667. https://doi.org/10.1093/brain/awab362

Lempel, A., & Ziv, J. (1976). On the complexity of finite sequences. *IEEE Transactions on Information Theory*, 22(1), 75–81. https://doi.org/10.1109/TIT.1976.1055501

Leung, A., Cohen, D., van Swinderen, B., & Tsuchiya, N. (2021). Integrated information structure collapses with anesthetic loss of conscious arousal in *Drosophila melanogaster*. *PLoS Computational Biology*, 17(2), e1008722. https://doi.org/10.1371/journal.pcbi.1008722

Lewis, L. D., Weiner, V. S., Mukamel, E. A., et al. (2012). Rapid fragmentation of neuronal networks at the onset of propofol-induced unconsciousness. *Proceedings of the National Academy of Sciences*, 109(49), E3377–E3386. https://doi.org/10.1073/pnas.1210907109

Li, N., Lu, D., Yang, L., et al. (2018). Nuclear spin attenuates the anesthetic potency of xenon isotopes in mice: implications for the mechanisms of anesthesia and consciousness. *Anesthesiology*, 129(2), 271–277. https://doi.org/10.1097/aln.0000000000002226

Li, N., You, Z., Ren, Y., et al. (2025). Microtubule-modulating drugs alter sensitivity to isoflurane in mice. *BMC Anesthesiology*, 25(1), 109. https://doi.org/10.1186/s12871-025-02956-9

Liang, Z., Fan, L., Zhang, B., et al. (2025). The changes in neural complexity and connectivity in thalamocortical and cortico-cortical systems after propofol-induced unconsciousness in different temporal scales. *NeuroImage*, 311, 121193. https://doi.org/10.1016/j.neuroimage.2025.121193

Lin, J., Cheng, X., Wang, H., et al. (2023). Activation of astrocytes in the basal forebrain in mice facilitates isoflurane-induced loss of consciousness and prolongs recovery. *BMC Anesthesiology*, 23(1), 213. https://doi.org/10.1186/s12871-023-02166-1

Liu, P.-C., Yao, W., Chen, X.-Y., et al. (2023). Parabrachial nucleus astrocytes regulate wakefulness and isoflurane anesthesia in mice. *Frontiers in Pharmacology*, 13, 991238. https://doi.org/10.3389/fphar.2022.991238

Liu, Q., Kidd, P. B., Dobosiewicz, M., & Bargmann, C. I. (2018). *C. elegans* AWA olfactory neurons fire calcium-mediated all-or-none action potentials. *Cell*, 175(1), 57–70.e17. https://doi.org/10.1016/j.cell.2018.08.018

Liu, Y. U., Ying, Y., Li, Y., et al. (2019). Neuronal network activity controls microglial process surveillance in awake mice via norepinephrine signaling. *Nature Neuroscience*, 22(11), 1771–1781. https://doi.org/10.1038/s41593-019-0511-3

Llinás, R., & Ribary, U. (1993). Coherent 40-Hz oscillation characterizes dream state in humans. *Proceedings of the National Academy of Sciences*, 90(5), 2078–2081. https://doi.org/10.1073/pnas.90.5.2078

Llinás, R., Ribary, U., Contreras, D., et al. (1998). The neuronal basis for consciousness. *Philosophical Transactions of the Royal Society of London. Series B: Biological Sciences*, 353(1377), 1841–1849. https://doi.org/10.1098/rstb.1998.0336

Lubke, G. H., Kerssens, C., Phaf, H., et al. (1999). Dependence of Explicit and Implicit Memory on Hypnotic State in Trauma Patients. *Anesthesiology*, 90(3), 670–680. https://doi.org/10.1097/00000542-199903000-00007

Luppi, A. I., Golkowski, D., Ranft, A., et al. (2025). General anaesthesia decreases the uniqueness of brain functional connectivity across individuals and species. *Nature Human Behaviour*, 9(5), 987–1004. https://doi.org/10.1038/s41562-025-02121-9

Lv, P., Xiao, Y., Liu, B., et al. (2016). Dose-dependent effects of isoflurane on regional activity and neural network function: a resting-state fMRI study of 14 rhesus monkeys. *Neuroscience Letters*, 611, 116–122. https://doi.org/10.1016/j.neulet.2015.11.037

Ma, C., Li, B., Silverman, D., et al. (2024). Microglia regulate sleep through calcium-dependent modulation of norepinephrine transmission. *Nature Neuroscience*, 27(2), 249–258. https://doi.org/10.1038/s41593-023-01548-5

Mainen, Z. F., & Sejnowski, T. J. (1995). Reliability of spike timing in neocortical neurons. *Science*, 268(5216), 1503–1506. https://doi.org/10.1126/science.7770778

Maquet, P., Dive, D., Salmon, E., et al. (1990). Cerebral glucose utilization during sleep-wake cycle in man determined by positron emission tomography and [18F]2-fluoro-2-deoxy-D-glucose method. *Brain Research*, 513(1), 136–143. https://doi.org/10.1016/0006-8993(90)91099-3

Markov, N. T., Ercsey-Ravasz, M. M., Ribeiro Gomes, A. R., et al. (2014). A weighted and directed interareal connectivity matrix for macaque cerebral cortex. *Cerebral Cortex*, 24(1), 17–36. https://doi.org/10.1093/cercor/bhs270

Maschke, C., O'Byrne, J., Colombo, M. A., et al. (2024). Critical dynamics in spontaneous EEG predict anesthetic-induced loss of consciousness and perturbational complexity. *Communications Biology*, 7, 946. https://doi.org/10.1038/s42003-024-06613-8

Mashour, G. A. (2013). Cognitive unbinding: a neuroscientific paradigm of general anesthesia and related states of unconsciousness. *Neuroscience & Biobehavioral Reviews*, 37(10 Pt 2), 2751–2759. https://doi.org/10.1016/j.neubiorev.2013.09.009

Mashour, G. A., & Hudetz, A. G. (2018). Neural correlates of unconsciousness in large-scale brain networks. *Trends in Neurosciences*, 41(3), 150–160. https://doi.org/10.1016/j.tins.2018.01.003

Mashour, G. A., Roelfsema, P., Changeux, J.-P., & Dehaene, S. (2020). Conscious processing and the global neuronal workspace hypothesis. *Neuron*, 105(5), 776–798. https://doi.org/10.1016/j.neuron.2020.01.026

Maslov, S., & Sneppen, K. (2002). Specificity and stability in topology of protein networks. *Science*, 296(5569), 910–913. https://doi.org/10.1126/science.1065103

Massimini, M., Ferrarelli, F., Huber, R., et al. (2005). Breakdown of cortical effective connectivity during sleep. *Science*, 309(5744), 2228–2232. https://doi.org/10.1126/science.1117256

McCarthy, M. M., Brown, E. N., & Kopell, N. (2008). Potential network mechanisms mediating electroencephalographic beta rhythm changes during propofol-induced paradoxical excitation. *Journal of Neuroscience*, 28(50), 13488–13504. https://doi.org/10.1523/JNEUROSCI.3536-08.2008

McCormick, D. A., & Bal, T. (1994). Sensory gating mechanisms of the thalamus. *Current Opinion in Neurobiology*, 4(4), 550–556. https://doi.org/10.1016/0959-4388(94)90056-6

Melnik, S., Hackett, A., Porter, M. A., et al. (2011). The unreasonable effectiveness of tree-based theory for networks with clustering. *Physical Review E*, 83(3), 036112. https://doi.org/10.1103/PhysRevE.83.036112

Merikle, P. M., & Daneman, M. (1996). Memory for Unconsciously Perceived Events: Evidence from Anesthetized Patients. *Consciousness and Cognition*, 5(4), 525–541. https://doi.org/10.1006/ccog.1996.0031

Meyer, H. (1899). Zur Theorie der Alkoholnarkose. *Archiv für experimentelle Pathologie und Pharmakologie*, 42(2–4), 109–118. https://doi.org/10.1007/BF01834479

Miao, A., Luo, T., Hsieh, B., et al. (2024). Brain clearance is reduced during sleep and anesthesia. *Nature Neuroscience*, 27(6), 1046–1050. https://doi.org/10.1038/s41593-024-01638-y. Author correction: *Nature Neuroscience*, 27(7), 1425. https://doi.org/10.1038/s41593-024-01698-0

Mihic, S. J., Ye, Q., Wick, M. J., et al. (1997). Sites of alcohol and volatile anaesthetic action on GABAA and glycine receptors. *Nature*, 389(6649), 385–389. https://doi.org/10.1038/38738

Moon, J.-Y., Lee, U., Blain-Moraes, S., & Mashour, G. A. (2015). General relationship of global topology, local dynamics, and directionality in large-scale brain networks. *PLoS Computational Biology*, 11(4), e1004225. https://doi.org/10.1371/journal.pcbi.1004225

Moore, J. T., Chen, J., Han, B., et al. (2012). Direct Activation of Sleep-Promoting VLPO Neurons by Volatile Anesthetics Contributes to Anesthetic Hypnosis. *Current Biology*, 22(21), 2008–2016. https://doi.org/10.1016/j.cub.2012.08.042

Nagumo, J., Arimoto, S., & Yoshizawa, S. (1962). An active pulse transmission line simulating nerve axon. *Proceedings of the IRE*, 50(10), 2061–2070. https://doi.org/10.1109/JRPROC.1962.288235

Neckelmann, D., & Ursin, R. (1993). Sleep stages and EEG power spectrum in relation to acoustical stimulus arousal threshold in the rat. *Sleep*, 16(5), 467–477. https://doi.org/10.1093/sleep/16.5.467

Nelson, L. E., Guo, T. Z., Lu, J., et al. (2002). The sedative component of anesthesia is mediated by GABAA receptors in an endogenous sleep pathway. *Nature Neuroscience*, 5(10), 979–984. https://doi.org/10.1038/nn913

Nelson, L. E., Lu, J., Guo, T., et al. (2003). The α2-Adrenoceptor Agonist Dexmedetomidine Converges on an Endogenous Sleep-promoting Pathway to Exert Its Sedative Effects. *Anesthesiology*, 98(2), 428–436. https://doi.org/10.1097/00000542-200302000-00024

Newman, M. E. J., & Ziff, R. M. (2000). Efficient Monte Carlo algorithm and high-precision results for percolation. *Physical Review Letters*, 85(19), 4104–4107. https://doi.org/10.1103/PhysRevLett.85.4104

Newman, M. E. J., Strogatz, S. H., & Watts, D. J. (2001). Random graphs with arbitrary degree distributions and their applications. *Physical Review E*, 64(2), 026118. https://doi.org/10.1103/PhysRevE.64.026118

Nir, Y., & Tononi, G. (2010). Dreaming and the brain: from phenomenology to neurophysiology. *Trends in Cognitive Sciences*, 14(2), 88–100. https://doi.org/10.1016/j.tics.2009.12.001

Nir, Y., Staba, R. J., Andrillon, T., et al. (2011). Regional Slow Waves and Spindles in Human Sleep. *Neuron*, 70(1), 153–169. https://doi.org/10.1016/j.neuron.2011.02.043

Noori, R., Park, D., Griffiths, J. D., et al. (2020). Activity-dependent myelination: A glial mechanism of oscillatory self-organization in large-scale brain networks. *Proceedings of the National Academy of Sciences*, 117(24), 13227–13237. https://doi.org/10.1073/pnas.1916646117

Orser, B. A., Pennefather, P. S., & MacDonald, J. F. (1997). Multiple mechanisms of ketamine blockade of N-methyl-D-aspartate receptors. *Anesthesiology*, 86(4), 903–917. https://doi.org/10.1097/00000542-199704000-00021

Overton, C. E. (1901). *Studien über die Narkose*. Jena: Gustav Fischer.

Pajevic, S., Basser, P. J., & Fields, R. D. (2014). Role of myelin plasticity in oscillations and synchrony of neuronal activity. *Neuroscience*, 276, 135–147. https://doi.org/10.1016/j.neuroscience.2013.11.007

Pajevic, S., Plenz, D., Basser, P. J., et al. (2023). Oligodendrocyte-mediated myelin plasticity and its role in neural synchronization. *eLife*, 12, e81982. https://doi.org/10.7554/elife.81982

Pal, D., Dean, J. G., Liu, T., et al. (2018). Differential role of prefrontal and parietal cortices in controlling level of consciousness. *Current Biology*, 28(13), 2145–2152.e5. https://doi.org/10.1016/j.cub.2018.05.025

Park, S. H., & Lefebvre, J. (2020). Synchronization and resilience in the Kuramoto white matter network model with adaptive state-dependent delays. *The Journal of Mathematical Neuroscience*, 10(1), 16. https://doi.org/10.1186/s13408-020-00091-y

Patel, A. J., Honoré, E., Lesage, F., et al. (1999). Inhalational anesthetics activate two-pore-domain background K+ channels. *Nature Neuroscience*, 2(5), 422–426. https://doi.org/10.1038/8084

Pavel, M. A., Petersen, E. N., Wang, H., Lerner, R. A., & Hansen, S. B. (2020). Studies on the mechanism of general anesthesia. *Proceedings of the National Academy of Sciences*, 117(24), 13757–13766. https://doi.org/10.1073/pnas.2004259117

Pigorini, A., Sarasso, S., Proserpio, P., et al. (2015). Bistability breaks-off deterministic responses to intracortical stimulation during non-REM sleep. *NeuroImage*, 112, 105–113. https://doi.org/10.1016/j.neuroimage.2015.02.056

Poskanzer, K. E., & Yuste, R. (2016). Astrocytes regulate cortical state switching in vivo. *Proceedings of the National Academy of Sciences*, 113(19), E2675–E2684. https://doi.org/10.1073/pnas.1520759113

Pouille, F., & Scanziani, M. (2001). Enforcement of temporal fidelity in pyramidal cells by somatic feed-forward inhibition. *Science*, 293(5532), 1159–1163. https://doi.org/10.1126/science.1060342

Proekt, A., & Hudson, A. E. (2018). A stochastic basis for neural inertia in emergence from general anaesthesia. *British Journal of Anaesthesia*, 121(1), 86–94. https://doi.org/10.1016/j.bja.2018.02.035

Purdon, P. L., Pierce, E. T., Mukamel, E. A., et al. (2013). Electroencephalogram signatures of loss and recovery of consciousness from propofol. *Proceedings of the National Academy of Sciences*, 110(12), E1142–E1151. https://doi.org/10.1073/pnas.1221180110

Purdon, P. L., Sampson, A., Pavone, K. J., et al. (2015). Clinical electroencephalography for anesthesiologists: Part I: Background and basic signatures. *Anesthesiology*, 123(4), 937–960. https://doi.org/10.1097/aln.0000000000000841

Radek, L., Kallionpää, R. E., Karvonen, M., et al. (2018). Dreaming and awareness during dexmedetomidine- and propofol-induced unresponsiveness. *British Journal of Anaesthesia*, 121(1), 260–269. https://doi.org/10.1016/j.bja.2018.03.014

Rampil, I. J. (1994). Anesthetic potency is not altered after hypothermic spinal cord transection in rats. *Anesthesiology*, 80(3), 606–610. https://doi.org/10.1097/00000542-199403000-00017

Rampil, I. J., Mason, P., & Singh, H. (1993). Anesthetic potency (MAC) is independent of forebrain structures in the rat. *Anesthesiology*, 78(4), 707–712. https://doi.org/10.1097/00000542-199304000-00014

Rao, R. P. N., & Ballard, D. H. (1999). Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2(1), 79–87. https://doi.org/10.1038/4580

Rasch, B., & Born, J. (2013). About Sleep's Role in Memory. *Physiological Reviews*, 93(2), 681–766. https://doi.org/10.1152/physrev.00032.2012

Redinbaugh, M. J., Phillips, J. M., Kambi, N. A., et al. (2020). Thalamus modulates consciousness via layer-specific control of cortex. *Neuron*, 106(1), 66–75.e12. https://doi.org/10.1016/j.neuron.2020.01.005

Rohden, M., Sorge, A., Timme, M., & Witthaut, D. (2012). Self-organized synchronization in decentralized power grids. *Physical Review Letters*, 109(6), 064101. https://doi.org/10.1103/PhysRevLett.109.064101

Rosanova, M., Fecchio, M., Casarotto, S., et al. (2018). Sleep-like cortical OFF-periods disrupt causality and complexity in the brain of unresponsive wakefulness syndrome patients. *Nature Communications*, 9, 4427. https://doi.org/10.1038/s41467-018-06871-1

Rudolph, U., & Antkowiak, B. (2004). Molecular and neuronal substrates for general anaesthetics. *Nature Reviews Neuroscience*, 5(9), 709–720. https://doi.org/10.1038/nrn1496

Sacha, M., Tesler, F., Cofre, R., & Destexhe, A. (2025). A computational approach to evaluate how molecular mechanisms impact large-scale brain activity. *Nature Computational Science*, 5(5), 405–417. https://doi.org/10.1038/s43588-025-00796-8

Sakaguchi, H., & Kuramoto, Y. (1986). A soluble active rotator model showing phase transitions via mutual entrainment. *Progress of Theoretical Physics*, 76(3), 576–581. https://doi.org/10.1143/PTP.76.576

Sanchez-Vives, M. V., Massimini, M., & Mattia, M. (2017). Shaping the default activity pattern of the cortical network. *Neuron*, 94(5), 993–1001. https://doi.org/10.1016/j.neuron.2017.05.015

Sanders, R. D., Tononi, G., Laureys, S., & Sleigh, J. W. (2012). Unresponsiveness ≠ unconsciousness. *Anesthesiology*, 116(4), 946–959. https://doi.org/10.1097/ALN.0b013e318249d0a7

Sanders, R. D., Gaskell, A., Raz, A., et al. (2017). Incidence of connected consciousness after tracheal intubation. *Anesthesiology*, 126(2), 214–222. https://doi.org/10.1097/ALN.0000000000001479

Sanz Leon, P., Knock, S. A., Woodman, M. M., et al. (2013). The Virtual Brain: a simulator of primate brain network dynamics. *Frontiers in Neuroinformatics*, 7, 10. https://doi.org/10.3389/fninf.2013.00010

Sarasso, S., Boly, M., Napolitani, M., et al. (2015). Consciousness and complexity during unresponsiveness induced by propofol, xenon, and ketamine. *Current Biology*, 25(23), 3099–3105. https://doi.org/10.1016/j.cub.2015.10.014

Schaefer, A., Kong, R., Gordon, E. M., et al. (2018). Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI. *Cerebral Cortex*, 28(9), 3095–3114. https://doi.org/10.1093/cercor/bhx179

Scheffer, M., Bascompte, J., Brock, W. A., et al. (2009). Early-warning signals for critical transitions. *Nature*, 461(7260), 53–59. https://doi.org/10.1038/nature08227

Schnider, T. W., Minto, C. F., Shafer, S. L., et al. (1999). The influence of age on propofol pharmacodynamics. *Anesthesiology*, 90(6), 1502–1516. https://doi.org/10.1097/00000542-199906000-00003

Self, M. W., Kooijmans, R. N., Supèr, H., Lamme, V. A., & Roelfsema, P. R. (2012). Different glutamate receptors convey feedforward and recurrent processing in macaque V1. *Proceedings of the National Academy of Sciences*, 109(27), 11031–11036. https://doi.org/10.1073/pnas.1119527109

Sepúlveda, P. O., Carrasco, E., Tapia, L. F., Ramos, M., et al. (2018). Evidence of hysteresis in propofol pharmacodynamics. *Anaesthesia*, 73(1), 40–48. https://doi.org/10.1111/anae.14009

Shiu, P. K., Sterne, G. R., Spiller, N., et al. (2024). A *Drosophila* computational brain model reveals sensorimotor processing. *Nature*, 634(8032), 210–219. https://doi.org/10.1038/s41586-024-07763-9

Siclari, F., Baird, B., Perogamvros, L., et al. (2017). The neural correlates of dreaming. *Nature Neuroscience*, 20(6), 872–878. https://doi.org/10.1038/nn.4545

Sitt, J. D., King, J.-R., El Karoui, I., et al. (2014). Large scale screening of neural signatures of consciousness in patients in a vegetative or minimally conscious state. *Brain*, 137(8), 2258–2270. https://doi.org/10.1093/brain/awu141

Solovey, G., Alonso, L. M., Yanagawa, T., et al. (2015). Loss of consciousness is associated with stabilization of cortical activity. *Journal of Neuroscience*, 35(30), 10866–10877. https://doi.org/10.1523/JNEUROSCI.4895-14.2015

Solt, K., Cotten, J. F., Cimenser, A., et al. (2011). Methylphenidate actively induces emergence from general anesthesia. *Anesthesiology*, 115(4), 791–803. https://doi.org/10.1097/ALN.0b013e31822e92e5

Steadman, P. E., Xia, F., Ahmed, M., et al. (2020). Disruption of Oligodendrogenesis Impairs Memory Consolidation in Adult Mice. *Neuron*, 105(1), 150–164.e6. https://doi.org/10.1016/j.neuron.2019.10.013

Steriade, M., McCormick, D. A., & Sejnowski, T. J. (1993b). Thalamocortical Oscillations in the Sleeping and Aroused Brain. *Science*, 262(5134), 679–685. https://doi.org/10.1126/science.8235588

Steriade, M., Nuñez, A., & Amzica, F. (1993a). A novel slow (< 1 Hz) oscillation of neocortical neurons in vivo: depolarizing and hyperpolarizing components. *Journal of Neuroscience*, 13(8), 3252–3265. https://doi.org/10.1523/JNEUROSCI.13-08-03252.1993

Steyn-Ross, M. L., Steyn-Ross, D. A., Sleigh, J. W., & Liley, D. T. J. (1999). Theoretical electroencephalogram stationary spectrum for a white-noise-driven cortex: Evidence for a general anesthetic-induced phase transition. *Physical Review E*, 60(6), 7299–7311. https://doi.org/10.1103/PhysRevE.60.7299

Stowell, R. D., Sipe, G. O., Dawes, R. P., et al. (2019). Noradrenergic signaling in the wakeful state inhibits microglial surveillance and synaptic plasticity in the mouse visual cortex. *Nature Neuroscience*, 22(11), 1782–1792. https://doi.org/10.1038/s41593-019-0514-0. Author correction: *Nature Neuroscience*, 23(1), 152. https://doi.org/10.1038/s41593-019-0558-1

Strogatz, S. H., & Mirollo, R. E. (1991). Stability of incoherence in a population of coupled oscillators. *Journal of Statistical Physics*, 63(3–4), 613–635. https://doi.org/10.1007/BF01029202

Struys, M. M. R. F., De Smet, T., Depoorter, B., et al. (2000). Comparison of plasma compartment versus two methods for effect compartment-controlled target-controlled infusion for propofol. *Anesthesiology*, 92(2), 399–406. https://doi.org/10.1097/00000542-200002000-00021

Su, C.-W., Zheng, L., Li, Y.-J., Zhou, H.-J., Wang, J., Huang, Z.-G., & Lai, Y.-C. (2020). Hysteresis in anesthesia and recovery: experimental observation and dynamical mechanism. *Physical Review Research*, 2(2), 023289. https://doi.org/10.1103/PhysRevResearch.2.023289

Supp, G. G., Siegel, M., Hipp, J. F., & Engel, A. K. (2011). Cortical hypersynchrony predicts breakdown of sensory processing during loss of consciousness. *Current Biology*, 21(23), 1988–1993. https://doi.org/10.1016/j.cub.2011.10.017

Suzuki, M., & Larkum, M. E. (2020). General anesthesia decouples cortical pyramidal neurons. *Cell*, 180(4), 666–676.e13. https://doi.org/10.1016/j.cell.2020.01.024

Tagliazucchi, E., Chialvo, D. R., Siniatchkin, M., et al. (2016). Large-scale signatures of unconsciousness are consistent with a departure from critical dynamics. *Journal of the Royal Society Interface*, 13(114), 20151027. https://doi.org/10.1098/rsif.2015.1027

Tarjan, R. E. (1975). Efficiency of a good but not linear set union algorithm. *Journal of the ACM*, 22(2), 215–225. https://doi.org/10.1145/321879.321884

Tasserie, J., Uhrig, L., Sitt, J. D., et al. (2022). Deep brain stimulation of the thalamus restores signatures of consciousness in a nonhuman primate model. *Science Advances*, 8(11), eabl5547. https://doi.org/10.1126/sciadv.abl5547

Tassi, P., & Muzet, A. (2000). Sleep inertia. *Sleep Medicine Reviews*, 4(4), 341–353. https://doi.org/10.1053/smrv.2000.0098

Taylor, N. E., Van Dort, C. J., Kenny, J. D., et al. (2016). Optogenetic activation of dopamine neurons in the ventral tegmental area induces reanimation from general anesthesia. *Proceedings of the National Academy of Sciences*, 113(45), 12826–12831. https://doi.org/10.1073/pnas.1614340113

Thrane, A. S., Rangroo Thrane, V., Zeppenfeld, D., et al. (2012). General anesthesia selectively disrupts astrocyte calcium signaling in the awake mouse cortex. *Proceedings of the National Academy of Sciences*, 109(46), 18974–18979. https://doi.org/10.1073/pnas.1209448109

Tian, Y., & Sun, P. (2022). Percolation may explain efficiency, robustness, and economy of the brain. *Network Neuroscience*, 6(3), 765–790. https://doi.org/10.1162/netn_a_00246

Toker, D., Pappas, I., Lendner, J. D., et al. (2022). Consciousness is supported by near-critical slow cortical electrodynamics. *Proceedings of the National Academy of Sciences*, 119(7), e2024455119. https://doi.org/10.1073/pnas.2024455119

Tong, C. D., Wang, Y., Yang, Z. J., et al. (2026). Tensor decomposition reveals discrete cortical reorganization under graded propofol anesthesia. *Frontiers in Neuroscience*, 20, 1950745. https://doi.org/10.3389/fnins.2026.1950745

Tononi, G., & Cirelli, C. (2014). Sleep and the price of plasticity: from synaptic and cellular homeostasis to memory consolidation and integration. *Neuron*, 81(1), 12–34. https://doi.org/10.1016/j.neuron.2013.12.025

Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450–461. https://doi.org/10.1038/nrn.2016.44

Tot, O. K., Mrđenović, S., Ivić, V., et al. (2024). Age-Related Effects of Inhalational Anesthetics in B4galnt1-Null and Cuprizone-Treated Mice: Clinically Relevant Insights into Demyelinating Diseases. *Current Issues in Molecular Biology*, 46(8), 8376–8394. https://doi.org/10.3390/cimb46080494

Tunstall, M. E. (1977). Detecting wakefulness during general anaesthesia for caesarean section. *British Medical Journal*, 1(6072), 1321. https://doi.org/10.1136/bmj.1.6072.1321-a

Turrigiano, G. G. (2008). The self-tuning neuron: synaptic scaling of excitatory synapses. *Cell*, 135(3), 422–435. https://doi.org/10.1016/j.cell.2008.10.008

U.S.-Canada Power System Outage Task Force. (2004). *Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations*. U.S. Department of Energy and Natural Resources Canada.

Uhrig, L., Sitt, J. D., Jacob, A., et al. (2018). Resting-state dynamics as a cortical signature of anesthesia in monkeys. *Anesthesiology*, 129(5), 942–958. https://doi.org/10.1097/ALN.0000000000002336

van Blooijs, D., Blok, S., Huiskamp, G. J. M., et al. (2024). The effect of propofol on effective brain networks. *Clinical Neurophysiology*, 161, 222–230. https://doi.org/10.1016/j.clinph.2024.01.012

VanRullen, R. (2016). Perceptual cycles. *Trends in Cognitive Sciences*, 20(10), 723–735. https://doi.org/10.1016/j.tics.2016.07.006

VanRullen, R., & Koch, C. (2003). Is perception discrete or continuous? *Trends in Cognitive Sciences*, 7(5), 207–213. https://doi.org/10.1016/S1364-6613(03)00095-0

Varela, F., Lachaux, J.-P., Rodriguez, E., & Martinerie, J. (2001). The brainweb: phase synchronization and large-scale integration. *Nature Reviews Neuroscience*, 2(4), 229–239. https://doi.org/10.1038/35067550

Velly, L. J., Rey, M. F., Bruder, N. J., et al. (2007). Differential dynamic of action on cortical and subcortical structures of anesthetic agents during induction of anesthesia. *Anesthesiology*, 107(2), 202–212. https://doi.org/10.1097/01.anes.0000270734.99298.b4

Veselis, R. A., Reinsel, R. A., Feshchenko, V. A., & Wroński, M. (1997). The comparative amnestic effects of midazolam, propofol, thiopental, and fentanyl at equisedative concentrations. *Anesthesiology*, 87(4), 749–764. https://doi.org/10.1097/00000542-199710000-00007

Vijayan, S., Ching, S., Purdon, P. L., et al. (2013). Thalamocortical Mechanisms for the Anteriorization of Alpha Rhythms during Propofol-Induced Unconsciousness. *Journal of Neuroscience*, 33(27), 11070–11075. https://doi.org/10.1523/jneurosci.5670-12.2013

Vinck, M., Oostenveld, R., van Wingerden, M., et al. (2011). An improved index of phase-synchronization for electrophysiological data in the presence of volume-conduction, noise and sample-size bias. *NeuroImage*, 55(4), 1548–1565. https://doi.org/10.1016/j.neuroimage.2011.01.055

Vlajkovic, G. P., & Sindjelic, R. P. (2007). Emergence delirium in children: many questions, few answers. *Anesthesia & Analgesia*, 104(1), 84–91. https://doi.org/10.1213/01.ane.0000250914.91881.a8

Vogels, T. P., Sprekeler, H., Zenke, F., et al. (2011). Inhibitory Plasticity Balances Excitation and Inhibition in Sensory Pathways and Memory Networks. *Science*, 334(6062), 1569–1573. https://doi.org/10.1126/science.1211095

Vyazovskiy, V. V., Olcese, U., Lazimy, Y. M., et al. (2009). Cortical Firing and Sleep Homeostasis. *Neuron*, 63(6), 865–878. https://doi.org/10.1016/j.neuron.2009.08.024

Wallace, M. T., & Stevenson, R. A. (2014). The construct of the multisensory temporal binding window and its dysregulation in developmental disabilities. *Neuropsychologia*, 64, 105–123. https://doi.org/10.1016/j.neuropsychologia.2014.08.005

Wang, X., Yi, R., Liang, X., et al. (2024). Myelin modulates the process of isoflurane anesthesia through the regulation of neural activity. *CNS Neuroscience & Therapeutics*, 30(8), e14922. https://doi.org/10.1111/cns.14922

Warnaby, C. E., Sleigh, J. W., Hight, D., Jbabdi, S., & Tracey, I. (2017). Investigation of slow-wave activity saturation during surgical anesthesia reveals a signature of neural inertia in humans. *Anesthesiology*, 127(4), 645–657. https://doi.org/10.1097/ALN.0000000000001759

Waters, J. R., Madeira, B., Shadlen, M. N., & Shushruth, S. (2026). Neural correlates of ketamine-induced dilation of temporal expectation. *bioRxiv* (preprint). https://doi.org/10.64898/2026.10.12.758574

White, J. G., Southgate, E., Thomson, J. N., & Brenner, S. (1986). The structure of the nervous system of the nematode *Caenorhabditis elegans*. *Philosophical Transactions of the Royal Society B*, 314(1165), 1–340. https://doi.org/10.1098/rstb.1986.0056

Wieling, W., Thijs, R. D., van Dijk, N., Wilde, A. A. M., Benditt, D. G., & van Dijk, J. G. (2009). Symptoms and signs of syncope: a review of the link between physiology and clinical clues. *Brain*, 132(10), 2630–2642. https://doi.org/10.1093/brain/awp179

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158), 209–212. https://doi.org/10.1080/01621459.1927.10502953

Wilson, H. R., & Cowan, J. D. (1972). Excitatory and Inhibitory Interactions in Localized Populations of Model Neurons. *Biophysical Journal*, 12(1), 1–24. https://doi.org/10.1016/s0006-3495(72)86068-5

Womelsdorf, T., Schoffelen, J.-M., Oostenveld, R., et al. (2007). Modulation of neuronal interactions through neuronal synchronization. *Science*, 316(5831), 1609–1612. https://doi.org/10.1126/science.1139597

Xie, L., Kang, H., Xu, Q., et al. (2013). Sleep Drives Metabolite Clearance from the Adult Brain. *Science*, 342(6156), 373–377. https://doi.org/10.1126/science.1241224

Yamamoto, K., & Kawabata, H. (2011). Temporal recalibration in vocalization induced by adaptation of delayed auditory feedback. *PLoS ONE*, 6(12), e29414. https://doi.org/10.1371/journal.pone.0029414

Yanagawa, T., Chao, Z. C., Hasegawa, N., & Fujii, N. (2013). Large-scale information flow in conscious and unconscious states: an ECoG study in monkeys. *PLoS ONE*, 8(11), e80845. https://doi.org/10.1371/journal.pone.0080845

Yap, J. Y. Y., Keatch, C., Lambert, E., et al. (2020). Critical Review of Transcutaneous Vagus Nerve Stimulation: Challenges for Translation to Clinical Practice. *Frontiers in Neuroscience*, 14, 284. https://doi.org/10.3389/fnins.2020.00284

Yeh, Y. C. (1996). Triple-triple redundant 777 primary flight computer. In *1996 IEEE Aerospace Applications Conference Proceedings* (Vol. 1, pp. 293–307). https://doi.org/10.1109/AERO.1996.495891

Ying, S.-W., Abbas, S. Y., Harrison, N. L., et al. (2006). Propofol block of I_h contributes to the suppression of neuronal excitability and rhythmic burst firing in thalamocortical neurons. *European Journal of Neuroscience*, 23(2), 465–480. https://doi.org/10.1111/j.1460-9568.2005.04587.x

Yokawa, K., Kagenishi, T., Pavlovič, A., et al. (2018). Anaesthetics stop diverse plant organ movements, affect endocytic vesicle recycling and ROS homeostasis, and block action potentials in Venus flytraps. *Annals of Botany*, 122(5), 747–756. https://doi.org/10.1093/aob/mcx155

Zalucki, O. H., Menon, H., Kottler, B., et al. (2015). Syntaxin1A-mediated resistance and hypersensitivity to isoflurane in *Drosophila melanogaster*. *Anesthesiology*, 122(5), 1060–1074. https://doi.org/10.1097/ALN.0000000000000629

Zamani Esfahlani, F., Jo, Y., Faskowitz, J., et al. (2020). High-amplitude cofluctuations in cortical activity drive functional connectivity. *Proceedings of the National Academy of Sciences*, 117(45), 28393–28401. https://doi.org/10.1073/pnas.2005531117

Zelmann, R., Paulk, A. C., Tian, F., et al. (2023). Differential cortical network engagement during states of un/consciousness in humans. *Neuron*, 111(21), 3479–3495.e6. https://doi.org/10.1016/j.neuron.2023.08.007

Zhang, X., Boccaletti, S., Guan, S., & Liu, Z. (2015). Explosive synchronization in adaptive and multilayer networks. *Physical Review Letters*, 114(3), 038701. https://doi.org/10.1103/PhysRevLett.114.038701

Zhang, Y., Wang, Y., Cheng, H., et al. (2023). EEG spectral slope: a reliable indicator for continuous evaluation of consciousness levels during propofol anesthesia. *NeuroImage*, 283, 120426. https://doi.org/10.1016/j.neuroimage.2023.120426

Zhang, Z., Ai, L., Zheng, X., et al. (2026). Wide-field Ca2+ imaging reveals cortical dynamics across propofol anesthesia and emergence in adult mice. *Anesthesia & Analgesia*. Advance online publication. https://doi.org/10.1213/ANE.0000000000008228

Zhao, Y., Ou, M., Liu, J., et al. (2024). Astrocytes Modulate a Specific Paraventricular Thalamus→Prefrontal Cortex Projection to Enhance Consciousness Recovery from Anesthesia. *Journal of Neuroscience*, 44(34), e1808232024. https://doi.org/10.1523/jneurosci.1808-23.2024

Zhou, B., Li, Q., Su, M., et al. (2025). Astrocyte morphological remodeling regulates consciousness state transitions induced by inhaled general anesthesia. *Molecular Psychiatry*, 30(9), 4006–4022. https://doi.org/10.1038/s41380-025-02978-2

Zhou, D. W., Mowrey, D. D., Tang, P., & Xu, Y. (2015). Percolation model of sensory transmission and loss of consciousness under general anesthesia. *Physical Review Letters*, 115(10), 108103. https://doi.org/10.1103/PhysRevLett.115.108103

Appendix B: Computational reference simulator (percolation model)

B.1 Purpose

The percolation model implements Section 4 exactly as written. The studies below check that the model has the properties the paper attributes to it and calibrate its one fitted parameter. A simulation cannot show that the brain works this way; it shows that the stated mechanism is internally consistent and produces the claimed behaviour. Every parameter belongs to one of the four classes of Appendix A.1: measured, calibrated, assumed or free. The comments in `simulator/dpc_simulator.py` (Appendix B.8) label both timing values "Assumed (Section 4.7)"; the arousal comment still calls 0.5 "the midpoint of the tested range", whereas it is the third of the four arousal levels tested (0, 0.3, 0.5 and 0.7), and this paper's classification is authoritative. Generated reports keep the labels they were written with; where a table in Appendices B and C gives a different kind for a check, the paper's table is authoritative (Appendix A.5).

B.2 Algorithms and software

Component	Method	Reference
Homogeneous network	Erdős-Rényi $G(n, m)$ random graph	Erdős and Rényi (1959)
Hub-dominated network	Barabási-Albert preferential attachment	Barabási and Albert (1999)
Giant component $F(\pi)$	Newman-Ziff algorithm: edges added in random order, giving the giant-component fraction for every π in one pass	Newman and Ziff (2000)
Connected components	Union-find with union by size and path compression	Tarjan (1975)
Self-consistent state	Fixed-point iteration of $S = F(\pi(c, S))$ from the previous state, so that induction and emergence follow their own branches	Section 4.3
Dose-response	Hill equation	Hill (1910)
Quorum threshold formula	Generating functions of the degree distribution	Callaway et al. (2000)
Statistics	Mean and sample standard deviation across seeds; the number of seeds in which a check holds, with its Wilson 95% interval	Wilson (1927)
Software	CPython; the percolation model and its validation, calibration and glial studies use the standard library only; the quorum thresholds, sensitivity analysis, oscillator, congestion and plant studies also use NumPy, SciPy and PyArrow, at the versions recorded in each report	

The code, its tests and the figures were written with Claude Opus 5.5 (Anthropic) through Claude Code under the authors' direction, as stated in the Declarations.

B.3 Calibration of the reference feedback

Source: `simulator/results/calibration_feedback_full_20261018T210709Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-18T21:07:09.189545+00:00
Finished (UTC)	2026-10-18T21:09:13.015266+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.15.0; standard library only
Git commit	`0677161701738895512e2b047b40f3857406a5d5` (working tree dirty: False)
SHA-256 `calibrate_feedback.py`	`fcdf7a99af69780b32150dbd9c12868165ce3d578fa27a6fc4ad6817ff56536f`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `provenance.py`	`94071e252347139bab3c5f113d52a38f65feaeaa0c03738ec8d81090c9139cb2`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`

Feedback λ calibrated to published inertia ratios (N = 10,000, 10 seeds)

Arousal level	Target ratio	λ (ms)	Model ratio	Reproduced	Source
0	1.07	5.43	1.069	yes	Warnaby et al. (2017), propofol, humans
0	1.08	5.64	1.080	yes	Friedman et al. (2010), isoflurane, mice (reference target)
0	1.11	6.06	1.109	yes	Iwakiri et al. (2005), propofol, humans
0	1.63	8.43	1.625	yes	Sepúlveda et al. (2018), propofol, humans
0	2.85	8.53	2.045	no	Friedman et al. (2010), halothane, mice
0.3	1.07	6.89	1.070	yes	Warnaby et al. (2017), propofol, humans
0.3	1.08	7.18	1.079	yes	Friedman et al. (2010), isoflurane, mice (reference target)
0.3	1.11	7.93	1.109	yes	Iwakiri et al. (2005), propofol, humans
0.3	1.63	11.92	1.627	yes	Sepúlveda et al. (2018), propofol, humans
0.3	2.85	12.19	2.085	no	Friedman et al. (2010), halothane, mice
0.5	1.07	8.49	1.069	yes	Warnaby et al. (2017), propofol, humans
0.5	1.08	8.85	1.079	yes	Friedman et al. (2010), isoflurane, mice (reference target)
0.5	1.11	10.06	1.110	yes	Iwakiri et al. (2005), propofol, humans
0.5	1.63	16.52	1.623	yes	Sepúlveda et al. (2018), propofol, humans
0.5	2.85	17.07	2.213	no	Friedman et al. (2010), halothane, mice
0.7	1.07	11.32	1.070	yes	Warnaby et al. (2017), propofol, humans
0.7	1.08	12.08	1.079	yes	Friedman et al. (2010), isoflurane, mice (reference target)
0.7	1.11	14.13	1.109	yes	Iwakiri et al. (2005), propofol, humans
0.7	1.63	27.06	1.627	yes	Sepúlveda et al. (2018), propofol, humans
0.7	2.85	28.45	2.402	no	Friedman et al. (2010), halothane, mice

B.4 Validation across network sizes

Network sizes run from 302 nodes, the number of *C. elegans* neurons (White et al., 1986), and 1,000, the finest common cortical parcellation (Schaefer et al., 2018), to 1,000,000. Claims C1-C7 are properties the paper attributes to the model; observations O1-O5 and extension tests E1-E3 are reported without entering the verdict. Each check is labelled with one of the four kinds of Appendix A.1: a test, which the model could fail; a construction check, which could not fail at the settings chosen (C1, limitation 25); a consistency check, which follows from the model's equations (C2 to C7); or a post hoc analysis. The generated report labels C1, C4, C5 and C6 "test"; the kinds below are authoritative. Every count of seeds is given with its Wilson 95% confidence interval (Wilson, 1927); a check is said to hold only when it holds in every seed. Source: `simulator/results/validation_full_20261019T113035Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-19T11:30:35+00:00
Finished (UTC)	2026-10-19T11:42:10+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.15.0; standard library only
Git commit	`d8b06eb7b3df89d97b0e5aa7570ac537e4627cfe` (working tree dirty: True)
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`
SHA-256 `validation_study.py`	`41c52312248b55a561b4e95113c02cc99664085bbdcecc6520365cb99a723f58`
Wall time	695.5 s
Worker processes	10
Seeds per size	20 (base seed 20261017)

Claims (seeds in which the claim holds)

ID	Statement	Kind	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
C1	Omission-fault and timing-fault agents both abolish quorum (Section 6)	construction check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
C2	Without reference feedback (lambda = 0) there is no hysteresis (Section 4.3)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
C3	Hysteresis (neural inertia) grows with lambda; no recovery counts as unbounded (prediction 5)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
C4	Reference feedback sharpens the loss-of-consciousness transition (prediction 1)	consistency check	19/20 (76–99%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
C5	Hub lesions lower the loss concentration more than random lesions (prediction 6)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
C6	Reduced arousal delays emergence more than it shifts induction (Kelz et al., 2008)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
C7	Without the agent's action on arousal circuits, quorum is still lost but at a higher concentration (Moore et al., 2012; Eikermann et al., 2011)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)

Observations (seeds in which the observation occurs; not part of the verdict)

ID	Statement	Kind	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
O1	Trapped unconscious at zero concentration, lambda = 16.52 ms at the reference arousal level	test	4/20 (8–42%)	2/20 (3–30%)	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)
O2	Trapped unconscious at zero concentration after hub lesion, lambda = 8.85 ms	test	4/20 (8–42%)	4/20 (8–42%)	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)
O3	Sleep with population-level down-states at the rat share (0.095) loses quorum	test	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)
O4	Sleep with population-level down-states at the human share (0.25) loses quorum	test	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
O5	Sleep with one typical human slow wave silencing its measured share of regions (0.271) loses quorum (Nir et al., 2011)	test	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)

Extension tests (seeds in which the pre-specified test passes; not part of the verdict)

ID	Statement	Kind	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
E1	Sleep (no arousal, connection-level down-states, share 0.095, rat) has a wakeable stable unconscious state	test	15/20 (53–89%)	15/20 (53–89%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
E2	Deeper sleep (connection-level down-states, share 0.25, human upper estimate) has a wakeable stable unconscious state	test	18/20 (70–97%)	17/20 (64–95%)	18/20 (70–97%)	20/20 (84–100%)	20/20 (84–100%)
E3	Deep anaesthesia has only an unconscious state, so no perturbation restores quorum	test	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)

Thresholds (mean ± SD across seeds, relative concentration units)

Quantity	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
c_off, GABAergic-like	0.909 ± 0.035	0.895 ± 0.019	0.886 ± 0.008	0.887 ± 0.005	0.881 ± 0.004
c_on, GABAergic-like	0.823 ± 0.052	0.826 ± 0.035	0.824 ± 0.012	0.825 ± 0.008	0.827 ± 0.005
Hysteresis, GABAergic-like	0.085 ± 0.037	0.069 ± 0.041	0.061 ± 0.013	0.061 ± 0.010	0.054 ± 0.005
Transition width, GABAergic-like	0.291 ± 0.102	0.246 ± 0.031	0.240 ± 0.014	0.241 ± 0.005	0.232 ± 0.004
Transmission failure p at c_off, GABAergic-like	0.385 ± 0.035	0.371 ± 0.019	0.362 ± 0.008	0.363 ± 0.005	0.358 ± 0.004
Effective jitter (ms) at c_off, GABAergic-like	3.880 ± 0.231	3.853 ± 0.177	3.871 ± 0.167	3.864 ± 0.172	3.707 ± 0.148
Coherence probability π at c_off, GABAergic-like	0.186 ± 0.015	0.191 ± 0.011	0.193 ± 0.009	0.193 ± 0.009	0.202 ± 0.008

Quantity	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
Share of populations silent at once for sleep to lose quorum	0.168 ± 0.032	0.171 ± 0.013	0.155 ± 0.005	0.152 ± 0.003	0.152 ± 0.000
c_off, NMDA-antagonist-like	0.626 ± 0.020	0.624 ± 0.009	0.622 ± 0.004	0.621 ± 0.002	0.620 ± 0.000
c_off, hub graph intact	0.913 ± 0.043	0.918 ± 0.018	0.904 ± 0.008	0.903 ± 0.004	0.900 ± 0.000
c_off, random lesion	0.890 ± 0.038	0.865 ± 0.019	0.858 ± 0.010	0.855 ± 0.005	0.860 ± 0.002
c_off, hub lesion	0.545 ± 0.136	0.512 ± 0.073	0.442 ± 0.029	0.451 ± 0.010	0.447 ± 0.004
c_on, hub lesion	0.521 ± 0.115 (16/20)	0.459 ± 0.056 (16/20)	0.429 ± 0.027	0.450 ± 0.011	0.447 ± 0.005
Seconds per seed	0.036 ± 0.007	0.073 ± 0.013	0.882 ± 0.086	15.717 ± 1.188	319.851 ± 24.713

Verdict: At least one claim failed; see the claims table.

B.5 Quorum thresholds of empirical connectomes

Source: `simulator/results/percolation_full_20261020T011339Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T01:13:39.951591+00:00
Finished (UTC)	2026-10-20T01:14:31.518881+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1, pyarrow 26.0.0
Git commit	`38e86cbaa5c084c7a9d05a0da2bf4d08648ada80` (working tree dirty: False)
SHA-256 `calibration.py`	`f0760a3875a9007af2640417c43eb9c99b6a2898e63ac5ce1a5b6c915a502f8d`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `percolation_thresholds.py`	`68e2a92cf39b833e3f277c3039404d0a04c51c36d22ce28bf36e63b3633a7bd7`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`1b04ef76f1fa0860e6a7daca5ea54dab72234bd9e11855efa616f248ad91c935`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`

Percolation and quorum thresholds (connectomes treated as undirected)

Organism	Nodes	Connections	Mean degree	Onset π_c (formula)	Quorum π_q, $q^* = 0.5$ (formula)	Quorum π_q (measured)	Critical failure p_q (formula / measured)	Critical jitter σ_q, ms (formula / measured)
C. elegans	300	3,465	23.1	0.0326	0.0568	0.0579 ± 0.0054	0.852 / 0.849	21.1 / 20.6
Drosophila	138,639	2,511,789	36.2	0.0053	0.0415	0.0438 ± 0.0002	0.892 / 0.886	28.8 / 27.3
Human	998	17,865	35.8	0.0239	0.0368	0.0381 ± 0.0022	0.904 / 0.901	32.5 / 31.4

Quorum threshold on the directed wiring (largest strongly connected component)

Organism	Quorum π_q, undirected	Quorum π_q, directed (strong component)	Critical failure p_q	Critical jitter σ_q, ms
C. elegans	0.0579	0.1058 ± 0.0079	0.724	11.3
Drosophila	0.0438	0.1503 ± 0.0006	0.608	7.9

Organism	Quorum π_q, undirected	Quorum π_q, directed (strong component)	Critical failure p_q	Critical jitter σ_q, ms
Human	0.0381	0.0495 ± 0.0015	0.871	24.2

Notes. The measured quorum threshold lies 2.0%, 5.5% and 3.4% above the generating-function estimate on the *C. elegans*, *Drosophila* and human connectomes (10 draws each); the gap is within sampling error for *C. elegans* and the human connectome (0.7 and 1.8 standard errors) and not for *Drosophila*. The ratio π_q/π_c is 1.7, 7.9 and 1.5, so the quorum threshold at $q^* = 0.5$ is not the percolation critical point (Section 4.2). The coefficient of variation of the undirected degree, $\sqrt{(1/\pi_c + 1)/\langle k \rangle - 1}$ from the columns above, is 0.61, 2.1 and 0.44: only the fly connectome is strongly heavy-tailed. Each undirected link here has one failure draw and one timing draw; the readout of Appendix C.12 instead gave each human link two (one per stored direction, either sufficing), which is why its human result differs from the static threshold (Appendix C.13).

B.6 Sensitivity to assumed and free parameters

Each assumed or free parameter is varied with the others held at their reference values. Source: `simulator/results/sensitivity_full_20261020T110144Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T11:01:44.960633+00:00
Finished (UTC)	2026-10-20T11:38:55.373135+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1
Git commit	`7ce71a8c1e3d7b2becfe0d5eeb5a91593a3f3702` (working tree dirty: False)
SHA-256 `calibration.py`	`5cdcad39d08837ed395d1f6dbd4c5d935fe24b8632b9814c1b672ee8c4146bb1`
SHA-256 `comparisons.py`	`340b044083ffa2e8f98988da8fb400fa25c961db5f08c4bce42e5681441b92c8`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`621859b0bca3cc34cedd82e1559e1097bf90142d698e8c7d95b3e0907417c452`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`e9dac3d0f0c2c9baa8e813f65ea71615e65a027edd4d3115e86e556203ece030`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`78a8c5e399d134fdcf01a2af3be99396ff3a3eb25ebb62915bcb27ec8635fc37`
SHA-256 `sensitivity_study.py`	`daea6c16bd09a62045dc06d6a25dcd796cb82dae7776925b5d06e0b841592911`
SHA-256 `species_study.py`	`423902f7037b6735db60c2826cfe212cf0c664859aeeda83317d9ea3421217ba`
SHA-256 `validation_study.py`	`70b5ecf19ef90acbd8d2b4729a130debe15094411a6c01f8bdbd2ca69a21f050`

Percolation model: claims holding across seeds (N = 10,000)

Variation	C1	C2	C3	C4	C5	C6	C7
Quorum fraction = 0.3	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Quorum fraction = 0.5 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Quorum fraction = 0.7	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	0/10 (0–28%)	10/10 (72–100%)	10/10 (72–100%)
Arousal level (λ recalibrated) = 0	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	0/10 (0–28%)	n/a	n/a
Arousal level (λ recalibrated) = 0.3	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	4/10 (17–69%)	10/10 (72–100%)	10/10 (72–100%)
Arousal level (λ recalibrated) = 0.5 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Arousal level (λ recalibrated) = 0.7	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)

Variation	C1	C2	C3	C4	C5	C6	C7
Feedback λ (ms), one per reproduced inertia ratio = 8.49	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Feedback λ (ms), one per reproduced inertia ratio = 8.85 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Feedback λ (ms), one per reproduced inertia ratio = 10.06	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	4/10 (17–69%)	10/10 (72–100%)	10/10 (72–100%)
Feedback λ (ms), one per reproduced inertia ratio = 16.52	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	0/10 (0–28%)	0/10 (0–28%)	10/10 (72–100%)
Hill coefficient = 2	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Hill coefficient = 4 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Hill coefficient = 8	10/10 (72–100%)	10/10 (72–100%)	9/10 (60–98%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Mean degree = 4	0/10 (0–28%)	10/10 (72–100%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)
Mean degree = 8 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Mean degree = 16	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Integration window W (ms) = 1	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)
Integration window W (ms) = 3 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Integration window W (ms) = 10	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Baseline jitter (ms) = 1	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Baseline jitter (ms) = 3 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Baseline jitter (ms) = 10	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)
Maximal transmission failure, GABAergic-like = 0.6	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	0/10 (0–28%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Maximal transmission failure, GABAergic-like = 0.8	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)
Maximal transmission failure, GABAergic-like = 0.95 (reference)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)	10/10 (72–100%)

Phase-oscillator model, human connectome: pillars holding across seeds

Variation	T2a	T2b	T3
Synaptic coupling K = 2	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)
Synaptic coupling K = 4 (reference)	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)
Synaptic coupling K = 8	5/5 (57–100%)	5/5 (57–100%)	5/5 (57–100%)
Natural-frequency spread = 0.25	0/5 (0–43%)	5/5 (57–100%)	0/5 (0–43%)
Natural-frequency spread = 0.5 (reference)	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)
Natural-frequency spread = 1	5/5 (57–100%)	5/5 (57–100%)	2/5 (12–77%)
Noise intensity D = 0.02	5/5 (57–100%)	5/5 (57–100%)	4/5 (38–96%)
Noise intensity D = 0.05 (reference)	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)
Noise intensity D = 0.1	5/5 (57–100%)	5/5 (57–100%)	4/5 (38–96%)
Quorum fraction = 0.3	5/5 (57–100%)	5/5 (57–100%)	5/5 (57–100%)
Quorum fraction = 0.5 (reference)	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)
Quorum fraction = 0.7	5/5 (57–100%)	5/5 (57–100%)	5/5 (57–100%)
Jitter correlation time τ = 0.25	5/5 (57–100%)	5/5 (57–100%)	4/5 (38–96%)
Jitter correlation time τ = 1 (reference)	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)

Variation	T2a	T2b	T3
Jitter correlation time τ = 4	5/5 (57–100%)	5/5 (57–100%)	3/5 (23–88%)

B.7 Glial regulation (postulate P9)

The glial extensions of Section 4.6 run on the same networks and seeds. The checks were fixed before the multi-seed run; a single exploratory seed at N = 1,000 was run beforehand to confirm that the code executes, and its results were seen before the checks were written, so the checks are not blind to the direction of the effects. Source: `simulator/results/glia_full_20261020T011431Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T01:14:31+00:00
Finished (UTC)	2026-10-20T01:14:38+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.15.0; standard library only
Git commit	`38e86cbaa5c084c7a9d05a0da2bf4d08648ada80` (working tree dirty: False)
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `glial_regulation.py`	`6a83e08c5caeb89942041d31bb0384932187e405f266d06ba78c9df07ac81e86`
SHA-256 `glial_study.py`	`00f4ea0fe4c4c671ffa2c6b56e3c3afdb81233603809840fa449c3462f141295`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`
SHA-256 `validation_study.py`	`41c52312248b55a561b4e95113c02cc99664085bbdcecc6520365cb99a723f58`
Wall time	6.4 s
Worker processes	10
Seeds per size	20 (base seed 20261017)

Pre-specified tests (seeds in which the test passes)

ID	Statement	Kind	N = 302	N = 1,000	N = 10,000	N = 100,000
M1	Damaged myelin lowers the loss concentration (integrity 1.0 > 0.75 > 0.5)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
G1	Removing the microglial loop hastens emergence (direction of Cao et al., 2023, P2Y12 loss)	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
G2	Removing the microglial loop does not hasten induction	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
G3	Neural inertia grows with the strength of the microglial loop	consistency check	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)	20/20 (84–100%)
A1	Re-specified after the down-state model changed: with global down-states (0.5 of populations), sleep loses quorum, and silencing astrocytes' share (0.5) of them restores it	construction check (could not pass)	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)	0/20 (0–16%)

Thresholds (mean ± SD across seeds, relative concentration units)

Quantity	N = 302	N = 1,000	N = 10,000	N = 100,000
c_off, myelin integrity 1.0	0.909 ± 0.035	0.895 ± 0.019	0.886 ± 0.008	0.887 ± 0.005
c_off, myelin integrity 0.75	0.834 ± 0.041	0.827 ± 0.022	0.818 ± 0.009	0.821 ± 0.002
c_off, myelin integrity 0.5	0.751 ± 0.054	0.745 ± 0.025	0.739 ± 0.012	0.743 ± 0.005

Quantity	N = 302	N = 1,000	N = 10,000	N = 100,000
c_off, microglial gain 0.0	0.909 ± 0.035	0.895 ± 0.019	0.886 ± 0.008	0.887 ± 0.005
c_on, microglial gain 0.0	0.823 ± 0.052	0.826 ± 0.035	0.824 ± 0.012	0.825 ± 0.008
c_off, microglial gain 0.25	0.901 ± 0.038	0.886 ± 0.020	0.875 ± 0.008	0.875 ± 0.005
c_on, microglial gain 0.25	0.782 ± 0.061	0.785 ± 0.045	0.775 ± 0.018	0.773 ± 0.011
c_off, microglial gain 0.5	0.891 ± 0.037	0.878 ± 0.019	0.865 ± 0.008	0.866 ± 0.005
c_on, microglial gain 0.5	0.732 ± 0.074	0.731 ± 0.063	0.714 ± 0.023	0.709 ± 0.014
Seconds per seed	0.011 ± 0.004	0.017 ± 0.005	0.149 ± 0.019	2.665 ± 0.246

Verdict: At least one pre-specified test failed; see the table.

B.8 Listing: `simulator/dpc_simulator.py`

```
"""Reference simulator for the Distributed Perceptual Consensus (DPC) hypothesis.

Implements the formal model of Section 4 of the white paper:

    pi(c, S)  = (1 - p(c)) * erf(W / (2 * sqrt(2) * sigma_eff))
    sigma_eff = sqrt(sigma_b^2 + sigma_agent(c)^2 + sigma_ref(c, S)^2)
    sigma_ref = lambda * (1 - S) * (1 - A(c, S))
    S         = F(pi(c, S))

The three timing errors are independent, so their standard deviations add in quadrature.
F is the giant-component fraction of the network under bond percolation, computed exactly
for a fixed set of random edge draws with the Newman-Ziff algorithm, so every fixed-point
evaluation is a lookup rather than a new Monte Carlo run. Sleep down-states silence whole
populations (Nir et al., 2011, report that slow waves are often regional), so they remove
nodes rather than individual connections.

The simulator shows whether the model behaves as the paper claims. It is not evidence about
the brain. Each parameter below is labelled assumed, calibrated or free.

Standard library only. Run with: python3 dpc_simulator.py
"""

from __future__ import annotations

import bisect
import math
import random
from dataclasses import dataclass, replace
from enum import Enum
from functools import lru_cache
from typing import Callable, Protocol, Sequence, TypeVar

Edge = tuple[int, int]


# ---------------------------------------------------------------------------
# Reference parameters
#
# Assumed: set from a published estimate (source given), not measured for this model.
# Calibrated: fitted to one published measurement (target given), then held fixed.
# Free: not measurable at this level of description; results are reported over a range.
# ---------------------------------------------------------------------------

REFERENCE_SEED = 20261017
# Free: illustrative random topology; empirical connectomes are used in the species study.
REFERENCE_NODE_COUNT = 2_000
REFERENCE_MEAN_DEGREE = 8.0
REFERENCE_EDGES_PER_NEW_NODE = 4  # preferential attachment; gives mean degree 8

# Assumed (Section 4.7): the window for integrating thalamic input in rodent barrel cortex spans 1-10 ms
# (Gabernet et al., 2005), and it is under 2 ms in hippocampal pyramidal cells (Pouille & Scanziani,
# 2001); 3 ms lies within that range, and the sensitivity analysis covers 1-10 ms.
REFERENCE_INTEGRATION_WINDOW_MS = 3.0
# Assumed (Section 4.7): in area MT of alert macaques, responses to time-varying stimuli are rate-modulated with
# a precision of about 3 ms (Buracas et al., 1998);
# single neurons in vitro reach under 1 ms (Mainen & Sejnowski, 1995). No trial-to-trial timing
# variability between cortical areas has been measured, so the sensitivity analysis covers 1-10 ms.
```

```
REFERENCE_BASELINE_JITTER_MS = 3.0
# Free: share of the timing reference sustained by wake-active arousal systems. The measured
# inertia ratios from 1.07 to 1.63 are reproduced at every level tested (calibrate_feedback.py), so
# they do not determine it; the midpoint of the tested range is used and results are reported across it.
REFERENCE_AROUSAL_LEVEL = 0.5
# Measured: loss-to-recovery concentration ratio for isoflurane in mice (Friedman et al., 2010), the
# reference target; the other published ratios give the sensitivity range (calibrate_feedback.py).
REFERENCE_INERTIA_RATIO = 1.08
# Calibrated: λ reproducing REFERENCE_INERTIA_RATIO at each arousal level tested (calibrate_feedback.py;
# test_dpc_simulator checks that the stored values still reproduce the ratio).
CALIBRATED_FEEDBACK_MS_BY_AROUSAL = {0.0: 5.64, 0.3: 7.18, 0.5: 8.85, 0.7: 12.08}
REFERENCE_FEEDBACK_JITTER_MS = CALIBRATED_FEEDBACK_MS_BY_AROUSAL[REFERENCE_AROUSAL_LEVEL]
# Free.
REFERENCE_QUORUM_FRACTION = 0.5

# Concentrations are expressed relative to each agent's half-maximal effective concentration (EC50).
REFERENCE_MAX_RELATIVE_CONCENTRATION = 3.0
REFERENCE_CONCENTRATION_STEP = 0.01

REFERENCE_LESION_FRACTION = 0.05
# No feedback, then the values calibrated at the reference arousal level to each published
# inertia ratio (calibrate_feedback.py): 1.07 (Warnaby et al., 2017), 1.08 (Friedman et al., 2010,
# isoflurane), 1.11 (Iwakiri et al., 2005) and 1.63 (Sepulveda et al., 2018). The halothane ratio of
# 2.85 (Friedman et al., 2010) is not reproduced at any arousal level: before the mean ratio reaches
# it, some seeds no longer recover at zero concentration.
REFERENCE_FEEDBACK_SERIES_MS = (0.0, 8.49, 8.85, 10.06, 16.52)
# Free: share of arousal drive left after orexinergic neurons are lost (Kelz et al., 2008);
# other arousal systems remain, so the share is partial rather than zero.
OREXIN_ABLATION_REMAINING_SHARE = 0.5
# Free: natural sleep modelled as complete withdrawal of the arousal drive, with no agent.
SLEEP_AROUSAL_LEVEL = 0.0
# Derived from reported means: OFF periods of 85.8 ms between ON periods of 815.5 ms in rat
# NREM sleep (Vyazovskiy et al., 2009), so 85.8 / (85.8 + 815.5) of the time is spent in OFF periods.
SLEEP_DOWN_STATE_FRACTION = 0.095
# Upper estimate for human NREM sleep, combining two studies: down-states of about 400 ms
# (Cash et al., 2009) at 37.3 slow waves per minute (Nir et al., 2011).
HUMAN_SLEEP_DOWN_STATE_FRACTION_UPPER = 0.25
# A concentration well beyond loss of quorum, used to test whether a perturbation can restore it.
DEEP_ANAESTHESIA_CONCENTRATION = 2.0

# Transition width: concentration span over which S falls from near-awake to near-collapsed.
TRANSITION_UPPER_FRACTION = 0.9
TRANSITION_LOWER_FRACTION = 0.1

SOLVER_TOLERANCE = 1e-9
SOLVER_MAX_ITERATIONS = 500


# ---------------------------------------------------------------------------
# Network
# ---------------------------------------------------------------------------


@dataclass(frozen=True)
class Graph:
    """Undirected graph of neural populations (nodes) and functional connections (edges)."""

    node_count: int
    edges: tuple[Edge, ...]

    def degrees(self) -> list[int]:
        degrees = [0] * self.node_count
        for source, target in self.edges:
            degrees[source] += 1
            degrees[target] += 1
        return degrees

    def without_nodes(self, removed: set[int]) -> Graph:
        """Lesioned nodes stay in the count, so they remain outside any coherent component."""
        kept = tuple(edge for edge in self.edges if edge[0] not in removed and edge[1] not in removed)
        return Graph(self.node_count, kept)


class GraphBuilder(Protocol):
```

```
    node_count: int

    def build(self, rng: random.Random) -> Graph: ...

    def resized(self, node_count: int) -> GraphBuilder: ...

    def with_mean_degree(self, mean_degree: float) -> GraphBuilder: ...


@dataclass(frozen=True)
class RandomGraphBuilder:
    """Erdos-Renyi G(n, m) graph with homogeneous (Poisson) degree distribution."""

    node_count: int
    mean_degree: float

    def build(self, rng: random.Random) -> Graph:
        target_edge_count = round(self.node_count * self.mean_degree / 2)
        edges: set[Edge] = set()
        while len(edges) < target_edge_count:
            source, target = rng.randrange(self.node_count), rng.randrange(self.node_count)
            if source != target:
                edges.add((min(source, target), max(source, target)))
        return Graph(self.node_count, tuple(sorted(edges)))

    def resized(self, node_count: int) -> RandomGraphBuilder:
        return replace(self, node_count=node_count)

    def with_mean_degree(self, mean_degree: float) -> RandomGraphBuilder:
        return replace(self, mean_degree=mean_degree)


@dataclass(frozen=True)
class PreferentialAttachmentGraphBuilder:
    """Barabasi-Albert graph: heavy-tailed degrees with a small set of hubs."""

    node_count: int
    edges_per_new_node: int

    def build(self, rng: random.Random) -> Graph:
        seed_size = self.edges_per_new_node + 1
        edges = [(source, target) for source in range(seed_size) for target in range(source + 1, seed_size)]
        endpoint_pool = [node for edge in edges for node in edge]
        for new_node in range(seed_size, self.node_count):
            targets: set[int] = set()
            while len(targets) < self.edges_per_new_node:
                targets.add(rng.choice(endpoint_pool))
            for target in targets:
                edges.append((target, new_node))
                endpoint_pool.extend((target, new_node))
        return Graph(self.node_count, tuple(edges))

    def resized(self, node_count: int) -> PreferentialAttachmentGraphBuilder:
        return replace(self, node_count=node_count)

    def with_mean_degree(self, mean_degree: float) -> PreferentialAttachmentGraphBuilder:
        """Each new node brings this many edges, so the mean degree is about twice the value."""
        return replace(self, edges_per_new_node=max(1, round(mean_degree / 2)))


# ---------------------------------------------------------------------------
# Percolation
# ---------------------------------------------------------------------------


class DisjointSet:
    """Union-find that tracks the size of the largest component."""

    def __init__(self, element_count: int) -> None:
        self._parent = list(range(element_count))
        self._size = [1] * element_count
        self.largest_component_size = 1 if element_count else 0

    def find(self, element: int) -> int:
```

```
        root = element
        while self._parent[root] != root:
            root = self._parent[root]
        while self._parent[element] != root:
            self._parent[element], element = root, self._parent[element]
        return root

    def union(self, first: int, second: int) -> None:
        first_root, second_root = self.find(first), self.find(second)
        if first_root == second_root:
            return
        if self._size[first_root] < self._size[second_root]:
            first_root, second_root = second_root, first_root
        self._parent[second_root] = first_root
        self._size[first_root] += self._size[second_root]
        self.largest_component_size = max(self.largest_component_size, self._size[first_root])


class PercolationCurve:
    """Exact giant-component fraction F(pi) for one fixed draw of edge thresholds (Newman-Ziff).

    Each edge receives a uniform threshold u. At occupation probability pi the
    coherent edges are those with u < pi, so F is deterministic and monotone in pi.
    """

    def __init__(self, sorted_thresholds: Sequence[float], giant_fractions: Sequence[float]) -> None:
        self._sorted_thresholds = list(sorted_thresholds)
        self._giant_fractions = list(giant_fractions)

    @classmethod
    def from_graph(cls, graph: Graph, rng: random.Random) -> PercolationCurve:
        threshold_by_edge = sorted((rng.random(), edge) for edge in graph.edges)
        components = DisjointSet(graph.node_count)
        giant_fractions = [components.largest_component_size / graph.node_count]
        for _, (source, target) in threshold_by_edge:
            components.union(source, target)
            giant_fractions.append(components.largest_component_size / graph.node_count)
        return cls([threshold for threshold, _ in threshold_by_edge], giant_fractions)

    def giant_fraction(self, occupation_probability: float) -> float:
        occupied_edge_count = bisect.bisect_left(self._sorted_thresholds, occupation_probability)
        return self._giant_fractions[occupied_edge_count]


# ---------------------------------------------------------------------------
# Pharmacodynamics and timing
# ---------------------------------------------------------------------------


@dataclass(frozen=True)
class AnaestheticAgent:
    """Circuit-level effect of an anaesthetic class (postulate P4), with a Hill dose-response.

    arousal_suppression is the share of the agent's effect that also acts on wake-promoting
    arousal circuits (1: fully; isoflurane and sevoflurane reduce c-Fos in orexinergic neurons,
    Kelz et al., 2008, and isoflurane depolarises sleep-promoting VLPO neurons, Moore et al., 2012).
    """

    name: str
    max_transmission_failure: float
    max_added_jitter_ms: float
    half_effect_concentration: float = 1.0
    hill_coefficient: float = 4.0
    arousal_suppression: float = 1.0

    def effect_fraction(self, concentration: float) -> float:
        if concentration <= 0:
            return 0.0
        scaled = (concentration / self.half_effect_concentration) ** self.hill_coefficient
        return scaled / (1 + scaled)

    def transmission_failure(self, concentration: float) -> float:
        return self.max_transmission_failure * self.effect_fraction(concentration)
```

```
    def added_jitter_ms(self, concentration: float) -> float:
        return self.max_added_jitter_ms * self.effect_fraction(concentration)


GABAERGIC_LIKE_AGENT = AnaestheticAgent("GABAergic-like (omission faults)", max_transmission_failure=0.95,
max_added_jitter_ms=1.0)
NMDA_ANTAGONIST_LIKE_AGENT = AnaestheticAgent("NMDA-antagonist-like (timing faults)", max_transmission_failure=0.2,
max_added_jitter_ms=40.0)


@dataclass(frozen=True)
class TimingParameters:
    """Integration window W, intrinsic jitter and reference feedback lambda (P2, P5)."""

    integration_window_ms: float
    baseline_jitter_ms: float
    feedback_jitter_ms: float


class DownStates(Enum):
    """What a slow-oscillation down-state silences.

    POPULATIONS: whole populations fall silent together, as down-states are population-wide
    OFF periods (Vyazovskiy et al., 2009) that are often regional (Nir et al., 2011).
    CONNECTIONS: each connection fails independently, the alternative in which signals stop
    crossing between regions (Section 5.6).
    """

    POPULATIONS = "populations"
    CONNECTIONS = "connections"


@dataclass(frozen=True)
class BrainState:
    """Physiological state independent of any agent (P7).

    arousal_level: share of the timing reference that wake-active arousal systems sustain on
    their own, independent of cortical coherence (0 none, 1 the reference never weakens).
    down_state_fraction: share of populations (or connections) silenced by slow-oscillation
    down-states at any moment (0 in wakefulness).
    """

    arousal_level: float
    down_state_fraction: float = 0.0
    down_states: DownStates = DownStates.POPULATIONS

    def silenced_connection_share(self) -> float:
        return self.down_state_fraction if self.down_states is DownStates.CONNECTIONS else 0.0

    def silenced_population_share(self) -> float:
        return self.down_state_fraction if self.down_states is DownStates.POPULATIONS else 0.0


def sustained_arousal(arousal_level: float, agent: AnaestheticAgent, concentration: float) -> float:
    """Share of the reference sustained by arousal, suppressed by the agent (Kelz et al., 2008; Moore et al., 2012).

    Shared by the percolation and phase-oscillator models so that both use one definition (P7).
    """
    return arousal_level * (1 - agent.arousal_suppression * agent.effect_fraction(concentration))


def within_window(window_ms: float, jitter_ms: float) -> float:
    """Probability that a Gaussian timing error of SD jitter_ms falls within ±window_ms / 2."""
    return math.erf(window_ms / (2 * math.sqrt(2) * jitter_ms))


class CoherenceModel:
    """Probability that a functional connection is timing-coherent (Section 4.1)."""

    def __init__(self, timing: TimingParameters, agent: AnaestheticAgent, state: BrainState) -> None:
        self._timing = timing
        self._agent = agent
        self._state = state
```

```
    def arousal(self, concentration: float) -> float:
        """Arousal drive, suppressed by the agent, as anaesthetics suppress orexinergic neurons (Kelz et al.,
2008)."""
        return sustained_arousal(self._state.arousal_level, self._agent, concentration)

    def arousal_at(self, concentration: float, coherent_fraction: float) -> float:
        """Arousal drive in a given network state; independent of the state here, overridden by extensions."""
        return self.arousal(concentration)

    def reference_error_ms(self, concentration: float, coherent_fraction: float) -> float:
        """Timing error added by a weakened reference: grows as the giant component shrinks, unless arousal sustains
it."""
        unsustained = 1 - self.arousal_at(concentration, coherent_fraction)
        return self._timing.feedback_jitter_ms * (1 - coherent_fraction) * unsustained

    def effective_jitter_ms(self, concentration: float, coherent_fraction: float) -> float:
        """Independent timing errors combine in quadrature."""
        return math.hypot(
            self._timing.baseline_jitter_ms,
            self._agent.added_jitter_ms(concentration),
            self.reference_error_ms(concentration, coherent_fraction),
        )

    def coherence_probability(self, concentration: float, coherent_fraction: float) -> float:
        jitter_ms = self.effective_jitter_ms(concentration, coherent_fraction)
        transmits = (1 - self._agent.transmission_failure(concentration)) * (1 -
self._state.silenced_connection_share())
        return transmits * within_window(self._timing.integration_window_ms, jitter_ms)


# ----------------------------------------------------------------------------
# Self-consistent giant component and concentration sweeps
# ----------------------------------------------------------------------------


class CoherentCoreSolver:
    """Solves S = F(pi(c, S)) by fixed-point iteration from a given starting state.

    Because F and pi are both monotone in S, iteration from the previous state
    converges to the nearest stable branch, which is what produces hysteresis.
    """

    def __init__(self, curve: PercolationCurve, coherence: CoherenceModel) -> None:
        self._curve = curve
        self._coherence = coherence

    def solve(self, concentration: float, initial_fraction: float) -> float:
        fraction = initial_fraction
        for _ in range(SOLVER_MAX_ITERATIONS):
            occupation = self._coherence.coherence_probability(concentration, fraction)
            next_fraction = self._curve.giant_fraction(occupation)
            if abs(next_fraction - fraction) < SOLVER_TOLERANCE:
                return next_fraction
            fraction = next_fraction
        raise RuntimeError(f"Giant component did not converge at concentration {concentration:.3f}")


@dataclass(frozen=True)
class ConcentrationGrid:
    maximum: float
    step: float

    def ascending(self) -> list[float]:
        point_count = round(self.maximum / self.step)
        return [index * self.step for index in range(point_count + 1)]


class HasConcentration(Protocol):
    concentration: float


Point = TypeVar("Point", bound=HasConcentration)
```

```
def loss_concentration(induction: Sequence[Point], is_conscious: Callable[[Point], bool]) -> float | None:
    """Highest concentration on induction before quorum is first lost (c_off); None if never lost or never held.

    Both thresholds are the last conscious grid point on their branch, so a system without
    bistability reports exactly zero hysteresis. Shared by every model in this directory.
    """
    first_unconscious = next((index for index, point in enumerate(induction) if not is_conscious(point)), None)
    if first_unconscious is None or first_unconscious == 0:
        return None
    return induction[first_unconscious - 1].concentration


def recovery_concentration(emergence: Sequence[Point], is_conscious: Callable[[Point], bool]) -> float | None:
    """Highest concentration on emergence at which quorum has returned (c_on)."""
    return next((point.concentration for point in emergence if is_conscious(point)), None)


def hysteresis_width(loss: float | None, recovery: float | None) -> float | None:
    return None if loss is None or recovery is None else loss - recovery


@dataclass(frozen=True)
class SweepPoint:
    concentration: float
    coherent_fraction: float
    is_conscious: bool


@dataclass(frozen=True)
class SweepResult:
    induction: tuple[SweepPoint, ...]
    emergence: tuple[SweepPoint, ...]

    @property
    def loss_concentration(self) -> float | None:
        return loss_concentration(self.induction, lambda point: point.is_conscious)

    @property
    def recovery_concentration(self) -> float | None:
        return recovery_concentration(self.emergence, lambda point: point.is_conscious)

    @property
    def hysteresis_width(self) -> float | None:
        return hysteresis_width(self.loss_concentration, self.recovery_concentration)

    @property
    def transition_width(self) -> float | None:
        """Induction span between the last S >= 0.9 and the first S < 0.1; small means sharp."""
        last_high = max(
            (point.concentration for point in self.induction if point.coherent_fraction >=
TRANSITION_UPPER_FRACTION),
            default=None,
        )
        first_low = next(
            (point.concentration for point in self.induction if point.coherent_fraction < TRANSITION_LOWER_FRACTION),
            None,
        )
        if last_high is None or first_low is None:
            return None
        return first_low - last_high


class AnaestheticSweep:
    """Raises concentration from zero (induction), then lowers it again (emergence)."""

    AWAKE_FRACTION = 1.0

    def __init__(self, solver: CoherentCoreSolver, quorum_fraction: float) -> None:
        self._solver = solver
        self._quorum_fraction = quorum_fraction

    def run(self, grid: ConcentrationGrid) -> SweepResult:
        ascending = grid.ascending()
        induction = self._follow_branch(ascending, self.AWAKE_FRACTION)
```

```
        emergence = self._follow_branch(list(reversed(ascending)), induction[-1].coherent_fraction)
        return SweepResult(tuple(induction), tuple(emergence))

    def _follow_branch(self, concentrations: Sequence[float], initial_fraction: float) -> list[SweepPoint]:
        points: list[SweepPoint] = []
        fraction = initial_fraction
        for concentration in concentrations:
            fraction = self._solver.solve(concentration, fraction)
            points.append(SweepPoint(concentration, fraction, fraction >= self._quorum_fraction))
        return points


# ---------------------------------------------------------------------------
# Scenarios and experiments
# ---------------------------------------------------------------------------


@dataclass(frozen=True)
class Scenario:
    """Everything needed to run one sweep, grouped to keep call sites small."""

    graph_builder: GraphBuilder
    timing: TimingParameters
    agent: AnaestheticAgent
    quorum_fraction: float
    grid: ConcentrationGrid
    seed: int
    state: BrainState = BrainState(REFERENCE_AROUSAL_LEVEL)


REFERENCE_SCENARIO = Scenario(
    graph_builder=RandomGraphBuilder(REFERENCE_NODE_COUNT, REFERENCE_MEAN_DEGREE),
    timing=TimingParameters(REFERENCE_INTEGRATION_WINDOW_MS, REFERENCE_BASELINE_JITTER_MS,
REFERENCE_FEEDBACK_JITTER_MS),
    agent=GABAERGIC_LIKE_AGENT,
    quorum_fraction=REFERENCE_QUORUM_FRACTION,
    grid=ConcentrationGrid(REFERENCE_MAX_RELATIVE_CONCENTRATION, REFERENCE_CONCENTRATION_STEP),
    seed=REFERENCE_SEED,
)

HUB_SCENARIO = replace(
    REFERENCE_SCENARIO,
    graph_builder=PreferentialAttachmentGraphBuilder(REFERENCE_NODE_COUNT, REFERENCE_EDGES_PER_NEW_NODE),
)

GraphTransform = Callable[[Graph, random.Random], Graph]
CoherenceFactory = Callable[[TimingParameters, AnaestheticAgent, BrainState], CoherenceModel]


@dataclass(frozen=True)
class ModelVariant:
    """Changes to the reference model: a lesion of the graph and/or a different coherence model."""

    transform_graph: GraphTransform | None = None
    coherence: CoherenceFactory = CoherenceModel


REFERENCE_MODEL = ModelVariant()


# A million-node graph and its curve take a few hundred megabytes, so each worker keeps only the
# most recent ones. The many experiments of one seed share them, which removes repeated builds.
GRAPH_CACHE_SIZE = 1
CURVE_CACHE_SIZE = 2


@lru_cache(maxsize=GRAPH_CACHE_SIZE)
def built_graph(builder: GraphBuilder, seed: int) -> tuple[Graph, object]:
    """The seed's graph and the random-generator state right after building it."""
    rng = random.Random(seed)
    graph = builder.build(rng)
    return graph, rng.getstate()
```

```
@lru_cache(maxsize=CURVE_CACHE_SIZE)
def percolation_curve(builder: GraphBuilder, seed: int, transform_graph: GraphTransform | None, silenced_share:
float) -> PercolationCurve:
    """Draws exactly as building the graph afresh would, so a cached graph gives identical results."""
    graph, state = built_graph(builder, seed)
    rng = random.Random()
    rng.setstate(state)
    if transform_graph is not None:
        graph = transform_graph(graph, rng)
    if silenced_share > 0:
        graph = remove_random_nodes(silenced_share)(graph, rng)
    return PercolationCurve.from_graph(graph, rng)


def build_curve(scenario: Scenario, transform_graph: GraphTransform | None = None) -> PercolationCurve:
    """The scenario's percolation curve; down-states silence a random share of whole populations."""
    return percolation_curve(scenario.graph_builder, scenario.seed, transform_graph,
scenario.state.silenced_population_share())


def solver_on(curve: PercolationCurve, scenario: Scenario, variant: ModelVariant = REFERENCE_MODEL) ->
CoherentCoreSolver:
    return CoherentCoreSolver(curve, variant.coherence(scenario.timing, scenario.agent, scenario.state))


def build_solver(scenario: Scenario, variant: ModelVariant = REFERENCE_MODEL) -> CoherentCoreSolver:
    return solver_on(build_curve(scenario, variant.transform_graph), scenario, variant)


def run_on_curve(curve: PercolationCurve, scenario: Scenario, variant: ModelVariant = REFERENCE_MODEL) ->
SweepResult:
    """A sweep on an already-built curve, so that variants of one network share its percolation draws."""
    return AnaestheticSweep(solver_on(curve, scenario, variant), scenario.quorum_fraction).run(scenario.grid)


def run_scenario(scenario: Scenario, variant: ModelVariant = REFERENCE_MODEL) -> SweepResult:
    return run_on_curve(build_curve(scenario, variant.transform_graph), scenario, variant)


@dataclass(frozen=True)
class StableStates:
    """Steady coherent fractions reached from an awake and from a collapsed starting state."""

    concentration: float
    from_awake: float
    from_collapsed: float
    quorum_fraction: float

    @property
    def holds_quorum(self) -> bool:
        """The state reached from wakefulness keeps quorum."""
        return self.from_awake >= self.quorum_fraction

    @property
    def can_be_woken(self) -> bool:
        """Both a conscious and an unconscious steady state exist, so a strong perturbation can restore quorum."""
        return self.from_awake >= self.quorum_fraction > self.from_collapsed

    @property
    def only_unconscious(self) -> bool:
        return self.from_awake < self.quorum_fraction and self.from_collapsed < self.quorum_fraction


COLLAPSED_FRACTION = 0.0


def stable_states(scenario: Scenario, concentration: float, variant: ModelVariant = REFERENCE_MODEL) -> StableStates:
    solver = build_solver(scenario, variant)
    return StableStates(
        concentration=concentration,
        from_awake=solver.solve(concentration, AnaestheticSweep.AWAKE_FRACTION),
        from_collapsed=solver.solve(concentration, COLLAPSED_FRACTION),
        quorum_fraction=scenario.quorum_fraction,
    )
```

```
def with_arousal(scenario: Scenario, arousal_level: float) -> Scenario:
    return replace(scenario, state=replace(scenario.state, arousal_level=arousal_level))


def arousal_ablation(base: Scenario, remaining_share: float) -> dict[str, SweepResult]:
    """Kelz et al. (2008): losing orexinergic arousal delays emergence without changing induction."""
    return {
        "intact arousal": run_scenario(base),
        "reduced arousal": run_scenario(with_arousal(base, base.state.arousal_level * remaining_share)),
    }


def active_shutdown_removed(base: Scenario) -> dict[str, SweepResult]:
    """The agent acting on the network only, not on arousal circuits: the effect of an acute
    lesion of the sleep-promoting circuits that anaesthetics recruit (Moore et al., 2012)."""
    passive_agent = replace(base.agent, arousal_suppression=0.0)
    return {"with active shutdown": run_scenario(base), "network only": run_scenario(replace(base,
agent=passive_agent))}


def sleep_versus_anaesthesia(base: Scenario, sleep: BrainState, deep_concentration: float) -> dict[str,
StableStates]:
    """Sleep as a brain state at zero concentration, compared with deep anaesthesia in the awake state."""
    return {
        "sleep": stable_states(replace(base, state=sleep), concentration=0.0),
        "deep anaesthesia": stable_states(base, concentration=deep_concentration),
    }


@dataclass(frozen=True)
class LossPoint:
    """What the agent is doing to connections at the last concentration before quorum is lost."""

    concentration: float
    transmission_failure: float
    agent_jitter_ms: float
    effective_jitter_ms: float
    coherence_probability: float


def loss_point(scenario: Scenario, result: SweepResult, variant: ModelVariant = REFERENCE_MODEL) -> LossPoint | None:
    if result.loss_concentration is None:
        return None
    point = next(p for p in result.induction if p.concentration == result.loss_concentration)
    model = variant.coherence(scenario.timing, scenario.agent, scenario.state)
    return LossPoint(
        concentration=point.concentration,
        transmission_failure=scenario.agent.transmission_failure(point.concentration),
        agent_jitter_ms=scenario.agent.added_jitter_ms(point.concentration),
        effective_jitter_ms=model.effective_jitter_ms(point.concentration, point.coherent_fraction),
        coherence_probability=model.coherence_probability(point.concentration, point.coherent_fraction),
    )


SILENCED_SHARE_BISECTION_STEPS = 8


def smallest_crossing(crosses: Callable[[float], bool], steps: int) -> float:
    """Smallest value in [0, 1] at which crosses becomes true, by bisection; 1.0 if it never does.

    crosses must be false below the crossing and true above it.
    """
    low, high = 0.0, 1.0
    for _ in range(steps):
        middle = (low + high) / 2
        low, high = (low, middle) if crosses(middle) else (middle, high)
    return high


def silenced_share_for_sleep(base: Scenario, sleep: BrainState) -> float:
    """Smallest share of silenced populations at which the sleeping network, started awake, loses quorum."""
```

```
    def loses_quorum(share: float) -> bool:
        state = replace(sleep, down_state_fraction=share, down_states=DownStates.POPULATIONS)
        return not stable_states(replace(base, state=state), concentration=0.0).holds_quorum

    return smallest_crossing(loses_quorum, SILENCED_SHARE_BISECTION_STEPS)


def remove_highest_degree_nodes(fraction: float) -> GraphTransform:
    def transform(graph: Graph, _: random.Random) -> Graph:
        degrees = graph.degrees()
        ranked = sorted(range(graph.node_count), key=lambda node: degrees[node], reverse=True)
        return graph.without_nodes(set(ranked[: round(fraction * graph.node_count)]))

    return transform


def remove_random_nodes(fraction: float) -> GraphTransform:
    def transform(graph: Graph, rng: random.Random) -> Graph:
        return graph.without_nodes(set(rng.sample(range(graph.node_count), round(fraction * graph.node_count))))

    return transform


def feedback_series_for(arousal_level: float) -> tuple[float, ...]:
    """The calibrated λ series, rescaled to an arousal level at which λ itself has been calibrated."""
    if arousal_level not in CALIBRATED_FEEDBACK_MS_BY_AROUSAL:
        raise ValueError(f"No calibrated feedback for arousal level {arousal_level}")
    scale = CALIBRATED_FEEDBACK_MS_BY_AROUSAL[arousal_level] /
CALIBRATED_FEEDBACK_MS_BY_AROUSAL[REFERENCE_AROUSAL_LEVEL]
    return tuple(value * scale for value in REFERENCE_FEEDBACK_SERIES_MS)


def feedback_strength_series(base: Scenario, feedback_values_ms: Sequence[float]) -> list[tuple[float, SweepResult]]:
    """Prediction 5: hysteresis width (neural inertia) grows with reference feedback lambda."""
    return [
        (feedback_ms, run_scenario(replace(base, timing=replace(base.timing, feedback_jitter_ms=feedback_ms))))
        for feedback_ms in feedback_values_ms
    ]


def agent_comparison(base: Scenario, agents: Sequence[AnaestheticAgent]) -> list[tuple[AnaestheticAgent,
SweepResult]]:
    """Section 6.2: omission-fault and timing-fault agents both abolish quorum."""
    return [(agent, run_scenario(replace(base, agent=agent))) for agent in agents]


def lesion_comparison(hub_scenario: Scenario, lesion_fraction: float) -> dict[str, SweepResult]:
    """Prediction 6: losing hubs lowers the loss concentration more than losing random nodes."""
    return {
        "intact": run_scenario(hub_scenario),
        "random lesion": run_scenario(hub_scenario,
ModelVariant(transform_graph=remove_random_nodes(lesion_fraction))),
        "hub lesion": run_scenario(hub_scenario,
ModelVariant(transform_graph=remove_highest_degree_nodes(lesion_fraction))),
    }


# ---------------------------------------------------------------------------
# Reporting
# ---------------------------------------------------------------------------


def format_concentration(value: float | None) -> str:
    return "not reached" if value is None else f"{value:.2f}"


def print_sweep_profile(title: str, result: SweepResult, sample_every: int) -> None:
    print(f"\n{title}")
    print(f"{'concentration':>14} {'S induction':>12} {'S emergence':>12}")
    emergence_by_concentration = {round(point.concentration, 6): point for point in result.emergence}
    for point in result.induction[::sample_every]:
        emergence_point = emergence_by_concentration[round(point.concentration, 6)]
        print(f"{point.concentration:>14.2f} {point.coherent_fraction:>12.3f}
{emergence_point.coherent_fraction:>12.3f}")
```

```
def print_threshold_row(label: str, result: SweepResult) -> None:
    print(
        f"{label:>38}  c_off={format_concentration(result.loss_concentration):>11}"
        f"  c_on={format_concentration(result.recovery_concentration):>11}"
        f"  hysteresis={format_concentration(result.hysteresis_width):>11}"
        f"  width={format_concentration(result.transition_width):>11}"
    )


PROFILE_SAMPLE_EVERY = 10


def main() -> None:
    print("DPC reference simulator: illustrative parameters, relative concentration units")

    print_sweep_profile(
        "1. Threshold and hysteresis (GABAergic-like agent, random graph)",
        run_scenario(REFERENCE_SCENARIO),
        PROFILE_SAMPLE_EVERY,
    )

    print("\n2. Neural inertia versus reference feedback lambda (prediction 5)")
    for feedback_ms, result in feedback_strength_series(REFERENCE_SCENARIO, REFERENCE_FEEDBACK_SERIES_MS):
        print_threshold_row(f"lambda = {feedback_ms:g} ms", result)

    print("\n3. Molecularly distinct agents converge on loss of quorum (Section 6.2)")
    for agent, result in agent_comparison(REFERENCE_SCENARIO, (GABAERGIC_LIKE_AGENT, NMDA_ANTAGONIST_LIKE_AGENT)):
        print_threshold_row(agent.name, result)

    print(f"\n4. Hub versus random lesion of {REFERENCE_LESION_FRACTION:.0%} of nodes (prediction 6)")
    for label, result in lesion_comparison(HUB_SCENARIO, REFERENCE_LESION_FRACTION).items():
        print_threshold_row(label, result)


if __name__ == "__main__":
    main()
```

B.9 Listing: `simulator/validation_study.py`

```
"""Size-scaling validation study for the DPC reference simulator.

Runs every reference experiment across several network sizes and independent
random seeds in parallel, checks each qualitative claim of the paper against
the model, and writes a JSON and a Markdown report that record the machine,
software versions, source hashes and timestamps of the run.

Usage:
    python3 validation_study.py           # full study
    python3 validation_study.py --quick   # small smoke run

Exit status is non-zero if any claim fails for any seed at any size.
"""

from __future__ import annotations

import argparse
import math
import os
import sys
import time
from dataclasses import asdict, dataclass, replace
from datetime import datetime, timezone
from typing import Callable, Sequence

import comparisons
import dpc_simulator as dpc
import provenance
import reporting

# Network sizes, chosen to bracket the scales at which the hypothesis could apply:
# 302 = C. elegans connectome (White et al., 1986); 1,000 = finest common cortical
```

```
# parcellation (Schaefer et al., 2018); 10^4 to 10^6 = mesoscale population models,
# used to check that thresholds converge as finite-size effects shrink.
FULL_NODE_COUNTS = (302, 1_000, 10_000, 100_000, 1_000_000)
FULL_SEEDS_PER_SIZE = 20
QUICK_NODE_COUNTS = (302, 1_000)
QUICK_SEEDS_PER_SIZE = 3




# ---------------------------------------------------------------------------
# Measurements
# ---------------------------------------------------------------------------


# Measured spatial extent of human slow waves (Nir et al., 2011, depth electrodes): on average a slow wave
# involves 27.1% of monitored regions (25.0% late in the night). That late-night share is a spatial extent, not the
share of time
# behind HUMAN_SLEEP_DOWN_STATE_FRACTION_UPPER, but the two coincide at 0.25, so O4 already tests it. Used as the
share of populations silent together during one slow wave.
HUMAN_SLOW_WAVE_EXTENT = 0.271
POPULATION_SLEEP_SHARES = (
    ("sleep, populations", dpc.SLEEP_DOWN_STATE_FRACTION),
    ("deep sleep, populations", dpc.HUMAN_SLEEP_DOWN_STATE_FRACTION_UPPER),
    ("slow wave, populations", HUMAN_SLOW_WAVE_EXTENT),
)

@dataclass(frozen=True)
class StudyDesign:
    label: str
    node_counts: tuple[int, ...]
    seeds_per_size: int
    base_seed: int = dpc.REFERENCE_SEED

    def tasks(self) -> list[SeedTask]:
        return [
            SeedTask(node_count, self.base_seed + seed_index)
            for node_count in self.node_counts
            for seed_index in range(self.seeds_per_size)
        ]


@dataclass(frozen=True)
class SeedTask:
    node_count: int
    seed: int


@dataclass(frozen=True)
class ThresholdSummary:
    loss_concentration: float | None
    recovery_concentration: float | None
    hysteresis_width: float | None
    transition_width: float | None

    @classmethod
    def of(cls, result: dpc.SweepResult) -> ThresholdSummary:
        return cls(
            result.loss_concentration,
            result.recovery_concentration,
            result.hysteresis_width,
            result.transition_width,
        )


@dataclass(frozen=True)
class SeedMeasurement:
    node_count: int
    seed: int
    reference: ThresholdSummary
    reference_loss_point: dpc.LossPoint | None
    silenced_share_for_sleep: float
    # Ordered from no feedback to the strongest calibrated feedback, as (λ in ms, thresholds) pairs.
    feedback_series: tuple[tuple[float, ThresholdSummary], ...]
```

```
    by_agent: dict[str, ThresholdSummary]
    by_lesion: dict[str, ThresholdSummary]
    by_arousal: dict[str, ThresholdSummary]
    by_shutdown: dict[str, ThresholdSummary]
    arousal_level: float
    stable_states: dict[str, dpc.StableStates]
    duration_seconds: float


ScenarioTransform = Callable[[dpc.Scenario], dpc.Scenario]


def unchanged(scenario: dpc.Scenario) -> dpc.Scenario:
    return scenario


def sized(scenario: dpc.Scenario, task: SeedTask) -> dpc.Scenario:
    return replace(scenario, graph_builder=scenario.graph_builder.resized(task.node_count), seed=task.seed)


def measure_seed(task: SeedTask, transform: ScenarioTransform = unchanged) -> SeedMeasurement:
    """Every experiment for one seed; transform alters both base scenarios (sensitivity analysis)."""
    started = time.perf_counter()
    reference = transform(sized(dpc.REFERENCE_SCENARIO, task))
    hub = transform(sized(dpc.HUB_SCENARIO, task))
    feedback = dpc.feedback_strength_series(reference, dpc.feedback_series_for(reference.state.arousal_level))
    agents = dpc.agent_comparison(reference, (dpc.GABAERGIC_LIKE_AGENT, dpc.NMDA_ANTAGONIST_LIKE_AGENT))
    lesions = dpc.lesion_comparison(hub, dpc.REFERENCE_LESION_FRACTION)
    arousal = dpc.arousal_ablation(reference, dpc.OREXIN_ABLATION_REMAINING_SHARE)
    shutdown = dpc.active_shutdown_removed(reference)
    connection_sleep = dpc.BrainState(dpc.SLEEP_AROUSAL_LEVEL, dpc.SLEEP_DOWN_STATE_FRACTION,
dpc.DownStates.CONNECTIONS)
    states = dpc.sleep_versus_anaesthesia(reference, connection_sleep, dpc.DEEP_ANAESTHESIA_CONCENTRATION)
    deep_sleep = replace(connection_sleep, down_state_fraction=dpc.HUMAN_SLEEP_DOWN_STATE_FRACTION_UPPER)
    states["deep sleep"] = dpc.stable_states(replace(reference, state=deep_sleep), concentration=0.0)
    for name, share in POPULATION_SLEEP_SHARES:
        population_sleep = dpc.BrainState(dpc.SLEEP_AROUSAL_LEVEL, share, dpc.DownStates.POPULATIONS)
        states[name] = dpc.stable_states(replace(reference, state=population_sleep), concentration=0.0)
    reference_result = dpc.run_scenario(reference)
    return SeedMeasurement(
        node_count=task.node_count,
        seed=task.seed,
        reference=ThresholdSummary.of(reference_result),
        reference_loss_point=dpc.loss_point(reference, reference_result),
        silenced_share_for_sleep=dpc.silenced_share_for_sleep(reference, dpc.BrainState(dpc.SLEEP_AROUSAL_LEVEL)),
        feedback_series=tuple((feedback_ms, ThresholdSummary.of(result)) for feedback_ms, result in feedback),
        by_agent={agent.name: ThresholdSummary.of(result) for agent, result in agents},
        by_lesion={label: ThresholdSummary.of(result) for label, result in lesions.items()},
        by_arousal={label: ThresholdSummary.of(result) for label, result in arousal.items()},
        by_shutdown={label: ThresholdSummary.of(result) for label, result in shutdown.items()},
        arousal_level=reference.state.arousal_level,
        stable_states=states,
        duration_seconds=time.perf_counter() - started,
    )


# ---------------------------------------------------------------------------
# Claims checked against the model
# ---------------------------------------------------------------------------


Claim = reporting.Check


def all_agents_lose_quorum(measurement: SeedMeasurement) -> bool:
    return all(summary.loss_concentration is not None for summary in measurement.by_agent.values())


def without_feedback(measurement: SeedMeasurement) -> ThresholdSummary:
    return measurement.feedback_series[0][1]


def strongest_feedback(measurement: SeedMeasurement) -> ThresholdSummary:
    return measurement.feedback_series[-1][1]
```

```
def no_hysteresis_without_feedback(measurement: SeedMeasurement) -> bool:
    width = without_feedback(measurement).hysteresis_width
    return width is not None and comparisons.equals(width, 0)


def inertia_width(summary: ThresholdSummary) -> float | None:
    """Hysteresis width, where never recovering (even at zero concentration) counts as unbounded inertia."""
    if summary.loss_concentration is None:
        return None
    return math.inf if summary.recovery_concentration is None else summary.hysteresis_width


def is_trapped_unconscious(summary: ThresholdSummary) -> bool:
    return summary.loss_concentration is not None and summary.recovery_concentration is None


def hysteresis_grows_with_feedback(measurement: SeedMeasurement) -> bool:
    widths = [inertia_width(summary) for _, summary in measurement.feedback_series]
    if any(width is None for width in widths):
        return False
    never_shrinks = all(comparisons.at_most(earlier, later) for earlier, later in zip(widths, widths[1:]))
    return never_shrinks and comparisons.exceeds(widths[-1], widths[0])


def feedback_sharpens_transition(measurement: SeedMeasurement) -> bool:
    width_without = without_feedback(measurement).transition_width
    width_with = strongest_feedback(measurement).transition_width
    return width_without is not None and width_with is not None and width_with < width_without


def hub_lesion_lowers_threshold_more(measurement: SeedMeasurement) -> bool:
    hub = measurement.by_lesion["hub lesion"].loss_concentration
    random_lesion = measurement.by_lesion["random lesion"].loss_concentration
    return hub is not None and random_lesion is not None and comparisons.exceeds(random_lesion, hub)


def reduced_arousal_delays_emergence_more_than_induction(measurement: SeedMeasurement) -> bool | None:
    """Undefined (None) without arousal, since there is then nothing to reduce."""
    if measurement.arousal_level == 0:
        return None
    intact, reduced = measurement.by_arousal["intact arousal"], measurement.by_arousal["reduced arousal"]
    values = (intact.loss_concentration, reduced.loss_concentration, intact.recovery_concentration,
reduced.recovery_concentration)
    if any(value is None for value in values):
        return False
    induction_shift = abs(intact.loss_concentration - reduced.loss_concentration)
    emergence_shift = intact.recovery_concentration - reduced.recovery_concentration
    return comparisons.exceeds(emergence_shift, induction_shift)


def active_shutdown_accelerates_but_is_not_required(measurement: SeedMeasurement) -> bool | None:
    """Undefined (None) without arousal, since the agent then has no arousal action to remove."""
    if measurement.arousal_level == 0:
        return None
    with_shutdown = measurement.by_shutdown["with active shutdown"].loss_concentration
    network_only = measurement.by_shutdown["network only"].loss_concentration
    return with_shutdown is not None and network_only is not None and comparisons.exceeds(network_only,
with_shutdown)


CLAIMS = (
    Claim("C1", "Omission-fault and timing-fault agents both abolish quorum (Section 6.2)", all_agents_lose_quorum),
    Claim("C2", "Without reference feedback (lambda = 0) there is no hysteresis (Section 4.3)",
no_hysteresis_without_feedback, reporting.CONSISTENCY_CHECK),
    Claim(
        "C3",
        "Hysteresis (neural inertia) grows with lambda; no recovery counts as unbounded (prediction 5)",
        hysteresis_grows_with_feedback,
        reporting.CONSISTENCY_CHECK,
    ),
    Claim("C4", "Reference feedback sharpens the loss-of-consciousness transition (prediction 1)",
feedback_sharpens_transition),
```

```
    Claim("C5", "Hub lesions lower the loss concentration more than random lesions (prediction 6)",
hub_lesion_lowers_threshold_more),
    Claim(
        "C6",
        "Reduced arousal delays emergence more than it shifts induction (Kelz et al., 2008)",
        reduced_arousal_delays_emergence_more_than_induction,
    ),
    Claim(
        "C7",
        "Without the agent's action on arousal circuits, quorum is still lost but at a higher concentration (Moore et
al., 2012; Eikermann et al., 2011)",
        active_shutdown_accelerates_but_is_not_required,
        reporting.CONSISTENCY_CHECK,
    ),
)

# Pre-specified tests of the sleep extension (Section 9). Reported, not part of the verdict:
# they test a proposed extension of the hypothesis, not a property the paper already claims.
EXTENSION_TESTS = (
    Claim(
        "E1",
        f"Sleep (no arousal, connection-level down-states, share {dpc.SLEEP_DOWN_STATE_FRACTION}, rat) has a wakeable
stable unconscious state",
        lambda m: m.stable_states["sleep"].can_be_woken,
    ),
    Claim(
        "E2",
        f"Deeper sleep (connection-level down-states, share {dpc.HUMAN_SLEEP_DOWN_STATE_FRACTION_UPPER}, human upper
estimate) has a wakeable stable unconscious state",
        lambda m: m.stable_states["deep sleep"].can_be_woken,
    ),
    Claim(
        "E3",
        "Deep anaesthesia has only an unconscious state, so no perturbation restores quorum",
        lambda m: m.stable_states["deep anaesthesia"].only_unconscious,
    ),
)


# Reported for transparency; they do not affect the verdict.
OBSERVATIONS = (
    Claim(
        "O1",
        f"Trapped unconscious at zero concentration, lambda = {dpc.REFERENCE_FEEDBACK_SERIES_MS[-1]:g} ms at the
reference arousal level",
        lambda m: is_trapped_unconscious(strongest_feedback(m)),
    ),
    Claim(
        "O2",
        f"Trapped unconscious at zero concentration after hub lesion, lambda = {dpc.REFERENCE_FEEDBACK_JITTER_MS:g}
ms",
        lambda m: is_trapped_unconscious(m.by_lesion["hub lesion"]),
    ),
    Claim(
        "O3",
        f"Sleep with population-level down-states at the rat share ({dpc.SLEEP_DOWN_STATE_FRACTION}) loses quorum",
        lambda m: not m.stable_states["sleep, populations"].holds_quorum,
    ),
    Claim(
        "O4",
        f"Sleep with population-level down-states at the human share ({dpc.HUMAN_SLEEP_DOWN_STATE_FRACTION_UPPER})
loses quorum",
        lambda m: not m.stable_states["deep sleep, populations"].holds_quorum,
    ),
    Claim(
        "O5",
        f"Sleep with one typical human slow wave silencing its measured share of regions ({HUMAN_SLOW_WAVE_EXTENT})
loses quorum (Nir et al., 2011)",
        lambda m: not m.stable_states["slow wave, populations"].holds_quorum,
    ),
)


def size_label(node_count: int) -> str:
```

```
    return f"N = {node_count:,}"


def evaluate_claims(
    claims: Sequence[Claim], measurements: Sequence[SeedMeasurement], node_counts: Sequence[int]
) -> list[reporting.CheckOutcome]:
    groups = {size_label(n): [m for m in measurements if m.node_count == n] for n in node_counts}
    return reporting.evaluate(claims, groups)


# ---------------------------------------------------------------------------
# Descriptive statistics
# ---------------------------------------------------------------------------


Distribution = reporting.Distribution
SummarySelector = Callable[[SeedMeasurement], float | None]

REPORTED_QUANTITIES: tuple[tuple[str, SummarySelector], ...] = (
    ("c_off, GABAergic-like", lambda m: m.reference.loss_concentration),
    ("c_on, GABAergic-like", lambda m: m.reference.recovery_concentration),
    ("Hysteresis, GABAergic-like", lambda m: m.reference.hysteresis_width),
    ("Transition width, GABAergic-like", lambda m: m.reference.transition_width),
    ("Transmission failure p at c_off, GABAergic-like", lambda m: m.reference_loss_point and
m.reference_loss_point.transmission_failure),
    ("Effective jitter (ms) at c_off, GABAergic-like", lambda m: m.reference_loss_point and
m.reference_loss_point.effective_jitter_ms),
    ("Coherence probability π at c_off, GABAergic-like", lambda m: m.reference_loss_point and
m.reference_loss_point.coherence_probability),
    ("Share of populations silent at once for sleep to lose quorum", lambda m: m.silenced_share_for_sleep),
    ("c_off, NMDA-antagonist-like", lambda m: m.by_agent[dpc.NMDA_ANTAGONIST_LIKE_AGENT.name].loss_concentration),
    ("c_off, hub graph intact", lambda m: m.by_lesion["intact"].loss_concentration),
    ("c_off, random lesion", lambda m: m.by_lesion["random lesion"].loss_concentration),
    ("c_off, hub lesion", lambda m: m.by_lesion["hub lesion"].loss_concentration),
    ("c_on, hub lesion", lambda m: m.by_lesion["hub lesion"].recovery_concentration),
    ("Seconds per seed", lambda m: m.duration_seconds),
)


def summarise(measurements: Sequence[SeedMeasurement], node_counts: Sequence[int]) -> dict[str, dict[int,
Distribution]]:
    return {
        label: {
            node_count: Distribution.of([select(m) for m in measurements if m.node_count == node_count])
            for node_count in node_counts
        }
        for label, select in REPORTED_QUANTITIES
    }


# ---------------------------------------------------------------------------
# Provenance
# ---------------------------------------------------------------------------


def collect_provenance(design: StudyDesign, worker_count: int) -> dict[str, object]:
    return provenance.collect(provenance.loaded_sources()) | {
        "design": {
            "label": design.label,
            "node_counts": list(design.node_counts),
            "seeds_per_size": design.seeds_per_size,
            "base_seed": design.base_seed,
            "worker_processes": worker_count,
            "reference_scenario": asdict(dpc.REFERENCE_SCENARIO),
            "hub_scenario_graph_builder": asdict(dpc.HUB_SCENARIO.graph_builder),
            "agents": [asdict(dpc.GABAERGIC_LIKE_AGENT), asdict(dpc.NMDA_ANTAGONIST_LIKE_AGENT)],
            "feedback_series_ms": list(dpc.REFERENCE_FEEDBACK_SERIES_MS),
            "lesion_fraction": dpc.REFERENCE_LESION_FRACTION,
        },
    }


# ---------------------------------------------------------------------------
# Reports
```

```
# ----------------------------------------------------------------------------


@dataclass(frozen=True)
class StudyReport:
    started_utc: datetime
    finished_utc: datetime
    provenance: dict[str, object]
    measurements: list[SeedMeasurement]
    statistics_by_quantity: dict[str, dict[int, Distribution]]
    claim_outcomes: list[reporting.CheckOutcome]
    observation_outcomes: list[reporting.CheckOutcome]
    extension_outcomes: list[reporting.CheckOutcome]

    @property
    def all_claims_hold(self) -> bool:
        return all(outcome.holds for outcome in self.claim_outcomes)

    @property
    def wall_seconds(self) -> float:
        return (self.finished_utc - self.started_utc).total_seconds()


def to_document(report: StudyReport) -> dict[str, object]:
    document = {
        "started_utc": report.started_utc.isoformat(),
        "finished_utc": report.finished_utc.isoformat(),
        "wall_seconds": report.wall_seconds,
        "all_claims_hold": report.all_claims_hold,
        "provenance": report.provenance,
        "claims": [asdict(outcome) | {"holds": outcome.holds} for outcome in report.claim_outcomes],
        "observations": [asdict(outcome) for outcome in report.observation_outcomes],
        "extension_tests": [asdict(outcome) for outcome in report.extension_outcomes],
        "statistics": {
            label: {str(node_count): asdict(distribution) for node_count, distribution in by_size.items()}
            for label, by_size in report.statistics_by_quantity.items()
        },
        "measurements": [asdict(measurement) for measurement in report.measurements],
    }
    return document


def to_markdown(report: StudyReport) -> str:
    design = report.provenance["design"]
    node_counts = design["node_counts"]
    sizes = [size_label(n) for n in node_counts]
    timestamps = {"started_utc": report.started_utc.isoformat(timespec="seconds"), "finished_utc":
report.finished_utc.isoformat(timespec="seconds")}
    study_rows = [
        f"| Wall time | {report.wall_seconds:.1f} s |",
        f"| Worker processes | {design['worker_processes']} |",
        f"| Seeds per size | {design['seeds_per_size']} (base seed {design['base_seed']}) |",
    ]
    lines = [
        *reporting.run_metadata_section(timestamps, [*provenance.markdown_rows(report.provenance), *study_rows]),
        "### Claims (seeds in which the claim holds)",
        "",
        *reporting.outcome_table(report.claim_outcomes, sizes),
        "",
        "### Observations (seeds in which the observation occurs; not part of the verdict)",
        "",
        *reporting.outcome_table(report.observation_outcomes, sizes),
        "",
        "### Extension tests (seeds in which the pre-specified test passes; not part of the verdict)",
        "",
        *reporting.outcome_table(report.extension_outcomes, sizes),
    ]
    lines += [
        "",
        "### Thresholds (mean ± SD across seeds, relative concentration units)",
        "",
        *reporting.markdown_table(
            ["Quantity", *sizes],
            [[label, *[by_size[n].describe() for n in node_counts]] for label, by_size in
```

```
report.statistics_by_quantity.items()],
        ),
    ]
    verdict = "All claims hold for every seed at every size." if report.all_claims_hold else "At least one claim
failed; see the claims table."
    lines += ["", f"**Verdict:** {verdict}", ""]
    return "\n".join(lines)


# ---------------------------------------------------------------------------
# Entry point
# ---------------------------------------------------------------------------


def claims_markdown(measurements: Sequence[SeedMeasurement], node_counts: Sequence[int]) -> str:
    sections = [
        (title, evaluate_claims(checks, measurements, node_counts))
        for title, checks in (("Claims", CLAIMS), ("Observations", OBSERVATIONS), ("Extension tests",
EXTENSION_TESTS))
    ]
    return reporting.outcome_sections(sections, [size_label(n) for n in node_counts])


def run_study(design: StudyDesign, worker_count: int) -> StudyReport:
    started = datetime.now(timezone.utc)
    provenance = collect_provenance(design, worker_count)
    tasks = design.tasks()
    live = reporting.ParallelRun("validation", worker_count, live_markdown=lambda done: claims_markdown(done,
design.node_counts))
    measurements = reporting.run_in_parallel(measure_seed, tasks, live)
    finished = datetime.now(timezone.utc)
    return StudyReport(
        started_utc=started,
        finished_utc=finished,
        provenance=provenance,
        measurements=measurements,
        statistics_by_quantity=summarise(measurements, design.node_counts),
        claim_outcomes=evaluate_claims(CLAIMS, measurements, design.node_counts),
        observation_outcomes=evaluate_claims(OBSERVATIONS, measurements, design.node_counts),
        extension_outcomes=evaluate_claims(EXTENSION_TESTS, measurements, design.node_counts),
    )


def parse_design(arguments: Sequence[str]) -> StudyDesign:
    parser = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
    parser.add_argument("--quick", action="store_true", help="small smoke run instead of the full study")
    options = parser.parse_args(arguments)
    if options.quick:
        return StudyDesign(reporting.RunLabel.QUICK, QUICK_NODE_COUNTS, QUICK_SEEDS_PER_SIZE)
    return StudyDesign(reporting.RunLabel.FULL, FULL_NODE_COUNTS, FULL_SEEDS_PER_SIZE)


def main(arguments: Sequence[str]) -> int:
    provenance.record_launch()
    design = parse_design(arguments)
    worker_count = os.cpu_count() or 1
    report = run_study(design, worker_count)
    markdown = to_markdown(report)
    reporting.publish("validation", to_document(report) | {"label": design.label}, markdown)
    return 0 if report.all_claims_hold else 1


if __name__ == "__main__":
    sys.exit(main(sys.argv[1:]))
```

B.10 Listing: `simulator/glial_regulation.py`

```
"""Glial regulation of timing coherence (postulate P9): slower loops nested around the integration cycle.

Three glial cell types are represented as parameters that change slowly compared with the
millisecond integration cycle, so each is held fixed during a concentration sweep (separation
of timescales):
```

```
- Oligodendrocytes (myelin): set conduction timing; damaged myelin adds conduction-time
  variability (Kato et al., 2020), modelled as a higher baseline jitter.
- Microglia: under anaesthesia, low noradrenaline activates microglia, which lower
  noradrenaline further (Liu et al., 2019; Stowell et al., 2019; Ma et al., 2024); genetic
  loss of their P2Y12 receptor speeds recovery (Cao et al., 2023). Drug depletion (PLX5622)
  also hastens emergence, but that drug speeds anaesthetic clearance (Cao et al., 2026).
  Modelled as an arousal drive that weakens further as the giant component collapses.
- Astrocytes: trigger slow-oscillation states (Poskanzer & Yuste, 2016); modelled as the share
  of sleep down-states that depends on astrocytic signalling.

Standard library only, like dpc_simulator.py, which it extends through its model variants.
"""

from __future__ import annotations

from dataclasses import replace
from functools import partial

import dpc_simulator as dpc

# Free parameters: no published measurement fixes them, so the study reports their effect over a range.
# Complete loss of myelin doubles the baseline conduction-time jitter.
DEMYELINATION_JITTER_GAIN = 1.0
MYELIN_INTEGRITY_SERIES = (1.0, 0.75, 0.5)
# 0 represents removing the microglial loop entirely.
MICROGLIAL_DEPLETED_GAIN = 0.0
MICROGLIAL_GAIN_SERIES = (MICROGLIAL_DEPLETED_GAIN, 0.25, 0.5)
STRONGEST_MICROGLIAL_GAIN = max(MICROGLIAL_GAIN_SERIES)
# Share of sleep down-states lost when astrocytic signalling is silenced.
ASTROCYTIC_DOWN_STATE_SHARE = 0.5

# Operational definition: a slow wave is global when it is detected at more than half of the
# recording locations (Nir et al., 2011). Half the populations is therefore the least a global
# down-state silences, and is used as its share.
GLOBAL_DOWN_STATE_SHARE = 0.5
GLOBAL_SLEEP = dpc.BrainState(dpc.SLEEP_AROUSAL_LEVEL, GLOBAL_DOWN_STATE_SHARE)


def demyelinated(scenario: dpc.Scenario, myelin_integrity: float) -> dpc.Scenario:
    """Scenario with baseline jitter raised in proportion to the share of myelin lost."""
    added_share = DEMYELINATION_JITTER_GAIN * (1 - myelin_integrity)
    jitter_ms = scenario.timing.baseline_jitter_ms * (1 + added_share)
    return replace(scenario, timing=replace(scenario.timing, baseline_jitter_ms=jitter_ms))


def astrocytes_silenced(sleep: dpc.BrainState) -> dpc.BrainState:
    """Sleep state without the down-states that depend on astrocytic signalling."""
    return replace(sleep, down_state_fraction=sleep.down_state_fraction * (1 - ASTROCYTIC_DOWN_STATE_SHARE))


class MicroglialCoherenceModel(dpc.CoherenceModel):
    """Coherence model in which a microglial loop weakens arousal as the giant component collapses.

    The loop is weak in wakefulness, where the coherent fraction is close to but below 1, grows as the
    giant component shrinks, and in the collapsed state removes the share microglial_gain of the arousal
    drive, which stabilises the unconscious state.
    With gain 0 it reproduces dpc.CoherenceModel exactly.
    """

    def __init__(
        self, timing: dpc.TimingParameters, agent: dpc.AnaestheticAgent, state: dpc.BrainState, microglial_gain:
float
    ) -> None:
        super().__init__(timing, agent, state)
        self._microglial_gain = microglial_gain

    def arousal_at(self, concentration: float, coherent_fraction: float) -> float:
        return self.arousal(concentration) * (1 - self._microglial_gain * (1 - coherent_fraction))


def with_microglia(microglial_gain: float) -> dpc.ModelVariant:
    return dpc.ModelVariant(coherence=partial(MicroglialCoherenceModel, microglial_gain=microglial_gain))
```

Appendix C: Phase-model, congestion and plant-tissue studies

C.1 Methods

The phase-oscillator model (`simulator/phase_dynamics.py`), a Kuramoto-Sakaguchi network (Kuramoto, 1984; Sakaguchi and Kuramoto, 1986) of the kind used for large-scale brain dynamics (Cabral et al., 2011; Breakspear, 2017), is run on three empirical connectomes (`simulator/connectomes.py`): *C. elegans* (OpenWorm c302 edge list of the complete hermaphrodite wiring diagram, derived from Cook et al., 2019; neurons only, 300 of the 302, since CANL and CANR have no synapses in the list, with each gap junction counted once in each direction), adult *Drosophila* (FlyWire v783, Dorkenwald et al., 2024, with synapses detected by Buhmann et al., 2021, and Heinrich et al., 2018, and transmitters predicted by Eckstein et al., 2024; connections of at least five synapses) and a 998-region human structural connectome (Hagmann et al., 2008; Sanz Leon et al., 2013). Each dataset is checked against a pinned SHA-256 digest before use. All parameters, with their basis, are listed in `simulator/calibration.py`. The congestion test uses Wilson-Cowan excitatory-inhibitory populations (Wilson and Cowan, 1972) with the local parameters and inhibitory plasticity rule of Abeysuriya et al. (2018), after Vogels et al. (2011), on the human connectome; synchrony is measured with the index of Golomb and Rinzel (1993) with measured conduction delays (`simulator/excitatory_inhibitory.py`). Plant tissue is a sheet of FitzHugh-Nagumo cells (FitzHugh, 1961; Nagumo et al., 1962) coupled through junctions (`simulator/plant_tissue.py`); on a square lattice the bond-percolation threshold is exactly 1/2 (Kesten, 1980). Pre-specified tests were fixed before the runs that evaluate them; post hoc analyses are labelled as such and do not enter any verdict. Timing jitter in the phase model persists for a correlation time τ before it changes (an Ornstein-Uhlenbeck process), so that it acts as a timing error during each exchange; jitter redrawn at every integration step would average to a uniform weakening of coupling, indistinguishable from transmission failure. As a null model, the GABAergic-like and sensory-stimulus conditions are repeated on rewired copies of the *C. elegans* and human connectomes made by degree-preserving double-edge swaps (Maslov and Sneppen, 2002), which keep every node's number of inputs and outputs exactly and create no self-connection or duplicate (conditions N1 and N2). Every parameter is listed below with its basis, in the four classes of Appendix A.1 (a "measured range" in earlier drafts is an assumed value). The sensitivity analysis (`simulator/sensitivity_study.py`, Appendix B.6) does not vary every parameter that is not measured. It varies, one at a time with the others at their reference values, in the percolation model at 10,000 nodes: the quorum fraction q^* (0.3, 0.5, 0.7), the arousal level with λ recalibrated (0, 0.3, 0.5, 0.7), λ (8.49, 8.85, 10.06 and 16.52 ms, one per reproduced inertia ratio), the Hill coefficient (2, 4, 8), the mean degree (4, 8, 16), W (1, 3, 10 ms), the baseline jitter σ_b (1, 3, 10 ms) and the GABAergic-like agent's maximal transmission failure (0.6, 0.8, 0.95); and in the phase model on the human connectome: the coupling K (2, 4, 8), the natural-frequency spread (0.25, 0.5, 1), the noise intensity D (0.02, 0.05, 0.1), q^* (0.3, 0.5, 0.7) and the jitter correlation time τ (0.25, 1, 4). It does not vary: the 40 Hz reference frequency, the 10 Hz fallback rate of P8, the maximal effects of the NMDA-antagonist-like agent (40 ms of added timing error, 20% transmission failure), the GABAergic-like agent's added jitter (1 ms), the arousal share a of the agent's effect, the learning time τ_L (Appendix C.10), the sensory stimulus (strength and frequency), the 5% lesion fraction of the hub test, the glial parameters γ, g and s_a (scanned separately in Appendix B.7), the oscillator feedback λ_{osc} (scanned in Appendices C.2 and C.12, not in the sensitivity analysis), the conduction velocity (measured), W and σ_b in the phase model, and the integration settings (checked in Appendix C.4).

Parameter	Value	Basis	Source
Baseline timing jitter (ms)	`3.0`	assumed	within-neuron value used between areas: precision of rate modulation in macaque MT in vivo, Buracas et al. (1998); swept 1-10
Integration window W (ms)	`3.0`	assumed	within-cell windows used between areas: 1-10 ms in rodent barrel cortex, Gabernet et al. (2005); under 2 ms in hippocampus, Pouille and Scanziani (2001); swept 1-10
Oscillation frequency (Hz)	`40.0`	assumed	within the gamma band, Fries (2015); Llinás and Ribary (1993)
Jitter correlation time τ	`1.0`	free	sensitivity analysis
Percolation feedback λ (ms)	`8.85`	calibrated	ratio 1.08, Friedman et al. (2010), isoflurane
Oscillator feedback λ	`0.45`	free	calibration to ratio 1.08 failed: inertia within noise
Conduction velocity (m/s)	`3.9`	measured	Lemaréchal et al. (2022)
Arousal level	`0.5`	free	the third of four levels tested (0, 0.3, 0.5, 0.7); inertia ratios 1.07-1.63 reproduced at every level tested
Fallback reference rate, P8 (Hz)	`10.0`	assumed	centre of the measured frontal alpha band, 8-12 Hz, Purdon et al. (2013)

Parameter	Value	Basis	Source
Synaptic coupling K	`4.0`	free	sensitivity analysis
Natural-frequency spread	`0.5`	free	sensitivity analysis
Noise intensity D	`0.05`	free	sensitivity analysis
Quorum fraction	`0.5`	free	sensitivity analysis
GABAergic-like agent	`{'name': 'GABAergic-like (omission faults)', 'max_transmission_failure': 0.95, 'max_added_jitter_ms': 1.0, 'half_effect_concentration': 1.0, 'hill_coefficient': 4.0, 'arousal_suppression': 1.0}`	free	maximal failure swept 0.6-0.95; failure at loss of quorum compared with 62% inhibition of glutamate release by 0.7 mM isoflurane (about 2 MAC) in cultured rat hippocampal neurons, Baumgart et al. (2015)
NMDA-antagonist-like agent	`{'name': 'NMDA-antagonist-like (timing faults)', 'max_transmission_failure': 0.2, 'max_added_jitter_ms': 40.0, 'half_effect_concentration': 1.0, 'hill_coefficient': 4.0, 'arousal_suppression': 1.0}`	free	maximal effects not measurable at this scale
Sensory stimulus	`{'strength': 1.0, 'angular_frequency': 0.3}`	free	see comment in calibration.py
FitzHugh-Nagumo a, b, ε	`[0.7, 0.8, 0.08]`	standard model values	FitzHugh (1961)
Plant junction conductance	`[0.5, 1.0, 2.0, 4.0]`	free	reported across the range

Numerics. Phases are integrated with the Euler-Maruyama scheme at a time step of 0.05 model time units, with independent noise $\sqrt{2D\,\Delta t}\,\xi_i$ on each population at each step. A sweep starts from identical phases (all zero), raises the concentration step by step and then lowers it again, each concentration starting from the phases the previous one ended with. At each concentration the first 400 steps (20 time units) are discarded and the next 838 steps, two cycles of the sensory stimulus of angular frequency 0.3, are measured; tests K1c and its diagnostics measure over eight cycles (3,351 steps). Timing error is converted to a phase error with standard deviation $2\pi f\sigma$, with f = 40 Hz and σ in seconds, and follows an Ornstein-Uhlenbeck process with correlation time τ = 1 time unit, so its persistence per step is $e^{-\Delta t/\tau}$. Model time is dimensionless: natural frequencies, coupling and the stimulus are not mapped to seconds, so the phase error has a physical scale and the time course does not. The integration window W in the table above is used by the percolation model and by the readout of Appendix C.12; the oscillator dynamics themselves do not use it (Section 4.7).

Drosophila settings. To keep each *Drosophila* run to between half an hour and an hour, those runs settle for 200 integration steps at each concentration rather than 400, and the concentration changes in steps of 0.1 EC50 rather than 0.05 EC50. Their loss concentrations are therefore resolved only to 0.1 and often coincide across seeds, so a standard deviation of zero across *Drosophila* seeds reflects the grid, not identical dynamics.

C.2 Oscillator feedback calibration

The attempt to calibrate the oscillator model's feedback to the measured inertia ratio is reported in full. Source: `simulator/results/calibration_phase_feedback_full_20261019T114211Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-19T11:42:11.362852+00:00
Finished (UTC)	2026-10-19T11:45:00.718733+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1
Git commit	`d8b06eb7b3df89d97b0e5aa7570ac537e4627cfe` (working tree dirty: True)

Field	Value
SHA-256 `calibrate_phase_feedback.py`	`357acbe814c3697094f2fd3b094253a6c509a6fdcd654625065b7c4b69213b9e`
SHA-256 `calibration.py`	`2fba7af8d15b3ce31231dc5bcf9c38ba429e740252ec8250f51fcc910f268915`
SHA-256 `connectomes.py`	`b6c0dfb53639d036ca42b1f9710afa944d5c29bb620ec7e584185ab363e35127`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`a50afa829fd68ec4d78524ff7e34fb7b9f7a6615b9108eb0f6906054e81a7041`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`

Oscillator feedback λ against the inertia ratio (human connectome, target 1.08)

λ	Mean ratio	Ratio per seed
0	0.990	1.000, 0.913, 1.000, 1.045
0.3	1.000	1.000, 1.000, 1.000, 1.000
0.45	1.000	1.000, 1.032, 1.000, 0.968
0.6	0.987	0.974, 1.000, 1.000, 0.972
0.75	0.986	n/a, 1.038, 1.000, 0.920

C.3 Cross-species study

Source: `simulator/results/species_full_20261020T103719Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T10:37:19.773458+00:00
Finished (UTC)	2026-10-20T10:37:19.784052+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1, pyarrow 26.0.0, matplotlib 3.11.2
Git commit	`ca2968fa252b68b06ddf08bfc97ee530bb2939ef` (working tree dirty: False)
SHA-256 `calibration.py`	`f0760a3875a9007af2640417c43eb9c99b6a2898e63ac5ce1a5b6c915a502f8d`
SHA-256 `comparisons.py`	`340b044083ffa2e8f98988da8fb400fa25c961db5f08c4bce42e5681441b92c8`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`621859b0bca3cc34cedd82e1559e1097bf90142d698e8c7d95b3e0907417c452`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`e9dac3d0f0c2c9baa8e813f65ea71615e65a027edd4d3115e86e556203ece030`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`78a8c5e399d134fdcf01a2af3be99396ff3a3eb25ebb62915bcb27ec8635fc37`
SHA-256 `species_study.py`	`423902f7037b6735db60c2826cfe212cf0c664859aeeda83317d9ea3421217ba`
Dataset `celegans_openworm`	MIT (OpenWorm c302); SHA-256 `142693f17556148d7f962835b18ac6dd5af18b7467eef61815ebc1dd5474c0ca`
Dataset `drosophila_flywire783`	CC BY 4.0 (FlyWire, Zenodo record 10676866); SHA-256 `24f960ae3e7d4f8cd30db3b62e99fb5179cc3d1e76d8c155bfb441e9737d3faf`
Dataset `human_tvb998`	GPL-3.0 (The Virtual Brain data); used, not redistributed; SHA-256 `31c65d4b6bf1760d06e0c4ab95c71c8dd499e48814e47bf2284e62a2b9301afd`

Re-evaluation: the runs of `species_full_20261019T123532Z.json` (run 2026-10-19T12:35:32.250135+00:00 to 2026-10-19T20:24:41.282006+00:00), unchanged, with every verdict recomputed by the code recorded above. No simulation was run.

Pillars (seeds in which the pillar holds)

ID	Statement	Kind	C. elegans	Drosophila	Human
T1	Coherence fluctuations surge at the collapse (paradoxical excitation as critical fluctuation)	test	7/20 (18–57%)	0/6 (0–39%)	5/20 (11–47%)
T2a	GABAergic-like agent abolishes quorum	construction check	20/20 (84–100%)	6/6 (61–100%)	20/20 (84–100%)
T2b	NMDA-antagonist-like agent abolishes quorum	construction check	20/20 (84–100%)	6/6 (61–100%)	20/20 (84–100%)
T3	Stimulated sensory area keeps processing while broadcast collapses	test	17/20 (64–95%)	0/6 (0–39%)	16/20 (58–92%)

Extension tests (pre-specified; not part of the verdict)

ID	Statement	Kind	C. elegans	Drosophila	Human
D1	With measured conduction delays, the GABAergic-like agent still abolishes quorum	test	not run	not run	20/20 (84–100%)
SG1	With GABA connections signed inhibitory, the GABAergic-like agent still abolishes quorum	test	not run	6/6 (61–100%)	not run
SG2	With GABA and glutamate connections signed inhibitory, the GABAergic-like agent still abolishes quorum	test	not run	6/6 (61–100%)	not run
K1	Ketamine pathway hypothesis: quorum persists while the stimulus stops propagating	test	12/20 (39–78%)	2/6 (10–70%)	10/20 (30–70%)
F1	Adaptive reference rate (P8) raises the loss concentration, GABAergic-like agent, paired by seed	test	20/20 (84–100%)	6/6 (61–100%)	20/20 (84–100%)
F2	Adaptive reference rate (P8) raises the loss concentration, NMDA-antagonist-like agent, paired by seed	test	19/20 (76–99%)	6/6 (61–100%)	18/20 (70–97%)
F3	The adaptive rate protects more against the jitter-type than the failure-type agent, paired by seed	test	0/20 (0–16%)	0/6 (0–39%)	0/20 (0–16%)

Secondary analyses (not part of the verdict)

ID	Statement	Kind	C. elegans	Drosophila	Human
T1b	Post hoc: fluctuations rise at least tenfold on the conscious side of the threshold	post hoc analysis	18/20 (70–97%)	6/6 (61–100%)	20/20 (84–100%)
T3b	Post hoc: sensory area retains at least 80% of its entrainment while broadcast collapses	post hoc analysis	17/20 (64–95%)	0/6 (0–39%)	16/20 (58–92%)
H0	No hysteresis without reference feedback	test	10/20 (30–70%)	6/6 (61–100%)	9/20 (26–66%)
N1	Null model: on a rewired connectome with the same degrees, the GABAergic-like agent abolishes quorum	test	20/20 (84–100%)	not run	20/20 (84–100%)
N2	Null model: on a rewired connectome, the stimulated area keeps processing while broadcast collapses	test	19/20 (76–99%)	not run	20/20 (84–100%)
H1	Hysteresis is never negative with reference feedback	test	15/20 (53–89%)	6/6 (61–100%)	17/20 (64–95%)
S1	Spontaneous local coherence outlasts global coherence (uniform jitter)	test	0/20 (0–16%)	0/6 (0–39%)	0/20 (0–16%)
S2	Spontaneous local coherence outlasts global coherence (conduction-length jitter)	test	not run	not run	0/20 (0–16%)
S3	Hysteresis is strictly positive with reference feedback	test	2/20 (3–30%)	0/6 (0–39%)	5/20 (11–47%)
S4	NMDA-antagonist-like agent abolishes quorum with fixed timing offsets instead of fluctuating jitter	test	20/20 (84–100%)	6/6 (61–100%)	20/20 (84–100%)

Measurements (mean ± SD across seeds; concentrations relative to EC50)

Quantity	Condition	C. elegans	Drosophila	Human
c_off	GABAergic-like	1.385 ± 0.059	1.400 ± 0.000	1.565 ± 0.046
c_off, adaptive reference rate (P8)	GABAergic-like, adaptive reference rate	1.518 ± 0.067	1.500 ± 0.000	1.738 ± 0.043
c_on	GABAergic-like	1.393 ± 0.054	1.400 ± 0.000	1.560 ± 0.031
Hysteresis	GABAergic-like	-0.008 ± 0.029	0.000 ± 0.000	0.005 ± 0.032
Susceptibility surge (peak / awake)	GABAergic-like	124.455 ± 62.656	294.912 ± 189.249	838.379 ± 320.712

Quantity	Condition	C. elegans	Drosophila	Human
c_off	NMDA-antagonist-like	0.650 ± 0.000	0.600 ± 0.000	0.695 ± 0.015
Broadcast ratio, awake	GABAergic-like, sensory stimulus	0.506 ± 0.131	0.063 ± 0.007	0.272 ± 0.175
Broadcast ratio, first unconscious	GABAergic-like, sensory stimulus	0.149 ± 0.047	0.057 ± 0.007	0.052 ± 0.028
Sensory entrainment, awake	GABAergic-like, sensory stimulus	0.426 ± 0.071	0.832 ± 0.003	0.184 ± 0.028
Sensory entrainment, first unconscious	GABAergic-like, sensory stimulus	0.785 ± 0.043	0.830 ± 0.004	0.753 ± 0.095
Local minus global coherence above chance	NMDA-antagonist-like	-0.059 ± 0.004	-0.007 ± 0.000	-0.091 ± 0.006
Seconds per run	GABAergic-like	15.271 ± 0.251	1910.615 ± 4.933	67.681 ± 0.279

Plant tissue (cellular control)

Claim	Holds
Propagation is blocked in every run	yes
Block never occurs below the bond-percolation threshold (1/2)	yes
Block point falls as junction conductance rises	yes

Junction conductance	Open-junction fraction at propagation block
0.5	0.827 ± 0.019
1	0.765 ± 0.016
2	0.707 ± 0.021
4	0.653 ± 0.022

C.4 Numerical convergence of the phase model

The GABAergic-like and sensory-stimulus conditions are repeated with twice the simulated time at each concentration and with half the integration step, on the human connectome (5 seeds) and on two *Drosophila* seeds, whose reference runs settle for only 200 steps. A result that does not move under either change does not depend on the numerical settings, and a hysteresis that survives twice the settling time is not an artefact of changing the concentration too fast. The reading rule below was written at 00:56 UTC on 19 October 2026, before the convergence study started (01:13 UTC), and committed at 01:19 UTC; after the code corrections recorded in Appendix D, the study was run again under the same rule: a loss concentration has converged when its mean across seeds changes by less than one step of the concentration grid (0.05 times the EC50 for the human connectome, 0.1 for *Drosophila*) under both changes; a broadcast ratio has converged when its mean changes by less than its standard deviation across seeds in the reference runs; and a hysteresis is an artefact of the settling time if it shrinks by more than half with twice the simulated time. A result that fails this rule is reported with a caveat in the main text. Source: `simulator/results/convergence_full_20261019T225958Z.md`.

Summary (moved from Section 4.5). The concentration is raised and lowered in steps, and at each step the network is integrated for a fixed time before it is measured. If that time were too short for the network to settle, the lag alone could produce an apparent hysteresis; the convergence study doubles the simulated time at every step and halves the integration step to check this, on the human connectome and on two *Drosophila* seeds (Appendix C.4). By the rule stated in Appendix C.4 before the study's first run, the loss concentrations converged on both connectomes: their means moved by at most 0.02 under either change. The phase model shows no hysteresis beyond one grid step on either connectome. One-step hysteresis appeared in two human reference runs and in one *Drosophila* seed with a stimulus and disappeared with twice the simulated time, so by the same rule it is an artefact of the settling time; with twice the time, one other human seed showed a one-step hysteresis instead. The *broadcast ratio* (the entrainment of the rest of the network to a stimulus, relative to that of the stimulated sensory populations) did not converge, except awake on the human connectome. Twice the simulated time also doubles the window over which broadcast is measured, which lowers its floor of chance locking (limitation 31), so this check mixes settling time with window length. With twice the simulated time, the broadcast left after quorum was lost fell further relative to the awake value: from 0.22 to 0.13 of it on the human connectome, and from 0.82 to 0.36 on *Drosophila*, where the awake broadcast itself more than doubled, from 0.06 to 0.15. The reference settings therefore understate the collapse of broadcast, most of all in the fly, and absolute broadcast ratios depend on the integration time; test T3 in the fly (Section 6) is read with this caveat.

Run metadata

Field	Value
Started (UTC)	2026-10-19T22:59:58.084085+00:00
Finished (UTC)	2026-10-20T01:13:27.351832+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)

Field	Value
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1, pyarrow 26.0.0
Git commit	`f71c3697cd844b603ce53bbf7dcf405b5e5f8d92` (working tree dirty: False)
SHA-256 `calibration.py`	`f0760a3875a9007af2640417c43eb9c99b6a2898e63ac5ce1a5b6c915a502f8d`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `convergence_study.py`	`cacef9c2f486a92f1160e76587408321d67f7ae713c51f5fb96f075b02f6ca8a`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`1b04ef76f1fa0860e6a7daca5ea54dab72234bd9e11855efa616f248ad91c935`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`
SHA-256 `species_study.py`	`614fcb5caba1cbd4534dbd00467c79b68f1c9d1def17877508922315609b11fd`

Phase model on the Human connectome: reference settings against longer and finer integration

Quantity (mean ± SD across seeds)	reference	twice the simulated time	half the time step
c_off	1.540 ± 0.055	1.550 ± 0.035	1.560 ± 0.022
Broadcast ratio, awake	0.341 ± 0.199	0.237 ± 0.102	0.334 ± 0.220
Broadcast ratio, first unconscious	0.077 ± 0.029	0.031 ± 0.010	0.045 ± 0.019
Hysteresis (c_off - c_on)	-0.010 ± 0.042	0.000 ± 0.000	0.000 ± 0.000

Phase model on the Drosophila connectome: reference settings against longer and finer integration

Quantity (mean ± SD across seeds)	reference	twice the simulated time	half the time step
c_off	1.400 ± 0.000	1.400 ± 0.000	1.400 ± 0.000
Broadcast ratio, awake	0.062 ± 0.001	0.146 ± 0.003	0.062 ± 0.002
Broadcast ratio, first unconscious	0.051 ± 0.000	0.053 ± 0.013	0.048 ± 0.000
Hysteresis (c_off - c_on)	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000

C.5 Congestion and graceful degradation (exploratory run)

Source: `simulator/results/congestion_full_20261020T011438Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T01:14:38.594732+00:00
Finished (UTC)	2026-10-20T01:32:36.430633+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1
Git commit	`38e86cbaa5c084c7a9d05a0da2bf4d08648ada80` (working tree dirty: False)
SHA-256 `calibration.py`	`f0760a3875a9007af2640417c43eb9c99b6a2898e63ac5ce1a5b6c915a502f8d`
SHA-256 `congestion_study.py`	`f6d346699defb083b07fbd6bed0cb4339053262712857274028c6d862e74e50a`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `excitatory_inhibitory.py`	`8a9b47e6f8a66f76cd22a46262f540d7d1182ac7a969d3ed69a58aa16666c074`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`

Field	Value
SHA-256 `plant_tissue.py`	`1b04ef76f1fa0860e6a7daca5ea54dab72234bd9e11855efa616f248ad91c935`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`

Pre-specified tests (runs in which the test passes)

ID	Statement	Kind	none, C=0.5	none, C=1	none, C=2	slow, C=0.5	slow, C=1	slow, C=2	fast, C=0.5	fast, C=1	fast, C=2
W1	Hub activity rises ≥10% before collapse, no compensation	test	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	not run	not run	not run	not run	not run	not run
W4	The rise is larger at high sensory load (and reaches 10%)	test	0/10 (0–28%)	0/10 (0–28%)	1/10 (2–40%)	0/10 (0–28%)	1/10 (2–40%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)
W2	Hub activity rises ≥10% before collapse, slow compensation	test	not run	not run	not run	0/10 (0–28%)	0/10 (0–28%)	1/10 (2–40%)	not run	not run	not run
W3	Hub activity rises ≥10% before collapse, fast compensation	test	not run	not run	not run	not run	not run	not run	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)

Post hoc analyses (not part of the verdict)

ID	Statement	Kind	none, C=0.5	none, C=1	none, C=2	slow, C=0.5	slow, C=1	slow, C=2	fast, C=0.5	fast, C=1	fast, C=2
X1	Post hoc: synchrony surges to ≥1.5 times awake before collapse	post hoc analysis	4/10 (17–69%)	10/10 (72–100%)	0/10 (0–28%)	8/10 (49–94%)	9/10 (60–98%)	0/10 (0–28%)	6/10 (31–83%)	10/10 (72–100%)	1/10 (2–40%)
X2	Post hoc: the synchrony surge is larger at high sensory load	post hoc analysis	5/10 (24–76%)	5/10 (24–76%)	7/10 (40–89%)	2/10 (6–51%)	6/10 (31–83%)	3/10 (11–60%)	2/10 (6–51%)	3/10 (11–60%)	6/10 (31–83%)

Graceful degradation, P8 (pre-specified; reported separately from the verdict)

ID	Statement	Kind	none, C=0.5	none, C=1	none, C=2	slow, C=0.5	slow, C=1	slow, C=2	fast, C=0.5	fast, C=1	fast, C=2
B1	Batching: burstiness rises to ≥1.5 times awake at or beyond the synchrony surge	test	0/10 (0–28%)	7/10 (40–89%)	1/10 (2–40%)	2/10 (6–51%)	8/10 (49–94%)	1/10 (2–40%)	2/10 (6–51%)	8/10 (49–94%)	1/10 (2–40%)
B2	Backoff: bursts are further apart at the deepest step that has them than at the surge	test	n/a	n/a	4/7 (25–84%)	2/2 (34–100%)	n/a	5/6 (44–97%)	1/1 (21–100%)	n/a	1/4 (5–70%)

Measurements (mean ± SD across seeds)

Condition	Peak hub activity / awake, low load	Same, high load	Synchrony surge, low load	Same, high load	Collapse concentration
none, C=0.5	1.009 ± 0.016	1.011 ± 0.014	1.551 ± 0.570	1.562 ± 0.509	1.288 ± 0.514
none, C=1	1.013 ± 0.019	1.009 ± 0.012	3.094 ± 0.718	3.195 ± 1.662	1.125 ± 0.000
none, C=2	1.025 ± 0.029	1.024 ± 0.039	1.204 ± 0.181	1.396 ± 0.396	1.000 ± 0.000
slow, C=0.5	1.011 ± 0.013	1.015 ± 0.019	1.931 ± 0.659	1.599 ± 0.474	1.150 ± 0.079
slow, C=1	1.023 ± 0.024	1.033 ± 0.038	3.084 ± 0.934	3.100 ± 1.224	1.125 ± 0.000
slow, C=2	1.029 ± 0.040	1.021 ± 0.029	1.115 ± 0.164	1.340 ± 0.663	0.988 ± 0.040
fast, C=0.5	1.000 ± 0.000	1.000 ± 0.002	2.060 ± 0.951	1.857 ± 1.037	1.137 ± 0.040
fast, C=1	1.000 ± 0.000	1.000 ± 0.000	3.872 ± 0.848	3.295 ± 0.738	1.125 ± 0.000
fast, C=2	1.000 ± 0.000	1.000 ± 0.000	1.188 ± 0.511	1.259 ± 0.347	0.975 ± 0.053

C.6 Synchrony surge: replication run on fresh seeds (no pass rule fixed)

No pass rule was fixed for this run; it repeats the exploratory study of Appendix C.5 on new seeds and reports counts. The generated report calls it confirmatory and heads its first table "Confirmatory tests"; that table is shown here as replicated tests, without a verdict.

Source: `simulator/results/congestion_confirmatory_20261020T013236Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T01:32:36.901705+00:00
Finished (UTC)	2026-10-20T01:50:32.632551+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1
Git commit	`38e86cbaa5c084c7a9d05a0da2bf4d08648ada80` (working tree dirty: False)
SHA-256 `calibration.py`	`f0760a3875a9007af2640417c43eb9c99b6a2898e63ac5ce1a5b6c915a502f8d`
SHA-256 `congestion_study.py`	`f6d346699defb083b07fbd6bed0cb4339053262712857274028c6d862e74e50a`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `excitatory_inhibitory.py`	`8a9b47e6f8a66f76cd22a46262f540d7d1182ac7a969d3ed69a58aa16666c074`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`1b04ef76f1fa0860e6a7daca5ea54dab72234bd9e11855efa616f248ad91c935`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`

Replicated tests (runs in which the test holds; no pass rule)

ID	Statement	Kind	none, C=0.5	none, C=1	none, C=2	slow, C=0.5	slow, C=1	slow, C=2	fast, C=0.5	fast, C=1	fast, C=2
X1	synchrony surges to ≥1.5 times awake before collapse	test	4/10 (17–69%)	9/10 (60–98%)	2/10 (6–51%)	5/10 (24–76%)	10/10 (72–100%)	3/10 (11–60%)	2/10 (6–51%)	8/10 (49–94%)	2/10 (6–51%)
X2	the synchrony surge is larger at high sensory load	test	5/10 (24–76%)	8/10 (49–94%)	5/10 (24–76%)	7/10 (40–89%)	7/10 (40–89%)	2/10 (6–51%)	5/10 (24–76%)	7/10 (40–89%)	4/10 (17–69%)

Activity tests (secondary in this mode)

ID	Statement	Kind	none, C=0.5	none, C=1	none, C=2	slow, C=0.5	slow, C=1	slow, C=2	fast, C=0.5	fast, C=1	fast, C=2
W1	Hub activity rises ≥10% before collapse, no compensation	test	0/10 (0–28%)	0/10 (0–28%)	1/10 (2–40%)	not run	not run	not run	not run	not run	not run
W4	The rise is larger at high sensory load (and reaches 10%)	test	0/10 (0–28%)	1/10 (2–40%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)
W2	Hub activity rises ≥10% before collapse, slow compensation	test	not run	not run	not run	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)	not run	not run	not run
W3	Hub activity rises ≥10% before collapse, fast compensation	test	not run	not run	not run	not run	not run	not run	0/10 (0–28%)	0/10 (0–28%)	0/10 (0–28%)

Graceful degradation, P8 (pre-specified; reported separately from the verdict)

ID	Statement	Kind	none, C=0.5	none, C=1	none, C=2	slow, C=0.5	slow, C=1	slow, C=2	fast, C=0.5	fast, C=1	fast, C=2
B1	Batching: burstiness rises to ≥1.5 times awake at or beyond the synchrony surge	test	1/10 (2–40%)	6/10 (31–83%)	4/10 (17–69%)	3/10 (11–60%)	7/10 (40–89%)	7/10 (40–89%)	1/10 (2–40%)	5/10 (24–76%)	4/10 (17–69%)
B2	Backoff: bursts are further apart at the deepest step that has them than at the surge	test	2/2 (34–100%)	n/a	4/6 (30–90%)	n/a	n/a	1/1 (21–100%)	2/2 (34–100%)	n/a	1/2 (9–91%)

Measurements (mean ± SD across seeds)

Condition	Peak hub activity / awake, low load	Same, high load	Synchrony surge, low load	Same, high load	Collapse concentration
none, C=0.5	1.009 ± 0.016	1.020 ± 0.025	1.482 ± 0.281	1.698 ± 0.738	1.125 ± 0.000
none, C=1	1.014 ± 0.016	1.023 ± 0.040	2.571 ± 1.025	3.502 ± 0.978	1.125 ± 0.000
none, C=2	1.043 ± 0.086	1.014 ± 0.023	1.308 ± 0.424	1.215 ± 0.252	1.000 ± 0.000
slow, C=0.5	1.005 ± 0.006	1.019 ± 0.018	1.836 ± 0.990	1.882 ± 0.734	1.113 ± 0.040
slow, C=1	1.017 ± 0.029	1.008 ± 0.012	3.116 ± 0.986	3.050 ± 0.892	1.125 ± 0.000
slow, C=2	1.015 ± 0.021	1.014 ± 0.025	1.455 ± 0.444	1.341 ± 0.569	0.975 ± 0.079
fast, C=0.5	1.000 ± 0.000	1.000 ± 0.000	1.378 ± 0.300	1.565 ± 0.771	1.137 ± 0.040
fast, C=1	1.000 ± 0.000	1.000 ± 0.001	2.868 ± 0.958	3.498 ± 0.644	1.125 ± 0.000
fast, C=2	1.000 ± 0.000	1.000 ± 0.000	1.251 ± 0.299	1.224 ± 0.410	0.988 ± 0.040

C.7 Pre-specified tests K2 and T1c, on seeds not used in the study that suggested T1c

The text below is quoted exactly as it was fixed before the run. It uses the report identifiers P1b and P1c, which this paper shows as T1b and T1c (Appendix C.9), and its feedback shares were computed on the earlier *C. elegans* graph: on the corrected graph of neurons only, 23% of *C. elegans* connections are feedback, and the *Drosophila* share is unchanged at 27%. In the quoted text, "the sensory stimulus of P3" means the stimulus of test T3, not postulate P3, and "seeds that no earlier run used" means seeds not used in the cross-species study that suggested T1c; they had been run once with the pre-correction code (see the last paragraph before the source line).

The tests below, their conditions and their pass rule were written into this paper before the study that evaluates them was run (`simulator/prespecified_study.py`). Before that, the code was checked with one shortened run (one *C. elegans* seed, with a quarter of the integration time); it is not part of the results. The study uses seeds that no earlier run used (from the base seed plus 2000): 20 for *C. elegans*, 6 for *Drosophila* and 20 for the human connectome. A test passes for an organism when it holds in at least 80% of that organism's seeds.

Feedback connections. A connection is classed as feedback when its target lies fewer steps from the sensory populations than its source, counting steps along directed connections from the nearest sensory population (breadth-first search). Connections from populations that no sensory population reaches are not classed. By this rule 21% of *C. elegans* connections and 27% of *Drosophila* connections are feedback. The human connectome is not used for K2: diffusion imaging gives no direction, so every connection appears in both directions and the classification would be arbitrary.

K2 (ketamine's timing faults on feedback connections). The NMDA-antagonist-like agent adds its timing error only on feedback connections; baseline timing error and transmission failure act as before, and the sensory stimulus of P3 is applied.

- K2a: quorum is lost within the concentration range.
- K2b: at the first concentration without quorum, the stimulus still spreads: the broadcast ratio is at least half its awake value. Seeds in which quorum is never lost count as failing.
- K2c (control, paired by seed): with the same added timing error on an equal number of connections drawn at random, quorum is lost at a higher concentration than with feedback faults, or not at all, while feedback faults abolish it.

What we expect, stated before the run: the phase model has no rule that agreement requires closed loops, and forward and lateral connections alone can keep populations in step. Test K1 showed that timing error confined to a minority of connections left quorum intact. K2 may therefore fail in this model. A failure would show that the model cannot decide where ketamine acts; it would not refute the proposal, which then rests on prediction 7 (Section 8) and on directional connectivity measured in patients (U. Lee et al., 2013). A pass would show that, in these two wiring diagrams, feedback connections are where timing faults break agreement.

P1c (replication of P1b). P1b, found post hoc in Appendix C.3, states that under the GABAergic-like agent the susceptibility at the last conscious concentration is at least ten times its awake value. P1c applies exactly this criterion to the fresh seeds on all three connectomes. A rise of fluctuations near a threshold is expected of threshold models in general, so P1c tests whether the model behaviour repeats, not whether DPC is right.

A first launch of the study was stopped after its *C. elegans* runs, before any of its results were read, so that agent names and comments in the simulator could be brought into line with the paper's terms (omission and timing faults; giant component). No equation or parameter changed. The study was then run again from the start on the same seeds; the report below is from that run. Worker processes of the stopped launch kept running for about an hour, sharing the processor with the first *Drosophila* runs of the new launch; this lengthened their reported run times but cannot change their results, which depend only on the code and the seed. While the run was in progress, an automated reviewer read the interim *C. elegans* results; nothing was changed after that, since the tests and pass rule were already fixed. The order of these events cannot be checked from the repository alone: the tests were written into the paper in the working copy before the run started (17:49 UTC on 18 October 2026), but the first commit that contains them was made at 18:25 UTC, after the run had started, so the reader has the authors' account of the order. The tests were not registered in any public registry. A hostile review and a second code audit on 19 October 2026 then found errors in the phase-

oscillator code (timing errors added rather than combined in quadrature, and entrainment measured over 1.43 instead of a whole number of stimulus cycles) and in how the connectomes were loaded (Appendix D). The study was therefore run again with the corrected code and the same seeds, tests and pass rule; the report below is from that run, and the first report is kept in `simulator/results/archive/`.

Source: `simulator/results/prespecified_full_20261019T202442Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-19T20:24:42.209041+00:00
Finished (UTC)	2026-10-19T22:59:49.502980+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1, pyarrow 26.0.0, matplotlib 3.11.2
Git commit	`3f06e46337f6f6fb5efc703dd7b9851a195c82f2` (working tree dirty: False)
SHA-256 `calibration.py`	`f0760a3875a9007af2640417c43eb9c99b6a2898e63ac5ce1a5b6c915a502f8d`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`07f3b3f1828859c628f596f543d65e33c45403483d5696bd355d8f48fd856fdf`
SHA-256 `phase_dynamics.py`	`7470f3f061d933b64e51bb060752914bbcfa166b3653d1dc89beeb4fc032241a`
SHA-256 `plant_tissue.py`	`1b04ef76f1fa0860e6a7daca5ea54dab72234bd9e11855efa616f248ad91c935`
SHA-256 `prespecified_study.py`	`3a736f1d66d9b9f307138f039ebe145cf44c74c8441a8c623b9d834dc84f866b`
SHA-256 `provenance.py`	`f320d7cc34706f1a0a92ea109c270872557e999139b9832304eef1fc5032b122`
SHA-256 `reporting.py`	`de80a53d63e90383ed325db5098aea06b71b3f846d8368ddeb12347ce3c4470b`
SHA-256 `species_study.py`	`614fcb5caba1cbd4534dbd00467c79b68f1c9d1def17877508922315609b11fd`

Pre-specified tests (seeds in which the test holds; passes at 80% or more)

ID	Statement	C. elegans	Drosophila	Human
K2a	Timing faults on feedback connections alone abolish quorum	1/20 (1–24%), fails	0/6 (0–39%), fails	not run
K2b	Where they do, the stimulus still spreads (broadcast at least half its awake value)	1/20 (1–24%), fails	0/6 (0–39%), fails	not run
T1c	Replication of P1b: fluctuations rise at least tenfold on the conscious side of the threshold	16/20 (58–92%), passes	6/6 (61–100%), passes	20/20 (84–100%), passes
K2c	Control: as many random connections need a higher concentration, or never abolish quorum	1/20 (1–24%), fails	0/6 (0–39%), fails	not run

Feedback connections (share of all connections)

Organism	Feedback share
C. elegans	23.3%
Drosophila	27.0%

Measurements (mean ± SD across seeds; concentrations relative to EC50)

Quantity	Condition	C. elegans	Drosophila	Human
Loss concentration	NMDA-antagonist-like on feedback connections, sensory stimulus	2.350 ± 0.000 (1/20)	not reached (0/6)	not reached (0/0)
Loss concentration	NMDA-antagonist-like on as many random connections, sensory stimulus	not reached (0/20)	not reached (0/6)	not reached (0/0)
Broadcast ratio, awake	NMDA-antagonist-like on feedback connections, sensory stimulus	0.524 ± 0.261	0.052 ± 0.008	not reached (0/0)
Broadcast ratio, first unconscious	NMDA-antagonist-like on feedback connections, sensory stimulus	0.486 ± 0.000 (1/20)	not reached (0/6)	not reached (0/0)

Quantity	Condition	C. elegans	Drosophila	Human
Coherence at the highest concentration, relative to awake	NMDA-antagonist-like on feedback connections, sensory stimulus	0.750 ± 0.064	0.875 ± 0.005	not reached (0/0)
Coherence at the highest concentration, relative to awake	NMDA-antagonist-like on as many random connections, sensory stimulus	0.914 ± 0.023	0.910 ± 0.002	not reached (0/0)
Loss concentration	GABAergic-like	1.388 ± 0.058	1.400 ± 0.000	1.560 ± 0.038

C.8 Figures

Figures 1 to 4 are shown in Section 6.

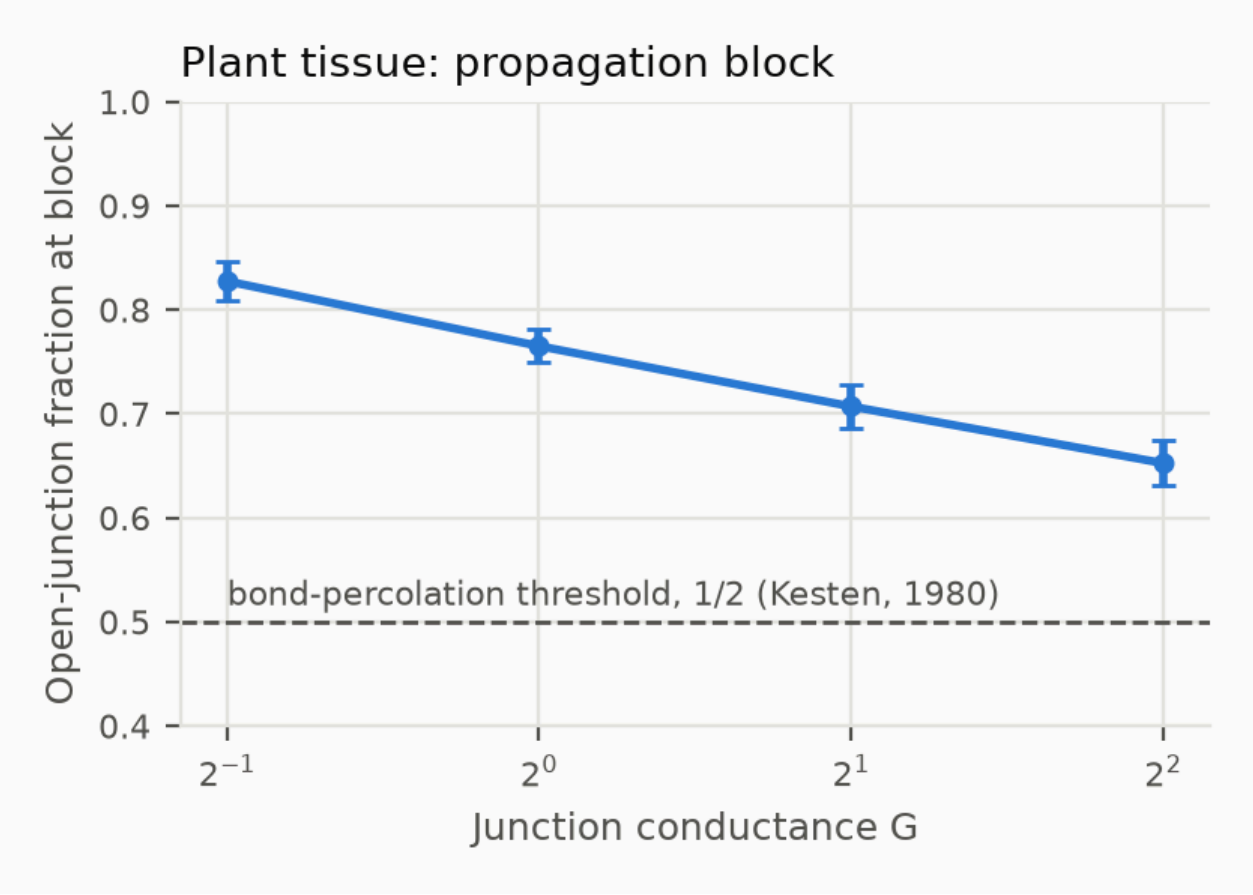


Figure 5. Open-junction fraction at which action-potential propagation fails in plant tissue, against junction conductance.

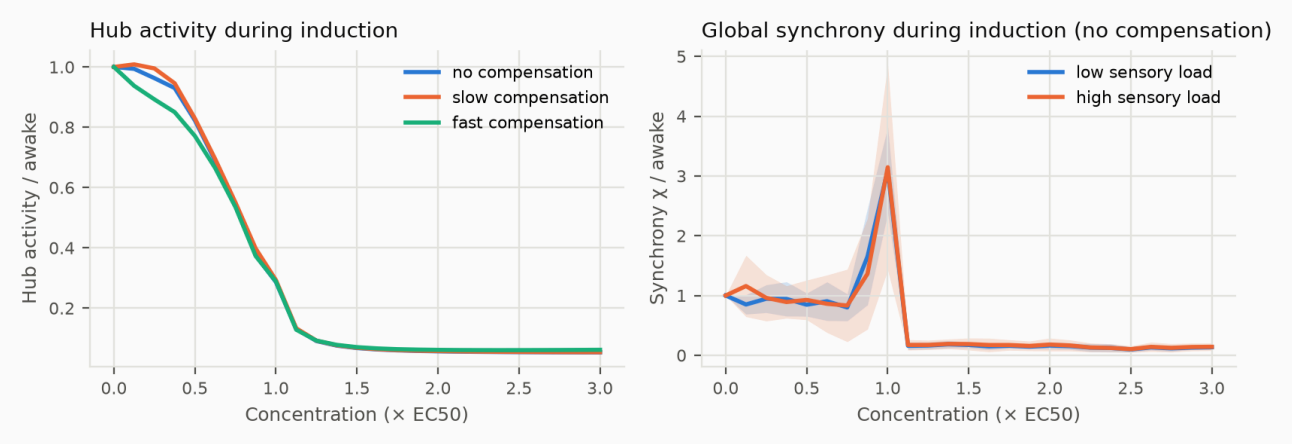


Figure 6. Hub activity and global synchrony during induction in the Wilson-Cowan model of the human connectome.

C.9 Reproducing every result

```
cd simulator
python3.13 -m venv ../.venv && ../.venv/bin/pip install -r requirements.txt
```

```
../.venv/bin/python -m unittest discover -s tests -t .
python3 calibrate_feedback.py && python3 validation_study.py && python3 glial_study.py
../.venv/bin/python percolation_thresholds.py
../.venv/bin/python calibrate_phase_feedback.py
../.venv/bin/python sensitivity_study.py
../.venv/bin/python congestion_study.py && ../.venv/bin/python congestion_study.py --confirm
../.venv/bin/python species_study.py
../.venv/bin/python convergence_study.py
../.venv/bin/python prespecified_study.py
../.venv/bin/python learned_timing_study.py
../.venv/bin/python disconnection_study.py
../.venv/bin/python dpc_readout_study.py
../.venv/bin/python pair_rule_diagnostic.py && ../.venv/bin/python dpc_pair_rule_study.py
../.venv/bin/python prediction15_study.py
../.venv/bin/python directed_threshold_estimate.py
../.venv/bin/python reciprocity_estimate.py
../.venv/bin/python fresh_seed_validation.py
../.venv/bin/python figures.py
python3 paper_appendices.py
../.venv/bin/python explainer_export.py && python3 results_catalogue.py
```

Some reports name tests with labels used elsewhere: P1 to P3b, which are also the labels of postulates P1 to P3 although the tests do not correspond to them, and the congestion study's G1 and G2, which are also the labels of the glial study's microglia checks. The paper and the explainer show these tests as T1 to T3b and B1 and B2; the reports keep the identifiers they were written with:

Study	Report identifier	Label in the paper
species and pre-specified	P1	T1
species and pre-specified	P1b	T1b
species and pre-specified	P1c	T1c
species and pre-specified	P2a	T2a
species and pre-specified	P2b	T2b
species and pre-specified	P3	T3
species and pre-specified	P3b	T3b
congestion	G1	B1
congestion	G2	B2

The open-data analyses of Appendix E are reproduced from the repository root with the commands of Appendix E.11 (`analysis/open_data/run.py`, with `run.py group` for the confirmatory verdicts), and the exploratory analyses with `analysis/open_data/deviations.py` and `analysis/open_data/exploratory_group.py` (Appendix E.13); their results are in `analysis/open_data/results/2026-10-21/`. The DPC readout and pair-rule reports are `simulator/results/dpc_readout_full_20261020T224553Z` and `simulator/results/dpc_pair_rule_full_20261021T105142Z` (Appendices C.12 and C.13). Interpreters: the calibration, validation and glial studies (Appendices B.3, B.4 and B.7) ran on CPython 3.15.0 with the standard library only; every other simulator report and every open-data result ran on CPython 3.13.2, as each report records. The C.14 reports are `simulator/results/prediction15_full_20261021T201526Z`, `directed_threshold_estimate_full_20261021T201513Z`, `reciprocity_estimate_full_20261021T201522Z` and `validation_fresh_seeds_full_20261021T201644Z`, all run on CPython 3.13.2 with the commands of Appendix C.14; the E.15 results are in `analysis/open_data/results/2026-10-21/exploratory_snr/` and `exploratory_monkey/`, reproduced with the commands of Appendix E.15.5, and the Michigan rerun with one generator per subject (Dev8) is in `analysis/open_data/results/2026-10-21/exploratory_reseeded/`, reproduced with the commands of Appendix E.15.

Every report records the machine, operating system, Python and package versions, Git commit and the SHA-256 hash of every simulator module that ran. Since the code corrections of 19 October 2026 (Appendix D), each study records its sources when it starts, so that the record describes the code that ran; earlier reports recorded them when the run ended. The table below is generated from the reports and the repository each time the appendices are built: for every report cited in Appendices B and C it compares the hash of each simulator file the report recorded with that file in the commit the report names, and, where they differ, finds the later commit in which that code was committed unchanged. Every earlier report is kept, unedited, in `simulator/results/archive/`, with a README for each set naming its successor and the reason.

Report	Commit	Sources recorded	Simulator files	Hashes against the commit
`calibration_feedback_full_20261018T210709Z`	`0677161`	at the end of the run	4	every file matches
`calibration_phase_feedback_full_20261019T114211Z`	`d8b06eb`	at launch	8	run from uncommitted changes to 4 file(s), committed unchanged in `f8aafa2`

Report	Commit	Sources recorded	Simulator files	Hashes against the commit
congestion_confirmatory_20261020T013236Z	38e86cb	at launch	9	every file matches
congestion_full_20261020T011438Z	38e86cb	at launch	9	every file matches
convergence_full_20261019T225958Z	f71c369	at launch	9	every file matches
directed_threshold_estimate_full_20261021T201513Z	74f9cb0	at launch	11	every file matches
disconnection_full_20261020T163516Z	7662889	at launch	11	every file matches
dpc_pair_rule_full_20261021T105142Z	b370495	at launch	38	every file matches
dpc_readout_full_20261020T224553Z	adc63f6	at launch	31	every file matches
glia_full_20261020T011431Z	38e86cb	at launch	6	every file matches
learned_timing_full_20261020T130252Z	07dc10e	at launch	11	every file matches
percolation_full_20261020T011339Z	38e86cb	at launch	8	every file matches
prediction15_full_20261021T201526Z	74f9cb0	at launch	17	every file matches
prespecified_full_20261019T202442Z	3f06e46	at launch	9	every file matches
reciprocity_estimate_full_20261021T201522Z	74f9cb0	at launch	17	every file matches
sensitivity_full_20261020T110144Z	7ce71a8	at launch	11	every file matches
species_full_20261020T103719Z	ca2968f	at launch	9	every file matches
validation_fresh_seeds_full_20261021T201644Z	74f9cb0	at launch	13	every file matches
validation_full_20261019T113035Z	d8b06eb	at launch	4	run from uncommitted changes to 2 file(s), committed unchanged in f8aafa2

C.10 Pre-specified test of learned arrival times (P4b)

The tests below, their conditions and their pass rule were written into this paper and committed before the study that evaluates them was run (`simulator/learned_timing_study.py`). Before that, the code was checked with one shortened run whose test outcomes were not computed or read. The study uses seeds that no earlier run used (from the base seed plus 3000): 20 for *C. elegans* and 20 for the human connectome. *Drosophila* is not run: with 2.7 million directed connections its runs would take most of a day. A test passes for an organism when it holds in at least 80% of that organism's seeds.

Learned arrival times. P4b proposes that the brain integrates inputs against learned expectations of when they arrive, so that a constant delay is harmless and only variability matters (Section 3.3). The models have no such learning, and in them constant offsets are as harmful as jitter. Here each connection $j \to i$ carries a learned expected arrival offset $\hat{c}_{ji}$, which enters the coupling as $\sin(\theta_j - \theta_i - \alpha_{ji} + \hat{c}_{ji})$ and moves as $\tau_L \, d\hat{c}_{ji}/dt = -\sin(\theta_j - \theta_i - \alpha_{ji} + \hat{c}_{ji})$, where α_{ji} is the connection's phase error (fixed lag plus jitter). The rule is driven only by what the receiving population can observe, the phase of the arriving signal relative to its own. It is adapted from the delay plasticity of Park and Lefebvre (2020), in which conduction delays change with the phase difference of the populations they connect; applying it to the receiver's expectation instead of the delay is our step. The learning time τ_L is ten jitter correlation times, so that the learning averages timing variability rather than tracking it, and about a sixth of the time held at each concentration, so that it settles within one concentration step; it is a free parameter and is not varied. The learned offsets start at zero in the awake state and carry over from one concentration to the next. Each test compares conditions with and without learning on the same seed; a concentration step is 0.05 times the EC50.

- L1: with learning, the NMDA-antagonist-like agent with fixed timing offsets (the condition of test S4) does not abolish quorum at any concentration up to three times the EC50.
- L2: with learning, the NMDA-antagonist-like agent with timing jitter of the same size still abolishes quorum, at most one concentration step later than without learning.
- L3 (human connectome only, which has tract lengths): with learning, the GABAergic-like agent with measured conduction delays abolishes quorum at most one concentration step below its loss without delays and without learning.
- L4 (safety): with learning, the GABAergic-like agent keeps at least 90% of the awake global coherence and abolishes quorum within one concentration step, either way, of its loss without learning. Adapting delays in phase models can stop a network from becoming fully in-phase (Park and Lefebvre, 2020) and can in principle cancel the coupling itself (Pajevic et al., 2014); L4 checks that the learning does not do so here.

What we expect, stated before the run: L1 and L3 test whether the learning absorbs constant offsets as intended, and should pass if it settles within each concentration step. L2 is the informative test: jitter that persists for a correlation time could be partly tracked by the learning, and if L2 fails, the distinction P4b draws between constant and variable timing errors does not hold even with learning. If L4 fails, learned expectations damage agreement in the model, and P4b would need a mechanism that this rule does not capture. Passing all four would show that a simple learning rule makes the models behave as P4b states; it would not show that the brain learns arrival times, which prediction 8 (Section 8) tests.

Source: `simulator/results/learned_timing_full_20261020T130252Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T13:02:52.282850+00:00
Finished (UTC)	2026-10-20T13:49:19.720634+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1, pyarrow 26.0.0, matplotlib 3.11.2
Git commit	`07dc10efe4c7ecd55177b7f3af2c0ad62fece747` (working tree dirty: False)
SHA-256 `calibration.py`	`927c086c7d329d4fbeb895412f083b07e776c05d83fd7be0a807076107bfefea`
SHA-256 `comparisons.py`	`340b044083ffa2e8f98988da8fb400fa25c961db5f08c4bce42e5681441b92c8`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `dpc_simulator.py`	`621859b0bca3cc34cedd82e1559e1097bf90142d698e8c7d95b3e0907417c452`
SHA-256 `learned_timing_study.py`	`466818c79d2cd6498e3f2af7577d69520ab5c02dcb83829871453d4cce04240d`
SHA-256 `phase_dynamics.py`	`256d7f68896a7b587732757eab90ba1d931a6eb9873db312f987773cdc0718fc`
SHA-256 `plant_tissue.py`	`e9dac3d0f0c2c9baa8e813f65ea71615e65a027edd4d3115e86e556203ece030`
SHA-256 `prespecified_runs.py`	`f1109431136f11f2a15c4c840ff674f722c2424ff44dc2f7772b033db7e93dae`
SHA-256 `provenance.py`	`4bf9f706eb10c32c3878843146193c79bbafa7ba37e2486d2fbdaa1c2b156350`
SHA-256 `reporting.py`	`661056b7bab4548101eb2b1008146146ed2b0977893b0b570a00a38fa0c8b0f2`
SHA-256 `species_study.py`	`ecf45bb37fa24ec88570c5846ac80f9b74df736f1c0e3e82e21f508872348c0f`

Pre-specified tests of learned arrival times (seeds in which the test holds; passes at 80% or more)

ID	Statement	C. elegans	Human
L1	With learned arrival times, fixed timing offsets never abolish quorum	0/20 (0–16%), fails	0/20 (0–16%), fails
L2	With learned arrival times, timing jitter of the same size still abolishes quorum, at most one step later	20/20 (84–100%), passes	20/20 (84–100%), passes
L3	With learned arrival times, conduction delays no longer lower the loss concentration by more than one step	not run	0/20 (0–16%), fails
L4	Learning keeps the awake state (at least 90% of its coherence) and moves the GABAergic-like loss by at most one step	0/20 (0–16%), fails	0/20 (0–16%), fails

Measurements (mean ± SD across seeds; concentrations relative to EC50)

Quantity	Condition	C. elegans	Human
Loss concentration	NMDA-antagonist-like, fixed timing offsets	0.600 ± 0.000	0.650 ± 0.000
Loss concentration	NMDA-antagonist-like, fixed timing offsets, learned arrival times	0.092 ± 0.034	0.118 ± 0.024
Loss concentration	NMDA-antagonist-like	0.650 ± 0.000	0.698 ± 0.011
Loss concentration	NMDA-antagonist-like, learned arrival times	0.075 ± 0.026	0.123 ± 0.038
Loss concentration	GABAergic-like	1.400 ± 0.063	1.578 ± 0.047
Loss concentration	GABAergic-like, learned arrival times	0.077 ± 0.026	0.123 ± 0.038
Loss concentration	GABAergic-like, conduction delays	not reached (0/0)	1.205 ± 0.043
Loss concentration	GABAergic-like, conduction delays, learned arrival times	not reached (0/0)	0.735 ± 0.206

Note on L2. L2 passed in every seed only because, with learning, quorum was lost below 0.13 times the EC50 under every condition, so a loss "at most one step later" than without learning was guaranteed; it is not support for P4b, and Appendix A.5 lists it with that caveat.

C.11 Pre-specified test of sensory disconnection with the broadcast floor removed (K1c)

The test below, its conditions and its pass rule were written into this paper and committed before the study that evaluates it was run (`simulator/disconnection_study.py`). Before that, the code was checked with one shortened run whose outcome was not computed or read.

Why this test. Test K1 (Appendix C.3) held in about half the seeds. A post hoc diagnostic then found that the broadcast measure, locking over two stimulus cycles, has a floor of chance locking (Section 10, limitation 31); with that floor subtracted, K1's criterion held in 16 of 20 *C. elegans* seeds. K1c tests that reading on fresh seeds. It is therefore not independent of the diagnostic that suggested it, but it can still fail.

Design. The condition is K1's: the NMDA-antagonist-like agent adds its timing error only on connections leaving the sensory populations, a periodic stimulus drives the sensory populations, and every other parameter keeps its reference value (Appendix C.1). Two things change. Broadcast is measured over eight stimulus cycles instead of two, and every seed is run twice, with the stimulus applied and with a phantom stimulus of the same rhythm and zero strength, whose locking measures the floor. The study uses seeds that no earlier run used (from the base seed plus 4000): 20 for *C. elegans* and 20 for the human connectome. *Drosophila* is not run: each run takes about an hour, and its awake broadcast, 0.05 to 0.06, is likely mostly floor.

- Gate: a seed is applicable only if its awake locking to the applied stimulus is at least twice its awake locking to the phantom; otherwise there is no broadcast to lose, and the seed is reported as not applicable.
- K1c: in an applicable seed, quorum holds at the highest concentration (three times the EC50; global coherence at least half its awake value), and the broadcast above the floor there, applied minus phantom locking, is at most half of the awake broadcast above the floor.
- Pass rule: K1c passes for an organism when it holds in at least 80% of that organism's applicable seeds, and it is evaluated only if at least half of the organism's seeds are applicable; otherwise it is reported as not evaluable. K1's original criterion at the longer window is reported alongside, without a verdict.

What we expect, stated before the run: on *C. elegans*, the post hoc diagnostic suggests that K1c will pass; on the human connectome, awake broadcast barely exceeded the floor with the shorter window, so K1c may not be evaluable there. A pass would make the disconnection of the senses from a surviving agreement a model result on *C. elegans*; it would not explain the preserved complexity and dreams of ketamine anaesthesia (Section 5.6), which prediction 7 tests in patients. A failure would leave ketamine's disconnected consciousness unexplained by the model, as Section 6 already states.

Run note. The study ran on 20 October 2026 from 16:35 to 17:10 UTC, from the commit that added this appendix (made at 16:35:00 UTC, before the run started at 16:35:16), with no uncommitted changes. The report's design record lists the study-wide integration settings, with 838 measurement steps; the runs themselves measured over 3,351 steps (eight stimulus cycles), as the condition names, the recorded window of eight cycles and the median run times (3.4 and 3.3 times those of the same condition in Appendix C.3, for *C. elegans* and the human connectome) show. After the run, the code was changed to record the window actually used; the report itself is unchanged. The design text above, fixed before the run, gives the fly's awake broadcast as 0.05 to 0.06; across the earlier studies it was 0.04 to 0.07 (0.052 to 0.074 in Appendix C.3), which does not change the reason for leaving the fly out.

Source: `simulator/results/disconnection_full_20261020T163516Z.md`.

Run metadata

Field	Value
Started (UTC)	2026-10-20T16:35:16.515774+00:00
Finished (UTC)	2026-10-20T17:09:58.140628+00:00
Machine	Mac14,9, Apple M2 Pro
CPU cores	10 (6 performance, 4 efficiency)
Memory	32 GiB
Operating system	macOS 26.7.1 (build 25G241), arm64
Python	CPython 3.13.2; numpy 2.5.4, scipy 1.18.1, pyarrow 26.0.0, matplotlib 3.11.2
Git commit	`76628896b357146e8d86648df46b3fe9ccfa14b6` (working tree dirty: False)
SHA-256 `calibration.py`	`da8a964155ad650d0c5cbdf570f95a5334a8fc3e8eacbcc83d59d9e5c60c5b3a`
SHA-256 `comparisons.py`	`340b044083ffa2e8f98988da8fb400fa25c961db5f08c4bce42e5681441b92c8`
SHA-256 `connectomes.py`	`b13446025b96ad86be7f4121b4c2b227dc58a4f38dc1c4db866c8b3bff86f3d1`
SHA-256 `disconnection_study.py`	`1a6644d5521c0bf32dd64cc32eaecb8a60116d1d8d756bd7d49258de8eadd9d6`
SHA-256 `dpc_simulator.py`	`621859b0bca3cc34cedd82e1559e1097bf90142d698e8c7d95b3e0907417c452`
SHA-256 `phase_dynamics.py`	`256d7f68896a7b587732757eab90ba1d931a6eb9873db312f987773cdc0718fc`
SHA-256 `plant_tissue.py`	`e9dac3d0f0c2c9baa8e813f65ea71615e65a027edd4d3115e86e556203ece030`

Field	Value
SHA-256 `prespecified_runs.py`	`a57ee64819832757fb2f45e357d2dd9b533e5ad97120a590c4dc8cb2b13c4ebb`
SHA-256 `provenance.py`	`4bf9f706eb10c32c3878843146193c79bbafa7ba37e2486d2fbdaa1c2b156350`
SHA-256 `reporting.py`	`661056b7bab4548101eb2b1008146146ed2b0977893b0b570a00a38fa0c8b0f2`
SHA-256 `species_study.py`	`c94753269f35b8b68dbf5f93173b244eedbfda7a5e97f3cb66ae6ac24d592ca9`

Pre-specified test K1c (seeds in which it holds, of the seeds that pass the awake gate; passes at 80% or more)

ID	Statement	C. elegans	Human
K1c	Ketamine-like faults on the sensory pathways keep quorum while the broadcast above the floor falls to half or less	20/20 (84–100%), passes	2/2 (34–100%), not evaluable (fewer than half the seeds have an awake broadcast above the floor)

Context, no verdict (medians across seeds of the locking value of the non-sensory populations)

Organism	Seeds passing the awake gate	Original K1 criterion at this window	Awake, applied	Awake, phantom	Highest dose, applied	Highest dose, phantom
C. elegans	20/20	20/20	0.163	0.036	0.033	0.027
Human	2/20	5/20	0.034	0.027	0.024	0.023

C.12 Pre-specified test of the connectome results with a DPC readout

The tests below, their conditions, seeds and pass rule were written into this paper and committed before the study that evaluates them was run (`simulator/dpc_readout_study.py`, with `simulator/dpc_readout.py` and `simulator/connectome_inertia.py`). Before that, the code was checked with unit tests and with one shortened run (`--quick`: one seed per organism and condition, a quarter of the settling time and about an eighth of the measurement time). Its outcomes were computed and read while the code was checked; with one seed per condition and shortened integration they are not part of the results.

Why this study. A hostile review found, and we confirmed, that the phase-oscillator model behind every connectome result in this appendix implements neither P2 nor P3. A timing error weakens a connection's coupling smoothly (by the factor $e^{-s^2/2}$, Section 4.5), the integration window W is never used, and quorum is defined as global coherence at least half its awake value (`InductionEmergenceRecord.is_conscious` in `simulator/phase_dynamics.py`). The same review found three further problems. The shared reference of P5 pulls every population towards the phase of the global mean field with no transmission failure, no timing error and no connections, so neither fault reaches the reference, and the model is effectively mean-field, which may be why the rewired null models matched the real wiring. The awake network is close to full synchrony (global coherence 0.93, 0.88 and 0.96 on *C. elegans*, *Drosophila* and the human connectome). And the percolation model's feedback λ was calibrated and its inertia tested only on random graphs of mean degree 8. The results of Appendices C.3 to C.11 therefore show that oscillator synchrony is lost under the two agents, which any coupled-oscillator model would show; they do not test whether timing-coherent connections stop forming a quorum. This study reads the same simulations as P2 and P3 are worded, adds the missing conditions, and repeats the main tests on fresh seeds.

The DPC readout

The dynamics are unchanged: the readout only looks at quantities the model already computes. A unit test checks that every measurement of a sweep is identical, to the last bit, with and without it, and the reference model's dynamics were checked to be identical, to the last bit, to those of the code that produced Appendices C.3 to C.11.

- *Timing error (P2).* At a sampled moment, a connection $j \to i$ that transmits ($m_{ji} = 1$) carries a signal whose timing error relative to the shared rhythm is, in radians of the reference cycle, $\delta_{ji} = \mathrm{wrap}(\theta_j - \psi) - \alpha_{ji}$, where θ_j is the phase of the sending population, ψ the phase of the network's mean rhythm, α_{ji} the connection's phase error in that same integration step (conduction lag, zero here, plus its current jitter draw, the same α_{ji} that enters the coupling term), and wrap maps an angle to $[-\pi, \pi)$. A population can lead or lag the rhythm by at most half a cycle, so its offset is wrapped; the delay error is not, so a signal a whole cycle late is late, as in the percolation model and as P2 states ("signals outside the window are not integrated").
- *Counting connection.* The connection counts in that cycle when it transmits and $|\delta_{ji}| \le W/2$, with W = 3 ms converted to phase at the reference rate in force ($\pi f W$ either side; 0.377 rad at 40 Hz).
- *Quorum (P3).* In each sampled cycle the counting connections are taken as undirected links, as in the percolation model, and the share of populations in their largest connected component is computed; a population with no counting connection is a component of one. Quorum holds in the cycle when that share is at least q^* = 0.5.
- *Cycles.* The model's time is dimensionless: only timing errors are converted from milliseconds to phase, so the model does not fix how many time units one 25 ms cycle lasts. One cycle is sampled per jitter correlation time τ, the time over which a

connection's timing error renews, so that successive cycles see largely fresh errors; if one time unit is read as one reference cycle, which puts the natural-frequency spread of 0.5 at about 3 Hz, this is also one sample per cycle. The measurement window of 838 steps of 0.05 gives 42 cycles at each concentration (168 with the eight-cycle window of K1d and K1e).

- *Quorum at a concentration.* Quorum holds at a concentration when it holds in at least half its cycles. The DPC loss concentration is the last concentration of induction at which it holds before it is first lost, and the DPC recovery concentration the highest concentration of emergence at which it has returned, as for the coherence criterion.
- *Static baseline.* With awake coherence near 0.96 the populations sit close to the rhythm, so the readout could largely reproduce the static percolation draw. Two baselines are therefore recorded at every concentration: the analytic prediction of Section 4.1, $\pi(c) = (1 - p(c))\,\mathrm{erf}(W/(2\sqrt{2}\,\sigma_{\mathrm{eff}}))$, evaluated for each transmitting connection with its own timing error and summed over all connections; and a *static reading*, the same giant-component readout with $\delta_{ji} = -\alpha_{ji}$, that is, static bond percolation on the model's own timing draws, ignoring how far each population has drifted from the rhythm. Tests T0a-D and T0b-D ask where the oscillator readout departs from the static one.
- *Secondary readings, reported without a verdict.* Timing measured at the receiver, with everything wrapped, $|\,\mathrm{wrap}(\theta_j - \theta_i - \alpha_{ji})| \le \pi f W$: the quantity whose sine the coupling term takes (communication through coherence); because α_{ji} is wrapped, an error far beyond a cycle leaves a chance floor of Wf (12%) of transmitting connections counting. And the largest strongly connected component of the directed counting connections, reported only for *C. elegans* and *Drosophila*: the human connectome is structurally symmetric (its 35,730 arcs are its 17,865 undirected links in both directions, since diffusion imaging gives no direction), so it enters no directed analysis. P2's wording ("relative to a shared phase reference") and Section 4.1 favour the shared rhythm, which is therefore primary. The coherence criterion is also reported for every run, for comparison.
- *Coupling left at each concentration.* The synaptic effective coupling $K_{\mathrm{eff}} = K \sum_{\mathrm{transmitting}} |w_{ji}|\, e^{-s_{ji}^2/2} / \sum_{\mathrm{all}} |w_{ji}|$, with s_{ji} the connection's phase-error SD (the mean-field factor by which Gaussian timing error weakens sine coupling; a fixed lag only shifts the phase and is ignored), and the coherence-dependent strength of the reference, $\lambda_{\mathrm{osc}} (1 - A(c))$. They are compared with the critical coupling of the all-to-all Kuramoto model with Gaussian natural frequencies of SD σ and phase noise of intensity D (Strogatz and Mirollo, 1991), $K_c = 2/\int g(\omega)\, D/(D^2 + \omega^2)\, d\omega = 2\sigma\sqrt{2\pi}/(\pi\,\mathrm{erfcx}(D/(\sigma\sqrt{2})))$, which without noise is Kuramoto's $\sigma\sqrt{8/\pi}$ (Kuramoto, 1984): 0.798 without noise and 0.863 with the model's D = 0.05 for σ = 0.5. The formula assumes all-to-all coupling, infinitely many oscillators and unimodal frequencies; the connectomes are sparse and finite, so K_c is a reference value, not the model's own threshold.

New conditions

- *Reference carried by the connections (referee point 3).* Each population's reference is the mean phase of the signals arriving from its transmitting inputs, $R_i e^{i\psi_i} = \sum_j w_{ji} m_{ji} e^{i(\theta_j - \alpha_{ji})} / s_i$, with the same failures m_{ji} and phase errors α_{ji} as the synaptic term in the same step and the same intact in-strength s_i, so that failures weaken the reference as they weaken coupling. Its pull keeps the reference model's form, $\lambda_{\mathrm{osc}}[R_i + A(c)(1 - R_i)]\sin(\psi_i - \theta_i)$, so the arousal share is kept; a population with no transmitting input receives none. The global reference is kept as the comparison condition, and every earlier study keeps it by default.
- *Weaker coupling (referee point 4).* Target, fixed before any run: awake global coherence between 0.5 and 0.7. For each connectome, the GABAergic-like condition is held at zero concentration only, with the synaptic coupling K = 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5 and 3, on 4 calibration seeds (from the base seed plus 5600); the chosen K is the one whose mean awake coherence lies in the target range and closest to 0.6. If none does, the connectome is reported as not calibrated and its weaker-coupling tests are not run.

Seeds

Every test uses seeds that no earlier run used, from the base seed plus 5000: 20 for *C. elegans*, 6 for *Drosophila* and 20 for the human connectome in the phase model, and 20 for each connectome in the percolation model. Calibrations use separate fresh blocks: the phase-model feedback scan and the percolation-model calibration from the base seed plus 5500 (10 seeds each), the weaker coupling from the base seed plus 5600 (4 seeds). Every calibrated value is tested on the first block only, so no seed both calibrates and tests a value. Checking the earlier studies for this found two overlaps. The percolation feedback was calibrated (`calibrate_feedback.py`) on the base seed plus 0 to 9 at 10,000 nodes, and the validation study (`validation_study.py`) uses the base seed plus 0 to 19 at every size, including 10,000 nodes, with the same graph builder: 10 of its 20 seeds at that size are the calibration networks themselves. The oscillator calibration (`calibrate_phase_feedback.py`) used the base seed plus 0 to 3 on the human connectome, which are also the first four human seeds of the cross-species study. Neither overlap changes a verdict reported above, but the validation at 10,000 nodes is not independent of the calibration.

Pass rule

One rule for every test: a test passes for an organism when the lower bound of the Wilson 95% interval (Wilson, 1927) for the share of seeds in which it holds lies above one half. We prefer it to the 80% rule of the earlier studies because it is a statement about the model's realisations (most of them, with 95% confidence) and because it is stricter where seeds are few: 15 of 20, 9 of 10 and 6 of 6 seeds pass; 14 of 20, 8 of 10 and 5 of 6 do not. With 20 seeds it is slightly weaker than the 80% rule (15 rather than 16 seeds). Seeds

are realisations of one wiring diagram, not independent brains (Section 10, item 28), and no correction for multiple comparisons is made; this study has about 30 tests per connectome.

Every other parameter keeps its reference value (Appendix C.1); concentrations run from 0 to 3 times the EC50 in steps of 0.05 (0.1 for *Drosophila*) in the phase model and 0.01 in the percolation model. A seed in which DPC quorum does not hold even at zero concentration counts as failing every test that needs a loss.

Tests: agents, null model and the wired reference

Conditions: the GABAergic-like and the NMDA-antagonist-like agent on the three connectomes, each with the global reference and with the reference carried by the connections; and all four on degree-preserving rewired copies of the *C. elegans* and human connectomes (Maslov and Sneppen, 2002), one rewiring per seed. The fly's null model is not run, as before, for cost.

- T2a-D, T2b-D: the GABAergic-like (T2a-D) and NMDA-antagonist-like (T2b-D) agent abolishes DPC quorum within the concentration range (a DPC loss concentration exists).
- T2a-DN, T2b-DN (null model): the same on the rewired connectomes.
- W2a, W2b: the same with the reference carried by the connections.
- WNa, WNb (wiring, paired by seed): with the reference carried by the connections, the DPC loss concentration on the connectome and on its rewired copy differ by more than one concentration step (a seed in which only one of them loses quorum counts as differing). GNa, GNb: the same comparison with the global reference.

Tests: where the readout departs from static percolation, and what kind of transition the loss is

- T0a-D, T0b-D: under each agent (global reference), DPC quorum is lost at a lower concentration than the static reading on the same timing draws loses it, or the static reading never loses it. A pass means that the populations' drift from the rhythm, not the timing draws alone, decides where quorum is lost. Reported alongside, without a verdict: the share of counting connections over the analytic $\pi(c)$ at the DPC loss.
- SP1 (paired by seed, per connectome): at their DPC losses the two agents leave more similar shares of counting connections than effective couplings K_{eff}, compared as absolute log ratios. This is what a percolation transition predicts: the same share of counting connections, however it is reached.
- SP2 (paired by seed, per connectome): at their coherence-criterion losses the two agents leave more similar effective couplings than shares of counting connections. This is what a synchronisation transition predicts.
- Reported without a verdict: K_{eff}/K_c at both losses on every connectome, which shows whether the coherence loss happens near a common effective coupling across connectomes.

Tests: fluctuations at the threshold (T1d, T1e, T1f)

Condition: the GABAergic-like agent with the global reference. The threshold is the last concentration with DPC quorum. T1b compared the susceptibility $N\,\mathrm{var}(R)$ there with its awake value, but $N\,\mathrm{var}(R)$ rises whenever coherence falls, even for independent phases, so a rise from the awake level does not show collective fluctuations. T1d compares it instead with what independent phases with the same mean coherence and the same spread around the rhythm would give, $\chi_{\text{ind}} = \frac{1}{2}\big(1 + \langle\cos 2(\theta - \psi)\rangle\big) - \langle R\rangle^2$, the variance of one population's projection on the mean rhythm; the ratio χ/χ_{ind} is about 1 plus $N-1$ times the mean correlation between populations' fluctuations, and equals 1 for independent phases. The approximation holds while $\langle R\rangle$ is well above chance ($1/\sqrt{N}$).

- T1d: at the threshold, $\chi/\chi_{\text{ind}} \geq 2$.
- T1e: at the threshold, the autocorrelation of global coherence R at a lag of one sampled cycle is higher than at zero concentration (critical slowing down; Scheffer et al., 2009).
- T1f (finite size): for the same seed, χ/χ_{ind} at the threshold is larger on the human connectome (998 populations) than on *C. elegans* (300), and larger on *Drosophila* (138,639) than on the human connectome; evaluated for each pair of neighbouring sizes, under the larger connectome.

A seed without a DPC threshold, or with an undefined quantity, counts as failing. T1b is reported again on these seeds without a verdict.

Tests: weaker coupling (T2a-M, T2b-M)

On each calibrated connectome, at its calibrated K: the GABAergic-like (T2a-M) and NMDA-antagonist-like (T2b-M) agent abolishes DPC quorum within the concentration range. Awake global coherence is reported for every seed.

Tests: sensory disconnection with its control (K1d, K1e)

Conditions: those of K1c (Appendix C.11): the NMDA-antagonist-like agent adds its timing error only on connections leaving the sensory populations, a sensory stimulus is applied, broadcast is measured over eight stimulus cycles, and each seed is run with the stimulus and with a phantom of zero strength. The control repeats both with the agent's timing error on as many connections

drawn at random for each seed (460 of 5,764 in *C. elegans*, 267 of 35,730 in the human connectome), from a random stream of their own. The awake gate and the evaluability rule are K1c's: a seed is applicable only if its awake locking to the stimulus is at least twice its awake locking to the phantom, and a test is judged for an organism only if at least half its seeds are applicable.

- K1d: in an applicable seed, DPC quorum holds at the highest concentration (three times the EC50) and the broadcast above the floor there is at most half the awake broadcast above the floor.
- K1e (control, paired by seed; applicable when both the sensory and the random runs pass the gate): K1d holds with the error on the sensory pathways and does not hold with it on as many random connections. Without K1e, a pass of K1 or K1c could come from construction alone: faults on a minority of connections leave quorum intact, and the stimulus may stop spreading for reasons unrelated to where they fall.

Tests: neural inertia in the phase model (I1, I2)

The reference term $\lambda_{\text{osc}} R \sin(\psi - \theta_i)$ is all-to-all Kuramoto coupling of strength λ_{osc}, so its locking threshold is K_c = 0.863 above. The calibration of Appendix C.2 stopped at 0.75, below it. Conditions: the GABAergic-like agent with λ_{osc} = 0, 0.5, 0.75, 1, 1.25 and 1.5 times K_c (0, 0.431, 0.647, 0.863, 1.078 and 1.294), on *C. elegans* and the human connectome, 10 seeds each from the calibration block.

- I1 (at each scanned λ_{osc}): DPC quorum returns on emergence more than one concentration step below the concentration at which it was lost. Inertia in the DPC readout is supported if I1 passes at at least one scanned value; with six values per connectome, one pass could arise by chance.
- Calibration rule: for each connectome, among the scanned values at which every calibration seed loses DPC quorum and regains it above zero concentration, the value whose mean ratio of loss to recovery concentration is closest to the reference ratio of 1.08 (Friedman et al., 2010). If no value qualifies, the connectome is reported as not calibrated and I2 is not run for it.
- I2 (held-out, 20 seeds from the test block at the calibrated value): the seed's ratio of DPC loss to DPC recovery concentration lies within the measured range that the percolation model reproduces, 1.07 to 1.63.

Tests: neural inertia in the percolation model on the connectome graphs (PI1 to PI4)

The percolation model, with every reference parameter unchanged (arousal level 0.5, the GABAergic-like agent, q^* = 0.5, W = 3 ms, baseline jitter 3 ms), runs on the undirected graphs of the three connectomes (`percolation_thresholds.undirected_graph`); each seed varies only the percolation draws. Calibration: on each of 10 calibration seeds, λ is scanned from 0 to 30 ms in steps of 0.1 ms; for each measured ratio (1.07, 1.08, 1.11, 1.63), the chosen λ is the scanned value at which every calibration seed loses quorum and regains it, with the mean ratio closest to the target; the target counts as reproduced when that mean lies within 0.01 of it, as in `calibrate_feedback.py`. Held-out tests, 20 seeds from the test block, at λ = 0 and at each reproduced target's λ:

- PI1: at the λ calibrated to 1.08, quorum is lost and regained at a strictly lower concentration.
- PI2 (C3's form): the hysteresis width never shrinks along the series of calibrated λ and is larger at its end than without feedback; never recovering counts as unbounded inertia.
- PI3 (C2's form, a consistency check): without feedback the hysteresis width is exactly zero.
- PI4: at the λ calibrated to 1.08, the seed's ratio lies within 1.07 to 1.63.

PI1 and PI4 are not run for a connectome on which 1.08 is not reproduced; PI2 needs at least one reproduced target.

What we expect, stated before the run

Before the referee's latest points, we ran the readout once on one seed outside every block (seed 99, not derived from the base seed) with shortened integration and a coarse grid of 0.25, to check that it works; the figures below come from that probe and are a single realisation. No other run informed these expectations.

- T2a-D should pass on all three connectomes: up to 95% transmission failure removes counting connections however well the remaining ones are timed (in the probe, DPC quorum was lost at 1.5 to 1.75 times the EC50, one or two probe steps later than coherence). As limitation 25 says of T2a, this cannot fail by construction; what it adds is where the loss happens. T2a-DN and W2a should pass for the same reason.
- T2b-D may fail on the human connectome. Its connections are dense (a mean undirected degree of about 36), and at three times the EC50 about 2.4% of them still counted in the probe, enough for an undirected giant component of 65% of the populations. On *C. elegans* (mean undirected degree 23) the giant component fell to about 20% in the probe, so T2b-D should pass there. The fly is not predicted. A failure would mean that timing faults alone do not abolish quorum in a densely wired network as DPC states it, which bears on P4(b) and on assigning ketamine to timing faults.
- T0a-D and T0b-D are uncertain. With the populations close to the rhythm while awake, the oscillator readout should follow the static reading at low concentrations; whether it departs before the loss depends on how far the populations drift as coupling weakens. If both fail, the DPC readout of the phase model is static percolation on its timing draws, and the oscillators add nothing to the quorum result.
- SP1 should pass and SP2 may pass: the DPC readout counts connections, so its loss should sit at a common share of counting connections, while the coherence criterion is a synchronisation measure. If SP2 fails as well, the coherence loss is not set by

the effective coupling either, and the two agents differ in more than coupling.

- WNa and WNb are the informative tests of the wired reference: if the global reference is why the null models matched the real wiring, GNa and GNb should fail and WNa or WNb pass. If all four fail, the match does not come from the reference.
- T1e should pass if the threshold behaves as a generic critical transition (in the probe, the lag autocorrelation was 0.93 at the threshold against about 0 awake on *C. elegans*). T1d is uncertain: in the probe the ratio at the *C. elegans* threshold was 2.2, just above the bar. T1f is uncertain, and its comparison is confounded: the three connectomes differ in wiring and density as well as size.
- T2a-M and T2b-M are uncertain. A weaker coupling leaves the populations further from the rhythm while awake, so fewer connections count; DPC quorum may already fail at zero concentration, which counts as failing.
- K1d should pass on *C. elegans*, where the sensory outputs are 8% of the connections, and is likely not evaluable on the human connectome, as K1c was. K1e is the informative test: if the broadcast collapse under K1 comes from the agent's uniform transmission failure and arousal suppression rather than from timing faults on the sensory pathways, K1e will fail, and the disconnection account of ketamine (Section 7.4) will lose its model support.
- I1 should fail at every value. For a unimodal distribution of natural frequencies, the onset of synchrony in the Kuramoto model is continuous, without the bistability that inertia needs (Kuramoto, 1984); above K_c the reference alone can hold the populations in step, so the coherence criterion may never register a loss while DPC quorum is lost to transmission failure, the disagreement that a unit test (`test_in_the_model_a_strong_reference_keeps_coherence_after_dpc_quorum_is_lost_Regression`) shows on a small network. If I1 fails, the calibration will pick the value whose ratio is nearest 1.08 from below, and I2 will fail; that would extend limitation 9 to the DPC readout. A pass of I1 and I2 would be the first phase-model support for P5.
- PI1 to PI4 are uncertain. The percolation model reproduces the measured ratios on random graphs, and its inertia comes from the self-consistency of the giant component with the reference, which does not depend on the graph's degree distribution; but the connectome graphs are denser than mean degree 8, so the transition is sharper and the calibrated λ may differ, or no λ may let every seed recover. A failure of PI4 on held-out seeds would show that the calibration does not generalise across seeds on a fixed wiring diagram.

Results

The code and this appendix were committed at 22:45:02 UTC on 20 October 2026 (commit `adc63f6`); the study started at 22:45:52 UTC and ended at 09:34 UTC on 21 October, after 10.8 hours (`simulator/results/dpc_readout_full_20261020T224553Z`). Counts are seeds in which a test holds, with the Wilson 95% interval; "not run" marks a pairing the design leaves out (the fly's null model and its size neighbour) and "not evaluable" the K1c gate.

Test	*C. elegans*	*Drosophila*	Human
T2a-D, transmission failure abolishes DPC quorum	20/20, passes	6/6, passes	20/20, passes
T2b-D, timing variability abolishes DPC quorum	20/20, passes	6/6, passes	0/20, fails
T2a-DN, T2b-DN, the same on the rewired connectome	20/20 and 20/20, pass	not run	20/20, passes; 0/20, fails
W2a, W2b, the same with the reference carried by the connections	20/20 and 20/20, pass	6/6 and 6/6, pass	20/20, passes; 0/20, fails
T2a-M, T2b-M, the same with weaker coupling	20/20 and 20/20, pass	6/6 and 6/6, pass	20/20, passes; 0/20, fails
T0a-D, transmission failure: the oscillators bring the loss forward	20/20, passes	6/6, passes	20/20, passes
T0b-D, timing variability: the oscillators bring the loss forward	9/20, fails	0/6, fails	0/20, fails
SP1, the loss sits at a common share of counting connections (a consistency check)	20/20, passes	6/6, passes	not run (no T2b-D loss to pair)
SP2, the coherence loss is synchronisation-like	20/20, passes	6/6, passes	20/20, passes
T1d, collective fluctuations at the threshold	17/20, passes	6/6, passes	16/20, passes
T1e, slower fluctuations at the threshold	20/20, passes	6/6, passes	20/20, passes
T1f, the excess grows with size	not run	6/6, passes	13/20, fails
WNa, WNb, GNa, GNb, the real wiring differs from its rewired copy	2/20 to 2/20, fail	not run	0/20 to 3/20, fail
K1d, sensory-pathway faults disconnect the senses with DPC quorum kept	20/20, passes	not run	not evaluable
K1e, control: not with as many random connections faulted	18/20, passes	not run	not evaluable
I1, phase-model inertia at any feedback up to 1.5 times the locking threshold	0/10 at every value, fails	not run	0/10 at every value, fails
I2, held-out phase-model inertia ratio within 1.07 to 1.63	0/20, fails	not run	0/20, fails
PI1, PI2, percolation-model inertia on the connectome graph	20/20 and 20/20, pass	not run (no λ reproduced a ratio)	20/20 and 20/20, pass

Test	*C. elegans*	*Drosophila*	Human
PI3, no hysteresis without feedback	20/20, passes	20/20, passes	20/20, passes
PI4, held-out ratio within 1.07 to 1.63	13/20, fails	not run	17/20, passes

What the results show, against the expectations stated before the run:

- *Transmission failure* abolishes DPC quorum on all three connectomes, with either reference and with weaker coupling, as expected and as limitation 25 says it must at the maximum effect chosen. The oscillator dynamics bring the loss forward: quorum is lost at 1.52, 1.40 and 1.82 times the EC50 on *C. elegans*, *Drosophila* and the human connectome, against 2.06, 1.90 and no loss at all for static percolation on the same timing draws (T0a-D).
- *Timing variability alone does not abolish DPC quorum on the human connectome*, in no seed of 20, under any condition: global reference, reference carried by the connections, rewired connectome or weaker coupling. This was named before the run as the expected failure. Global coherence falls below half its awake value at 0.69 times the EC50, but the giant component of counting connections stays above q^* up to three times the EC50: the dense human wiring keeps a quorum of timing-coherent links even when the agent's timing error exceeds a whole cycle (Appendix C.13 shows that this depends on counting each human connection as two independently timed arcs). On *C. elegans* and *Drosophila* timing variability does abolish quorum, at 1.21 and 1.10 times the EC50, but at the same concentration as static percolation on its own timing draws (T0b-D fails; at the loss the counting share is 0.98 and 0.97 of the static prediction), so there the oscillators add nothing. Clause (b) of P4 (timing faults; not the postulate P4b), as stated, therefore holds on the two invertebrate connectomes and fails on the human one in this model (Appendix C.13 shows that this depends on counting each human connection as two independently timed arcs).
- Where DPC quorum is lost by both agents, it is lost at a common share of counting connections rather than at a common effective coupling (SP1); the coherence criterion of the earlier appendices is instead lost at a common effective coupling (SP2), as in a synchronisation transition. SP1, like SP1-1 in Appendix C.13, could not have failed, since the effective coupling of the NMDA-antagonist-like agent at its loss is orders of magnitude below that of the GABAergic-like agent; it is a consistency check, not evidence that the transition is percolation rather than synchronisation.
- The fluctuation claim survives its corrected test. At the threshold, coherence fluctuates at least twice as much as independent phases with the same mean coherence would (T1d) and is more autocorrelated than awake (T1e) on all three connectomes; the growth with size (T1f) holds between the human and fly connectomes but not between *C. elegans* and the human one.
- The specific wiring does not matter: on degree-preserving rewired copies, DPC quorum is lost at the same concentration, within one step, in almost every seed, whether the reference is global or carried by the connections. Carrying the reference over the connections did not create a dependence on the wiring, so the match of the null models in Appendix C.3 is not explained by the global reference. The threshold is set by the degree distribution (Appendix B.5), not by which populations connect.
- The sensory disconnection of ketamine (K1c) survives its control on *C. elegans*: with the agent's timing error on the sensory pathways, DPC quorum holds while broadcast above the floor falls to half or less in every seed (K1d), and the same error on as many randomly chosen connections does not do this in 18 of 20 seeds (K1e). The human connectome again could not be evaluated.
- Inertia remains a property of the percolation model only. In the phase model read with DPC's rule there is no hysteresis at any feedback strength scanned, up to 1.5 times the locking threshold (I1, I2), as expected. In the percolation model on the connectome graphs, a calibrated feedback produces inertia on the *C. elegans* and human graphs (PI1, PI2), and on held-out seeds the human ratio stays within the measured range (PI4, 17 of 20) while the *C. elegans* ratio does not (13 of 20); on the fly graph no feedback up to 30 ms produced any inertia.

Notes added in the third review round (21 October 2026), after the run.

- *The wrapped reading.* Timed at the receiver with everything wrapped, the secondary reading defined above (the communication-through-coherence reading), the NMDA-antagonist-like agent abolishes quorum on no connectome: 0 of 20 *C. elegans*, 0 of 6 *Drosophila* and 0 of 20 human seeds, under the global reference, the reference carried by the connections, the rewired connectomes and weaker coupling alike (`simulator/results/dpc_readout_full_20261020T224553Z.md`). However large the timing error, at least $Wf \approx 12\%$ of transmitting connections count by chance once errors are wrapped onto one cycle. Timing variability therefore abolishes quorum in this model only if a signal arriving a whole cycle or more late counts as lost rather than as on time in a later cycle, which is P2's rule: the content a signal carries belongs to the cycle in which it was sent (Section 3.2), a rule for which no evidence is offered. Under wrapping a signal 0.6 of a cycle late is already lost; the two rules differ only for signals arriving within $W/2$ of a whole number of cycles late.
- *The arousal term.* The reference term $\lambda_{\text{osc}}[R + A(c)(1-R)]\sin(\psi - \theta_i)$ has two parts. Only the first, $\lambda_{\text{osc}} R \sin(\psi - \theta_i)$, is all-to-all Kuramoto coupling, and K_c = 0.863 is the locking threshold of that part alone. The second, $\lambda_{\text{osc}} A(c)(1-R)\sin(\psi - \theta_i)$, pulls every population towards the mean phase whatever the coherence, which breaks the symmetry of the incoherent state; with it the onset of synchrony is not the Kuramoto transition, and the scan in multiples of K_c is a reference scale, not a scan across the model's own threshold.
- *Kinds.* SP1 could not have failed (above), so it is listed as a consistency check, a reclassification made after the run (Appendix A.5). T2a-D, T2a-DN, W2a and T2a-M could not fail at the maximal transmission failure chosen (limitation 25) and are

construction checks.

- *Fluctuation measures.* T1d and T1e are computed on global coherence R, not on the giant component S, whose variance and autocorrelation were not computed. The T1d bar, $\chi/\chi_{\text{ind}} \geq 2$, corresponds to a mean pairwise correlation of fluctuations of only $1/(N-1)$. T1e samples one cycle per jitter correlation time, over which the Ornstein-Uhlenbeck jitter keeps an autocorrelation of e^{-1}. Neither shows a peak at the threshold, and T1f compares different connectomes, not sizes of one network, so it is not a finite-size test.

Commands

From `simulator/`: the unit tests, `../.venv/bin/python -m unittest discover -s tests -t .`; a check that the code runs end to end in seconds, `../.venv/bin/python dpc_readout_study.py --smoke` (*C. elegans* only, one seed, two concentrations, the shortest integration); a shortened run of every organism, `../.venv/bin/python dpc_readout_study.py --quick`; the full study, `../.venv/bin/python dpc_readout_study.py`. The report is written to `simulator/results/dpc_readout_full_<start time>.md` and `.json`; smoke and quick reports (`_quick_`) are ignored by git and are not results.

C.13 Pre-specified test of the pair rule: one timing draw per anatomical connection

The tests below, their conditions, seeds and pass rule were written and committed before the study that evaluates them was run (`simulator/dpc_pair_rule_study.py`, with `simulator/connection_pairs.py` and `simulator/dpc_readout.py`), in `analysis/2026-10-21_pair_rule_prespecification.md` (commit `b370495`), and are reproduced here as committed. The only changes are to statements written before the run about the run itself, which are marked as pre-run statements, and the Results subsection, added after the run. Before that, the code was checked with unit tests and with one smoke run (*C. elegans* only, one seed, two concentrations, the shortest integration). Its outcomes were computed and read while the code was checked; with one seed and two concentrations they are not part of the results and did not inform the expectations below.

Why this study. The DPC readout of Appendix C.12 treats every directed arc of a connectome as a separate connection with its own timing draw, and links two populations in the undirected giant component if either arc between them is on time. DPC never defined when a connection between two populations counts, so this was a modelling choice, and it is not neutral across the three connectomes. The human connectome comes from diffusion MRI, which records no direction, so each of its 17,865 fibres is stored as two arcs, and in C.12 every fibre had two independent chances to be on time in each cycle. On the other two connectomes the arcs carry measured direction, and most pairs of populations are linked one way only:

Connectome	Populations	Arcs	Anatomical connections	Linked both ways	Chances per connection in C.12
C. elegans	300	5,764	3,465	1,579 (46%)	1.66
Drosophila	138,639	2,700,513	2,511,789	188,724 (7.5%)	1.08
Human	998	35,730	17,865	17,865 (100%)	2.00

An anatomical connection is an unordered pair of populations linked by at least one arc. In *C. elegans*, 720 of the 5,764 arcs repeat a direction already present between the same two neurons (a chemical synapse and a gap junction in the same direction), so some pairs had three chances in C.12.

T2b-D, the test that timing variability alone abolishes DPC quorum, failed on the human connectome in C.12 (0 of 20 seeds) while it passed on *C. elegans* (20 of 20) and *Drosophila* (6 of 6). After that result had been seen, a post hoc diagnostic (`simulator/pair_rule_diagnostic.py`, seed 7, 20 draws) asked whether the failure could come from the counting alone. It runs no oscillators: every arc is on time independently with the static probability of C.12's baseline at three times the EC50 of the NMDA-antagonist-like agent, $(1-p)\,\mathrm{erf}(W/(2\sqrt{2}\sigma_{\text{eff}}))$ with σ_{eff} = 39.6 ms, which is 0.024, and the giant component is read under each rule below:

Connectome	One draw per connection	Both directions	Either direction (C.12)
C. elegans	0.05	0.02	0.17
Drosophila	0.30	0.28	0.32
Human	0.10	0.00	0.66

On the human connectome the share of populations in the giant component is 0.66 when every fibre has two chances and 0.10 when it has one, well below q^* = 0.5; on the other two it stays below q^* under every rule. The diagnostic was first run, with a different draw sequence, as a one-off command whose figures (0.64 and 0.09) were those seen when the rule was chosen; the script reproduces it. It is static: it leaves out the oscillator dynamics, which in C.12 did not reliably bring the timing loss forward (T0b-D failed on all three connectomes), so it estimates the counting effect, not the phase model's loss. If each arc is on time with probability p and the two arcs of a fibre are independent, a fibre counts with probability $1-(1-p)^2$ rather than p; when few connections are on time this almost doubles the share of counting fibres, which in the dense human wiring (a mean of 35.8 connections per population) is enough to keep a giant component. The human failure of T2b-D may therefore come from how the readout counts a diffusion-MRI fibre rather than from the network.

What this study is, and what it is not. The authors chose the primary rule below after the diagnostic had been seen. This is a pre-specified test on fresh seeds informed by a post hoc finding, and it is weaker evidence than a blind test: the expected result on the human connectome is largely a prediction from the diagnostic, so a pass would confirm that diagnosis on new realisations, not support P4(b) independently. The verdicts of Appendix C.12 stand as reported. In particular, T2b-D fails on the human connectome under the readout that was fixed before C.12 ran, and that verdict is reported beside this one, not replaced by it.

Why neither "every fibre two-way" nor "every fibre one-way" can be assumed. In macaque cortex, where interareal connections were measured with tract tracing rather than inferred from diffusion, 66% of the possible connections between the areas studied exist, and unidirectional connections are unexpectedly common (Markov et al., 2014). Diffusion MRI cannot tell which human fibres carry axons in one direction or both. A rule that gives every human fibre two independent chances assumes that every fibre is two independent connections; a rule that reads only the arcs of one direction assumes that none is. The primary rule counts what the data record, one fibre as one connection, and the two bounds below bracket the possibilities that the data cannot separate.

The pair rules

The dynamics are unchanged, and so is every reading of C.12: the readout only adds three readings of the same counting connections. A unit test checks that every earlier reading of a sweep is identical, to the last bit, to that of the code that produced C.12 (commit `adc63f6`), and another that the either-direction rule reproduces C.12's giant-component reading exactly at every concentration. A connection's arcs are timed exactly as in C.12: an arc $j \to i$ is on time in a cycle when it transmits and $|\delta_{ji}| \le W/2$, with δ_{ji} its timing error relative to the shared rhythm.

- *One timing draw per anatomical connection (primary).* Each anatomical connection is read on one canonical arc: the first arc, in the connectome's order, from the lower-indexed to the higher-indexed population when such an arc exists, otherwise the first arc the other way. The connection counts in a cycle when that arc transmits and is on time, so each connection has one failure draw and one timing draw, as in the percolation model and in the threshold theory of Appendix B.5. The choice of canonical arc is fixed by the wiring, not by which arc happens to be on time.
- *Both directions on time (lower bound).* A connection with arcs both ways counts only when it is on time both ways; one with arcs one way only counts when that arc is on time. Each direction is read on its first arc, so a repeated arc in one direction gives no extra chance.
- *Either direction on time (upper bound).* A connection counts when any of its arcs transmits and is on time: the rule C.12 used.

In every cycle the connections counting under the both-directions rule are among those counting under the primary rule, which are among those counting under the either-direction rule, so the giant component can only be smaller under the stricter rules (checked by a unit test). Quorum, the quorum at a concentration and the loss concentration are defined as in C.12, with the counting connections of the rule in force taken as undirected links. The share of counting connections under a rule is the share of the anatomical connections that count.

Conditions, seeds and pass rule

Conditions: the GABAergic-like and the NMDA-antagonist-like agent with the global reference, on *C. elegans*, *Drosophila* and the human connectome, every other parameter at its reference value (Appendix C.1), with C.12's integration (838 measurement steps of 0.05, one cycle sampled per jitter correlation time, 42 cycles per concentration) and concentration grid (0 to 3 times the EC50 in steps of 0.05, 0.1 for *Drosophila*). No rewired, wired or weaker-coupling condition is run.

Seeds: from the base seed plus 6000 (seeds 20,267,017 onwards), 20 for *C. elegans*, 6 for *Drosophila* and 20 for the human connectome. Earlier studies used offsets 0 to 19, 1000, 2000, 3000 and 4000, and C.12 used 5000, 5500 and 5600, so no seed of this study has been run before.

Pass rule: C.12's. A test passes for an organism when the lower bound of the Wilson 95% interval for the share of seeds in which it holds lies above one half (15 of 20 and 6 of 6 seeds pass; 14 of 20 and 5 of 6 do not). A seed in which quorum under the primary rule does not hold even at zero concentration counts as failing T2a-1 and T2b-1. There are three tests per connectome, and no correction for multiple comparisons is made.

Tests

- T2a-1: the GABAergic-like agent abolishes DPC quorum under the primary rule within the concentration range (a loss concentration under the primary rule exists).
- T2b-1: the NMDA-antagonist-like agent abolishes DPC quorum under the primary rule within the concentration range.
- SP1-1 (paired by seed, per connectome): at their losses under the primary rule, the two agents leave more similar shares of counting anatomical connections than effective couplings K_{eff}, compared as absolute log ratios, as in SP1. Undefined, and not counted, in a seed in which either agent keeps quorum; not judged for a connectome on which no seed has both losses.

Reported for every connectome and agent, without a verdict: the loss concentration under each of the three rules; the share of anatomical connections counting at the loss under the primary rule; the share of populations in the giant component at three times the EC50 under each rule; the loss concentration of the coherence criterion; awake global coherence; and the run time.

What we expect, stated before the run

- T2b-1 should pass on the human connectome. This is the prediction of the post hoc diagnostic above (a giant component of 0.10 at three times the EC50 with one draw per fibre, in the static estimate), now tested in the phase model on fresh seeds. Under the either-direction rule the human connectome should again keep quorum up to three times the EC50, as in C.12, and the both-directions rule should lose it earliest. If T2b-1 fails, timing variability does not abolish quorum on the human connectome even with one chance per fibre, and C.12's conclusion holds under every rule considered here.
- T2a-1 should pass on all three connectomes, as T2a-D did (20 of 20, 6 of 6 and 20 of 20 seeds). Because the giant component under the primary rule is never larger than under C.12's rule, a seed that loses quorum under C.12's rule also loses it under the primary rule, at the same or a lower concentration, provided quorum holds while awake. It should: at the reference values about 38% of connections are timing-coherent while awake (Section 4.7), which leaves each population with about 9 (*C. elegans*), 14 (human) and 14 (*Drosophila*, on average) counting connections; in a random graph of that mean degree the giant component holds nearly every population. As limitation 25 says of T2a, this cannot fail by construction at the maximum transmission failure chosen.
- On *C. elegans* and *Drosophila*, T2a-1 and T2b-1 should pass as T2a-D and T2b-D did, for the same reason. The losses should come at the same concentration as under C.12's rule or below it: on *Drosophila*, where only 7.5% of connections are linked both ways (1.08 chances per connection in C.12), within about one step of it; on *C. elegans* (1.66 chances per connection) earlier than C.12's 1.52 and 1.21 times the EC50.
- SP1-1 should pass on *C. elegans* and *Drosophila*, as SP1 did (20 of 20 and 6 of 6): the readout counts connections, so where both agents lose quorum the loss should sit at a common share of counting connections. On the human connectome SP1 could not be run in C.12, since no seed lost quorum under the NMDA-antagonist-like agent; with the primary rule it becomes evaluable, and its outcome is uncertain.
- What would change the reading of P4(b). If T2b-1 passes on the human connectome while the either-direction rule keeps quorum there, P4(b) holds on all three connectomes in this model only if a diffusion-MRI fibre counts once, and its human verdict depends on the reciprocity of human cortical connections, which diffusion imaging cannot measure. The paper will then report both verdicts and state that dependence; it will not report the human failure of C.12 as resolved.

Results

The code and the pre-specification above were committed at 10:50:37 UTC on 21 October 2026 (commit `b370495`); the study started at 10:51:42 UTC from a clean working tree at that commit and ended at 14:54:42 UTC, after just over four hours (`simulator/results/dpc_pair_rule_full_20261021T105142Z`). Counts are seeds in which a test holds, with the Wilson 95% interval.

Test	*C. elegans*	*Drosophila*	Human
T2a-1, the GABAergic-like agent abolishes DPC quorum, one timing draw per anatomical connection	20/20 (84–100%), passes	6/6 (61–100%), passes	20/20 (84–100%), passes
T2b-1, the NMDA-antagonist-like agent abolishes DPC quorum, one timing draw per anatomical connection	20/20 (84–100%), passes	6/6 (61–100%), passes	20/20 (84–100%), passes
SP1-1, at their losses the two agents leave a more similar share of counting anatomical connections than effective coupling (a consistency check, reclassified after the run)	20/20 (84–100%), passes	6/6 (61–100%), passes	20/20 (84–100%), passes

Measurements reported without a verdict (mean ± SD across seeds; concentrations relative to the EC50; "giant component" is the share of populations in it):

Quantity	Agent	*C. elegans*	*Drosophila*	Human
DPC loss concentration, one timing draw per connection (primary)	GABAergic-like	1.375 ± 0.034	1.400 ± 0.000	1.565 ± 0.024
DPC loss concentration, both directions on time	GABAergic-like	1.193 ± 0.024	1.400 ± 0.000	0.950 ± 0.000
DPC loss concentration, either direction on time (C.12's rule)	GABAergic-like	1.530 ± 0.041	1.400 ± 0.000	1.833 ± 0.041
Giant component at three times the EC50, one timing draw per connection	GABAergic-like	0.017 ± 0.002	0.040 ± 0.001	0.009 ± 0.001
Giant component at three times the EC50, both directions on time	GABAergic-like	0.012 ± 0.001	0.027 ± 0.001	0.002 ± 0.000
Giant component at three times the EC50, either direction on time	GABAergic-like	0.030 ± 0.004	0.054 ± 0.001	0.035 ± 0.005
Share of anatomical connections counting at the primary loss	GABAergic-like	0.061 ± 0.003	0.055 ± 0.000	0.041 ± 0.002
Coherence-criterion loss concentration	GABAergic-like	1.427 ± 0.060	1.400 ± 0.000	1.552 ± 0.030
DPC loss concentration, one timing draw per connection (primary)	NMDA-antagonist-like	0.903 ± 0.011	1.000 ± 0.000	1.150 ± 0.000
DPC loss concentration, both directions on time	NMDA-antagonist-like	0.708 ± 0.018	1.000 ± 0.000	0.600 ± 0.000
DPC loss concentration, either direction on time (C.12's rule)	NMDA-antagonist-like	1.200 ± 0.016	1.100 ± 0.000	not reached (0/20)
Giant component at three times the EC50, one timing draw per connection	NMDA-antagonist-like	0.049 ± 0.003	0.296 ± 0.000	0.083 ± 0.006
Giant component at three times the EC50, both directions on time	NMDA-antagonist-like	0.022 ± 0.001	0.271 ± 0.000	0.002 ± 0.000
Giant component at three times the EC50, either direction on time	NMDA-antagonist-like	0.198 ± 0.011	0.319 ± 0.000	0.645 ± 0.005

Quantity	Agent	*C. elegans*	*Drosophila*	Human
Share of anatomical connections counting at the primary loss	NMDA-antagonist-like	0.063 ± 0.002	0.051 ± 0.000	0.039 ± 0.000
Coherence-criterion loss concentration	NMDA-antagonist-like	0.650 ± 0.000	0.600 ± 0.000	0.700 ± 0.000

The human failure of C.12 came from counting each connection inferred from diffusion MRI as two independent chances to be on time. Counted once, or with both directions required, timing variability alone abolishes DPC quorum on all three connectomes, so the human verdict of clause (b) of P4 (timing faults; not the postulate P4b) rests on how many human connections are reciprocal, how independently their two directions are timed and whether one direction suffices. Diffusion MRI cannot settle this, and directed human measurements, such as responses to single-pulse cortical stimulation (as in the dataset analysed in Appendix E), constrain it only indirectly. None of this shows that real brains lose consciousness this way.

Against the expectations stated before the run:

- *Timing variability on the human connectome.* T2b-1 passes the pre-specified test on the human connectome (20 of 20 seeds): with one timing draw per connection, the NMDA-antagonist-like agent abolishes DPC quorum at 1.15 times the EC50 in every seed. Under C.12's either-direction rule it does not in any seed up to three times the EC50, where the giant component is 0.645, and under the both-directions rule quorum is lost at 0.60 times the EC50, earliest, as expected. At three times the EC50 the giant component under the three rules (0.083, 0.002 and 0.645) is close to the static diagnostic's 0.10, 0.00 and 0.66. The human verdict of P4(b) therefore depends on the counting rule. It fails under the rule that counts a connection when either of two independently timed directions is on time, applied to a connectome in which every connection is two-way; intermediate degrees of reciprocity were not run. Whether one on-time direction should suffice to join two populations is itself open: P1's recurrent processing would favour requiring both, under which quorum is lost earliest. The human map is 100% reciprocal because diffusion MRI records no direction, not because reciprocity was measured; in macaque cortex, where tract tracing measures direction, unidirectional connections are unexpectedly common (Markov et al., 2014); and the two directions of a connection share the same rhythm, so fully independent timing is an upper bound. How reciprocal human cortical connections are is not known, and this study does not decide it. As stated before the run, both verdicts are reported, and the human failure of C.12 stands beside this result.
- *C. elegans and Drosophila.* T2a-1 and T2b-1 pass, with losses at or below C.12's, as expected. On *C. elegans* quorum is lost at 1.375 (GABAergic-like) and 0.903 (NMDA-antagonist-like) times the EC50 under the primary rule, earlier than C.12's 1.52 and 1.21 and than the 1.530 and 1.200 of the either-direction rule on these seeds. On *Drosophila* it is lost at 1.400 and 1.000, against C.12's 1.40 and 1.10 and the either-direction rule's 1.400 and 1.100 here: the same concentration for the GABAergic-like agent and one step lower for the NMDA-antagonist-like agent.
- *T2a-1* passes on all three connectomes (20 of 20, 6 of 6, 20 of 20), as expected. As the pre-specification said, it cannot fail by construction at the maximum transmission failure chosen (limitation 25); what it adds is where the loss happens, at 1.375, 1.400 and 1.565 times the EC50.
- *SP1-1* passes on all three connectomes (20 of 20, 6 of 6, 20 of 20), and is now evaluable on the human connectome. At the primary loss the share of anatomical connections that count is 0.061 and 0.063 on *C. elegans*, 0.055 and 0.051 on *Drosophila* and 0.041 and 0.039 on the human connectome (GABAergic-like and NMDA-antagonist-like agent), between 3.9% and 6.3% throughout. This could not have failed: the effective coupling of the NMDA-antagonist-like agent at its loss is 10^{-3} to 10^{-9} of the GABAergic-like agent's, because effective coupling collapses once timing error exceeds about half a cycle at 40 Hz, and a common counting share is what a giant-component readout of independently counted connections implies. SP1-1, like SP1, is a consistency check, not evidence that the transition is percolation rather than synchronisation.
- *Coherence criterion.* Under the NMDA-antagonist-like agent the coherence criterion of the earlier appendices is lost before DPC quorum on every connectome, at 0.650, 0.600 and 0.700 times the EC50, against 0.903, 1.000 and 1.150 under the primary rule. Under the GABAergic-like agent the two are lost close together (coherence at 1.427, 1.400 and 1.552, DPC quorum at 1.375, 1.400 and 1.565), and on *C. elegans* DPC quorum is lost first.

What the results do and do not show. This is a pre-specified test informed by a post hoc finding, and weaker evidence than a blind test: the human result was largely predicted by the static diagnostic before the run. C.12's verdict stands and is reported beside this one: under the readout fixed before C.12 ran, T2b-D fails on the human connectome (0 of 20 seeds). A pass shows that the model is consistent with P4(b) under a stated counting rule, not that brains work this way. The dynamics were not changed: in the phase model each human connection still couples its two populations through two arcs with independent timing, and the primary rule reads only one of them. A model in which the two directions share one timing draw was not run.

Notes added in the third review round (21 October 2026), after the run.

- *Counting and the threshold theory.* Section 4.1 and Appendix B.5 give each undirected link one failure draw and one timing draw. C.12's rule gave each human link two, one per stored direction, either sufficing, which contradicts that framing; the primary rule here restores it in the readout. The dynamics still couple each human pair through two independently timed arcs, so the phase model itself keeps two draws per link.
- *What else the human verdict depends on.* Besides reciprocity, the human verdict depends on the NMDA-antagonist-like agent's maximal added timing error, $\sigma_{\max}$ = 40 ms, a free parameter that the sensitivity analysis does not vary (Appendix C.1). Static estimates under the either-direction rule (Appendix C.14), post hoc in origin, place the crossing of q^* at three times the EC50

at a reciprocal share of about 0.62 and, with every connection two-way, at $\sigma_{\max}$ of about 49.5 ms; the other rules were not swept.
- *Kinds.* T2a-1 is a construction check (limitation 25); SP1-1 is a consistency check, reclassified after the run.

Commands

From `simulator/`: the post hoc diagnostic, `../.venv/bin/python pair_rule_diagnostic.py` (about 15 seconds); the unit tests, `../.venv/bin/python -m unittest discover -s tests -t .`; a check that the code runs end to end in seconds, `../.venv/bin/python dpc_pair_rule_study.py --smoke` (*C. elegans* only, one seed, two concentrations, the shortest integration); the full study, `../.venv/bin/python dpc_pair_rule_study.py`. The report is written to `simulator/results/dpc_pair_rule_full_<start time>.md` and `.json`; smoke reports (`_quick_`) are ignored by git and are not results. Before the run we stated that, from the run times of C.12's agent block and the measured cost of the added readings, the full study should take about two and a half hours on the machine that ran C.12, almost all of it on *Drosophila*; it took just over four hours (Results).

C.14 Pre-specified analyses requested in the third review round

The text from here to Results is reproduced as committed in `74f9cb0` (`analysis/2026-10-21_round3_simulator_prespecification.md`), without its draft header and its list of references to add, which are in the reference list; the Results subsection was added after the runs.

The four analyses below, their conditions, seeds and pass rules, or the statement that an analysis is reported without a verdict, were written and committed before any of them was run in full. The code is `simulator/prediction15_study.py` (analysis A), `simulator/directed_threshold_estimate.py` (B), `simulator/reciprocity_estimate.py` (C) and `simulator/fresh_seed_validation.py` (D), with the package `simulator/round3_study/`; the unit tests are `simulator/tests/test_round3_pulse.py` and `simulator/tests/test_round3_estimates.py`.

What these analyses are, and what they are not. Referees asked for all four after the results of Appendices C.12, C.13 and E.14 had been seen. They are therefore post hoc in origin, even though each is fixed here before its own run, and none of them is a blind test of DPC. Analyses A and D ask whether the percolation model has properties the paper attributes to it; analyses B and C are a deterministic computation and a static estimate, reported without a verdict. No verdict reported earlier is changed by them.

What had been seen. Before this text was written we had read every report cited in Appendices B and C, including the results of C.12, C.13 and E.14. The code was then checked with one smoke run per command. The first smoke runs used the first seed of the blocks then intended for analyses A and D (the base seed plus 7000 and 7500, at 302 nodes) and both generators then intended for C (7200 and 7201). Because their outcomes were read, those seeds were withdrawn and every block was moved to offsets that no run has used (below); smoke runs now use a block of their own, the base seed plus 7900 and 7901, which no analysis uses. What the smoke runs showed, on one seed or one draw each:

- A, at 302 nodes: on the withdrawn seed the window was one grid step wide (c_on 0.88, c_off 0.89) and both checks held; on the smoke seed it was 0.86 to 0.93, both checks held, and the smallest effective increases of the arousal drive were 0.061 inside the window and 0.368 above c_off.
- B is deterministic, so its smoke run computed the *C. elegans* row exactly: onset 0.0377, quorum estimate 0.0967, against 0.1058 measured in Appendix B.5. That row is known before the full run.
- C, with one assignment of direction and two draws: a giant component of 0.66 to 0.68 with every fibre two-way and 0.07 to 0.13 with every fibre one-way, and 0.66 to 0.67 at the reference timing error. These agree with the post hoc diagnostic of Appendix C.13 (0.66 under the either-direction rule, 0.10 with one draw per fibre).
- D, at 302 nodes: on the withdrawn seed every claim held except C4; on the smoke seed every claim held, with a loss-to-recovery ratio of 1.081.

Seeds

Earlier studies used the base seed (20,261,017) plus 0 to 19, 1000, 2000, 3000, 4000, 5000, 5500, 5600 and 6000, each block at most 20 seeds long. The seeds of this appendix are:

Analysis	Seeds (offset from the base seed)	Count
A, prediction 15	7100 to 7119, at each network size	20 per size, as in B.4
C, reciprocity and timing error	7300 (reciprocity sweep) and 7301 (timing-error sweep), one generator each	2
D, validation on fresh seeds	7600 to 7619, at each network size	20 per size, as in B.4
Smoke runs only	7900 and 7901	2
Withdrawn (seen in early smoke runs)	7000, 7200, 7201, 7500	4

Analysis B uses no random numbers. A unit test checks that every block of this appendix is disjoint from every earlier block, from the withdrawn seeds, from the smoke block and from the others, and in particular that D's seeds include none of the calibration seeds of `calibrate_feedback.py` and none of B.4's.

A. Prediction 15 in the percolation model

Why. Section 5.4 derives prediction 15 from P5 and P7: between the recovery and loss concentrations, where the percolation model has two stable states, a brief arousing stimulus should restore quorum that outlasts it, and above the loss concentration quorum should return only while the stimulus lasts. The section states that this follows from the model's equation and has not been simulated. A referee asked for the simulation, and for the width of the window in which the prediction applies, since an experiment has to hold the concentration inside it.

Model and procedure. The percolation model of Appendix B with every reference value (random graph of mean degree 8, GABAergic-like agent, W = 3 ms, baseline jitter 3 ms, arousal level A_0 = 0.5, λ = 8.85 ms, q^* = 0.5, concentration grid 0 to 3 times the EC50 in steps of 0.01), at the five network sizes of B.4 (302, 1,000, 10,000, 100,000 and 1,000,000 nodes), 20 seeds each. For each seed:

1. The reference sweep of B.4 gives the loss concentration c_{off} (the last concentration of induction with quorum) and the recovery concentration c_{on} (the highest concentration of emergence with quorum). The window is $[c_{\text{on}}, c_{\text{off}}]$; a seed without both, or with $c_{\text{off}} \leq c_{\text{on}}$, has no window.
2. Two test concentrations are fixed from the seed's own window: (i) its midpoint, $(c_{\text{on}} + c_{\text{off}})/2$; (ii) c_{off} + 0.1 times the EC50. The value 0.1 was chosen before any run of this analysis because it lies above c_{off} in every seed and, in every seed of B.4 at every size, below the loss concentration of the model without feedback (λ = 0), which is the state the full pulse produces (below): B.4's no-feedback loss exceeded c_{off} + 0.1 by at least 0.02 EC50 in every seed.
3. The starting state is the unconscious state that induction leaves: the network is taken up the induction branch to three times the EC50, where quorum is lost, then set to the test concentration and settled there by the reference solver. This is the state on the emergence branch, the one an animal is in when it is stimulated during emergence.
4. The model has no time other than the steps of its self-consistency map, $S \to F(\pi(c, S))$. The pulse raises the arousal drive $A(c)$ of Section 4.3 by a fixed amount, capped at 1, for 500 steps (the iteration limit within which the reference solver must converge, so that the network settles under the pulse); A then returns to baseline and the map is iterated to its steady state by the reference solver. The primary pulse raises A by 1, which takes it to its maximum at every concentration: during the pulse the reference does not weaken (σ_{ref} = 0), the strongest stimulus the model can represent. The secondary pulse raises A by A_0 = 0.5, which brings the drive back at least to its awake value at every concentration, since the agent removes at most A_0.
5. Quorum before the pulse, at its end and after it is read against q^* = 0.5.

Checks. Both are consistency checks, not tests, for a reason found while the code was written. With the primary pulse the pulsed map does not depend on S, so one step takes the network to the state of the model without feedback. Because the map is monotone in S and in A, that state lies at or above the conscious branch wherever the conscious branch exists, so after the pulse the network settles on it (P15a); above c_{off} no state with quorum exists, so after any pulse the network returns below q^* (P15b). P15a can fail only where the window is too narrow for the starting state to lie below q^* at its midpoint, and P15b only where the model without feedback has already lost quorum at c_{off} + 0.1. The checks confirm that the code implements the equations; they do not test them.

- P15a (consistency check): inside the window, at its midpoint, the network starts without quorum, the primary pulse restores quorum, and quorum persists after the pulse.
- P15b (consistency check): at c_{off} + 0.1 EC50, the network starts without quorum, the primary pulse restores quorum, and quorum is lost again after the pulse.

A seed without a window counts as failing both.

Pass rule. Appendix C.12's: a check passes at a network size when the lower bound of the Wilson 95% interval for the share of seeds in which it holds lies above one half (15 of 20 seeds pass, 14 do not). There are two checks per size and no correction for multiple comparisons.

Reported without a verdict. P15a and P15b with the secondary pulse (A raised by 0.5); whether each seed has a window; at each size, the mean and SD across seeds of c_{off}, c_{on}, the window width $c_{\text{off}} - c_{\text{on}}$ in EC50, its midpoint, and the precision an experiment would need, half the width as a share of the midpoint (aiming at the midpoint, the concentration must stay within that share of its target to remain inside the window); the giant component before, at the end of and after each pulse and the first step of the pulse at which quorum returned; and, by bisection on the increase of A to 1/1024, the smallest increase that leaves quorum persisting at the window's midpoint and the smallest that restores quorum during the pulse at c_{off} + 0.1. These two are the informative quantities: they say how strong an arousing stimulus must be, in the model's units, which the checks do not.

What we expect, stated before the run.

- P15a and P15b should pass at every size, for the reason given under Checks. In B.4 one window, at 1,000 nodes, was one grid step wide, and the no-feedback loss lay above c_{off} + 0.1 in every seed, so failures, if any, should be confined to a few seeds at 302 or 1,000 nodes.
- The window should be about as wide as B.4's hysteresis: 0.085 ± 0.037, 0.069 ± 0.041, 0.061 ± 0.013, 0.061 ± 0.010 and 0.054 ± 0.005 EC50 from 302 to 1,000,000 nodes, around a midpoint near 0.86 EC50. An experiment would then have to hold the con-

centration within about 5% of its target in small networks and about 3% at 1,000,000 nodes. This width belongs to random graphs of mean degree 8 with λ calibrated to the inertia ratio 1.08; on the connectome graphs of C.12 the calibrated λ differs, and the window there is not computed here.
- The secondary pulse should restore quorum at the midpoint in every seed and leave it persisting, since at the midpoint the conscious branch exists and the drive is restored above its awake value. At c_{off} + 0.1 it is marginal: a mean-field estimate for a random graph of mean degree 8 gives a giant component of about 0.6 during the pulse, so P15b with the secondary pulse may fail in some seeds because quorum does not return, not because it persists.
- The smallest increase at the midpoint depends on where the midpoint sits relative to c_{on} and should be small; the smallest increase that restores quorum during the pulse above c_{off} should be several times larger.

B. Directed generating-function estimate of the strong component

Why. Appendix B.5 measured the quorum threshold of each connectome on its directed wiring, as the occupation at which the largest strongly connected component reaches q^*, and compared the undirected measurement with the generating-function formula of Callaway et al. (2000), which it matched within 6%. A referee asked whether the directed measurement also has a theoretical counterpart.

Method. For an uncorrelated directed network with joint distribution p_{jk} of in-degree j and out-degree k, each arc kept with probability π (Newman, Strogatz and Watts, 2001; Dorogovtsev, Mendes and Samukhin, 2001):

$$\pi_c = \frac{\langle k \rangle}{\langle jk \rangle}, \qquad x = 1 - \pi + \frac{\pi}{\langle k \rangle} \sum_{jk} p_{jk}\, j\, x^{k}, \qquad y = 1 - \pi + \frac{\pi}{\langle k \rangle} \sum_{jk} p_{jk}\, k\, y^{j}, \qquad S_{\text{strong}} = \sum_{jk} p_{jk} \bigl(1 - x^{k}\bigr)\bigl(1 - y^{j}\bigr),$$

where x is the probability that an arc does not lead into the giant out-component and y that it does not come from the giant in-component, each the smallest solution in [0, 1], found by iteration from zero. π_c is the onset of a giant strongly connected component, and the quorum estimate is the smallest π at which S_{strong} reaches q^* = 0.5, by bisection to about 10^{-12}. Each population keeps its own pair (j, k); every arc counts, as in B.5's measurement, so a repeated arc is a second chance; self-connections are left out (there are none in the three connectomes). The estimates are computed for *C. elegans*, *Drosophila* and the human connectome and set beside the directed thresholds measured in B.5 (`simulator/results/percolation_full_20261020T011339Z.json`: 0.1058 ± 0.0079, 0.1503 ± 0.0006 and 0.0495 ± 0.0015), with the critical transmission failure and timing error they imply, as in B.5.

Reported without a verdict. The computation is deterministic. The estimate assumes an uncorrelated network, so it ignores every correlation between populations, among them reciprocity (46% of *C. elegans* connections and 7.5% of *Drosophila* connections are two-way, Appendix C.13, and every human connection is), degree assortativity and modularity. A difference from the measurement measures how much those correlations matter; it does not test DPC.

What we expect, stated before the run. The *C. elegans* row is known from the smoke run: the estimate, 0.0967, is 9% below the measurement. For the human connectome, in which every population has as many inputs as outputs, $\pi_c = \langle k \rangle / \langle k^2 \rangle$ = 35.8 / 1,536.3 = 0.0233 from B.5's degree moments, close to the undirected onset of 0.0239. Reciprocity creates two-arc cycles that an uncorrelated network lacks, which should lower the measured threshold below the estimate, while modularity and the other correlations can move it either way; *C. elegans* moved the other way. We therefore state no expectation for the direction of the difference on *Drosophila* or the human connectome, and expect agreement within about 20%, as the undirected formula agreed within 6%.

C. Static estimates of reciprocity and maximal timing error on the human connectome

Why. Appendix C.13 showed that the human verdict of clause (b) of P4 depends on how a connection inferred from diffusion MRI is counted: with every fibre two-way and either direction sufficient, timing variability alone does not abolish quorum; with one timing draw per fibre, it does. Diffusion MRI cannot say how many human fibres are two-way. A referee asked how much reciprocity the either-direction result needs, and how large the agent's timing error must be before even a fully reciprocal connectome loses quorum.

Method. Static, as the diagnostic of C.13 (`pair_rule_diagnostic.py`): no oscillators run, and every arc is on time independently with the probability of C.12's static baseline at three times the EC50 of the NMDA-antagonist-like agent, $(1 - p)\ \mathrm{erf}\bigl(W / (2\sqrt{2}\,\sigma_{\text{eff}})\bigr)$, with σ_{eff} combining the baseline jitter and the agent's added error (0.024 at its reference maximal error of 40 ms). The giant component is read under C.12's either-direction rule: a fibre counts when any of its arcs is on time.

- (i) Reciprocity sweep. For each reciprocal share r from 0 to 1 in steps of 0.05, a random share r of the 17,865 human fibres is made two-way and the rest one-way in a random direction (each direction with probability one half); the number of fibres never changes. Ten assignments of direction per value of r, 20 timing draws on each.
- (ii) Timing-error sweep. With every fibre two-way (r = 1), the agent's maximal added timing error is set to 20, 25, ..., 80 ms (its reference value is 40 ms), ten assignments of 20 draws each.

For each sweep, the mean giant component (over assignments and draws) is reported at every value, and the value at which it first falls below q^* = 0.5 is interpolated linearly between the two grid values around the crossing, reading r downwards from 1 and the

timing error upwards from 20 ms. A sweep that never crosses is reported as such.

Reported without a verdict. These are post hoc static estimates requested by referees. They leave out the oscillator dynamics, which in C.12 did not reliably bring the timing loss forward (T0b-D), and they assume that the two directions of a fibre are timed independently, an upper bound on the benefit of reciprocity.

What we expect, stated before the run. Under the either-direction rule with independent arcs, a fibre counts with probability $p + r\,p(1-p)$, and the direction of a one-way fibre does not matter, so sweep (i) is bond percolation on the undirected human graph at that occupation. Setting it equal to B.5's undirected quorum threshold (0.0368 by formula, 0.0381 measured) gives a crossing at r of about 0.53 to 0.59: quorum should survive the timing faults only if more than about half of the human fibres are two-way. For sweep (ii), $2p - p^2$ reaches the same thresholds when the maximal added timing error is about 50 to 52 ms, so a fully reciprocal human connectome should keep quorum at three times the EC50 unless the agent's maximal timing error exceeds about 50 ms, a quarter more than the 40 ms assumed.

D. The validation study on seeds disjoint from calibration and from B.4

Why. Appendix C.12 found that the percolation feedback was calibrated (`calibrate_feedback.py`) on the base seed plus 0 to 9 at 10,000 nodes, and that B.4 used the base seed plus 0 to 19 at every size with the same graph builder, so ten of B.4's twenty networks at 10,000 nodes were the calibration networks themselves. A referee asked for the validation to be repeated on networks that played no part in calibration.

Method. `validation_study.py` unchanged, with every claim (C1 to C7), observation (O1 to O5) and extension test (E1 to E3) of B.4, at B.4's five sizes with 20 seeds each, on the base seed plus 7600 to 7619. The calibrated λ is not recalibrated. Also reported, without a verdict, at every size: the loss-to-recovery concentration ratio $c_{\text{off}}/c_{\text{on}}$ of the reference sweep (mean ± SD across seeds) and the number of seeds in which it lies within the published range of 1.07 to 1.63.

Pass rule. B.4's: a claim holds at a size only when it holds in every seed, and the verdict is that every claim holds at every size, or that at least one failed. Observations and extension tests are reported, not judged.

What we expect, stated before the run. The code and design are B.4's, so B.4's outcomes should recur. C1 to C3 and C5 to C7 held in every seed at every size and should hold again. C4 held in 19 of 20 seeds at 302 nodes, so the every-seed rule may fail again there for that reason alone, as it did in B.4. The ratio should fall with size, as in B.4 (1.106 ± 0.050, 1.086 ± 0.055, 1.075 ± 0.017, 1.075 ± 0.013 and 1.066 ± 0.006 from 302 to 1,000,000 nodes, within the published range in 14, 10, 14, 15 and 9 of 20 seeds). At 10,000 nodes B.4's calibration seeds gave 1.079 ± 0.015 and its other ten seeds 1.071 ± 0.019, so on fresh seeds the ratio should be near 1.07, slightly below the target of 1.08 to which λ was calibrated, and near the lowest published ratio at 1,000,000 nodes. A ratio well below 1.07 on fresh seeds would show that the calibration fitted its own networks rather than the model.

Commands and run times

From `simulator/`: the unit tests, `../.venv/bin/python -m unittest discover -s tests -t .`; and for each analysis, a check that the code runs end to end in about a second (`--smoke`, on the smoke seeds only) and the full run:

Analysis	Full run	Estimated run time
A	`../.venv/bin/python prediction15_study.py`	about 2 minutes, at most 5
B	`../.venv/bin/python directed_threshold_estimate.py`	under 1 minute
C	`../.venv/bin/python reciprocity_estimate.py`	under 1 minute
D	`../.venv/bin/python fresh_seed_validation.py`	about 12 minutes

The estimates are for the machine that ran B.4 (Apple M2 Pro, 10 worker processes). D repeats B.4's design, which took 695.5 s. A needs one graph, one percolation curve and one sweep per seed, against about fifteen of each per seed in B.4, whose seeds took 320 s each at 1,000,000 nodes, so about 25 s per seed there and about one minute for its 20 seeds on 10 workers; the pulses and bisections add about 13,000 steps of the map per seed, each a single lookup. B needs about 8 s to load the *Drosophila* connectome, and its *C. elegans* row took under a second. C reads the giant component about 6,800 times on the human connectome, at about 0.5 ms a reading and 1.4 ms per assignment of direction as timed on that machine, about 5 s in all. Each report is written to `simulator/results/<kind>_full_<start time>.md` and `.json`, with kinds `prediction15`, `directed_threshold_estimate`, `reciprocity_estimate` and `validation_fresh_seeds`; smoke reports (`_quick_`) are ignored by git and are not results.

Results

All four analyses ran from a clean working tree at commit `74f9cb0`, made at 20:10:45 UTC on 21 October 2026 (reports `simulator/results/directed_threshold_estimate_full_20261021T201513Z`, `reciprocity_estimate_full_20261021T201522Z`, `prediction15_full_20261021T201526Z` and `validation_fresh_seeds_full_20261021T201644Z`, each as `.md` and `.json`). B ran from 20:15:13 to 20:15:22 UTC, C from 20:15:22 to 20:15:25, A from 20:15:26 to 20:16:43 and D from 20:16:44 to 20:28:45 (721.1 s on 10 worker processes). D's command exited with status 1 because at least one claim failed under B.4's every-seed rule; the run itself

completed. Counts are seeds in which a check or claim holds, with the Wilson 95% interval (20/20 is 84–100%); concentrations are relative to the EC50, as mean ± SD across seeds.

A. Prediction 15.

Check	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
P15a, primary pulse: inside the window (at its midpoint), quorum restored by the pulse persists after it (consistency check)	20/20, passes	20/20, passes	20/20, passes	20/20, passes	20/20, passes
P15b, primary pulse: at c_{off} + 0.1 EC50, quorum restored by the pulse is lost again after it (consistency check)	20/20, passes	20/20, passes	20/20, passes	20/20, passes	20/20, passes
P15a with the secondary pulse (A raised by 0.5), no verdict	20/20	20/20	20/20	20/20	20/20
P15b with the secondary pulse, no verdict	18/20 (70–97%)	20/20	20/20	20/20	20/20
The seed has a window, no verdict	20/20	20/20	20/20	20/20	20/20

Quantity (no verdict)	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
c_{off} (loss)	0.890 ± 0.035	0.895 ± 0.018	0.886 ± 0.007	0.885 ± 0.005	0.882 ± 0.004
c_{on} (recovery)	0.796 ± 0.056	0.810 ± 0.034	0.820 ± 0.013	0.826 ± 0.008	0.825 ± 0.005
Window width $c_{\text{off}} - c_{\text{on}}$	0.094 ± 0.040	0.084 ± 0.032	0.066 ± 0.013	0.059 ± 0.006	0.057 ± 0.007
Window midpoint	0.843 ± 0.042	0.852 ± 0.022	0.853 ± 0.008	0.855 ± 0.006	0.854 ± 0.003
Precision needed: half the width as a share of the midpoint	0.057 ± 0.025	0.050 ± 0.019	0.039 ± 0.008	0.035 ± 0.004	0.033 ± 0.004
Giant component at the midpoint before, at the end of and after the primary pulse (means)	0.158, 0.816, 0.766	0.101, 0.802, 0.751	0.019, 0.807, 0.746	0.006, 0.807, 0.745	0.001, 0.808, 0.748
Giant component at c_{off} + 0.1 before, at the end of and after the primary pulse (means)	0.058, 0.636, 0.058	0.021, 0.649, 0.021	0.004, 0.657, 0.004	0.001, 0.660, 0.001	0.000, 0.664, 0.000
Giant component at c_{off} + 0.1 at the end of the secondary pulse	0.567 ± 0.088	0.610 ± 0.018	0.631 ± 0.010	0.635 ± 0.009	0.639 ± 0.007
Smallest increase of A for quorum persisting at the midpoint	0.079 ± 0.030	0.067 ± 0.024	0.050 ± 0.011	0.044 ± 0.007	0.042 ± 0.004
Smallest increase of A for quorum during the pulse at c_{off} + 0.1	0.412 ± 0.062	0.367 ± 0.037	0.310 ± 0.012	0.298 ± 0.013	0.288 ± 0.012

Against the expectations stated before the run:

- *P15a and P15b* hold in every seed at every size and pass, as expected, with no failure at 302 or 1,000 nodes. They are consistency checks: with the primary pulse the reference does not weaken, so the outcome follows from the equations. They show that the code implements prediction 15 as Section 5.4 derives it, not that the prediction holds in brains, and they do not separate DPC from other accounts with two stable states (Section 8).
- *Window.* Every seed has one. Its width falls from 0.094 ± 0.040 EC50 at 302 nodes to 0.057 ± 0.007 at 1,000,000, around a midpoint of 0.843 to 0.855, close to B.4's hysteresis (0.085 ± 0.037 to 0.054 ± 0.005) and to the expected midpoint near 0.86. An experiment aiming at the midpoint would have to hold the concentration within 5.7 ± 2.5% of its target at 302 nodes and 3.3 ± 0.4% at 1,000,000 to stay inside the window, as expected (about 5% and 3%). These widths belong to random graphs of mean degree 8 with λ calibrated on such graphs; the window on the connectome graphs was not computed, and the EC50 here is that of the circuit effect, not a clinical concentration (limitation 23).
- *Pulse outcomes.* With the primary pulse quorum returned at the first step in every seed; the giant component reached 0.80 to 0.82 at the midpoint and 0.64 to 0.66 above c_{off}, and after the pulse it settled at 0.75 to 0.77 inside the window and fell back to its value before the pulse above c_{off}.
- *Secondary pulse* (no verdict). At the midpoint it restored quorum, which persisted, in every seed at every size, as expected. Above c_{off} quorum returned during the pulse in 18 of 20 seeds at 302 nodes and in every seed at larger sizes, and was lost again after the pulse wherever it had returned. The two seeds at 302 nodes failed because quorum did not return, the failure anticipated; the giant component during the pulse was 0.567 ± 0.088 at 302 nodes and 0.639 ± 0.007 at 1,000,000, near the mean-field estimate of about 0.6.
- *Stimulus strength.* The smallest increase of the arousal drive that leaves quorum persisting at the midpoint is 0.079 ± 0.030 at 302 nodes and 0.042 ± 0.004 at 1,000,000; the smallest that restores quorum during the pulse above c_{off} is 0.412 ± 0.062 and 0.288 ± 0.012. As expected, it is small inside the window and several times larger above it. Both are in the model's units (the awake arousal share A_0 is 0.5) and have no stated mapping to the intensity of a real stimulus.

B. Directed generating-function estimate. Deterministic, reported without a verdict.

Organism	Populations	Arcs	Onset π_c	Quorum threshold, estimate	Quorum threshold, measured (B.5)	Estimate / measured	Critical failure p_q (estimate)	Critical jitter σ_q, ms (estimate)
C. elegans	300	5,764	0.0377	0.0967	0.1058 ± 0.0079	0.91	0.747	12.3

Organism	Populations	Arcs	Onset π_c	Quorum threshold, estimate	Quorum threshold, measured (B.5)	Estimate / measured	Critical failure p_q (estimate)	Critical jitter σ_q, ms (estimate)
Drosophila	138,639	2,700,513	0.0095	0.1441	0.1503 ± 0.0006	0.96	0.624	8.3
Human	998	35,730	0.0233	0.0474	0.0495 ± 0.0015	0.96	0.876	25.2

- The estimate lies below the measured directed threshold on all three connectomes, at 0.91, 0.96 and 0.96 of it (9%, 4% and 4% below), within the about 20% expected. The *C. elegans* row equals the smoke run's, as stated before the run, and the human onset is the 0.0233 computed in advance.
- No direction of the difference was expected for *Drosophila* or the human connectome; on all three the measurement lies above the estimate. On the human connectome every arc has its reverse, and the uncorrelated estimate counts the arc back to the population a path came from as leading to a new population, which by itself places the estimate below the true threshold; the positive correlation of in- and out-reachability that reciprocity creates acts the other way. The theory of directed networks with reciprocal links (Boguñá and Serrano, 2005) would be the appropriate comparison and was not computed. The expectation stated before the run, that reciprocity alone should lower the measured threshold below the estimate, was therefore wrong as stated. The difference measures how much the ignored correlations matter; it does not test DPC.

C. Reciprocity and maximal timing error on the human connectome. Static estimates, post hoc in origin, reported without a verdict: either-direction rule, NMDA-antagonist-like agent at three times the EC50, an arc on time with probability 0.0242 at the reference maximal timing error of 40 ms, and the mean giant component over 10 assignments of direction of 20 timing draws each.

Reciprocal share r	Giant component	r	Giant component	r	Giant component
1	0.654	0.65	0.518	0.3	0.267
0.95	0.642	0.6	0.490	0.25	0.240
0.9	0.618	0.55	0.462	0.2	0.200
0.85	0.606	0.5	0.423	0.15	0.170
0.8	0.587	0.45	0.396	0.1	0.133
0.75	0.563	0.4	0.356	0.05	0.108
0.7	0.542	0.35	0.324	0	0.085

Maximal added timing error (ms), every fibre two-way	20	25	30	35	40	45	50	55	60	65	70	75	80
Giant component	0.903	0.854	0.795	0.728	0.656	0.573	0.492	0.393	0.306	0.218	0.146	0.117	0.084

- *Reciprocity.* The giant component is 0.654 with every fibre two-way and 0.085 with every fibre one-way, and first falls below q^* at a reciprocal share of 0.618 by linear interpolation (about 0.62), slightly above the 0.53 to 0.59 expected from B.5's undirected thresholds. The difference is small in occupation: at r = 0.618 a fibre counts with probability 0.0242 + 0.618 × 0.0242 × 0.9758 ≈ 0.0388, within 2% of B.5's measured 0.0381 ± 0.0022, and r magnifies it, since 0.01 in r moves the occupation by only 0.00024. The timing-error sweep crosses at the same occupation (about 0.0388 at 49.5 ms). Under the either-direction rule, timing faults at three times the EC50 leave quorum on the human connectome only if more than about 62% of its fibres are two-way, with their two directions timed independently, which is an upper bound on the benefit of reciprocity.
- *Maximal timing error.* With every fibre two-way the giant component falls below q^* at a maximal added timing error of 49.5 ms (49.521 by interpolation), slightly below the about 50 to 52 ms expected and about a quarter more than the 40 ms assumed. At 40 ms it is 0.656, close to the 0.645 of C.13's phase model and the 0.66 of its static diagnostic.
- *What these estimates do and do not show.* Under the either-direction rule the human verdict of clause (b) of P4 depends on two unmeasured quantities, the share of two-way human cortical connections and the agent's maximal timing error between areas. The estimates say where each would tip the verdict, not which way it goes; they leave out the oscillator dynamics, and both sweeps were requested after the result they qualify had been seen.

D. Validation on seeds disjoint from calibration and from B.4. Kinds are those of B.4 (the report labels C1, C4, C5 and C6 "test", as B.4's did).

Claim	Kind	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
C1, omission-fault and timing-fault agents both abolish quorum	Construction check	20/20	20/20	20/20	20/20	20/20
C2, no hysteresis without feedback	Consistency check	20/20	20/20	20/20	20/20	20/20
C3, hysteresis grows with λ	Consistency check	20/20	20/20	20/20	20/20	20/20
C4, feedback sharpens the transition	Consistency check	20/20	20/20	20/20	20/20	20/20

Claim	Kind	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
C5, hub lesions lower the loss concentration more than random lesions	Consistency check	**19/20 (76–99%), fails**	20/20	20/20	20/20	20/20
C6, reduced arousal delays emergence more than induction	Consistency check	20/20	20/20	20/20	20/20	20/20
C7, without action on arousal circuits quorum is still lost, at a higher concentration	Consistency check	20/20	20/20	20/20	20/20	20/20

Verdict under B.4's rule: at least one claim failed (C5 at 302 nodes).

Observation or extension test (no verdict; B.4's count in brackets where it differs)	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
O1, trapped unconscious at zero concentration, λ = 16.52 ms	2/20 (4)	1/20 (2)	0/20	0/20	0/20
O2, trapped unconscious at zero concentration after a hub lesion, λ = 8.85 ms	6/20 (4)	3/20 (4)	0/20	0/20	0/20
O3, sleep at the rat share of down-states (0.095) loses quorum	0/20	0/20	0/20	0/20	0/20
O4, sleep at the human share (0.25) loses quorum	19/20 (20)	20/20	20/20	20/20	20/20
O5, one typical human slow wave (0.271) loses quorum	20/20	20/20	20/20	20/20	20/20
E1, rat-share sleep has a wakeable stable unconscious state	13/20 (15)	12/20 (15)	20/20	20/20	20/20
E2, human-share sleep has a wakeable stable unconscious state	18/20	20/20 (17)	20/20 (18)	20/20	20/20
E3, deep anaesthesia has only an unconscious state	20/20	20/20	20/20	20/20	20/20

Quantity, GABAergic-like agent (no verdict)	N = 302	N = 1,000	N = 10,000	N = 100,000	N = 1,000,000
c_{off}	0.909 ± 0.030	0.900 ± 0.014	0.887 ± 0.006	0.884 ± 0.005	0.880 ± 0.002
c_{on}	0.827 ± 0.039	0.840 ± 0.032	0.828 ± 0.010	0.827 ± 0.007	0.826 ± 0.005
Hysteresis	0.082 ± 0.031	0.060 ± 0.031	0.058 ± 0.011	0.057 ± 0.007	0.054 ± 0.005
Loss-to-recovery ratio $c_{\text{off}}/c_{\text{on}}$	1.101 ± 0.042	1.072 ± 0.040	1.071 ± 0.014	1.069 ± 0.009	1.066 ± 0.007
Seeds with the ratio within the published range, 1.07 to 1.63	15/20 (53–89%)	10/20 (30–70%)	11/20 (34–74%)	13/20 (43–82%)	9/20 (26–66%)

Against the expectations stated before the run:

- *Claims.* C1 to C4, C6 and C7 hold in every seed at every size, as expected; C4 now also holds in every seed at 302 nodes, where it failed in one seed of B.4. C5, expected to hold everywhere, holds in 19 of 20 seeds at 302 nodes: in one seed (the base seed plus 7615) the hub-lesioned network has no loss concentration (the hub-lesion c_{off} is defined in 19 of 20 seeds at that size), so the claim does not hold there by its definition. Under B.4's every-seed rule the verdict is that at least one claim failed, as in B.4, though for a different claim. The failure is confined to the smallest networks, where hub lesions also leave 6 of 20 networks trapped unconscious at zero concentration (O2).
- *Observations and extension tests* differ from B.4 by at most three seeds in any cell, and only at 302 to 10,000 nodes.
- *Neural inertia.* The loss-to-recovery ratio falls with size, as expected: 1.101 ± 0.042 at 302 nodes, 1.071 ± 0.014 at 10,000 and 1.066 ± 0.007 at 1,000,000. λ = 8.85 ms was chosen so that ten networks of 10,000 nodes give the isoflurane ratio of 1.08; on networks that played no part in the choice, the same λ gives a mean ratio below 1.08 at every size from 1,000 nodes up, and at 100,000 and 1,000,000 nodes a mean (1.069 and 1.066) just below 1.07, the lowest published ratio, with only 13 and 9 of 20 seeds inside the published range. At 10,000 nodes, the calibration size, fresh seeds give 1.071 ± 0.014 (standard error about 0.003), the same as B.4's ten seeds that were not calibration seeds (1.071 ± 0.019), against 1.079 ± 0.015 on the calibration seeds, so part of the shortfall is the calibration fitting its own ten networks; the further fall to 1.066 at 1,000,000 nodes, which B.4 also gave (1.066 ± 0.006), is the size dependence a single-size calibration does not correct. Both are small. The reading fixed before the run (a ratio well below 1.07 would show that the calibration fitted its own networks) did not define "well below", so it is not applied.

Appendix D: Model revision history

The hypothesis and its models changed while this paper was written. Each change is listed here in the order it happened, with the result that prompted it. No failed test was removed: every one is still reported in Appendices B and C, and every superseded report is kept in `simulator/results/archive/` and can be opened in the explainer's results archive.

Topic	Initial assumption	What the simulations showed	What the literature added	Revision	New test
Model after internal review (17 October 2026)	Independent timing errors added; baseline jitter 1 ms; timing error redrawn at every integration step	Redrawn jitter averages to a uniform weakening of coupling, indistinguishable from transmission failure	In vivo cortical spike-timing precision is about 3 ms (Buracas et al., 1998)	Errors combined in quadrature; baseline jitter 3 ms; jitter that persists for a correlation time; feedback recalibrated; rewired null model and Wilson intervals added	All studies rerun; the first reports are kept in `archive/pre_revision_2026-10-17/`
Calibration target (18 October 2026)	λ calibrated to the human propofol ratio 1.11 (Iwakiri et al., 2005)	With λ calibrated to 1.11, the hub-lesion test C5 held in 9 of 20 seeds at 10,000 nodes; with λ calibrated to 1.08, in every seed at every size	Iwakiri et al. describe the two concentrations as similar; Friedman et al. (2010) show, in mice, a hysteresis not fully explained by pharmacokinetics	Reference ratio changed to 1.08 (isoflurane in mice), after the results at 1.11 had been seen; the other reported ratios are kept as a sensitivity range (Section 10, items 2 and 29)	λ for each reproduced ratio in the sensitivity analysis; the halothane ratio of 2.85 cannot be reproduced
Paradoxical excitation	Congestion: as connections fail, traffic concentrates on the surviving routes (Section 5.1)	Hub activity rose by at least 10% before collapse in at most 1 of 10 seeds in every setting (tests W1 to W4), in an exploratory and a confirmatory run	Neurons do not queue or resend signals; interneuron antisynchrony is the developed account (McCarthy et al., 2008)	Congestion is not supported and rests on prediction 4 alone. It is replaced by survivor lock-in with critical fluctuations (Section 5.1): fluctuations rising tenfold on the conscious side of the threshold (T1b, post hoc) replaced the pre-specified T1, which failed, and a synchrony surge before collapse (X1, post hoc) was partly reproduced on fresh seeds	T1c, the replication of T1b on fresh seeds (Appendix C.7): passed in 16 of 20 *C. elegans*, 6 of 6 *Drosophila* and 20 of 20 human seeds
Neural inertia (P5)	The shared reference, produced by the agreeing group, makes the network bistable, so consciousness returns at a lower concentration than the one at which it is lost	Percolation model: inertia appears with reference feedback and reproduces the reported ratios 1.07 to 1.63 once λ is calibrated, but cannot reach halothane's 2.85 before some networks stop recovering. Phase model: no graded inertia at any feedback strength (S3 in at most 5 of 20 seeds); global coherence instead fluctuates strongly near the threshold (T1b, T1c), with one stable state rather than switching between two	Noise-driven switching between metastable states (Proekt and Hudson, 2018) and adaptive feedback (Kim et al., 2018) also explain inertia	P5 kept as workable only with an all-or-nothing integration window (P2), part of its fit being a calibration. Lead noticed after the results: inertia and trapping as two sides of one feedback, with arousal setting how much inertia is safe (Section 5.4)	Prediction 5, in data; the arousal lead is untested

Topic	Initial assumption	What the simulations showed	What the literature added	Revision	New test
Ketamine	The senses are cut off from an internal agreement that survives (K1)	Internal agreement held but the stimulus still spread (K1 held in at most 2 of 20 seeds). After the code corrections K1 held in about half the seeds (12 of 20, 2 of 6, 10 of 20); a post hoc diagnostic (`simulator/results/diagnostics/`) traces the change in *C. elegans* mainly to the corrected measurement window and the removal of muscles. A second post hoc diagnostic (20 October 2026) found a floor of chance locking in the broadcast measure; with it subtracted, K1's criterion holds in 16 of 20 *C. elegans* seeds (limitation 31). The test fixed in advance with that floor removed and a longer window (K1c, Appendix C.11) then passed on *C. elegans* in 20 of 20 seeds and could not be evaluated on the human connectome	NMDA receptors carry recurrent more than feedforward signals in macaque V1 (Self et al., 2012); ketamine inhibits frontal-parietal feedback connectivity while feedforward connectivity is preserved (U. Lee et al., 2013)	Timing faults concentrated on feedback connections, so that loops cannot close; the sensory-pathway account is kept, shown on *C. elegans*	K2a to K2c, fixed before the run (Appendix C.7); next row; K1c (Appendix C.11)
Ketamine, feedback faults	Timing faults on feedback connections alone abolish quorum while the stimulus still spreads (K2a, K2b), at a lower concentration than on as many random connections (K2c)	Quorum was lost with feedback faults in only 1 of 20 *C. elegans* seeds and none of 6 *Drosophila* seeds, and never with random ones; feedback faults lowered coherence more (exploratory)	Nothing new; the measured loss of frontal-parietal feedback connectivity under ketamine (U. Lee et al., 2013) still stands	Kept only as a contributing factor: in the model, ketamine's timing faults abolish quorum reliably when they act across the network (T2b), and feedback placement deepens the loss of agreement without causing it	None in the model; the stronger form is not re-tested under another name
Ketamine, local islands	Spontaneous local coherence outlasts global coherence (S1, S2)	Failed in every seed	Not pursued further; the idea is not re-tested under another name	Dropped	None
Sleep down-states	Down-states make individual connections fail	Supported a wakeable sleep state at the rat share (E1, every seed from 10,000 nodes) and at the human upper estimate (E2, every seed from 100,000 nodes)	A down-state silences a whole local population (Vyazovskiy et al., 2009); slow waves are often regional (Nir et al., 2011)	Down-states silence whole populations; the connection-level version is kept as an alternative	O3, O4 and O5, the measured extent of a human slow wave (Appendix B)
Astrocytes and sleep	Removing the astrocytic share of down-states restores quorum in sleep (A1)	Re-specified when the down-state model changed, before its multi-seed run; then failed in every seed at every network size. Found on 20 October 2026: the test could not have passed, because the quarter of the populations it leaves silent is the human share at which quorum is lost anyway (O4)	Astrocytic effects on anaesthesia differ in sign between regions (Section 2.2)	The astrocytic route to down-states is untested in the model, not refuted	None
Microglia	Drug depletion of microglia hastens emergence, as reported with PLX5622	Reproduced by construction (consistency check G1)	PLX5622 also speeds the clearance of anaesthetics, and genetic depletion did not hasten emergence (Cao et al., 2026)	The firmer evidence is the genetic P2Y12 result (Cao et al., 2023); G1 is reported as a consistency check, not support	Prediction 12, with the 2026 result noted

Topic	Initial assumption	What the simulations showed	What the literature added	Revision	New test
Graceful degradation (P8)	A slower reference protects more against timing faults than against transmission failure	The slower rate raised the loss concentration (F1) but its distinctive prediction failed in every seed (F3)	Frontal alpha is attributed mainly to thalamic mechanisms (Ching et al., 2010; Vijayan et al., 2013)	P8 is kept as a proposal whose specific mechanism the model does not support	Prediction 9, in recorded data
Phase-oscillator code (hostile review, 19 October 2026)	The phase model implements Section 4 as written	An AI review of the code found that timing errors were added instead of combined in quadrature, that entrainment was measured over 1.43 stimulus cycles, so that a population not following the stimulus left a residue, that a floating-point comparison counted equal hysteresis widths as shrinking, and that each report recorded its source files at the end of the run rather than at its start	Not applicable	Code corrected, each defect with a regression test; entrainment measured over two whole stimulus cycles; sources recorded at launch; every phase-model study and the pre-specified tests rerun with the same seeds	All phase-model results reported here come from the corrected code; the first reports are archived in `simulator/results/archive/`
Connectome loading (second code audit, 19 October 2026)	The loaders and null model match the datasets	A second audit found that the *C. elegans* graph included muscles and other cells (448 nodes) and counted most gap junctions twice in each direction, that the null model's target shuffle broke reciprocal pairs and some degrees, that missing transmitter predictions in FlyWire were read as GABA, and that a single noisy window could count as a congestion collapse	Not applicable	The worm graph is neurons only (300 nodes) with each gap junction counted once per direction; the null model uses degree-preserving double-edge swaps (Maslov and Sneppen, 2002); missing predictions count as unknown; a collapse must be sustained; each with a regression test	Every affected study rerun on the same seeds; the earlier reports are archived
Verdict comparisons (20 October 2026)	Comparing two concentrations, gains or widths with a plain greater-than is exact	Two verdicts were decided by floating-point rounding: test F3 called two equal one-step gains unequal in every *Drosophila* seed (6 of 6, correctly 0 of 6), and test T1 counted 3 of 7 runs lying exactly on its window edge as outside (*C. elegans* 5 of 20, correctly 7; human 4 of 20, correctly 5)	Not applicable	One comparison rule for every verdict on grid values, equal within 10^{-9} counting as equal, with regression tests built from these values; the species verdicts were recomputed from the stored runs without rerunning any simulation; no other report contains such a tie	None; the recomputed report records the run it was computed from

Topic	Initial assumption	What the simulations showed	What the literature added	Revision	New test
Test labels (20 October 2026)	Test identifiers may reuse other labels	Tests P1 to P3b shared their names with postulates P1 to P3, although they do not test those postulates, and the congestion study's G1 and G2 shared theirs with the glial checks	Not applicable	Shown in this paper and the explainer as T1 to T3b and B1 and B2; the reports keep their identifiers (Appendix C.9)	None
Constant delays (20 October 2026)	A constant delay is harmless because the brain learns when its inputs arrive (P4b)	In the models constant offsets are not harmless: fixed timing offsets abolish quorum earlier than jitter of the same size (S4), and measured conduction delays lower the loss concentration (D1). With a learned expectation of arrival phase, coherence fell even at zero concentration and, without conduction delays, quorum was lost below 0.13 times the EC50 with every agent (L1, L3 and L4 held in no seed); with conduction delays, the learning first raised awake coherence from 0.41 to 0.62 in every seed	Adapting delays can stop phase models from becoming fully in-phase (Park and Lefebvre, 2020) and can cancel the coupling itself (Pajevic et al., 2014)	None: a receiver that sees only the arrival phase cannot separate a constant delay from the phase relation that coupling maintains, so P4b's claim about constant delays is untested in the models; prediction 8 tests its claim about variability in data	L1 to L4, fixed before the run (Appendix C.10)
Scope and structure (third review round, 21 October 2026)	P1 to P9 of equal standing; P3 stated for "integrated, reportable content"; consensus terms used as analogies	The human result of C.12 and C.13 and the wrapped reading of C.12 showed that timing variability abolishes quorum only under P2's rule that a late signal is lost	GNW models already contain delays, ignition and a vigilance input (Dehaene et al., 2003; Dehaene and Changeux, 2005); OFF-period accounts already link bistability to lost causality (Pigorini et al., 2015; Rosanova et al., 2018)	P3 pinned to one scope (any experience requires quorum; connected consciousness requires the sensory populations in it); core P1 to P5 and P7; P6, P8 and P9 auxiliary; P4b's variability claim merged into P4 and its constant-delay claim made an open question; consensus and quorum used in their everyday sense only	The analyses of Appendices C.14 and E.15
Pair rule (21 October 2026)	A connection between two populations counts when either of its arcs is on time (C.12's readout), so each human connection inferred from diffusion MRI, stored as two arcs, has two independent chances	Timing variability alone did not abolish DPC quorum on the human connectome (C.12: T2b-D, 0 of 20 seeds); a post hoc static diagnostic suggested it came from the counting	Diffusion MRI records no direction; in macaque cortex, tract tracing finds unidirectional connections unexpectedly common (Markov et al., 2014)	One timing draw per anatomical connection was adopted as the primary rule, with both directions and either direction as bounds, fixed before a study on fresh seeds; C.12's verdicts kept and reported beside it	T2a-1, T2b-1 and SP1-1 (Appendix C.13): pass on all three connectomes; T2b-1 passes on the human connectome in 20 of 20 seeds, so the human verdict depends on the counting rule

Topic	Initial assumption	What the simulations showed	What the literature added	Revision	New test
Postulates and predictions (fourth review round, 22 October 2026)	P3 necessary only; the claim that anaesthetic unconsciousness is loss of quorum stated in prose; P2's rule for late signals stated as a reason; the counting rule for a connection left implicit; prediction 1's edge rule a bound on the SD of single-trial latency	A falsifier stated against a necessary condition could be absorbed by disconnected or covert consciousness; the human verdict depends on the counting rule (C.12 to C.14) and on the rule for late signals (C.12)	One participant of the study behind the Michigan dataset followed commands covertly, by mental imagery, while unresponsive under propofol (Huang et al., 2018a)	P4c added to the core (anaesthetic unconsciousness is loss of quorum); P2's late-signal rule marked unsupported; P3 states that the counting rule is open and that no rule adopted after the C.12 failure is part of the hypothesis; prediction 1's edge rule restated as a per-trial share $\hat{\pi}$ and its falsifier restricted to established unconsciousness	None run; prediction 1 as restated

Appendix E. Pre-specified tests of the predictions on open data

E.1 Purpose and what was seen before the tests were fixed

Every result in Appendices B to D comes from simulation. This appendix fixes, before any analysis, how the predictions of Section 8 that open data can reach will be tested on recordings made by others, and what result would count against DPC. The tests, their measures, thresholds and pass rules were written and committed before any of the datasets was analysed.

Before the tests were fixed we read the datasets' documentation and file layouts, the concentration schedules and loss and return of responsiveness tables of the Michigan dataset (E.2), the condition markers of one Neurotycho session and its electrode map, the stimulation and channel tables of one PRIOS patient and the event table of one TMS-EEG participant, and we loaded the files of one Michigan subject (sub-02) and of 16 channels of one Neurotycho session to check that they parse, reporting only their sizes, the share of censored volumes and the number of bipolar derivations. While checking the Neurotycho file format, the first samples and the standard deviation of one raw ECoG channel were printed. No measure on which a test depends (giant component, coupling, lag, power, latency or any indicator built from them) was computed on real data and read. Before the authors ruled that no pipeline check may use real recordings, one dry run of the fMRI pipeline was completed on sub-02's files with the BOLD replaced by phase-randomised surrogates, which destroy all coupling; its output also held the concentrations at sub-02's real loss and return of responsiveness, and it was deleted without being opened. Two later dry runs, with placeholder event times, were stopped before finishing, and their outputs were deleted without being opened. The dry-run option was then removed from the code, which is proved on synthetic data only (E.11).

Three published results on the same data were known to us when the tests were fixed, and each is a prior against which a result must be read. On the Michigan dataset, functional connectivity fell gradually during induction while recovery was abrupt (Huang et al., 2021). On the Neurotycho recordings, cortical dynamics became more stable, not less, as consciousness was lost under every agent (Solovey et al., 2015), which argues against the critical slowing down of prediction 2. On the PRIOS recordings, the median latency of the earliest response component lengthened from 22.0 ms awake to 26.4 ms under propofol (van Blooijs et al., 2024), so a change in mean latency is already known.

E.2 Datasets checked

Each dataset was opened on 20 October 2026: its licence was read where it is published, its files listed and, where noted, one sample downloaded and read. Sizes are those reported by the archive.

Dataset	Licence and access	Size	What it contains	Use here
Michigan Human Anesthesia fMRI Dataset-1, OpenNeuro ds006623 v1.0.0 (Huang et al., 2021)	CC0, open	34.1 GB; 0.54 GB used	26 volunteers, 4 imagery runs and 2 rest runs (TR 0.8 s), XCP-D parcellated series; effect-site and plasma propofol concentration at every volume; loss of responsiveness (LOR) volume for 26, return (ROR) volume for 20 (6 marked N/A); two stepped protocols, 0.4 to 2.4 µg/ml (14 subjects) and 1.0 to 3.0 µg/ml (12), in steps of 0.4 µg/ml; LOR at 0.8 to 3.0 µg/ml	H1, H2a, H4
Neurotycho "Anesthesia and Sleep" task 78 (Yanagawa et al., 2013)	Copyright waived on neurotycho.org (CC0 wording), open	32.6 GB used (23 anaesthesia sessions, 5 electrode maps)	4 macaques, 128-channel subdural ECoG at 1 kHz; ketamine-medetomidine (11 sessions), propofol, ketamine, medetomidine (4 each), sleep (7, not used); markers for awake eyes open and closed, injection, start and end of the anaesthetised condition, recovery; one injection per session, no concentrations, no stimuli	H2b, H3
PRIOS, OpenNeuro ds004370 v1.0.2 (van Blooijs et al., 2024)	CC0, open	29.6 GB	7 patients in `participants.tsv` (the README says 6); ECoG at 2048 Hz; single-pulse stimulation awake (clinical session) and under propofol (operating room, on a later day); stimulated pair, current and notes per pulse; no concentrations; co-medication in the operating room not documented in the files checked	H5a (secondary)
Allen Institute cortical stimulation, DANDI 000458 (Claar et al., 2023)	CC BY 4.0, open	361.5 GB, 24 sessions	Mice; 30-channel EEG; Neuropixels LFP and sorted units where probes were used (the 18 larger sessions, by file size); single biphasic pulses to secondary motor cortex at 20, 50 and 100 µA, awake and under isoflurane (1 to 1.5% maintenance), induction epoch marked	Not pre-specified (E.9)
TMS-EEG under propofol sedation, OpenNeuro ds005620 v1.0.0 (Bajwa et al.)	CC0 in `dataset_description.json`, CC BY 4.0 in the README; treated as CC BY 4.0	83.0 GB	21 participants (1 excluded by the authors); 64-channel EEG at 5 kHz; TMS awake and in up to three sedated runs; about 185 pulses per run in the participant checked; no concentrations in the files checked	Not pre-specified (E.9)
MGH propofol EEG, DANDI 001285	CC BY 4.0, open	1.19 GB	One NWB file holding one 64 × 2,316,950 array at 250 Hz (2.6 h) and nothing else: no channel names, no stimulus or response events, no concentrations, no LOR or ROR times	Unusable
VitalDB (Lee et al., 2022)	CC BY-NC-SA 4.0 and a data use agreement (no disclosure to others, report misuse within 24 hours, no re-identification)	6,388 cases	3,344 cases with raw two-channel frontal EEG and propofol effect-site concentration, of which 3,342 also received remifentanil; no LOR markers	Not used
Cambridge propofol EEG (Chennu et al., 2016)	CC BY 2.0 UK as stated on the repository page	3.44 GB	20 volunteers; 91-channel resting EEG, preprocessed into 10 s epochs, at baseline, mild and moderate sedation and recovery, with plasma concentration and a two-choice task	Not used
Propofol fMRI with a story, OpenNeuro ds003171 v2.0.1 (Kandeepan et al., 2020)	CC0, open	4.8 GB	17 volunteers; auditory and rest runs awake, light, deep and recovery; no concentrations in the files checked	Not used

DANDI 001285 lacks everything a test needs. VitalDB records two frontal channels, which cannot form a network, nearly every case also received an opioid, there is no LOR marker, and its agreement binds each user, so the authors would have to accept it themselves. The Cambridge and Kandeepan datasets have four levels without a marked LOR; four points cannot separate a threshold from a smooth decline.

E.3 Which predictions open data can test

Prediction	Testable now?	Dataset and test	What limits it
1. Giant component collapses at LOR, sharper than local power	Partly	Michigan fMRI, H1	BOLD at 0.8 s resolves no millisecond timing and no stimulus-locked phase locking; it tests the shape of the collapse of an ongoing coupling network against concentration, not the network of content timing that prediction 1 names

Prediction	Testable now?	Dataset and test	What limits it
2. Critical slowing down before LOR	Partly	Michigan fMRI, H2a (slow stepped induction); Neurotycho ECoG, H2b (single injection)	fMRI order parameter from co-fluctuations; Neurotycho inductions are fast, not the slow induction the prediction requires, and the anaesthetised marker follows LOR
3. Lag variability, not mean lag, predicts LOR across agents	Partly	Neurotycho ECoG, H3	Ongoing activity, not responses across trials; awake and anaesthetised states, not the moment of LOR; propofol, ketamine, medetomidine and their combination, no sevoflurane; 4 monkeys
4. Induction excitation scales with sensory load	No	None	No open dataset varies sensory load at induction
5. Inertia tracks thalamocortical coupling	No	None	Needs manipulation of thalamic drive in animals
6. Hub disruption lowers the LOR concentration more	No	None	Needs focal inactivation
7. Ketamine spares internal more than stimulus coherence	No	None	No open recording delivers stimuli under ketamine; Neurotycho's ketamine sessions have none
8. Inter-areal latency variability rises before mean latency	Partly	PRIOS, H5a (secondary)	One deep propofol state against wakefulness, on different days; "just below loss of responsiveness" cannot be reached; mean latency is already known to change
9. Alpha-band phase locking preserved at alpha onset	Possible, not pre-specified	Neurotycho propofol inductions	Four sessions in two monkeys; left exploratory
10. Arousal threshold follows down-states	No	None	No open sleep recording with graded arousing stimuli was found
11 to 13. Myelin, microglia, astrocytes	No	None	Need animal manipulations
14. Sleep against anaesthesia at matched depth	No	None	Neurotycho has sleep and anaesthesia in the same monkeys but no graded arousal and no concentrations

Prediction 1 in the form stated, a stimulus-locked network during a slow ramp with documented LOR, and predictions 4 to 7 and 10 to 14 cannot be tested with any open dataset we found. The 2013 propofol EEG study with stimuli every 4 s (Purdon et al., 2013) would serve prediction 1, but its public deposit (DANDI 001285) contains none of the events.

E.4 Rules common to every test

- **Family and correction.** The primary tests are H1, H2a, H2b, H3 and H4. Their one-sided p-values are corrected together with Holm's method at a family-wise α of 0.05. Secondary analyses (E.9) are reported without a verdict.
- **Three outcomes.** A test *supports DPC* if its corrected p-value is below 0.05 and its effect estimate is at least the smallest effect size of interest (SESOI) stated with it. It *counts against DPC* if two one-sided tests (Lakens et al., 2018) show the effect to lie within ± SESOI, or if the estimate is at or beyond −SESOI. Otherwise it is *inconclusive*. Absence of an effect is therefore claimed only through equivalence, as Section 8 requires.
- **Seed and resampling.** Every random step uses seed 20261021 (`analysis/open_data/settings.py`).
- **Exclusions.** A subject or session is excluded from a test only by a rule stated below; the exclusion and its reason are written into the result file and listed in the group report.
- **Provenance.** Every result file records the commit, the SHA-256 of every analysis module, every setting and the software versions. Any change to the code or settings after the commit of this appendix makes the affected test exploratory, and it will be reported as such alongside the result of the committed version.
- **No peeking.** Group verdicts are computed once, by `run.py group`, after every per-subject file exists.

E.5 H1: is the collapse of the giant component a threshold? (prediction 1, Michigan fMRI)

Data. Imagery runs 1 to 4 of every subject in the XCP-D output without global signal regression, band-pass filtered, in the 156-region parcellation (`4S156Parcels`). Run 1 precedes the infusion (concentration zero throughout) and is the awake reference; run 2 is the stepped induction. Run 2 has under a minute at zero concentration (28 volumes in sub-02), too little for one window.

Preprocessing. Task-evoked activity is regressed out with the dataset's four instruction regressors (`Ideal_task{run}_{action,navi,squeeze,tennis}.xmat.1D`). Volumes with framewise displacement above 0.5 mm are censored; a window with more than 20% censored volumes is dropped.

Network and order parameter. Windows of 75 volumes (60 s), stepped by 25 volumes. In each window the Pearson correlation of every pair of regions is an edge if it exceeds the 99th percentile of correlations between independently phase-randomised copies

of the same window (20 surrogates, pooled over pairs). S is the share of regions in the largest connected component. Each window is assigned its mean effect-site concentration. The mean correlation and the mean standard deviation of the regional signals (local amplitude) are kept for each window.

Shape. A decreasing four-parameter logistic and a straight line are fitted to S against concentration; the steepness k of the logistic (its 10 to 90% width is $2\ln 9/k$) is bounded to widths between 0.05 and 20 µg/ml.

Thresholding control. Thresholding a network whose coupling declines smoothly already yields a sigmoid giant component, so the observed steepness is compared with the steepness the same pipeline produces from surrogate data whose coupling declines smoothly. Each surrogate window mixes a common-phase surrogate of a randomly chosen awake window (which keeps its correlations) with an independent surrogate (which removes them) in proportions that scale every correlation by a factor f. In null A, f makes mean coupling fall on a straight line in concentration from its awake level to its level in the top 20% of concentrations. Each null is realised 200 times and fitted like the data. In null B (secondary), f follows the observed mean coupling of each window exactly, which asks whether the network falls more abruptly than its mean coupling.

Statistic. For each subject, $x = \log(k_{\text{observed}}/\text{median } k_{\text{null A}})$. Group test: one-sided Wilcoxon signed-rank test of $x > 0$. Effect: the median of x. SESOI: $\log 1.5$ (a collapse at least 1.5 times as steep as the smooth-decline null).

Pass rule. Supports DPC as in E.4, and only if the median of x is also positive with the edge test at α of 0.001 and of 0.05 (null A recomputed at each). Counts against DPC as in E.4, or if the giant component stays intact through LOR: the median over subjects of mean S after LOR divided by mean awake S is at least 0.9 (the falsifier stated in Section 8). **Measurement gate:** if BOLD coupling does not fall with concentration (median over subjects of deep over awake mean coupling at least 0.9), the fMRI network does not register the drug and H1 is reported as not evaluable rather than as evidence either way.

Secondary, reported without verdict. The sigmoid against line comparison (number of subjects with a Bayesian information criterion lower by at least 10 for the sigmoid); steepness relative to null B; the ratio of the steepness of S to that of local amplitude (prediction 1's "sharper than local power"; SESOI 1.5); and the distance between the sigmoid midpoint and the concentration at LOR, tested for equivalence within ±0.4 µg/ml (one concentration step).

E.6 H2a and H2b: critical slowing down before LOR (prediction 2)

H2a, Michigan fMRI. The order parameter is computed at every volume from edge co-fluctuations (Zamani Esfahlani et al., 2020): each region's residual signal is standardised on the awake run, the product $z_i z_j$ of every pair is an edge if it exceeds the 99th percentile of the same products between independent surrogates of the awake run, and $S(t)$ is the giant component. The segment runs from the first volume with non-zero concentration in run 2 to the LOR volume. Censored volumes are interpolated; a subject with more than 20% of that segment censored is excluded.

H2b, Neurotycho ECoG. For each anaesthesia session, from the injection marker to the start of the anaesthetised condition. Bipolar derivations, filtering and the long-range pairs are as in E.7; in consecutive 2 s windows the gamma-band (30 to 45 Hz) phase-locking value of every long-range pair is an edge if it exceeds the 99th percentile of the same pairs with one member circularly shifted by at least 0.5 s (20 shifts), and $S(t)$ is the giant component over all derivations. The start of the anaesthetised condition was marked after LOC was established, so the segment can include a few minutes after LOC; this biases against finding a rise before LOC only if the rise is confined to the last minutes.

Indicators (both). Following Dakos et al. (2012): Gaussian detrending with a bandwidth of 10% of the segment, variance and lag-1 autocorrelation in a rolling window of half the segment, and Kendall's τ of each against time. Per subject or session, a one-sided p-value against 1,000 phase-randomised surrogates of the same segment is reported.

Statistic and pass rule. For each indicator, a one-sided Wilcoxon signed-rank test of $\tau > 0$ across subjects (H2a) or sessions (H2b). Both indicators must rise: the test's p-value is the larger of the two, and its effect is the smaller of the two median τ. SESOI: $\tau = 0.2$. Counts against DPC if both indicators are shown to lie within ±0.2, or if either median is at or below −0.2. A failure of H2b is weaker evidence than a failure of H2a, because the Neurotycho inductions are not slow; Section 8 states the falsifier for "carefully controlled slow induction".

E.7 H3: does lag variability, rather than mean lag or power, track loss of responsiveness across agents? (prediction 3, Neurotycho ECoG)

Sessions. The 23 anaesthesia sessions of task 78. Task 75 holds three recordings from two of the same days and monkeys and is not used. All channels present in every recording of the session are used. A session lacking the awake eyes-closed or anaesthetised markers is excluded.

Epochs. Awake blocks come from the awake eyes-closed condition and anaesthetised blocks from the anaesthetised condition. The monkeys wore an eye mask from the eyes-closed condition until recovery, so eye closure is matched.

Preprocessing. Notch filters at 50, 100 and 150 Hz (quality 30). Bipolar derivations between each electrode and its nearest neighbour on the session's electrode map (the map of the same monkey by the same experimenter). Pairs of derivations that share no

electrode and are further apart than the median distance between derivations are long-range pairs; only these enter the coupling measures, which limits volume conduction without discarding coupling near zero lag. Fourth-order Butterworth band-pass filtering, applied forwards and backwards, with 2 s margins, and the analytic signal.

Measures. In 30 s blocks, divided into 1 s windows, for every long-range pair: the circular mean phase difference in each window; across the 30 windows, the *lag variability* (circular standard deviation, in ms at the band's centre frequency) and the absolute *mean lag* (ms). Per block, these are averaged over pairs; *local power* is the mean log power of the derivations. The phase-locking value (lag-sensitive, includes zero lag) and the weighted phase lag index (Vinck et al., 2011; blind to zero lag) are computed alongside. Each measure is standardised against the session's awake blocks.

Muscle. A block is dropped if its 70 to 150 Hz power exceeds the median of blocks in the same state by more than five median absolute deviations (minimum deviation 0.05 in log power). The reference is the same state because ketamine raises high-frequency power across the cortex. The rule removes bursts, not tonic muscle activity; the beta band (13 to 30 Hz), which muscle contaminates less, is analysed as a secondary band for that reason.

Contrast. For each measure, the direction of its change from awake to anaesthetised is learned on the other agents and applied to the held-out agent, and the area under the ROC curve (AUC) separating that agent's awake from anaesthetised blocks is computed; the four agent AUCs are averaged. A measure that changes in opposite directions under different agents, as power does between propofol and ketamine, cannot predict LOR across agents. The statistic is $D = \mathrm{AUC}_{\text{lag variability}} - \max(\mathrm{AUC}_{\text{mean lag}}, \mathrm{AUC}_{\text{power}})$.

Test and pass rule. Primary band gamma (30 to 45 Hz), the band of the model's 40 Hz reference. A bootstrap over sessions, resampled within agent (2,000 resamples), gives a one-sided p-value (share of resamples with $D \le 0$) and a 95% interval. SESOI: 0.05 in AUC. Supports DPC as in E.4. Counts against DPC if mean lag or power predicts as well as or better than lag variability, that is if the share of resamples with $D \ge 0.05$ is below 0.05 (the falsifier stated in Section 8). Theta, alpha and beta bands, and the phase-locking value and weighted phase lag index, are reported without verdict. Monkey and agent are partly confounded (two monkeys received only the combination), which the report states.

Signal-to-noise. Lower power makes phase estimates noisier, which by itself raises lag variability. The gamma power falls under propofol and rises under ketamine, so a rise in lag variability under both agents is not explained by signal-to-noise alone; under propofol alone it could be, and the per-agent AUCs are reported so that this can be seen.

E.8 H4: hysteresis between induction and emergence (Michigan fMRI)

Data. As H1; the induction curve uses runs 1 and 2, the emergence curve runs 3 and 4 (run 3 alone for sub-29, which lacks run 4).

Measure. The sigmoid midpoint of S against concentration on induction ($c_{50,\mathrm{ind}}$) and on emergence ($c_{50,\mathrm{em}}$); network hysteresis $H = c_{50,\mathrm{ind}} / c_{50,\mathrm{em}}$. Behavioural hysteresis is the concentration at LOR divided by that at ROR, for the 20 subjects with an ROR volume.

Pharmacokinetic sensitivity. A wrong effect-site model creates hysteresis by itself. The effect-site series is therefore also recomputed from the dataset's plasma series with a first-order effect compartment at five equilibration rate constants, 0.2, 0.26, 0.456, 0.8 and 1.21 per minute, starting each run from the dataset's effect-site value. 0.2 and 1.21 per minute correspond to the equilibration half-lives of 3.5 min and 34 s compared by Struys et al. (2000), 0.456 per minute is the value of Schnider et al. (1999), and 0.26 and 0.8 fill the range between them. The pharmacokinetic model behind the dataset's own series is not stated in the files checked.

Statistic and pass rule. One-sided Wilcoxon signed-rank test of $\log H > 0$; SESOI $\log 1.05$. Supports DPC only if the largest p-value over the six effect-site series, after Holm correction, is below 0.05 and the smallest median $\log H$ is at least $\log 1.05$. Counts against DPC if, with the dataset's own effect-site series, $\log H$ is shown to lie within $\pm \log 1.05$ or its median is at or below $-\log 1.05$. Secondary: the Spearman correlation between network and behavioural hysteresis across subjects. The paper's own models disagree on hysteresis (Section 10, limitation 9), so a failure here bears on P5's percolation form, not on P1 to P3.

Emergence follows the end of the infusion, so its concentration falls continuously while induction rose in held steps; time at each concentration differs between the two curves, which the effect-site sensitivity analysis only partly addresses.

E.9 Secondary and deferred analyses

H5a, PRIOS (prediction 8, partial). For each patient, the awake and propofol recordings; both polarities of a stimulated pair pooled; pulses whose notes report a deviation from the requested current excluded; response electrodes at least 20 mm (in the dataset's MNI coordinates) from both stimulated electrodes and marked good. Trials from -500 to 120 ms around the pulse, baseline subtracted (median from -500 to -10 ms); a connection responds if the mean response from 9 to 100 ms exceeds 3.4 baseline standard deviations in both states. Trial counts are equalised by random subsampling (at least 6). Mean latency is that of the largest negative deflection of the mean response; single-trial latency is the shift, within ±20 ms and refined by parabolic interpolation, that best aligns each trial with the mean response, and its spread is the scaled median absolute deviation. Per connection, $r = \log(\mathrm{SD}_{\mathrm{prop}}/\mathrm{SD}_{\mathrm{awake}}) - \log(\mathrm{latency}_{\mathrm{prop}}/\mathrm{latency}_{\mathrm{awake}})$, positive when variability rises more than mean latency; per patient, the median r. One-sided Wilcoxon test over patients, SESOI $\log 1.2$, reported without verdict because 7 patients, one deep state and

two different days make it a weak test. Confounds: the propofol recordings were made in the operating room at electrode removal, with unknown co-medication; lower response amplitude under propofol makes single-trial latencies noisier.

Not pre-specified. DANDI 000458 could test the second half of prediction 8, that single-neuron timing stays precise while timing between areas varies, but its units table has not been inspected, so no analysis of it is fixed here. The TMS-EEG dataset needs source reconstruction to separate areas. Prediction 9 on the four Neurotycho propofol inductions is left exploratory. Any of these, if run, will be labelled exploratory unless a separate prespecification is committed first.

E.10 Confounds and how they are handled

Confound	Where	Handling
Thresholding a smooth decline	H1, H4	Null A and null B surrogates through the identical pipeline (E.5); three edge thresholds
Volume conduction	H2b, H3	Bipolar derivations; only long-range pairs sharing no electrode; wPLI reported beside the lag-sensitive measures
Signal-to-noise	H3	Cross-agent design (power falls under propofol, rises under ketamine); per-agent AUCs reported
Muscle	H3	Within-state outlier blocks dropped; beta band as secondary
Eye closure	H3	Awake reference is the eyes-closed condition, matched to the eye mask under anaesthesia
Head motion	H1, H2a, H4	Framewise displacement censoring at 0.5 mm; windows over 20% censored dropped
Task-evoked co-activation	H1, H2a, H4	Instruction regressors removed before any coupling
Pharmacokinetic lag	H4	Effect-site recomputed at five equilibration constants
Opioids	All	None in the Michigan or Neurotycho protocols as documented; PRIOS co-medication unknown, stated
Fast induction	H2b	Stated as weaker than H2a; the prediction's falsifier requires slow induction
Prior results on the same data	H1, H2b, H5a	Stated in E.1

E.11 Commands, downloads and run times

From the repository root, with the repository's Python 3.13 environment (NumPy 2.5.4, SciPy 1.18.1; nothing else is needed). `-I` isolates the interpreter so that no file in the data folders can be imported.

```
# 1. Downloads: each command lists files and sizes and writes nothing; add --execute to download.
.venv/bin/python -I analysis/open_data/download.py michigan      # 434 files, 0.54 GB
.venv/bin/python -I analysis/open_data/download.py neurotycho    # 28 archives, 32.6 GB (about twice that on disk
after extraction)
.venv/bin/python -I analysis/open_data/download.py prios         # 114 files, 29.6 GB

# 2. Loader checks: no statistic is computed.
SUBJECTS=$(tail -n +2 data_external/ds006623/derivatives/LOR_ROR_Timing.csv | cut -d, -f1 | sed 's/^/--subject /')
.venv/bin/python -I analysis/open_data/run.py michigan --root data_external/ds006623 $SUBJECTS --out unused --check-
only
for e in data_external/neurotycho/*_mat_ECoG128; do .venv/bin/python -I analysis/open_data/run.py neurotycho --
experiment "$e" --out unused --check-only; done

# 3. Confirmatory runs.
OUT=analysis/open_data/results/$(date -u +%Y-%m-%d)
.venv/bin/python -I analysis/open_data/run.py michigan --root data_external/ds006623 $SUBJECTS --out $OUT
for e in data_external/neurotycho/*_mat_ECoG128; do .venv/bin/python -I analysis/open_data/run.py neurotycho --
experiment "$e" --out $OUT; done
for s in data_external/ds004370/sub-*/ses-1; do .venv/bin/python -I analysis/open_data/run.py prios --session-dir
"$s" --out $OUT; done
.venv/bin/python -I analysis/open_data/run.py group --inputs $OUT --out $OUT/verdicts.json

# 4. Secondary bands for H3 (reported without verdict).
for b in theta alpha beta; do for e in data_external/neurotycho/*_mat_ECoG128; do
  .venv/bin/python -I analysis/open_data/run.py neurotycho --experiment "$e" --band $b --out $OUT; done; done

# Tests of the analysis code (synthetic data only).
.venv/bin/python -I analysis/open_data/run_tests.py
```

Run times on the machine that wrote the code (Apple M2 Pro, one core per process), measured on synthetic data of the real size, never on real recordings: Michigan, 1,123 s (19 minutes) for one subject of 156 regions with runs of 1,145, 2,270, 2,270 and 1,145 volumes at the full settings, measured while other work ran on the machine, so about 8 hours for 26 subjects; Neurotycho, about 3.4 s per 30 s block and 1.1 s per 2 s window of the induction on noise at full montage size (127 derivations, 3,903 long-range pairs), so about 8 minutes per session and 3 hours for 23; PRIOS, not measured. Downloads depend on the connection.

E.12 What open data cannot test

No open dataset combines repeated stimuli, a slow concentration ramp, documented loss and return of responsiveness and recordings dense enough for a network of phase locking between areas. Prediction 1 in its stated form, the stimulus-locked part of predictions 3 and 8, prediction 7 and every prediction that needs an intervention or a graded arousal stimulus (4, 5, 6, 10 to 14) therefore remain untested by this appendix. A pass of every test above would show that functional networks of ongoing activity behave as DPC predicts at the resolution of fMRI and ECoG; it would not show that the timing of content between areas fails as P2 to P4b propose.

Datasets. Huang, Z., et al. (2025). Michigan Human Anesthesia fMRI Dataset-1. OpenNeuro ds006623 v1.0.0, https://doi.org/10.18112/openneuro.ds006623.v1.0.0. van Blooijs, D., Blok, S., Huiskamp, G. J. M., & Leijten, F. S. S. (2023). PRIOS. OpenNeuro ds004370 v1.0.2, https://doi.org/10.18112/openneuro.ds004370.v1.0.2. Neurotycho Anesthesia and Sleep task, RIKEN Brain Science Institute, https://neurotycho.org/anesthesia-and-sleep-task. Already in the reference list: Huang et al. (2021), Lakens et al. (2018), Purdon et al. (2013), Solovey et al. (2015), van Blooijs et al. (2024). The Michigan dataset is cited, as its authors ask, with Huang et al. (2018), Huang et al. (2021, *Cell Reports*) and Huang et al. (2021, *NeuroImage*); Neurotycho with Yanagawa et al. (2013). All three are released under CC0; only derived results are published here.

E.13 Deviations from the committed plan

Labels and terms. Outside Appendices E.1 to E.12, which are committed and unchanged, the tests H1, H2a, H2b, H3, H4 and H5a are called OD1, OD2a, OD2b, OD3, OD4 and OD5, and the deviations D1 to D7 below are called Dev1 to Dev7, because H1 and D1 are also tests of Appendix C.3 (Appendix A.5). In E.1 to E.12, "Huang et al. (2021)" in E.1 is Huang et al. (2021b), "Huang et al., 2021" in E.2, the dataset's source, corresponds to Huang et al. (2021a, 2021b), and E.12's "Huang et al. (2018)" is Huang et al. (2018a); "LOC" in E.6 is the moment the Neurotycho experimenters marked, a loss of responsiveness; the effect-site and plasma propofol series of the Michigan dataset (E.2) are predictions of a pharmacokinetic model, not measurements; and "(Bajwa et al.)" in E.2 is the dataset Bajwa et al. (2024) under Data citations, where DANDI 000458 is Claar et al. (2023b), described in Claar et al. (2023a), and DANDI 001285 is Yousefi (2025); "(Claar et al., 2023)" in E.2 is the dataset Claar et al. (2023b). Abbreviations used in E.1 to E.15 (BOLD, TR, XCP-D, ROC, AUC, MNI, wPLI, ke0, NWB, DANDI, LOC) are defined in Appendix A.3.

Errata to the committed text. In E.8, "0.26 and 0.8 fill the range between them" misdescribes 0.26 per minute, which is the ke0 of the Marsh model as implemented in the Diprifusor pump (Iwakiri et al., 2005); 1.21 per minute is the value that combines Marsh kinetics with a time to peak effect of 1.6 min (Struys et al., 2000), and 0.456 per minute that of Schnider et al. (1999). Three of the five rate constants are therefore those of commonly used models; which, if any, matches the dataset's unstated model is unknown. In E.12, "All three are released under CC0" is corrected: the Michigan and PRIOS datasets are released under CC0, and Neurotycho's copyright is waived on the provider's site.

This appendix and its code were committed at 22:45:02 UTC on 20 October 2026 (commit `adc63f6`); the downloads began at 22:50 UTC. Four features of the real data, and one of the group step, were not anticipated, and the committed code refused them; D6 and D7 change only how the work was run. Section E.4 states that any change makes the affected test exploratory, to be reported alongside the committed version. The committed code was therefore run unchanged wherever it could run (confirmatory), and the changes below were made in a separate module, `analysis/open_data/deviations.py`, with its own tests, which replaces exactly the committed behaviours named and nothing else (exploratory).

Deviation	What the data or code showed	Change	Tests affected
D1	In 19 of 26 Michigan subjects, one to eight of the 156 regions of the parcellated signal are empty in every volume of at least one run; the committed loader refuses any run with a missing value	For each subject, the regions empty in any of its runs are removed from all its runs; a region missing in only some volumes is still refused (none was found)	H1, H2a, H4
D2	3 of 23 Neurotycho sessions carry more than one injection marker (top-up injections); the committed code requires exactly one	The first injection marks the start of induction	H2b, H3 for those 3 sessions
D3	A sigmoid fit that does not converge raises an error that the committed runner does not catch, ending the whole run	A null realisation whose fit does not converge is left out of the null distribution and counted (this makes the null steeper, so the test is harder to pass); a test whose observed fit does not converge is recorded as not evaluable for that subject	H1
D4	One of 7 PRIOS sessions (sub-PRIOS02) holds two awake stimulation runs; the committed code requires one	The earlier awake run is used	H5a for that patient

Deviation	What the data or code showed	Change	Tests affected
D5	The committed group step assumes at least one analysable subject for every primary test and fails when a dataset has none	A primary test with no analysable subject keeps its place in the Holm family with p = 1, so the family stays at five, and is reported as not evaluable	Group verdicts
D6	The Michigan analysis took about two hours per subject on one core, not the 19 minutes estimated on synthetic data	The subjects were split across nine processes, each seeding its random generator once with the committed seed and analysing its subjects in turn, as the committed runner does; a subject's random draws (edge-test surrogates and null realisations) therefore depend on its place within its process and, except for sub-02, differ from those of one run of all subjects. One subject (sub-28) was moved to a tenth process and started afresh; its first process, which had begun it 23 minutes earlier, was stopped and that partial analysis discarded	H1, H2a, H4 (procedure only)
D7	The exploratory inputs are spread over two folders, the committed code's results where no deviation applied (`confirmatory/`) and the deviations' results (`exploratory/`); the committed group step reads one folder	A separate module, `analysis/open_data/exploratory_group.py`, reads the files named in a list (`exploratory/group_inputs.txt`) in file-name order, as one folder would be read, and otherwise runs the committed group step with D1 to D5	Exploratory group verdicts (procedure only)

Three further facts bear on the exploratory results. First, `deviations.py` was extended as each deviation was found. A result file records the SHA-256 of the module as it was on disk when the file was written: `bfc26c8e…` for the three Neurotycho sessions of D2, `0b862f16…` for sub-PRIOS02 and `5e919a65…`, the version committed in `8971ca7`, for every Michigan subject. The module was last changed at 06:41 UTC, and the nine Michigan processes started at 09:37 UTC and the tenth at 16:20 UTC, so all 25 Michigan subjects ran that version. `deviations.py` was first committed in `8971ca7`, after the exploratory runs had started; its version is fixed by the SHA-256 recorded in each result file. Second, before D6, a single process analysed sub-02 to sub-05 in turn and was then stopped; the parallel processes analysed them again, and the second analyses are the ones used. That process had loaded an earlier version of the module (`bfc26c8e…`); its files record later hashes because the hash is taken when a file is written. The first analyses of sub-03 to sub-05 remain in the repository history (commit `b370495`); that of sub-02 was overwritten before it was committed. They show how much a subject's H1 result can move between analyses that differ in their random draws (and in the module version loaded): sub-05's null A p-value was 0.025 in the first analysis and 0.065 in the second, and its observed midpoint moved by 0.005 µg/ml. Third, the committed loader refused sub-29 under a check it already made: in its run 3 the effect-site series has 2,270 samples and the BOLD 1,417 volumes. No deviation was made for it, and the exploratory Michigan analysis has 25 subjects. The committed code's refusal of 3 Neurotycho sessions and of sub-PRIOS02 is an unplanned exclusion; the confirmatory n of 20 and 5 reflects it.

Process allocation (D6). The nine Michigan processes analysed, in this order, sub-02, sub-14 and sub-23; sub-03, sub-15 and sub-24; sub-04, sub-16 and sub-25; sub-05, sub-17 and sub-26; sub-06, sub-18 and sub-27; sub-07, sub-19 and sub-28 (sub-28 later moved); sub-11, sub-20 and sub-29 (refused); sub-12, sub-21 and sub-30; and sub-13 and sub-22. The tenth process analysed sub-28 alone (`provenance.arguments.subject` in each file of `analysis/open_data/results/2026-10-21/exploratory/`). A rerun in which each subject is seeded on its own (Dev8, Appendix E.15) left every verdict unchanged.

E.14 Results

Confirmatory (committed code, unchanged). The Michigan tests produced no result: the committed loader refused 19 of 26 subjects (D1), and the first of the remaining seven ended the run at a non-converging fit (D3). The Neurotycho tests ran on the 20 sessions with one injection marker, and the PRIOS analysis on the 6 sessions with one awake run; in one of them (sub-PRIOS09) no stimulated connection met the committed response criteria, leaving 5. Group verdicts were computed once, after every file existed, with D5 for the empty Michigan tests (`analysis/open_data/results/2026-10-21/confirmatory/verdicts_D5.json`).

Test	Data	n	Effect (SESOI)	p (Holm-adjusted)	Outcome under the E.4 rule (confirmatory)
H1, the collapse is a threshold	Michigan	0			Not evaluable
H2a, slowing before LOR	Michigan	0			Not evaluable
H2b, slowing before LOR	Neurotycho	20	smaller median τ 0.06 (0.2)	0.45 (1.0)	Inconclusive
H3, lag variability tracks LOR across agents	Neurotycho	20	D = 0.07 AUC (0.05), 95% interval −0.23 to 0.20	0.36 (1.0)	Inconclusive
H4, induction-emergence hysteresis	Michigan	0			Not evaluable
H5a, latency variability rises more than latency (secondary, no verdict)	PRIOS	5	median r = 1.33 (log 1.2)	0.031	Reported without verdict

In H2b the variance of the order parameter rose before LOR (median τ 0.45) but its autocorrelation hardly did (0.06), and the test requires both. In H3, lag variability separated awake from anaesthetised blocks better on average than mean lag or local power

(mean AUC 0.60, 0.53 and 0.34), strongly under ketamine (0.92) and ketamine-medetomidine (0.62) but not under medetomidine (0.39) or propofol (0.47); power changes in opposite directions under different agents, so a direction learned on the others fails on the held-out one. In H5a, propofol raised the trial-to-trial variability of response latency much more than the latency itself in all five analysable patients (per-patient median r from 0.64 to 1.63; group median 1.33, about a 3.8-fold relative rise); with five patients, 0.031 is the smallest p-value the test can give, and the comparison is between a deep propofol state and wakefulness on different days, not the graded approach to LOR that prediction 8 names.

Exploratory (with D1 to D7). Every subject and session was analysed: 25 Michigan subjects (sub-29 refused, E.13), the 23 Neurotycho sessions (20 unchanged, 3 with D2) and the 7 PRIOS sessions (6 unchanged, sub-PRIOS02 with D4). The group outcomes were computed once, after every file existed (`analysis/open_data/results/2026-10-21/exploratory/verdicts_exploratory.json`), and every group value was then recomputed independently from the per-subject files, with separate code, without a mismatch. In D3, 12 of the 20,000 null realisations (100 null distributions of 200) were left out because their fit did not converge. H4 could not be computed for three subjects: in sub-22 and sub-28 every window of run 3 exceeds the censoring limit, and in sub-19 no sigmoid fit converged.

Test	Data	n	Effect (SESOI)	p (Holm-adjusted)	Outcome under the E.4 rule (exploratory)
H1, the collapse is a threshold	Michigan	25	median log steepness ratio to null A 0.22 (log 1.5 = 0.41)	0.10 (0.50)	Counts against DPC: giant component intact through LOR (exploratory, Dev1 to Dev7; descriptive criterion fixed in advance; measure near ceiling)
H2a, slowing before LOR	Michigan	25	smaller median τ −0.04 (0.2)	0.42 (1.0)	Inconclusive
H2b, slowing before LOR	Neurotycho	23	smaller median τ −0.01 (0.2)	0.51 (1.0)	Inconclusive
H3, lag variability tracks LOR across agents	Neurotycho	23	D = 0.085 AUC (0.05), 95% interval −0.21 to 0.23	0.22 (0.88)	Inconclusive
H4, induction-emergence hysteresis	Michigan	22	median log H 0.78 with the dataset's effect-site series, 0.28 with the least favourable (log 1.05 = 0.049)	0.025; least favourable 0.44 (1.0)	Inconclusive
H5a, latency variability rises more than latency (secondary, no verdict)	PRIOS	6	median r = 1.28 (log 1.2)	0.031	Reported without verdict

H1 counts against DPC by the falsifier stated in E.5, not by its steepness test. The measurement gate was passed (median deep over awake BOLD coupling 0.79), so the network registered the drug, yet after LOR its giant component kept a median 0.95 of its awake size (range 0.86 to 1.03; at least 0.9 in 21 of 24 subjects; sub-16 has no window after its LOR). The steepness test alone would have been inconclusive (Holm-adjusted p 0.50), and a sigmoid was preferred over a line by the stated BIC margin in 1 of 25 subjects. As E.3 stated before any run, BOLD at 0.8 s measures an ongoing coupling network, not the stimulus-locked network of content timing that prediction 1 names, so this result counts against the ongoing-coupling form of prediction 1 and leaves the stimulus-locked form untested. Two observations made after the run, not part of the rule, bear on how far the BOLD giant component could fall: in every subject the median window held 0.86 to 0.98 of the regions in the giant component, so the network was close to fully connected throughout, and in 8 of 25 subjects the fitted transition was as narrow as the fit allows (0.05 µg/ml; a ninth, sub-20, at 0.05001), so its steepness was capped. Neither changes the outcome.

In H2a neither indicator rose by the smallest effect of interest (median τ −0.04 for variance, 0.09 for autocorrelation); in H2b, as in the confirmatory run, variance rose (0.42) and autocorrelation did not (−0.01). In H3 the three added sessions left the picture unchanged: mean AUC 0.62 for lag variability, 0.53 for mean lag and 0.35 for power; for lag variability, 0.92 under ketamine, 0.64 under ketamine-medetomidine, 0.44 under medetomidine and 0.47 under propofol. In H4, with the dataset's own effect-site series the network showed hysteresis (median log H 0.78, p 0.025 before correction), but the rule requires it under every effect-site series considered, and with the least favourable (ke0 = 0.2) the effect fell to 0.28 (p 0.44); network and behavioural hysteresis were unrelated (Spearman −0.01, 19 subjects with a recorded ROR). In H5a, sub-PRIOS02 (D4) is the one patient in whom latency variability did not rise more than latency (−0.10), giving a group median of 1.28 (about a 3.6-fold relative rise) and p 0.031 in six patients; the limits stated for the confirmatory H5a apply.

Notes added in the third and fourth review rounds (21 and 22 October 2026). None changes an outcome.

- *OD1 (H1) is exploratory and descriptive.* Its outcome rests on the cut-off of E.5, the giant component after LOR at least 0.9 of its awake size, which was fixed in advance but is a descriptive criterion. E.4 and Section 8 promised that a null result would be claimed only through an equivalence test, and this is not one. The measure was near its ceiling (the awake network close to fully connected), so it had little room to fall. Future tests of prediction 1 must fix the edge density so that the awake network lies near its own percolation threshold. The outcome reproduced when each subject was given its own random generator (Dev8, Appendix E.15).
- *Point-estimate rule.* E.4 also counts a test against DPC "if the estimate is at or beyond −SESOI", a rule on a point estimate without inference; it is a limitation of the committed plan.

- *Smallest effects of interest.* Only two have a stated basis: the ±0.4 µg/ml midpoint margin is one concentration step of the protocol, and the 0.9 cut is Section 8's falsifier. The others (log 1.5, τ = 0.2, 0.05 in AUC, log 1.05, log 1.2) are conventional choices. Minimal detectable effects were not computed, so the power of the inconclusive tests is not known; H3's interval (−0.21 to 0.23 exploratory) shows that it could not have reached either verdict, and its monkey-level interval (Appendix E.15) spans about 0.50 AUC, ten times its SESOI of 0.05, so neither outcome was reachable with four monkeys.
- *OD3 (H3) tests ongoing coupling, not evoked timing.* Its lag variability is the circular standard deviation across windows, $\sqrt{-2\ln \bar{R}}$, a monotonic transform of the across-window phase-locking value $\bar{R}$ of the window phase differences, so H3 tests the ongoing form of prediction 3, not the evoked form. Medetomidine, an α2-adrenoceptor agonist like dexmedetomidine, acts through arousal suppression (P7), which in the percolation model raises the effective timing error through $\sigma_{\mathrm{ref}} = \lambda(1-S)(1-A)$; whether DPC predicts quorum loss under it is not settled, and this reading was made after the results; its inclusion was fixed in advance. Agent-specific AUCs in the exploratory analysis: 0.92 under ketamine, 0.64 under ketamine-medetomidine, 0.47 under propofol and 0.44 under medetomidine.
- *Inference unit.* The Neurotycho tests resample sessions, but the 23 sessions come from 4 monkeys, so the unit of inference is the session, not the animal. A two-stage bootstrap over monkeys and then sessions (Appendix E.15) leaves the point estimates unchanged and widens the 95% intervals of OD3 (H3) to −0.25 to 0.22 (confirmatory) and −0.26 to 0.24 (exploratory); those of OD2b (H2b) span −0.51 to 0.52 and −0.47 to 0.39. Both remain inconclusive.
- *OD4 (H4) is not interpretable.* Its sigmoid midpoints come from giant-component curves that span only 0.86 to 0.98 of the regions, and three of the five equilibration rate constants are those of commonly used models (0.26 per minute, Marsh as implemented in the Diprifusor, Iwakiri et al., 2005; 0.456, Schnider et al., 1999; 1.21, Marsh kinetics with a time to peak effect of 1.6 min, Struys et al., 2000), while which, if any, matches the dataset's model, which the files do not state, is unknown (erratum in E.13); neither the dataset-series result nor the least favourable one can be read as network hysteresis.
- *OD5 (H5a) is not interpretable as timing variability* without a control for signal-to-noise: lower response amplitude under propofol makes single-trial latencies noisier by itself (E.9). In the signal-to-noise control of Appendix E.15 the SNR was not lower under propofol (group median 3.54 against 3.10 awake), and the SNR-matched comparison is uninformative by the rule fixed before the reanalysis, after E.14's results were known, because at the matched SNR (1.94) the estimator could not see a 2 ms jitter in simulation. At the data's own SNR (about 3) and 10 trials, a 2 ms jitter was measured as at most 0.14 ms (white noise), so the as-run statistic, not only the matched one, lies below the estimator's sensitivity; OD5 therefore remains not interpretable as timing variability.

E.15 Reanalyses requested in review after the results of E.14

The text from here to Results is reproduced as committed in `74f9cb0` (`analysis/2026-10-21_round3_open_data_prespecification.md`), without its draft header; the Results subsection was added after the runs.

E.15.1 Status and common rules

The referees asked for three reanalyses after the results of E.14 had been seen by them and by us. Each is therefore exploratory: it was fixed, and committed with its code, after E.14 and before its own run, and none of them changes a verdict of E.14. Their results are reported beside those of E.14, never in their place.

- **Seed.** Every random step uses the committed seed, 20261021 (`settings.SEED`), as stated below for each.
- **Numbers.** Every number on which a reanalysis depends that is not already in `settings.py` is in `analysis/open_data/round3_settings.py`.
- **Provenance.** Every result file records, as before, the commit, the SHA-256 of every analysis module, every setting and the software versions. The settings of this round are written into each of its result files.
- **Inputs.** R1 reads the PRIOS recordings again and R3 the Michigan recordings; R2 reads only the per-session result files of 21 October.

E.15.2 R1: H5a with a signal-to-noise control (PRIOS)

Concern. Lower evoked amplitude under propofol could by itself make single-trial latency estimates noisier, and so inflate the trial-to-trial spread on which H5a rests (E.9 stated this confound).

What was seen before this was fixed. The H5a results of E.14 (exploratory: median r 1.28 over six patients, p 0.031). No signal-to-noise ratio (SNR) of the data has been computed or read. While the code was being written, the committed estimator was run on synthetic responses (10 trials, a 2 ms true jitter, white noise). It showed that when each trial is aligned with a template that includes that trial, a low SNR pulls the single-trial shifts towards zero, because each trial then aligns best with its own noise at zero shift: below an SNR of about 6 the measured spread fell far below the true 2 ms instead of rising above it. With 200 trials per state the same estimator behaved as the referees expect, a lower SNR raising the spread. This is why the simulation below varies the number of trials and includes a null check of the matching.

Data and connections. The exploratory H5a analysis exactly as run for E.14: the committed code, the same seed and therefore the same connections and the same equalised trials, with deviation D4 for sub-PRIOS02. The H5a statistic is recomputed alongside; it must reproduce E.14 (median r 1.28, p 0.031). If it does not, the reanalysis is stopped and the discrepancy reported.

SNR of a connection in one state. On the baseline-corrected, equalised trials of H5a: the absolute value of the mean response at its N1 (the committed peak: the largest negative deflection from 9 to 100 ms), divided by the noise, the median over trials of the standard deviation of the de-meaned first 266 samples of the trial (from −500 to −370 ms at 2,048 Hz). 266 samples is the length of the span the latency estimator reads (about −11 to 119 ms around the pulse), so the noise is measured over the same length as it acts.

(a) Matching rule. For each connection, the state with the higher SNR is degraded to the SNR of the other. Each of its trials receives, over the span the estimator reads only, the de-meaned first 266 samples of the next trial of the same state (the last trial receives the first trial's), multiplied by $\sqrt{(\mathrm{SNR}_{\mathrm{high}}/\mathrm{SNR}_{\mathrm{low}})^2 - 1}$, which raises the noise of that state by the factor that equalises the two SNRs. The added noise is the state's own pre-pulse activity, so it has that state's spectrum, and no random draw is involved. Mean latency, single-trial latencies and their spread are then measured on the degraded trials by the committed estimator (`evoked_latency.connection_latency`, as `pipeline_prios.measure` calls it), and the matched relative change $r_{\mathrm{matched}} = \log(\mathrm{SD}_{\mathrm{prop}}/\mathrm{SD}_{\mathrm{awake}}) - \log(\mathrm{latency}_{\mathrm{prop}}/\mathrm{latency}_{\mathrm{awake}})$ is computed as in H5a. A connection whose matched spread is zero in either state has no r_{matched}. Per patient, the median over connections. Because the SNR is taken from the mean response, a larger true jitter, which flattens the mean response, also lowers the SNR of its state, and matching then adds noise to the other state. With many trials this can only remove part of a true rise. At H5a's trial counts, where a lower SNR can lower the measured spread (below), it can instead raise r_{matched}; the null check of (b) measures this.

(a′) Restriction (secondary). The H5a statistic as run, restricted to connections whose propofol SNR is at least the awake SNR divided by 1.25 (log SNR ratio at least −0.22). Connections whose SNR rose under propofol are kept, because under the referees' concern a higher SNR cannot inflate the spread; the simulation shows whether that holds at H5a's trial counts. Per patient, the median over the kept connections; a patient with no kept connection is left out of this row.

(b) Simulation. Synthetic cortico-cortical evoked potentials go through the committed H5a measure (`pipeline_prios.measure`: baseline correction, response test, N1 latency and single-trial alignment) at 2,048 Hz over the committed trial span (−500 to +120 ms). Each response is a negative Gaussian deflection, latency 22 ms (the awake median of van Blooijs et al., 2024, E.1), width (standard deviation) 5 ms, an assumed shape, with a Gaussian trial-to-trial jitter of the latency. Noise is Gaussian with unit standard deviation over each trial, white or pink (power falling as 1/f). The grid: nominal SNR (amplitude over noise standard deviation) from 0.5 to 16 in eleven half-octave steps; true jitter 0, 2 and 4 ms; 6 and 10 trials (the range and median of H5a's equalised trial counts) and 100 trials (to show the large-sample behaviour); 200 synthetic connections per cell; one generator seeded with the committed seed. Per cell: the median measured SNR (measured as for the data), the median latency spread over connections that pass the response test, and the share that pass. From these, for every pair of SNRs within a colour, jitter and trial count, the r an SNR change alone produces, $\log$ of the spread at the lower SNR over that at the higher (the mean latency is the same by construction). **Null check:** the matching rule itself on synthetic pairs of states that differ only in SNR (10 trials; awake SNR 4, 8 and 16; propofol SNR half or a quarter of it; every colour and jitter; 200 connections each), reporting the median r as run and after matching. Finally, the group step reads the simulation at the group median SNRs of the data (the median over patients of each patient's median awake and propofol SNR), interpolating log spread in log measured SNR, and reports for each colour, jitter and trial count the r the observed SNR change alone would produce, and whether the observed SNRs lie inside the simulated range.

(c) Test, reported without verdict. As H5a: one-sided Wilcoxon signed-rank test over patients of the per-patient median $r_{\mathrm{matched}} > 0$, with its median and the SESOI $\log 1.2$, beside the same test for the H5a statistic as run and for the restricted statistic. No verdict, as for H5a (E.9): six patients, one deep state, two days. With six patients the smallest one-sided p-value is 1/64 = 0.016, and 0.031 if the patient closest to zero goes the other way. The reading is fixed now. First, sensitivity: matching lowers one state's SNR, and the simulation may show that at the matched SNR the estimator no longer sees a true jitter. The matched SNR of a connection is the lower of its two SNRs; the group value is the median over patients of each patient's median. If, at that SNR, the simulated median spread of a 2 ms true jitter with 10 trials (interpolated in log measured SNR) is below 1 ms in either noise colour, the matched comparison cannot detect a rise, and its result is reported as uninformative rather than read either way. Otherwise: if the median r_{matched} is at least $\log 1.2$ with p below 0.05, the rise of latency variability over latency in H5a is not explained by a difference in SNR between the states; if it falls below $\log 1.2$ while the H5a statistic as run stays above it, the H5a result is not distinguishable from an SNR difference and will be reported as such in Section 8 and E.14.

Expectations, stated before running. (1) The simulation will show that at 6 and 10 trials a lower SNR lowers the measured spread over most of the simulated range, and raises it only at 100 trials, because with few trials each trial carries a large share of its own template (the development check above). (2) The null check will give a median r_{matched} within ±0.2 of zero in most conditions, because both states are then measured at the same SNR and trial count. (3) We cannot say in advance whether the matched comparison will be informative: that depends on the SNRs of the data, which have not been measured; in the development check a 2 ms jitter with 10 trials was seen only above an SNR of about 6. (4) We do not know which state has the higher SNR in these data; E.9 assumed lower amplitude under propofol, but no amplitude has been measured. If propofol lowers the SNR, matching adds noise to the awake trials, which at these trial counts lowers the awake spread, so we expect r_{matched} to be at least the r as run (median 1.28) and, if the comparison is informative, the reading above to find the rise not explained by SNR. If propofol raises the SNR in most connections, the propofol trials are degraded, which at these trial counts lowers the propofol spread, and r_{matched} will be lower than 1.28; by how much depends on SNRs that have not been measured, and in that case we make no prediction. (5) The restricted row will keep fewer connections and give a median close to the as-run median, if r does not depend on the SNR change.

E.15.3 R2: monkey-level bootstrap of H2b and H3 (Neurotycho)

Concern. The sessions of one monkey are not independent, so tests over sessions (the Wilcoxon test of H2b, the session bootstrap of H3) may overstate the evidence; the inference should be to monkeys.

Inference unit. The monkey. Four monkeys: Chibi and George received every agent; Kin2 and Su received only ketamine-medetomidine.

What is computed. From the per-session result files only, no recording: the confirmatory set (`results/2026-10-21/confirmatory/`, 20 sessions) and the exploratory set (`results/2026-10-21/exploratory/group_inputs.txt`, 23 sessions), each read exactly as the committed group step reads it (gamma band, sleep sessions left out, H2b exclusions as recorded). Beside the session-level results, recomputed with the committed group functions (they reproduce E.14), a two-stage bootstrap with 2,000 resamples and one generator seeded with the committed seed:

- **H2b.** Draw the monkeys with replacement (as many as there are), then, within each drawn monkey, its sessions with replacement (as many as it has); recompute the committed effect, the smaller of the two median Kendall τ.
- **H3.** Within each agent, draw with replacement the monkeys that received it, then each drawn monkey's sessions with that agent, with replacement; recompute $D = \mathrm{AUC}_{\text{lag variability}} - \max(\mathrm{AUC}_{\text{mean lag}}, \mathrm{AUC}_{\text{power}})$ with the committed leave-one-agent-out AUC. Resampling within agent keeps every agent in every resample, as the committed bootstrap does, so that D stays defined; a monkey that received two agents is drawn separately for each.

For each: the point estimate (identical to E.14), the 95% percentile interval, the share of resamples at or below zero and the share at or above the SESOI (0.2 in τ, 0.05 in AUC), each as $(1 + k)/(1 + B)$ as in E.7.

Rule. Reported without verdict, as a sensitivity analysis of H2b and H3; the verdicts of E.14 stand. With two monkeys for three of the four agents there are only three distinct monkey draws per agent, and a percentile bootstrap over n units understates the variance of a mean by a factor $(n - 1)/n$, a half for two monkeys. The monkey-level interval is therefore a lower bound on the uncertainty at the monkey level: it can show that the session-level results overstate the evidence, but it cannot show that they understate it.

Expectations, stated before running. The point estimates equal E.14 (H2b: 0.06 confirmatory, −0.01 exploratory; H3: D 0.07 and 0.085). For H2b, the share at or below zero will be near one half, because the smaller median τ is near zero. For H3, the 95% interval will include zero and be at least as wide as the session-level interval (−0.23 to 0.20 and −0.21 to 0.23), because the large ketamine AUC rests on two monkeys; H3 stays inconclusive.

E.15.4 R3: Michigan reanalysed with one generator per subject (deviation D8)

Concern. Under D6 each process seeded its generator once and analysed its subjects in turn, so a subject's random draws (edge-test surrogates and null realisations) depended on its place in its process, and no subject's result except sub-02's can be reproduced by analysing that subject alone.

Deviation D8. Each subject's generator is seeded with the committed seed plus the number in the subject's name (sub-02 with 20261023, sub-30 with 20261051), and the seed is written into its result file. Every other step is the exploratory analysis as run for E.14: the committed code with D1 to D5 (`deviations.py`, unchanged), the same 25 subjects (sub-29 still refused, E.13), the same settings (200 null realisations, 1,000 surrogates). D8 is implemented in a new module, `reseeded.py`, which replaces only the committed runner's single generator; `deviations.py` and `exploratory_group.py`, whose hashes are recorded in the earlier results, are not changed. Each subject is now independent, so the launcher (`launch_reseeded.py`) runs one process per subject, at most ten at a time, longest expected first, with linear-algebra libraries held to one thread each. It is a dry run unless `--execute` is given, checks the interpreter, the timing table and every subject's runs by file name before starting, writes one log per subject and a report with one line per subject and outcome, reports every failure with its subject, command, exit code and the end of its log, ends with a summary and exits non-zero if any check or subject failed. Re-running it skips every subject whose result exists, so an interrupted run resumes.

Group step. Once all 25 files exist, `reseeded.py listing` writes a list of inputs that is the list of E.14 (`exploratory/group_inputs.txt`) with each Michigan file replaced by its reseeded one; it refuses if any is missing, if any carries a seed other than its D8 seed, or if the folder holds a Michigan file not in the earlier list. The Neurotycho and PRIOS inputs are the same files as in E.14. The unchanged exploratory group step (D7) then computes the five primary verdicts with the committed rules and D5.

Rule. The D8 verdicts are exploratory and are reported beside those of E.14 (D6). Where they differ, both are reported and neither replaces the other; the difference measures how much the exploratory results depend on the random draws. Per subject, the H1 null-A p-value and steepness ratio, the H2a τ and the H4 $\log H$ are tabulated beside their D6 values.

Expectations, stated before running. The verdicts of E.14 will not change: H1 counts against DPC through the falsifier (giant component intact through LOR, median 0.95 of its awake size), because the edge threshold is the 99th percentile of 20 surrogates pooled over every pair of regions and varies little between draws; H2a and H4 stay inconclusive. Per-subject null-A p-values will move by amounts like sub-05's between its two D6-era analyses (0.025 and 0.065, E.13), and group medians by less.

E.15.5 Commands and run times

From the repository root, after the commit. Run times are for the machine that ran E.14 (Apple M2 Pro, one core per process): R3 from the per-subject times of the D6 run (0.76 to 4.42 h, 55.1 h in all, measured with nine or ten processes running at once), R1 and R2 from synthetic data of the real size.

```
OUT=analysis/open_data/results/$(date -u +%Y-%m-%d)
PREVIOUS=analysis/open_data/results/2026-10-21/exploratory/group_inputs.txt

# R1. Simulation (about 10 minutes), the seven PRIOS sessions (under a minute each), group step (seconds).
.venv/bin/python -E -s -m analysis.open_data.snr_control simulate --out $OUT/exploratory_snr/snr_simulation.json
for s in data_external/ds004370/sub-*/ses-1; do
  .venv/bin/python -E -s -m analysis.open_data.snr_control prios --session-dir "$s" --out $OUT/exploratory_snr; done
.venv/bin/python -E -s -m analysis.open_data.snr_control group --inputs $OUT/exploratory_snr \
  --simulation $OUT/exploratory_snr/snr_simulation.json --out $OUT/exploratory_snr/snr_group.json

# R2. Monkey-level bootstrap, confirmatory and exploratory inputs (about 10 seconds each).
.venv/bin/python -E -s -m analysis.open_data.monkey_bootstrap --inputs analysis/open_data/results/2026-10-
21/confirmatory \
  --out $OUT/exploratory_monkey/monkey_bootstrap_confirmatory.json
.venv/bin/python -E -s -m analysis.open_data.monkey_bootstrap --inputs $PREVIOUS \
  --out $OUT/exploratory_monkey/monkey_bootstrap_exploratory.json

# R3. Michigan with one generator per subject (D8).
# Dry run: plan, checks and expected wall time; starts nothing.
.venv/bin/python -E -s -m analysis.open_data.launch_reseeded --root data_external/ds006623 --previous $PREVIOUS --out
$OUT/exploratory_reseeded
# Staged first run on the shortest subject (about 45 minutes), to be checked before the rest.
.venv/bin/python -E -s -m analysis.open_data.launch_reseeded --root data_external/ds006623 --previous $PREVIOUS --out
$OUT/exploratory_reseeded --subject sub-26 --execute
# The other 24 subjects (about 5.8 hours on ten processes); sub-26 is skipped because its result exists.
.venv/bin/python -E -s -m analysis.open_data.launch_reseeded --root data_external/ds006623 --previous $PREVIOUS --out
$OUT/exploratory_reseeded --execute
# Group step, once all 25 files exist.
.venv/bin/python -E -s -m analysis.open_data.reseeded listing --previous $PREVIOUS --folder $OUT/exploratory_reseeded
\
  --out $OUT/exploratory_reseeded/group_inputs.txt
.venv/bin/python -E -s -m analysis.open_data.exploratory_group group --inputs
$OUT/exploratory_reseeded/group_inputs.txt \
  --out $OUT/exploratory_reseeded/verdicts_reseeded.json

# Tests of the analysis code (synthetic data only).
.venv/bin/python -I analysis/open_data/run_tests.py
```

R1 and R2 can run while R3 runs, on a core that R3 leaves free (with ten R3 processes on ten cores, wait for R3 or run R3 on nine).

Results

R1 and R2 were run after commit `74f9cb0` with the commands of E.15.5 and with the analysis code of that commit, unchanged. Each result file records the commit at the head of the repository when the file was written: `74f9cb0` for R2 (written at 20:15:56 and 20:16:15 UTC on 21 October 2026) and `f697899` for R1 (written from 20:32:52 to 20:33:41 UTC; that commit changed only the paper and the simulator result files). The R1 simulation took 995.4 s. The files are in `analysis/open_data/results/2026-10-21/exploratory_snr/` and `analysis/open_data/results/2026-10-21/exploratory_monkey/`. Every result below is exploratory, and none changes a verdict or outcome of E.14. R3 was run later and is reported in its own subsection below.

R1: H5a with a signal-to-noise control. The H5a statistic as run reproduced E.14 (median r 1.28 over six patients, p 0.031), so the reanalysis went ahead; sub-PRIOS09 again had no responding connection. Per patient (SNR is the median over connections; "propofol SNR higher" counts connections with a positive log SNR ratio; matched r was undefined for 3 connections of sub-PRIOS01 and 1 of sub-PRIOS04, whose matched spread was zero):

Patient	Connections	Propofol SNR higher	SNR awake	SNR propofol	Matched SNR	Median r as run	Median r, SNR-matched	Median r, SNR-restricted (connections kept)
sub-PRIOS01	395	190	2.61	2.49	1.66	1.24	1.03	1.05 (221)
sub-PRIOS02 (D4)	57	35	3.27	4.40	2.97	−0.10	−0.13	−0.20 (40)
sub-PRIOS03	209	133	2.93	4.33	2.23	0.64	0.45	0.55 (157)
sub-PRIOS04	138	35	3.35	1.26	0.84	1.51	0.40	1.18 (43)
sub-PRIOS05	224	175	1.31	2.75	1.10	1.33	1.74	1.32 (186)
sub-PRIOS06	13	6	5.64	4.45	3.23	1.63	1.42	1.49 (7)

Patient	Connections	Propofol SNR higher	SNR awake	SNR propofol	Matched SNR	Median r as run	Median r, SNR-matched	Median r, SNR-restricted (connections kept)
sub-PRIOS09	0							

Group (six patients)	Median r (SESOI log 1.2 = 0.18)	One-sided p
As run (E.14)	1.28	0.031
SNR-matched	0.74	0.031
SNR-restricted	1.12	0.031

Group median SNR (the median over patients of each patient's median): 3.10 awake and 3.54 under propofol; median log SNR ratio 0.03. Group matched SNR: 1.94.

Reading fixed in advance. At the group matched SNR of 1.94, the simulated median spread of a 2 ms true jitter with 10 trials is 0.04 ms in white noise and 0.13 ms in pink noise, below the 1 ms floor in both colours. By the rule of E.15.2(c), the matched comparison therefore cannot detect a rise and is reported as **uninformative**, not read either way; the rule's two further readings (that the rise is not explained by an SNR difference, or that it cannot be distinguished from one) do not apply. OD5 (H5a) remains not interpretable as timing variability, as E.14 stated.

The simulation read at the group SNRs of the data (awake 3.10, propofol 3.54, inside the simulated range in every cell) gives the r that this SNR change alone would produce. It is reported as pre-specified; the rule does not use it.

Noise	True jitter (ms)	6 trials	10 trials	100 trials
White	0	0.00	0.01	−0.04
White	2	0.12	0.15	−0.01
White	4	0.29	1.43	0.00
Pink	0	0.07	0.06	−0.16
Pink	2	0.20	0.27	−0.04
Pink	4	1.12	0.26	−0.02

Read after the run and not part of the rule: these values overlap the observed r (1.43 at 10 trials and 4 ms in white noise), and up to a measured SNR of 5.6 a 2 ms jitter was measured as at most 0.14 ms (10 trials, white noise), so the as-run statistic, not only the matched one, lies below the estimator's sensitivity at these trial counts.

Against the expectations stated before the run:

1. *The estimator at few trials.* As expected, at 6 and 10 trials a lower SNR lowered the measured spread over most of the simulated range, and only at 100 trials did it raise it. With 10 trials and a true jitter of 2 ms, the median measured spread was at most 0.14 ms (white) and 0.38 ms (pink) up to a measured SNR of 5.59 and 4.39, and reached 1.46 and 1.44 ms only at 7.71 and 6.09. With 100 trials in pink noise the spread fell as the SNR rose at every step.
2. *Null check.* After matching, the median r lay within ±0.2 of zero in 33 of 36 conditions, as expected. The three exceptions (−0.31, −0.34 and −0.42) all had a 4 ms jitter and a propofol SNR a quarter of the awake one. Without matching, the same synthetic pairs gave median r from −4.29 to 0.09: a lower SNR alone lowered the measured spread.
3. *Informativeness.* The matched comparison was not informative (above).
4. *Which state has the higher SNR.* E.9 assumed lower response amplitude under propofol; the SNR, which combines amplitude and noise, was not lower under propofol: its group median was 3.54 against 3.10 awake, it was higher under propofol in 574 of 1,036 connections, and the patient's median SNR was higher under propofol in three of the six patients. For propofol raising the SNR in most connections we expected r_{matched} below 1.28 and predicted no size; it is 0.74, but by the rule above it is not read.
5. *Restriction.* The restricted statistic kept 654 of the 1,036 connections and gave a median of 1.12, against 1.28 as run.

R2: monkey-level bootstrap of H2b and H3. Two-stage bootstrap (2,000 resamples), monkeys and then sessions; four monkeys in H2b, and in H3 two monkeys for each of ketamine, medetomidine and propofol and four for ketamine-medetomidine. Shares are of resamples, as $(1 + k)/(1 + B)$.

Set	Test	Point estimate (SESOI)	Session-level 95% interval (E.14)	Monkey-level 95% interval	Share at or below zero	Share at or above the SESOI
Confirmatory, 20 sessions	H2b, smaller median τ	0.06 (0.2)	none (Wilcoxon p 0.45)	−0.51 to 0.52	0.52	0.20
Confirmatory, 20 sessions	H3, D in AUC	0.07 (0.05)	−0.23 to 0.20	−0.25 to 0.22	0.47	0.39
Exploratory, 23 sessions	H2b, smaller median τ	−0.01 (0.2)	none (Wilcoxon p 0.51)	−0.47 to 0.39	0.57	0.08

Set	Test	Point estimate (SESOI)	Session-level 95% interval (E.14)	Monkey-level 95% interval	Share at or below zero	Share at or above the SESOI
Exploratory, 23 sessions	H3, D in AUC	0.085 (0.05)	−0.21 to 0.23	−0.26 to 0.24	0.42	0.40

Against the expectations stated before the run: the point estimates equal E.14's, as expected. For H2b the share of resamples at or below zero is near one half (0.52 and 0.57), as expected, and the monkey-level interval spans about −0.5 to 0.5. For H3 the monkey-level interval includes zero and is wider than the session-level one in both sets, as expected, and H3 stays inconclusive. Because three of the four agents rest on two monkeys, these intervals are a lower bound on the uncertainty at the level of the animal (E.15.3); the verdicts and outcomes of E.14 stand.

R3: Michigan reanalysed with one generator per subject (D8). R3 was run after commit `74f9cb0` with the commands of E.15.5 and with the analysis code of that commit, unchanged: every result file records for `reseeded.py` (`d0026164…`), `deviations.py` (`5e919a65…`, the version of E.14) and the Michigan pipeline the SHA-256 of the files of that commit, and its D8 seed, which in every file is the committed seed plus the subject's number. The commit at the head of the repository when a file was written is `ad4f822` for 23 subjects and `44ecc92` for sub-25, sub-27 and the group step; neither commit changed analysis code. The staged run of sub-26 took 2,137 s and ended at 20:50:51 UTC on 21 October 2026; the other 24 subjects then ran on ten processes from 20:51 UTC to 01:25 UTC on 22 October (4.6 h, against 5.8 h expected), 0.57 to 3.17 h per subject and 40.8 h in all (D6: 0.76 to 4.42 h and 55.1 h). The launcher's report (`logs/launch_report.ndjson`) records 25 subjects done (sub-26 in the staged run, the other 24 in the second), sub-26 skipped in the second because its result existed, and no failure. The group step (3.0 s) read the 25 reseeded Michigan files and the Neurotycho and PRIOS files of E.14. The files are in `analysis/open_data/results/2026-10-21/exploratory_reseeded/`. The medians and counts of H1, H2a and H4 below were recomputed from the per-subject files with separate code and match the group file. In D3, 12 of the 20,000 null realisations were left out, as in E.14.

Test	Data	n, D6 → D8	Effect (SESOI), D6 → D8	p, D6 → D8	Holm-adjusted p, D6 → D8	Outcome, D6 → D8
H1, the collapse is a threshold	Michigan	25 → 25	median log steepness ratio to null A 0.22 → 0.28 (log 1.5 = 0.41)	0.10 → 0.048	0.50 → 0.24	Counts against DPC (exploratory; descriptive cut-off, measure near ceiling), unchanged
H2a, slowing before LOR	Michigan	25 → 25	smaller median τ −0.04 → −0.05 (0.2)	0.42 → 0.41	1.0 → 1.0	Inconclusive → inconclusive
H2b, slowing before LOR	Neurotycho (same files)	23 → 23	smaller median τ −0.01 → −0.01 (0.2)	0.51 → 0.51	1.0 → 1.0	Inconclusive → inconclusive
H3, lag variability tracks LOR across agents	Neurotycho (same files)	23 → 23	D = 0.085 → 0.085 AUC (0.05)	0.22 → 0.22	0.88 → 0.88	Inconclusive → inconclusive
H4, induction-emergence hysteresis	Michigan	22 → 22	median log H 0.78 → 0.80 with the dataset's effect-site series, least favourable 0.28 → 0.26 (log 1.05 = 0.049)	0.025 → 0.023; least favourable 0.44 → 0.42	1.0 → 1.0	Inconclusive → inconclusive

H2b and H3 read the same Neurotycho files as E.14 and are unchanged by construction; the Holm family of five changes only through H1, H2a and H4. H4 excluded the same three subjects (sub-19: no sigmoid fit converged; sub-22 and sub-28: every window of run 3 exceeds the censoring limit). The secondary H1 rows, reported without verdict, moved little: null B median log ratio −0.03 → 0.00 (p 0.81 → 0.82), local amplitude 0.11 → 0.25 (p 0.26 → 0.19), midpoint minus LOR −0.47 µg/ml in both.

H1 per subject. The measurement gate was passed again (median deep over awake BOLD coupling 0.789 under both D6 and D8). After LOR the giant component kept a median 0.947 of its awake size under both (range 0.861 to 1.036, against 0.861 to 1.033 under D6), at least 0.9 in 21 of 24 subjects under both (sub-16 again has no window after its LOR). A sigmoid was preferred over a line by the BIC margin in 1 of 25 subjects (sub-18 under both). The null-A p-value was below 0.05 in the same 8 subjects (sub-02, sub-13, sub-14, sub-17, sub-23, sub-25, sub-26 and sub-30). The fitted transition was as narrow as the fit allows (0.05 µg/ml) in 6 of 25 subjects (sub-02, sub-04, sub-05, sub-06, sub-14 and sub-25), against 8 under D6, which also had sub-24 and sub-26 there; sub-20 (0.050004) and sub-18 (0.05007) lie just above.

Per subject, as pre-specified in E.15.4 (D6 value → D8 value; n/e, not evaluable, E.14):

Subject (D8 seed)	H1 null-A p	H1 steepness ratio to null A	H2a τ, variance	H2a τ, autocorrelation	H4 log H, dataset series
sub-02 (20261023)	0.020 → 0.025	47.66 → 48.49	−0.02 → −0.03	0.34 → 0.33	−1.13 → −1.13
sub-03 (20261024)	0.915 → 0.925	0.47 → 0.40	−0.26 → −0.26	−0.64 → −0.64	0.76 → 0.76
sub-04 (20261025)	0.075 → 0.159	1.63 → 1.45	0.76 → 0.76	0.59 → 0.54	−1.95 → −1.85

Subject (D8 seed)	H1 null-A p	H1 steepness ratio to null A	H2a τ, variance	H2a τ, autocorrelation	H4 log H, dataset series
sub-05 (20261026)	0.065 → 0.159	1.18 → 1.19	−0.09 → −0.07	0.09 → 0.05	1.65 → 1.64
sub-06 (20261027)	0.194 → 0.215	1.28 → 1.44	−0.61 → −0.62	−0.25 → −0.31	0.94 → 0.93
sub-07 (20261028)	0.428 → 0.413	1.91 → 2.09	0.32 → 0.31	−0.83 → −0.82	0.88 → 0.95
sub-11 (20261032)	0.940 → 0.985	0.17 → 0.17	−0.05 → −0.05	0.75 → 0.75	0.01 → 0.01
sub-12 (20261033)	1.000 → 0.990	0.11 → 0.11	0.22 → 0.22	−0.73 → −0.72	−26.54 → −18.43
sub-13 (20261034)	0.050 → 0.030	18.35 → 22.98	0.69 → 0.69	0.73 → 0.72	0.08 → 0.05
sub-14 (20261035)	0.035 → 0.035	27.12 → 28.25	0.50 → 0.50	0.48 → 0.49	0.30 → 0.30
sub-15 (20261036)	1.000 → 1.000	0.07 → 0.06	−0.48 → −0.48	−0.54 → −0.53	1.25 → 1.25
sub-16 (20261037)	0.900 → 0.925	0.40 → 0.37	0.71 → 0.71	0.09 → 0.11	1.20 → 1.20
sub-17 (20261038)	0.020 → 0.015	5.22 → 6.72	0.47 → 0.48	−0.61 → −0.62	0.74 → 0.79
sub-18 (20261039)	0.488 → 0.269	1.01 → 1.20	−0.62 → −0.62	−0.72 → −0.72	0.21 → 0.21
sub-19 (20261040)	0.871 → 0.642	0.62 → 0.86	−0.61 → −0.61	−0.50 → −0.47	n/e → n/e
sub-20 (20261041)	0.239 → 0.199	1.21 → 1.23	−0.42 → −0.41	−0.54 → −0.54	−0.79 → −0.78
sub-21 (20261042)	0.174 → 0.159	5.20 → 5.78	0.40 → 0.40	0.53 → 0.54	1.07 → 0.92
sub-22 (20261043)	0.945 → 0.355	0.02 → 1.32	−0.25 → −0.23	0.51 → 0.47	n/e → n/e
sub-23 (20261044)	0.010 → 0.035	59.43 → 55.18	0.09 → 0.08	0.26 → 0.28	0.80 → 0.80
sub-24 (20261045)	0.109 → 0.935	1.16 → 0.46	−0.04 → −0.05	−0.18 → −0.18	1.18 → 1.17
sub-25 (20261046)	0.035 → 0.040	1.24 → 1.22	0.89 → 0.88	0.73 → 0.74	1.07 → 1.07
sub-26 (20261047)	0.010 → 0.005	51.14 → 43.19	−0.45 → −0.44	−0.24 → −0.24	1.22 → 1.22
sub-27 (20261048)	0.060 → 0.075	6.70 → 5.87	−0.50 → −0.49	0.54 → 0.55	0.65 → 0.64
sub-28 (20261049)	0.945 → 0.945	0.46 → 0.54	0.75 → 0.75	0.60 → 0.61	n/e → n/e
sub-30 (20261051)	0.030 → 0.030	12.97 → 10.03	−0.72 → −0.73	0.56 → 0.55	1.03 → 1.03

Plain reading. With one generator per subject, so that any subject's result can be reproduced by analysing that subject alone, every outcome of E.14 is unchanged. The steepness test's p-value moved from 0.10 to 0.048 before correction and from 0.50 to 0.24 after Holm adjustment. This shows how sensitive that test is to the random draws: the edge-test surrogates set each subject's edge threshold, so a new draw changes the observed curve as well as its null. H1's outcome, however, is set by the falsifier of E.5, the share of the giant component kept after LOR, and that did not move (median 0.947 under both). As the rule of E.15.4 states, neither the D6 nor the D8 values replace the other.

Against the expectations stated before the run:

1. *Outcomes.* As expected, none changed: H1 counts against DPC through the falsifier (median 0.95 of the awake size), and H2a and H4 stay inconclusive.
2. *Per-subject and group movement.* Not as expected for the H1 null-A p-values. The median change was 0.015, but 7 of 25 subjects moved by more than sub-05's 0.04 between its two D6-era analyses, most of all sub-24 (0.109 → 0.935) and sub-22 (0.945 → 0.355), whose fitted transition widths also changed (0.05 → 0.13 µg/ml and 3.27 → 0.05 µg/ml). The group median log steepness ratio moved from 0.22 to 0.28; the median share of the giant component kept after LOR moved by less than 0.001, and the H2a τ of any subject by at most 0.06.

Commands. From the repository root; these are the commands of E.15.5 with the date of the run filled in.

```
PREVIOUS=analysis/open_data/results/2026-10-21/exploratory/group_inputs.txt
OUT=analysis/open_data/results/2026-10-21/exploratory_reseeded
# Staged run of sub-26, then the other 24 subjects (sub-26 is skipped because its result exists).
.venv/bin/python -E -s -m analysis.open_data.launch_reseeded --root data_external/ds006623 --previous $PREVIOUS --out
$OUT --subject sub-26 --execute
.venv/bin/python -E -s -m analysis.open_data.launch_reseeded --root data_external/ds006623 --previous $PREVIOUS --out
$OUT --execute
# Group step.
.venv/bin/python -E -s -m analysis.open_data.reseeded listing --previous $PREVIOUS --folder $OUT --out
$OUT/group_inputs.txt
.venv/bin/python -E -s -m analysis.open_data.exploratory_group group --inputs $OUT/group_inputs.txt --out
$OUT/verdicts_reseeded.json
# One subject alone, as the launcher runs it (here sub-05).
.venv/bin/python -E -s -m analysis.open_data.reseeded michigan --root data_external/ds006623 --subject sub-05 --out
$OUT
```

Appendix F. Detailed accounts moved from the main text

The restructure of 21 October 2026 shortened Sections 1 to 11. The detailed accounts it displaced are kept here, with their figures, corrected where the main text was corrected; nothing in them is new. Passages that only repeated the main text or Appendices B to E were removed on 22 October 2026; the rows taken out of the Section 6 table are in F.7. Test identifiers are used freely (Appendix A.5).

F.1 Molecular routes in full (Section 2.1)

Molecular route	Agents with evidence	Effect on ion flow	DPC fault type (P4) [H]	Sources
GABA-A receptor potentiation (prolonging channel opening)	Propofol, etomidate, barbiturates, isoflurane, sevoflurane	More Cl^- inhibitory current	Transmission failure	Mihic et al. (1997); Jurd et al. (2003)
NMDA receptor block: open-channel block together with fewer channel openings (ketamine), or binding at the glycine co-agonist site (xenon, isoflurane)	Ketamine, xenon, nitrous oxide; isoflurane	Less Na^+ and Ca^{2+} excitatory current	Timing variability and transmission failure	Orser et al. (1997); Franks et al. (1998); Jevtović-Todorović et al. (1998); Dickinson et al. (2007)
Two-pore-domain K^+ channel activation (TREK-1, TASK)	TREK-1: the volatile agents halothane, isoflurane, chloroform and diethyl ether, and the gases xenon, nitrous oxide and cyclopropane; TASK: of these agents only halothane and isoflurane, as the gases leave TASK-3 unaffected	More K^+ leak, hyperpolarisation	Transmission failure	Patel et al. (1999); Gruss et al. (2004); Heurteaux et al. (2004)
Inhibition of hyperpolarisation-activated cyclic-nucleotide-gated (HCN) channels	Volatile agents (subunit-specific); propofol; ketamine, whose hypnotic potency is strongly reduced in mice lacking HCN1	Less pacemaker cation current	Altered rhythm and timing	Chen et al. (2005); Ying et al. (2006); Chen et al. (2009)
Presynaptic Na^+ channel inhibition, reducing Ca^{2+} entry and vesicle release	Isoflurane	Less transmitter released per spike	Transmission failure	Herold and Hemmings (2012) for Na^+ channels; Baumgart et al. (2015) for reduced Ca^{2+} entry and release
α2-adrenoceptor agonism, acting through the endogenous sleep pathway	Dexmedetomidine	Indirect: inhibition of locus coeruleus neurons	Arousal suppression (P7)	Nelson et al. (2003)
Disruption of membrane lipid domains, releasing PLD2, which activates TREK-1	Chloroform, isoflurane	Indirect: more K^+ current	Transmission failure	Pavel et al. (2020)
Mitochondrial complex I (a sensitivity factor rather than an ion channel)	Volatile agents	Not a channel effect; complex I dysfunction increases sensitivity	Modulates susceptibility	Kayser et al. (1999)

Two further proposals are unconfirmed and are noted only for completeness. A photoactive anaesthetic analogue was found bound to β-tubulin in tadpoles (Emerson et al., 2013). A single dose of the microtubule stabiliser epothilone B delayed isoflurane-induced loss of righting in rats (Khan et al., 2024) and, by about 29 s on the following day, in mice (Huang et al., 2026), in studies from one laboratory. In contrast, two weeks of daily epothilone D, also a stabiliser, made mice more sensitive to isoflurane, as did the destabiliser vinblastine, while paclitaxel shifted sensitivity marginally the other way (Li et al., 2025). The direction of the effect differs between drugs and dosing schedules, and whether microtubules contribute to anaesthesia remains unsettled. A single study reported that xenon isotopes carrying nuclear spin are less potent in mice than spin-free isotopes (Li et al., 2018); no independent replication has been found.

F.2 Systems-level findings and prior models in full (Sections 2.2 and 2.3)

The findings below are established unless a sentence says otherwise: results from a single study, a review's reading, results that their own authors doubt and results later found to be confounded are marked as such where they appear.

- **Thalamocortical involvement.** Two volatile anaesthetics both suppressed regional thalamic and midbrain reticular activity in human volunteers (Alkire et al., 2000). Nicotine microinjected into the central medial thalamus reverses sevoflurane-induced LORR in rats (Alkire et al., 2007). Thalamic stimulation can restore arousal in anaesthetised macaques (Redinbaugh et al., 2020). Whether the thalamus or the cortex changes first is open: in patients with deep-brain electrodes, frontal EEG changed sharply at loss of consciousness while subcortical recordings changed little (Velly et al., 2007), whereas in stereo-EEG complexity in the anterior thalamic nucleus fell before that of prefrontal and temporal cortex (Fan et al., 2026).

- **Breakdown of cortical communication.** Propofol, sevoflurane and ketamine all disrupt frontal-parietal feedback connectivity (U. Lee et al., 2013). At the onset of propofol-induced unconsciousness, a slow oscillation appears and neuronal spiking becomes confined to brief phase windows that are asynchronous across cortex, fragmenting cortical networks (Lewis et al., 2012). A recent review reads this finding as a fragmentation of the timing and coordination of local activity across cortical areas rather than its disappearance, and describes activity under propofol and volatile agents as often more stereotyped, temporally restricted and poorly coordinated between regions (Kordas, 2026). Under anaesthesia in monkeys, the most frequent patterns of functional connectivity come to mirror anatomical connectivity, whereas wakefulness explores a richer repertoire unlike the anatomy (Barttfeld et al., 2015). Reviews of large-scale network studies conclude that unresponsiveness under anaesthetics is commonly accompanied by reduced activity in, or functional disconnection of, the lateral frontoparietal networks (Hudetz and Mashour, 2016), together with reduced complexity and a smaller repertoire of brain states (Mashour and Hudetz, 2018).
- **Agent-specific EEG signatures.** Propofol produces increased beta power during sedation, followed by coherent frontal alpha (8 to 12 Hz) together with slow-delta (0.1 to 4 Hz) oscillations at LOR (Purdon et al., 2013, 2015; Ching et al., 2010). Ketamine produces increased gamma power and does not produce the propofol-like alpha pattern (Akeju et al., 2016).
- **Cellular decoupling.** In mice, anaesthetics decouple the apical dendrites of layer 5 pyramidal neurons from their somata, which would interrupt cortico-cortical feedback integration if that feedback arrives mainly on the apical dendrites (Suzuki and Larkum, 2020).
- **Loss of coordination, not of activity, across species.** In *Caenorhabditis elegans*, isoflurane leaves individual head neurons active but collapses the correlations between them (Awal et al., 2020). In *Drosophila*, isoflurane reduces stimulus-locked power and coherence in the central brain, while increasing them at twice the stimulus frequency in the optic lobes for some stimuli (Cohen et al., 2016). It also impairs low-frequency feedback while leaving high-frequency feedforward connectivity intact (Cohen et al., 2018), and fragments the integrated structure of brain activity into separate clusters (Leung et al., 2021).
- **Loss of individual signature.** Under sevoflurane and propofol, functional connectivity becomes less distinctive between individuals, most strongly in transmodal association cortex, and anaesthetised human connectivity becomes more similar to that of the macaque (Luppi et al., 2025; a single study).
- **Complexity.** The perturbational complexity index (PCI) is high in wakefulness and low during propofol- and xenon-induced unconsciousness. It remains high under ketamine, consistent with the dream reports of healthy volunteers on emergence (Sarasso et al., 2015); the original validation of the index used midazolam, xenon and propofol (Casali et al., 2013).
- **Neural inertia.** In both insects and mammals, the concentration at which consciousness returns is lower than the concentration at which it is lost, and the difference is not fully explained by pharmacokinetics. This hysteresis is termed neural inertia; it was demonstrated in both mice and *Drosophila* (Friedman et al., 2010) and depends on excitability genes that act independently of induction sensitivity (Joiner et al., 2013; see also Kelz et al., 2008). Its measured size varies by agent. The ratio of induction to emergence concentration is about 1.08 for isoflurane and 2.85 for halothane in mice (Friedman et al., 2010). In humans it is about 1.07 for propofol effect-site concentrations (1.5 and 1.4 μg/ml at loss and recovery of behavioural responsiveness; Warnaby et al., 2017), whose authors found no consistent hysteresis in these values and identified inertia instead in slow-wave saturation. It is 1.11 in another study (Iwakiri et al., 2005), whose authors describe the concentrations as similar, although the difference was statistically significant. For measured propofol concentrations it is about 1.63 (Sepúlveda et al., 2018), whose authors caution that the gap could instead arise from systematic error in how effect-site concentrations were estimated, not from neural inertia. A prospective study designed to test neural inertia in humans, in 36 volunteers with stepwise increases and decreases of propofol or sevoflurane, each with or without remifentanil, found no difference between induction and recovery concentrations for propofol, and differences for sevoflurane at most endpoints, for some only with remifentanil; its authors concluded that the endpoint measured influences whether neural inertia is detected (Kuizenga et al., 2018). Whether neural inertia exists in humans is therefore contested, and DPC's account of it (P5) is conditional on it. Recovery proceeds through a sequence of discrete, metastable activity states (Hudson et al., 2014).
- **Glial cells.** Glia respond to anaesthetics, and manipulating them changes induction and emergence in mice. In awake mice, anaesthetics suppress calcium signals in cortical astrocytes even at doses too low to change how neurons respond when the whiskers are stimulated (Thrane et al., 2012). Inhaled agents also cause a reversible disruption of the fine astrocytic processes in somatosensory cortex (Zhou et al., 2025). Activating astrocytes in the basal forebrain speeds isoflurane induction and slows recovery (Lin et al., 2023), whereas activating astrocytes in the parabrachial nucleus or the paraventricular thalamus speeds emergence (Liu et al., 2023; Zhao et al., 2024). The effect therefore depends on the region. Depleting microglia with the drug PLX5622 reduces sensitivity to several anaesthetics and shortens their effect (Cao et al., 2023; He et al., 2023). The same group later found, however, that PLX5622 also activates a liver receptor that speeds the clearance of anaesthetics, and that depleting microglia genetically did not hasten emergence (Cao et al., 2026). The firmer evidence is that genetic loss of the microglial receptor P2Y12 speeds recovery (Cao et al., 2023). Myelin sets conduction timing. In auditory brainstem circuits the geometry of myelinated axons is tuned to set when action potentials arrive (Ford et al., 2015), and in mice a subtle myelin deficit spreads out conduction times and reduces the synchrony of motor cortex activity (Kato et al., 2020). We found no study of anaesthetic requirements in human demyelinating disease, only case reports. Two animal studies exist. Demyelinated mice took longer to recover from isoflurane, with effects that depended on the region demyelinated (Wang et al., 2024), and older demyelinated mice took longer to be induced at a fixed concentration, in small groups (Tot et al., 2024).

Conceptual frameworks not detailed in Section 2.3:

Framework	Central claim	Relationship to DPC
Perceptual cycles (VanRullen and Koch, 2003; VanRullen, 2016)	Perception proceeds in discrete cycles tied to brain oscillations; current evidence points to several rhythms rather than one	P1 adopts discrete cycles; DPC adds a timing condition on each exchange, a network-level quorum, and the proposal that anaesthetics act by breaking the cycles' timing
Recurrent processing (Lamme and Roelfsema, 2000; Lamme, 2006)	A feedforward sweep is probably insufficient for awareness; recurrent processing between areas is needed	P1 adopts recurrent cycles; DPC adds a timing condition on each exchange and a network-level quorum
Thalamocortical switch (Alkire et al., 2000)	The thalamus gates conscious state	DPC assigns the thalamocortical core the role of supplying a shared phase reference, not of acting as an on/off switch
Mean-field anaesthetic phase transition (Steyn-Ross et al., 1999; Bojak and Liley, 2005)	Anaesthetic modulation of inhibitory synaptic kinetics moves a mean-field model of cortex through distinct stationary regimes, suggesting a phase transition (Steyn-Ross et al., 1999); a related mean-field model accounts for anaesthetic changes in the EEG through ligand-gated synaptic responses (Bojak and Liley, 2005)	DPC also treats loss of consciousness as a transition, but in a network of discrete connections whose timing coherence percolates; it does not model the EEG spectrum, which these models do
Slow oscillation as the default cortical state (Sanchez-Vives et al., 2017)	Slow oscillations are proposed as the baseline, low-complexity pattern of cortical activity, which the brain must leave to regain the complexity that accompanies consciousness	Compatible with P6 and P7: in DPC the slow oscillation is what the network does once quorum is lost; DPC adds a criterion for when that happens
Dendritic integration theory (Aru et al., 2020, whose abstract calls it Dendritic Information Theory)	Cortical pyramidal cells gate the integration of bottom-up and top-down signals in their dendrites, and in unconscious states this integration is prevented; anaesthetics decouple the dendrites of layer 5 pyramidal cells from their cell bodies (Suzuki and Larkum, 2020)	Compatible: a cellular route by which feedback signals could fail to count, one candidate mechanism for the omission faults of P4

Quantitative models:

Prior model	What it established	What DPC adds
Breskin et al. (2006)	In living neuronal cultures, chemically weakening synapses makes the giant connected component disintegrate at a percolation transition	DPC applies percolation to whole connectomes and to timing as well as transmission
Arese Lucini et al. (2019)	Human functional brain networks are modelled as passing from conscious to subliminal perception by k-core percolation: the outer k-shells of the network switch off, while its innermost core stays active	DPC applies percolation to anaesthetic loss of consciousness and defines occupation by a timing criterion rather than by functional activation
Zhou et al. (2015)	A percolation model of thalamocortical and cortico-cortical transmission, with one transmission probability p, reproduces EEG features of anaesthesia; information transfer changes sharply near a critical p	Zhou et al. already placed every agent on one transmission parameter; DPC adds timing variability as a second, independent route into the same occupation probability
Su et al. (2020)	Measured an induction-recovery hysteresis loop in the burst-suppression ratio of local field potentials (two mice, isoflurane) and explained it by percolation on self-similar multilayer networks	DPC attributes hysteresis to feedback through the shared rhythm rather than to self-similar network structure; for feedback-driven hysteresis see Kim et al. (2018), below
Tian and Sun (2022)	On the fruit-fly connectome, the formation of functional connectivity is described as a percolation process controlled by the balance of excitation and inhibition	DPC uses percolation on the same connectome for the loss of quorum under anaesthesia, with occupation set by transmission and timing
Du and Huang (2025)	A recurrent network model fitted to macaque electrocorticography across wakefulness, anaesthesia and recovery: its response function is suppressed and its dynamics more chaotic under anaesthesia	Unlike this model, DPC's models are not fitted to recordings; Section 8 proposes the measurements that would test them
Kim et al. (2016)	The conditions for explosive synchronisation are met in human EEG networks just after anaesthetic-induced loss of consciousness	DPC tests a percolation transition, not explosive synchronisation, on empirical connectomes of three species with a single parameter set
Kim et al. (2018)	Stuart-Landau oscillators with adaptive feedback on a 78-region connectome reproduce hysteresis of functional connectivity seen in EEG under ketamine and sevoflurane	The feedback mechanism in DPC (P5) is closely related, and Kim et al. already linked the size of the hysteresis to anaesthetic potency; DPC attributes the feedback to a shared timing reference and calibrates its strength to reported inertia ratios
Moon et al. (2015); H. Lee et al. (2013)	Network topology determines phase lead and lag between hubs and periphery, and the front-to-back direction of information flow disappears under sevoflurane; the hub structure of EEG networks is reconfigured after propofol-induced unconsciousness	Hub disruption under anaesthesia has already been reported. DPC explains why hubs are vulnerable, by percolation on networks whose degree distribution is heavy-tailed (Albert et al., 2000), which among the three connectomes holds mainly for *Drosophila* (Section 4.2), and predicts that disrupting hub connections lowers the concentration needed for loss of responsiveness more than disrupting the same number of random connections (prediction 6)

Prior model	What it established	What DPC adds
Proekt and Hudson (2018); Hudson et al. (2014)	Recovery of consciousness passes through discrete metastable states (Hudson et al., 2014), and neural inertia can be explained by noise-driven transitions on a multi-well landscape, compared with inertia data from mice and *Drosophila* (Proekt and Hudson, 2018)	DPC offers a network-level origin for such barriers; the two accounts are compatible

F.3 Distributed computing and engineering analogues (Section 3.1)

In distributed computing, a set of processes reaches *consensus* when it agrees on a single value despite some processes failing. Fischer, Lynch and Paterson (1985) proved that deterministic consensus is impossible in a fully asynchronous system if even one process can crash. Dwork, Lynch and Stockmeyer (1988) showed that consensus becomes achievable under *partial synchrony*, meaning that message delays and relative processing speeds are bounded, even if the bounds are not known in advance. Systems built on these results commonly require a *quorum*, a minimum fraction of participating processes, before committing a value. Faults are classified by how a process deviates from its specification (Cristian, 1991): it may stop (a *crash*), fail to deliver a message (an *omission*), or deliver it outside the agreed time (a *timing* failure). Timing bounds are not the only route to agreement: randomised protocols reach it without any bound on delays, provided a majority of processes keep working (Ben-Or, 1983), and so do protocols that use unreliable failure detectors (Chandra and Toueg, 1996). DPC takes the timing-bounded route as its analogy; whether neural circuits use anything resembling randomised agreement or failure detection is unknown.

The analogy, which is not a claim of literal equivalence, is that the brain must repeatedly settle on a single coherent interpretation of multisensory and internal signals that arrive at different areas at different times. Lamport (1978) showed that causality only partially orders the events of a distributed system. He also showed that a consistent total order can be obtained with a system of logical clocks, with synchronised physical clocks added so that the order agrees with what users observe. We propose that the brain faces a comparable constraint.

The terms used in this paper come from engineered systems that keep many distributed units in agreement despite delays, noise and failures, a problem the brain is proposed to face as well. The table lists the systems that motivated each postulate, with references to their specifications, and states where each analogy fails.

Engineered system	How it works	Counterpart in DPC	Where the analogy fails
Fly-by-wire flight control: the triple-triple redundant primary flight computer of the Boeing 777 (Yeh, 1996) and the command/monitor computers of the Airbus A320 family (Briere and Traverse, 1993)	Redundant channels compute the same command; disagreeing channels are outvoted or disconnected, and the aircraft falls back to a simpler control law when too few agree	Quorum (P3); fall-back to default dynamics when quorum is lost (P6)	Neural populations do not compute identical copies of one value, and there is no designated voter
Time-triggered architecture (Kopetz and Bauer, 2003), used in safety-critical vehicles	Nodes transmit in fixed time slots that repeat in a cycle (a round-robin of slots) against a synchronised global clock; a message outside its slot is discarded, and a node that loses synchrony is excluded from membership	Integration cycles (P1); the window W (P2); late signals not integrated	The brain has no fixed slots and no global clock; its timing reference emerges from oscillation
Controller Area Network bus (ISO 11898-1:2024; ISO, 2024)	Every node must sample each bit within the same bit time, so signal propagation delay limits the product of bus length and bit rate	A bounded-delay condition that limits network size and speed (P2)	A CAN bus is one shared medium; the brain is a sparse network of point-to-point connections
Precision Time Protocol (IEEE Std 1588-2019; IEEE, 2020)	A reference clock is distributed through the network, with path delays measured and compensated	The shared thalamocortical phase reference (P5)	PTP has a designated master; the neural reference is emergent and itself depends on coherence
Synchronous power grids (Filatrella et al., 2008; Rohden et al., 2012)	Generators must remain frequency-synchronised; their dynamics are modelled with Kuramoto-like oscillator equations, and loss of synchrony can cascade into a blackout (U.S.-Canada Power System Outage Task Force, 2004)	The same mathematics as the oscillator model in Appendix C; collapse of global coherence	Grid protection disconnects equipment by design; DPC makes no claim that the neural transition is designed or protective

Engineered system	How it works	Counterpart in DPC	Where the analogy fails
Distributed consensus (Lamport et al., 1982; Fischer et al., 1985; Dwork et al., 1988)	Deterministic agreement is impossible without timing bounds when a process can crash (randomised protocols and failure detectors avoid this), and is achievable under partial synchrony (Lamport et al., 1982, give the tolerance bounds)	Motivation for P2 and for the two failure modes in P4; no formal result is transferred, since partial synchrony concerns delay bounds that either exist but are not known in advance or are known but hold only after an unknown time, whereas W is a fixed window	The brain does not run a consensus algorithm, and no impossibility or tolerance bound applies to it; these results motivated P2 and P4 and constrain nothing

Signals arrive at many speeds. The brain receives its inputs with widely different delays, set by biology rather than by the physics of the stimulus. Responses in human primary auditory cortex peak about 15 ms after a sound (Celesia, 1976), whereas the earliest visual response recorded over occipital cortex begins at about 56 ms (Foxe and Simpson, 2002), even though light travels far faster than sound. Peripheral nerve fibres span about two orders of magnitude in conduction velocity, from unmyelinated fibres at about 0.5 to 2 m/s to large myelinated fibres at about 80 to 120 m/s (values tabulated by Yap et al., 2020, from a standard textbook). The brain compensates: once people have spent several minutes exposed to sounds and images offset by an unchanging delay, the point at which they judge the two to coincide moves towards that delay (Fujisaki et al., 2004). Its rhythms are also nested, the phase of slower rhythms modulating the amplitude of faster ones, delta modulating theta and theta modulating gamma (Lakatos et al., 2005; Canolty and Knight, 2010). DPC proposes [H] that these compensations work only while timing slack (Section 3.3) remains positive; prediction 8 (Section 8) states what would falsify it.

F.4 Postulates in detail (Sections 3.2 to 3.4)

The statements below are those of the earlier drafts, kept for their evidence. P3's scope is now the one stated in Section 3.2; P4b's variability claim is part of P4, and its constant-delay claim is the open question of Section 3.4; P6, P8 and P9 are auxiliary (Section 3.3).

- **P1 (Integration cycles).** Conscious content is generated in repeated integration cycles, during which distributed populations converge on a shared state. The cycles adopt two earlier proposals (Section 2.3): that perception proceeds in discrete epochs (VanRullen and Koch, 2003), and that awareness needs recurrent processing between areas rather than a single feedforward sweep (Lamme and Roelfsema, 2000; Lamme, 2006). Related accounts tie such loops to re-entry between cortex and thalamus (Edelman and Gally, 2013), to dynamic links established by synchronisation across frequency bands (Varela et al., 2001), to thalamocortical binding by temporal coincidence (Llinás et al., 1998) and to the comparison of feedback predictions with feedforward errors (Rao and Ballard, 1999). The evidence points to several perceptual rhythms rather than one (VanRullen, 2016), so DPC's single phase reference (P5) is a simplification (Section 10). Several studies report that anaesthesia disrupts this recurrent component more than the feedforward one, though not uniformly. Under propofol, directed feedback connectivity from frontal to parietal cortex weakens as consciousness is lost and recovers when the participants again respond to spoken commands, while earlier animal data had indicated that feedforward processing may continue (Lee et al., 2009). In rats, halothane and isoflurane reduced both feedforward and feedback information transfer at 50 Hz, the feedback transfer from frontal to posterior regions most, while increasing both at 30 Hz (Imas et al., 2005). In mice, anaesthesia decouples the apical dendrites of layer 5 pyramidal neurons, the presumed target of feedback input, from their somata (Suzuki and Larkum, 2020). DPC proposes [H] that a cycle that loses quorum partway is abandoned rather than half-completed.
- **P4 (Two failure modes).** Anaesthetics reduce the fraction of functional connections that are timing-coherent in one or both of two ways. (a) They increase the probability that transmission fails (omission faults, signals that never arrive: for example enhanced GABA-A-mediated inhibition and reduced glutamate release; Baumgart et al., 2015). (b) They increase the trial-to-trial variability of signal timing (timing faults, signals that arrive outside their window: proposed here for NMDA antagonism by ketamine [H]). The hypothesis is that the molecular route affects the network only through these two quantities and through the share of the shared rhythm sustained by arousal (P7) [H]. Counting enhanced inhibition as an omission fault is a modelling simplification: an inhibitory signal does arrive, but the excitatory transmission the connection would otherwise carry does not. Networks with signed, inhibitory connections are tested separately (Appendix C, tests SG1 and SG2). The assignment of ketamine to timing faults is an assumption of the model, not a finding. We found no study showing that ketamine increases timing variability between areas. Nor can the assignment rest on NMDA antagonism alone: xenon, also an NMDA antagonist (Section 2.1), lowers cortical complexity as propofol does, whereas ketamine leaves it wakefulness-like (Sarasso et al., 2015). A circuit effect of NMDA receptor inhibition that is established is different: in awake rats it mainly reduced the activity of putative inhibitory interneurons and, with a delay, increased the firing of most pyramidal neurons, which suggests disinhibition (Homayoun and Moghaddam, 2007).
- **P4b (Network timing, a hypothesis not yet tested in data).** The brain integrates inputs against learned expectations of when they will arrive. What fails under anaesthesia is proposed to be the timing of signals *between* areas relative to the shared rhythm, which becomes more variable as that rhythm weakens (P5), and not the precision of single neurons. A constant delay is harmless under this hypothesis, because it can be anticipated; only variability matters. The models contain no such anticipation, and in them constant timing errors are not harmless. In the phase model, fixed timing offsets abolish quorum (S4) one or two concentration steps earlier than fluctuating jitter of the same size in 20 of 20 *C. elegans* and 18 of 20 human seeds, and at the same step in all 6 *Drosophila* seeds; with measured conduction delays, quorum on the human connectome is lost at a mean of 1.16 rather than 1.57 times the EC50, the concentration at which the agent has half its maximal effect (D1;

both comparisons post hoc; concentrations were sampled every 0.05 EC50). This part of P4b therefore depends on a compensation mechanism that the models lack. A test fixed before its run (Appendix C.10) gave each connection a learned expectation of its arrival time, driven by the arriving signal's phase relative to the receiver's own, and it failed: coherence fell even at zero concentration with every agent, most likely because the learning also absorbed the small phase differences that keep populations with different natural frequencies in step (this mechanism was not measured directly), and without conduction delays quorum was lost below 0.13 times the EC50 with every agent, against 0.60 to 1.58 without learning (tests L1, L3 and L4 held in no seed). With measured conduction delays, the learning first did what P4b proposes, raising awake coherence from 0.41 to 0.62 in every seed, before the same drift took quorum away at a mean of 0.735 times the EC50, against 1.205 without learning. A receiver that sees only the arrival phase cannot tell a constant delay from the phase relation that coupling maintains, so how the brain would make constant delays harmless remains open. At the behavioural scale, people adapt to the mean of a variable delay about as well as to a constant one (Yamamoto and Kawabata, 2011; Beech et al., 2025, 2026); those delays, of 100 to 230 ms on average, are far longer than the window proposed here, so they do not test whether variability between areas is compensated. The few local measurements agree that single-neuron timing does not become noisier. Deepening ketamine-xylazine anaesthesia lengthened first-spike latency in the gerbil auditory brainstem without changing spike jitter (Jing et al., 2021). Propofol made the spike timing of rat insular pyramidal neurons more synchronous, by strengthening shared inhibition (Koyanagi et al., 2021). The timing regularity of bursting neurons was maintained under pentobarbital (Kato et al., 2016). We found no study of the trial-to-trial latency variability of responses between cortical areas under anaesthesia. The closest measured the trial-to-trial variability of auditory responses with fMRI: the reduction in variability that a repeated stimulus normally produces was abolished under propofol sedation and anaesthesia (Huang et al., 2018b), which concerns the amplitude of responses within regions, not the timing of signals between them. In the nearest intracranial study, responses of distant areas to single electrical pulses were more variable under propofol than in wakefulness, most of all in prefrontal cortex, while the latency of the first response peak did not change with state (Zelmann et al., 2023). That variability was the spread of the response signal, not of its timing. It also rose in spontaneous activity, and it was measured after loss of consciousness rather than across graded concentrations. The hypothesis predicts that the trial-to-trial latency variability of responses between areas rises at concentrations just below loss of responsiveness, before their mean latency changes appreciably (Section 8, prediction 8).

- **P9 (Glial cascade control; an extension, on which P1 to P8 do not depend).** The integration cycle, which runs in milliseconds, is regulated by slower glial loops that maintain the conditions for timing coherence, each on its own timescale. Myelin sets conduction timing (Section 2.2), and in mouse motor cortex a subtle myelin deficit spreads out thalamocortical arrival times (Kato et al., 2020). Blocking the formation of new oligodendrocytes in adult mice also prevents the learning-driven coupling of hippocampal and cortical rhythms (Steadman et al., 2020). That oligodendrocytes adjust conduction delays over time so that signals from distant regions arrive together is proposed by models of adaptive myelination (Pajevic et al., 2014, 2023; Noori et al., 2020). Astrocytes regulate cortical state over seconds to hours: their activity precedes and can trigger the slow oscillation (Poskanzer and Yuste, 2016), astrocytic calcium tracks the need for sleep (Ingiosi et al., 2020), and blocking astrocytic vesicle release reduces gamma oscillations (Lee et al., 2014). Microglia take part in a loop with noradrenaline: low noradrenergic tone, as under anaesthesia, increases microglial surveillance (Liu et al., 2019; Stowell et al., 2019), and microglial signalling lowers noradrenaline further (Ma et al., 2024). P9 proposes that anaesthetics disrupt these loops as well as the integration cycle, and that the microglial loop is one biological source of the reference feedback of P5, holding the unconscious state stable once it is reached. Regulation of fast neural activity by slower processes is an established principle (Turrigiano, 2008); what P9 adds is that the glial loops act on timing coherence. P1 to P8 do not depend on P9.

F.5 Model details (Section 4)

Although conscious *access* is modelled as a threshold on S, the order parameter itself is continuous. This is consistent with graded sedation, with the progressive EEG changes seen with increasing concentration, and with individual variability in thresholds. Recordings disagree on how abrupt the transition is. During slow propofol induction, functional connectivity measured with fMRI fell gradually while subcortical-cortical connectivity recovered abruptly (Huang et al., 2021b); in stereo-EEG, complexity in the anterior nucleus of the thalamus fell before loss of consciousness and earlier than in prefrontal and temporal cortex, so local and inter-regional changes were staggered in time (Fan et al., 2026); and frontal slow-wave coupling changed in a discrete step while posterior alpha coupling declined smoothly (Tong et al., 2026). DPC predicts an abrupt change in the giant component of timing-coherent connections, not in every measure of brain activity; a gradual decline of that giant component through loss of responsiveness would refute it (prediction 1). Several measures in fact do not fall abruptly or at all. In stereo-EEG, inter-regional permutation mutual information increased or stayed stable after propofol-induced loss of consciousness (Liang et al., 2025); under sevoflurane the magnitude of EEG functional connectivity was larger during loss of consciousness than when conscious (Dong et al., 2023); during an ultra-slow propofol infusion the EEG spectral slope tracked response time continuously (Zhang et al., 2023); and in monkeys functional connectivity kept declining with isoflurane dose beyond loss of consciousness (Lv et al., 2016). Measures of the magnitude of coupling can rise when activity becomes hypersynchronous (Section 7.8), so they do not test DPC; the order in which regions change, which these studies report, is not predicted by DPC as stated.

What the phase model represents. In the phase-oscillator implementation (Appendix C), the phase θ_i of population i evolves as

$$\dot{\theta}_i = \omega_i + \frac{K}{s_i} \sum_j w_{ji}\, m_{ji}\, \sin\!\big(\theta_j - \theta_i - \alpha_{ji}\big) + \lambda_{\text{osc}} \big[R + A(c)(1 - R)\big] \sin(\psi - \theta_i) + \xi_i,$$

where ω_i is its natural frequency, K the coupling, s_i its total input strength, w_{ji} the weight of the connection from j, m_{ji} equal to 0 if that connection has failed and 1 otherwise, α_{ji} its phase error (conduction lag plus timing jitter), R and ψ the magnitude and phase of the network's mean rhythm, λ_{osc} the dimensionless strength of the reference in this model (distinct from λ of the percolation model, which is in milliseconds) and ξ_i noise; in the stimulus conditions, the sensory populations also receive a periodic drive. In the human connectome each node is a neural population, and its phase is the phase of that population's collective rhythm, the quantity reflected in local field potentials and EEG, not the state of any single neuron. In the *C. elegans* and *Drosophila* connectomes each node is a single neuron, whose phase stands for the timing of its activity; for them the model tests the wiring, not the cellular dynamics (Section 10, item 17). Models of that kind exist: a leaky integrate-and-fire model of the whole *Drosophila* brain, built from an earlier release of the same connectome and its predicted neurotransmitters, predicted neurons of the feeding circuit that were then confirmed experimentally, and recovered the known circuit for antennal grooming (Shiu et al., 2024). Reducing an oscillating neural system to a phase is standard for weakly coupled oscillators near a periodic regime (Kuramoto, 1984; Brown et al., 2004), and phase-oscillator networks on connectomes are an established phenomenological description of large-scale brain synchrony (Cabral et al., 2011; Breakspear, 2017). The model is therefore phenomenological: it represents timing and coupling between populations and deliberately leaves out spikes, synaptic kinetics, the generation of the rhythm itself (Section 10, item 18) and neuromodulation. Its results concern the network conditions for agreement, not the cellular mechanisms of any agent. Time in the model is dimensionless, its scale set by the coupling K. Timing errors and conduction delays, which are given in milliseconds, enter only as phase errors, $2\pi f\sigma$ at the reference frequency f, so timing affects the results only through its ratio to the cycle length. The convergence of these settings is summarised in Appendix C.4, and the *Drosophila* run settings in Appendix C.1.

In the phase-oscillator implementation (Appendix C), a timing error of σ milliseconds becomes a phase error with standard deviation $2\pi f\sigma$, where f is the frequency of the reference rhythm. Under P8 the reference slows as the agent takes effect,

$$f(c) = f_{\text{awake}} - \big(f_{\text{awake}} - f_{\text{fallback}}\big)\, h(c),$$

from $f_{\text{awake}} = 40$ Hz, within the gamma band of 30 to 90 Hz (Fries, 2015), towards $f_{\text{fallback}} = 10$ Hz, the centre of the 8 to 12 Hz frontal alpha seen at propofol-induced loss of consciousness (Purdon et al., 2013). Constant conduction delays are converted to phase at the same rate. Because the effective coupling of a connection with Gaussian timing error of phase standard deviation s is reduced by the factor $e^{-s^2/2}$, a slower reference preserves coupling for the same error in milliseconds. The same fallback rate is used for every agent, although ketamine does not produce frontal alpha (Akeju et al., 2016); this is a simplification. In the cross-species study the slower rate raised the loss concentration of the GABAergic-like agent in every seed on all three connectomes (test F1), but its distinctive prediction, that it protects more against timing faults than against transmission failure, failed in every seed (test F3: 0 of 20, 0 of 6 and 0 of 20). P8 is kept as a proposal whose specific mechanism the model does not support (Appendix D).

Because the glial loops change slowly compared with the integration cycle, each is represented by a parameter held fixed during a concentration sweep, a separation of timescales. Damaged myelin raises the agent-independent baseline jitter σ_b (3 ms; Section 4.7):

$$\sigma_b(\mu) = \sigma_b \left(1 + \gamma\,(1 - \mu)\right)$$

The microglial loop weakens the arousal drive as the giant component collapses:

$$A(c, S) = A(c)\left(1 - g\,(1 - S)\right)$$

so that it is weak in wakefulness, where S is close to but below 1, grows as the giant component shrinks and removes the share g of the arousal drive in the collapsed state; because it already acts during induction, removing it slightly delays the loss of quorum as well as hastening emergence (Appendix B.7); g = 0 represents microglial depletion. Silencing astrocytes removes their share of sleep down-states, $d \to d\,(1 - s_a)$. No published measurement fixes γ, g or s_a; they are free parameters, tested at γ = 1, $\mu \in$ {1, 0.75, 0.5}, $g \in$ {0, 0.25, 0.5} and s_a = 0.5. Because λ was calibrated without the microglial loop, the tests concern the direction of each effect, not its size. Five checks were specified before the multi-seed run (Appendix B): four consistency checks, which follow from the equations (M1, G1 to G3), and one test, A1, which the model could fail. A1 failed in every seed at every network size: silencing the astrocytic share of global down-states did not restore quorum in sleep (Appendix B). The test could not have passed, however: the share it leaves silent, a quarter of the populations, is the human share at which test O4 already loses quorum, so A1 says nothing about astrocytes, and the astrocytic route is untested rather than refuted. Because the loop is silent in wakefulness, it cannot reduce sensitivity at induction, as drug depletion of microglia does in mice (Cao et al., 2023), a result confounded by faster drug clearance (Cao et al., 2026); check G2 can therefore show only that depletion does not hasten induction, and the model does not reproduce the reduced sensitivity.

F.6 Clinical mapping in detail (Section 5)

Low concentrations of GABAergic agents can cause disinhibition, agitation or euphoria, accompanied by increased beta and gamma power (Brown et al., 2010; Purdon et al., 2013). A network-modelling account proposes that low-dose propofol, through an interaction between GABA-A currents and the slow potassium M-current, switches interneurons from synchrony to antisynchrony, which the authors propose as a network mechanism of the beta-band changes of paradoxical excitation (McCarthy et al., 2008).

DPC proposes a **complementary** contribution. As π falls towards π_c, remaining coherent paths carry a greater share of recurrent drive, so activity becomes more intense on fewer routes, a form of load concentration. This account makes a distinguishing prediction (Section 8, prediction 4): the magnitude of induction-phase excitation should scale with the sensory and recurrent load at the time of induction, which a purely interneuron-based account does not require.

What the simulations showed. In the Wilson-Cowan excitatory-inhibitory model on the human connectome (Appendix C.5), this load concentration did not appear. Hub activity rose by at least 10% before collapse in at most 1 of 10 seeds in each of nine settings (three global couplings, each with no, slow or fast homeostatic inhibitory plasticity), in both an exploratory run and a confirmatory run on new seeds, and the rise was not larger at high sensory load. The exploratory run did show, post hoc, a surge of synchrony to at least 1.5 times the awake level before collapse, almost only at the middle of the three global couplings tested (29 of 30 seeds at coupling 1, 18 of 30 at the weaker and 1 of 30 at the stronger coupling). The confirmatory run reproduced it at coupling 1 (8 to 10 of 10 seeds, depending on the plasticity) but not reliably at the weaker or stronger one (2 to 5 of 10); all three couplings are free parameters, so no coupling is privileged, and the surge was not larger at high sensory load (2 to 8 of 10 seeds, close to chance). The congestion account is therefore not supported by the model and rests on prediction 4 alone. The pre-collapse synchrony surge is a model behaviour to compare with the rise in beta and gamma power during induction, not evidence for it. A related model behaviour found post hoc, coherence fluctuations rising at least tenfold on the conscious side of the threshold (T1b, Appendix C.3), was then tested as T1c, fixed before the run, on seeds not used in the study that suggested it (they had been run once with the pre-correction code), and it replicated in 16 of 20 *C. elegans* seeds, 6 of 6 *Drosophila* seeds and 20 of 20 human seeds (Appendix C.7).

Revised account [H]. DPC therefore replaces congestion with two behaviours the models do show as the network approaches the percolation threshold: coherence fluctuates collectively before the network fragments for good (collective fluctuations, T1b to T1d), and in the Wilson-Cowan model synchrony rises just before the network breaks apart while hub activity falls to about 30% of its awake value (survivor lock-in, the post hoc synchrony surge above, reproduced on fresh seeds only at the middle coupling). On this account the excitation of induction would reflect an unstable, over-synchronised remnant of the network rather than extra load on its surviving routes, which links it to the hypersynchrony discussed in Section 7.8. The model reproduces the rising synchrony but not raised activity, so it does not yet account for the excitation itself. Two limits apply. Rising fluctuations near a threshold are expected of threshold models in general, so T1c shows that the model behaviour repeats, not that DPC is right; and the surge was not robust across couplings. The account is tested in data by prediction 2 (rising variance before loss of responsiveness).

At $\pi < \pi_c$ the giant component disappears. The thalamic reticular nucleus (TRN), a sheet of GABAergic neurons that gates thalamocortical transmission, is proposed here to amplify this transition [H], and thalamocortical models show how propofol produces coherent frontal alpha through thalamic circuits (Ching et al., 2010). In EEG from 17 surgical patients at graded propofol target concentrations, frontal slow-wave self-coupling (bicoherence) changed in a discrete step during loss of consciousness, whereas posterior alpha coupling declined smoothly with concentration (Tong et al., 2026). A stepwise change is what a threshold predicts, although it does not show that the threshold is a percolation transition. Under DPC, the resulting cortical activity is the network's default dynamics (P6): the slow oscillation with fragmented, phase-limited spiking for propofol (Lewis et al., 2012), and a different, gamma-rich pattern for ketamine (Akeju et al., 2016). The transition is not purely passive. In mice, isoflurane directly depolarises sleep-promoting neurons of the ventrolateral preoptic nucleus (VLPO), halothane and isoflurane increase their activity, and destroying these neurons produces an acute resistance to isoflurane that had reversed by 24 days (Moore et al., 2012); sedation by several agents runs through the brain's own sleep pathway (Nelson et al., 2002, 2003); and a hypothalamic population activated by diverse anaesthetics deepens anaesthesia when stimulated and shortens it when silenced optogenetically (Jiang-Xie et al., 2019). VLPO neurons are not required, however: rats still become anaesthetised with isoflurane after VLPO ablation, and the resulting sleep loss even increases their sensitivity (Eikermann et al., 2011). DPC therefore states this in a testable form, as postulate P7: anaesthetics also act on dedicated arousal-regulating circuits, which accelerates the loss of quorum without being necessary for it. In the model, removing the agent's action on arousal raises the loss concentration but does not prevent the loss (Appendix B, check C7); this follows from the model's equations and is a consistency check, not a test. DPC makes no claim that this shutdown evolved to protect neural tissue.

Under propofol sedation, temporal-lobe responses to speech persist at every level of sedation, while the extra responses that sentences containing ambiguous words evoke in frontal and posterior temporal cortex are lost at light sedation (Davis et al., 2007). Intracranial recordings agree: after propofol-induced loss of responsiveness, spiking and field-potential responses in primary auditory cortex persisted, responses in higher-order auditory regions were variable with spiking largely attenuated, and responses in association cortices were attenuated (Krom et al., 2020). Hippocampal neurons of anaesthetised patients still encode grammatical and semantic features of natural speech and predict features of upcoming words (Katlowitz et al., 2026). DPC interprets these findings as local processing inside small clusters of coherent populations that fall below quorum. Their content is compu-

ted, and local state can even change, but DPC proposes [H] that it is not globally broadcast or available for later report. Hippocampal neurons of anaesthetised patients showed plasticity: their discrimination of oddball tones grew over about 10 minutes (Katlowitz et al., 2026), so the claim is not that nothing is stored locally; it is that nothing reaches a global, reportable state.

The isolated forearm technique shows something different. A cuff keeps the muscle relaxant out of one forearm, and some patients move that hand on command during apparently adequate anaesthesia (Tunstall, 1977; Sanders et al., 2012): after tracheal intubation, 12 of 260 patients (4.6%) did so in a multicentre study (Sanders et al., 2017). A patient who follows a command is responsive and connected, so DPC reads these cases not as local processing but as quorum retained at a dose judged adequate [H], and predicts that a giant component above quorum would be found in such a patient.

The same distinction separates two kinds of memory. Implicit memory, such as priming, depends on plasticity within the populations that processed the input and does not require awareness; explicit (episodic) memory is memory for events that were consciously experienced. Under anaesthesia, patients can show implicit memory for words presented during surgery with no explicit recall, which declined as hypnosis deepened and was still present at bispectral index values of 40 to 60 (Lubke et al., 1999), on an EEG-derived scale of hypnotic depth that runs from 100 (awake) to 0; a meta-analysis pooling 44 studies found memory for specific material presented during anaesthesia, provided that patients were tested within 36 hours (Merikle and Daneman, 1996), although individual studies conflict, and in one study implicit memory appeared only for words played during surgical stimulation, although that group did not differ significantly from the group tested before stimulation (Deeprose et al., 2004). DPC proposes an account of this pattern [H]: local learning continues while populations process their inputs, but an episode can only be written for a state that reached quorum, and under anaesthesia no such state exists. On emergence, local traces persist, but no episode exists to recall. At equally sedating concentrations, propofol and midazolam impair memory strongly, thiopental mildly and fentanyl not at all (Veselis et al., 1997), so memory and responsiveness can be affected separately; we do not claim that loss of quorum is the *only* cause of anaesthetic amnesia.

As concentration falls, the system remains on the low-S branch until c_{on} is reached (Section 4.3), which can reproduce the hysteresis observed by Friedman et al. (2010) once λ is fitted to it (Section 4.3); this is a calibration, not an independent explanation. In the model, recovery requires that σ_{eff} falls below the value at which the high-S branch becomes the only solution. Emergence is an active process that involves arousal pathways, including orexinergic signalling (Kelz et al., 2008): methylphenidate actively induces emergence from isoflurane in rats (Solt et al., 2011), optogenetic activation of dopamine neurons of the ventral tegmental area restores righting during continuous isoflurane in mice (Taylor et al., 2016), and cholinergic stimulation of prefrontal, but not parietal, cortex restores wake-like behaviour despite continuous sevoflurane in rats (Pal et al., 2018); stimulating the central lateral thalamus in anaesthetised macaques restored arousal and wake-like neural processing (Redinbaugh et al., 2020), and thalamic stimulation reversed the electrophysiological features of propofol-induced unconsciousness in macaques, among them the fall in coherence above 4 Hz (Bastos et al., 2021). In these experiments consciousness returns while the agent's concentration, and so its action on connections, is unchanged. DPC accounts for this through P7 [H]: the arousal drive sustains part of the shared rhythm, so raising it at a fixed concentration lowers the error of a weakened reference, $\sigma_{\text{ref}} = \lambda(1 - S)(1 - A)$ in Section 4.3, more connections fall inside the window, and S can rise above q^* with $p(c)$ and $\sigma_0(c)$ unchanged. This follows from the model's equation; simulated, it holds in every seed at every network size, as a consistency check must (Appendix C.14). It gives a sharper prediction (prediction 15): between c_{on} and c_{off}, where the percolation model has two stable states, a brief arousing stimulus should restore quorum that outlasts the stimulus, whereas above c_{off} quorum should return only while the stimulus lasts. The phase model showed no robust inertia (Section 4.3), so this prediction rests on the percolation model alone. It passes through discrete metastable states (Hudson et al., 2014), and early return of consciousness is associated with subcortical and limbic activation before full frontal-parietal recovery (Långsjö et al., 2012). DPC proposes [H] that these intermediate states correspond to giant components of intermediate size. Recovery of responsiveness does not mean that every measure of coordination has recovered: in mice, the inter-regional correlation of cortical calcium activity stayed below its level before anaesthesia for at least 1.5 hours after behavioural recovery, while local activity recovered (Zhang et al., 2026). DPC requires only that the giant component of timing-coherent connections exceed q^* [H], so it predicts that this component, unlike the mean correlation, returns with responsiveness.

What the simulations showed. P5 was proposed to explain neural inertia: the shared reference is produced by the agreeing group, so a shrinking group weakens it, which shrinks the group further, and recovery has to rebuild the reference from a weak start. In the percolation model this produces inertia. Without reference feedback there is none (consistency check C2, every seed at every network size), and with it quorum is lost at a higher concentration than the one at which it returns: on random networks of up to 1,000,000 nodes, $c_{\text{off}} \approx 0.88$ and $c_{\text{on}} \approx 0.83$ times the EC50. With λ calibrated, the model reproduces every reported ratio from 1.07 to 1.63 at every arousal level tested (Appendix B.3), although the lowest, 1.07, comes from a study that reported no consistent hysteresis (Warnaby et al., 2017). Four limits qualify this result. First, reproducing a ratio is a calibration, and a feedback loop acting through a threshold is expected to have two stable states, so the result shows that the mechanism works, not that the brain uses it. Second, the model has a ceiling: the halothane ratio of 2.85 (Friedman et al., 2010) could not be reached, because before the mean ratio approached it some networks no longer recovered at zero concentration; the largest ratio reached rose with the arousal level, from 2.04 at level 0 to 2.40 at 0.7. Third, the phase-oscillator model, in which coupling is graded rather than all-or-nothing, showed no graded inertia at any feedback strength tried (Appendix C.2): each run's ratio of loss to recovery concentration stayed within run-to-run noise, 0.91 to 1.05, except one of four runs at the strongest feedback, for which no ratio could be computed (no loss or no recovery within the concentration range). A post hoc scan on the *C. elegans* connectome found no inertia beyond run-to-run noise at arousal level 0 or at feedback strengths between 0.6 and 0.75, and every run recovered (`simulator/results/diagnostics/`). At the reference strength, hysteresis averaged −0.0075 ± 0.029, 0.000 and 0.005 ± 0.032 EC50 on

the *C. elegans*, *Drosophila* and human connectomes, and it was strictly positive in only 2 of 20, 0 of 6 and 5 of 20 seeds (test S3, Appendix C.3). That network instead fluctuated collectively near the threshold, its fluctuations rising at least tenfold before collapse (T1b, then T1c on seeds not used in the study that suggested it). Fourth, other mechanisms also explain inertia, among them noise-driven switching between metastable states (Proekt and Hudson, 2018) and adaptive feedback in oscillator networks (Kim et al., 2018). We therefore keep P5 as a hypothesis that the models show to be workable under one condition, an all-or-nothing integration window (P2), not as an established explanation; prediction 5 tests it in data.

A lead noticed after the results [H]. The feedback that produces inertia can also trap the network: with the strongest calibrated feedback, some small networks stayed unconscious with no agent present (test O1: 4 of 20 seeds at 302 nodes and 2 of 20 at 1,000, none in larger networks), and the larger networks did not; whether the arousal drive of P7 prevents this was not tested at that feedback strength. Inertia and the risk of not waking may therefore be two sides of one feedback, with the arousal level setting how much inertia is safe. This was noticed after the results were seen and has not been tested. It would predict that losing arousal pathways makes trapping more frequent and emergence slower, as the loss of orexinergic neurons slows emergence but leaves induction unaffected (Kelz et al., 2008), and that a stronger arousal drive allows larger inertia without trapping.

Microglia are a candidate biological contribution to neural inertia (P9). Genetic loss of their P2Y12 receptor speeds recovery (Cao et al., 2023), the pattern expected if they help stabilise the unconscious state; the faster emergence after drug depletion of microglia (He et al., 2023) is confounded, because the drug also speeds the clearance of anaesthetics (Cao et al., 2026). We know of no study showing that microglia signal that recovery is safe; after isoflurane they boost neuronal activity once the anaesthetic has gone, by shielding inhibitory synapses (Haruwaka et al., 2024). Appendix B checks whether removing the microglial loop hastens emergence in the model (consistency check G1).

Emergence delirium (agitation and confusion in the immediate recovery period) is classified separately from postoperative delirium, which is defined as delirium occurring in hospital up to one week after the procedure or until discharge (Evered et al., 2018). Emergence delirium is described mostly in children, its incidence rose with the newer inhaled anaesthetics, and anxiety before surgery and pain after it are among its suspected contributors (Vlajkovic and Sindjelic, 2007). DPC proposes that emergence delirium partly reflects **asynchronous recovery** of subsystems: limbic and subcortical components regain coherence before frontal-parietal components (consistent with Långsjö et al., 2012), producing a transient state in which affective processing is integrated but executive and contextual processing is not. We do not adopt the metaphor of "corrupted memory" because we know of no evidence that anaesthesia corrupts stored information. As stated, this account does not distinguish emergence delirium from agitation driven by pain, and it is not tested here.

Natural sleep also abolishes consciousness, without any drug, and is neurophysiologically distinct from anaesthesia (Akeju and Brown, 2017). Under P7, non-rapid-eye-movement (NREM) sleep combines withdrawal of the arousal drive with slow-oscillation down-states. A down-state is a period in which a whole local population falls silent together (Vyazovskiy et al., 2009), and slow waves are often regional, the widespread ones being typical of early NREM sleep (Nir et al., 2011). The model therefore treats a down-state as silencing whole populations and asks what share of populations must be silent at the same moment for the sleeping network to lose quorum (Appendix B). In the model that share is about 15% (0.152 in the largest networks) at the reference parameters; it moves with the arousal level, the integration window and the mean degree (Appendix B.6). The measured share of time spent in down-states gives a scale for comparison: 0.095 in rat NREM sleep (silent periods of at least 50 ms, averaging 85.8 ms, between active periods averaging 815.5 ms; Vyazovskiy et al., 2009; shorter silences are not counted, so this underestimates the share), and an upper estimate of about 0.25 for human NREM sleep (down-states of about 400 ms, the duration Cash et al., 2009, report for both K-complexes and slow-oscillation down-states, at 37.3 slow waves per minute at each recording site, Nir et al., 2011). In human sleep, a slow wave involves on average 27.1% of the monitored regions, and 85% of slow waves reach fewer than half of the recording sites (Nir et al., 2011). A slow wave detected at more than half of the recording sites is classed as global (Nir et al., 2011), so a global down-state silences a larger share than either average. The rat share therefore abolishes quorum in no seed at any network size (test O3, which asks whether it does), whereas the human upper estimate abolishes it in every seed (test O4); the sleep result rests on that upper estimate. Appendix B also reports an alternative account in which down-states make individual connections fail. It fits the finding that, after transcranial magnetic stimulation, activity in wakefulness travels on to linked cortical regions some centimetres distant, whereas in NREM sleep the response fades quickly and stays confined to the site of stimulation (Massimini et al., 2005).

Why sleep should pause the integration process is a further hypothesis of the authors [H]. Global agreement is proposed to be the most costly process the brain runs, so that pausing it frees resources for maintenance that does not need broadcast. In a small study, cerebral glucose use fell by about 44% in slow-wave sleep and returned to waking levels in rapid-eye-movement (REM) sleep (Maquet et al., 1990), the stage in which vivid dreaming is most often reported; and the synaptic homeostasis hypothesis proposes that the brain disconnects from the environment during sleep so that spontaneous activity can renormalise synaptic strength and restore cellular homeostasis (Tononi and Cirelli, 2014). Memory consolidation during slow-wave sleep coordinates hippocampus and neocortex (Diekelmann and Born, 2010). On this view, waking from sleep resumes a process whose machinery stayed intact, whereas recovery from anaesthesia usually waits for the agent to clear, although arousal pathways can drive emergence during continuous anaesthetic (Section 5.4), and shows neural inertia. Resumption is not instantaneous: performance remains below par for a while after being woken suddenly, more so from slow-wave sleep, and unless the sleeper was severely sleep-deprived beforehand this period seldom lasts beyond 30 minutes (sleep inertia; Tassi and Muzet, 2000). The view predicts that, at matched depth of slow-wave activity, responsiveness returns faster and with less hysteresis on arousal from sleep than on emergence from anaesthesia (Section 8, prediction 14).

Astrocytes are a candidate source of the down-states (P9): their activity precedes and can trigger the slow oscillation (Poskanzer and Yuste, 2016), astrocytic calcium falls during sleep, and disrupting an astrocytic calcium pathway increases microarousals (Bojarskaite et al., 2020). Appendix B tests whether, with global down-states, removing the astrocyte-dependent share of them restores quorum, that is, wakes the sleeper (test A1). The test was re-specified when the down-state model changed, before its multi-seed run, and it failed in every seed at every network size: removing the astrocytic share of global down-states did not restore quorum. That outcome was set by the test's design rather than by the astrocytes: the quarter of the populations left silent is the human share at which quorum is lost anyway (test O4), so the astrocytic route to down-states remains untested in the model.

Dreaming poses the opposite question: why sleep, with low arousal, is sometimes accompanied by vivid experience. Dreams are reported from both REM and NREM sleep and are associated with local reductions of low-frequency activity in posterior cortex (Siclari et al., 2017). DPC proposes that during dreaming the populations reach agreement on internally generated activity, such as reactivated memories, while sensory input is gated at the thalamus (McCormick and Bal, 1994; Steriade et al., 1993b): the brain, disconnected from the environment, generates experience by itself (Nir and Tononi, 2010). This is disconnected consciousness in the sense of Section 1.3. Sleep supports the consolidation of memories, which are reactivated and redistributed from hippocampus to neocortex during slow-wave sleep (Diekelmann and Born, 2010; Rasch and Born, 2013); DPC does not explain that function, but places the experience of dreaming in the internal agreement that may accompany it. Whether sleep also increases the clearance of metabolites from the brain is disputed: an increase was reported during sleep and ketamine-xylazine anaesthesia (Xie et al., 2013), and, measured with a different method, a reduction during sleep and anaesthesia (Miao et al., 2024).

The same account applies to ketamine. On emerging from ketamine anaesthesia, volunteers described lengthy and vivid dreams with no connection to their surroundings, and cortical complexity stayed close to that of wakefulness (Sarasso et al., 2015). DPC first proposed that ketamine disrupts the coupling of cortex to its sensory input more than the agreement among internally generated activity, so that sensory broadcast fails while internal agreement survives (the ketamine pathway hypothesis, test K1 in Appendix C). As specified, the phase model did not support it. With the agent's timing error confined to connections leaving the sensory populations, internal agreement held in every seed (on average 96% to 100% of its awake value at the highest concentration on the three connectomes), but broadcast fell to half its awake value or less in only 12 of 20 *C. elegans*, 2 of 6 *Drosophila* and 10 of 20 human seeds (test K1), and its median to about half (0.47, 0.52 and 0.55 of the awake value). A post hoc check later found that much of this remaining broadcast is a floor of chance locking rather than propagation: with that floor subtracted, the criterion holds in 16 of 20 *C. elegans* seeds (20 of 20 with a longer window), a post hoc reading (limitation 31). A test of it, fixed before its run on fresh seeds with a window of eight stimulus cycles (K1c, Appendix C.11), then passed on *C. elegans*: quorum held in all 20 seeds, at 94% of its awake value or more, while the broadcast above the floor fell to a median of 13% of its awake value and at most 31%. On the human connectome the awake broadcast reached twice the floor in only 2 of 20 seeds, so the test could not be evaluated there. Before the code corrections the stimulus had spread almost unimpeded (2 of 20, 0 of 6 and 0 of 20 seeds); a post hoc diagnostic traces the change in *C. elegans* mainly to the corrected measurement window and the removal of muscles from the graph (Appendix D). That first result prompted the revision below; the corrected K1 held in about half the seeds, and K1c, described above, then supported the pathway account on *C. elegans*. In macaque V1, blocking NMDA receptors lowered figure-ground modulation, an index of recurrent processing that relies on signals sent back from higher visual areas, whereas blocking AMPA-type glutamate receptors cut feedforward-driven activity without altering that index (Self et al., 2012). Under ketamine, frontal-to-parietal (feedback) connectivity fell after loss of consciousness while parietal-to-frontal (feedforward) connectivity was preserved, as it is under propofol and sevoflurane (U. Lee et al., 2013). The revised proposal was that ketamine's timing faults fall mainly on feedback connections, so that loops cannot close while signals still travel forward. Because propofol and sevoflurane also inhibit feedback connectivity, this alone would not explain why complexity and dreaming are preserved under ketamine; that remains open (Section 7.2). We fixed a test of the revision (K2) and its pass rule before running it, together with the expectation that it might fail in this model (Appendix C.7). It failed. With the agent's timing error confined to feedback connections, quorum was lost within the concentration range in only 1 of 20 *C. elegans* seeds, at 2.35 EC50, and in none of 6 *Drosophila* seeds, and the same error on as many randomly chosen connections never abolished it. Feedback faults did lower agreement more than random ones: at the highest concentration, coherence fell to 75% of its awake value against 91% in *C. elegans*, and to 87% against 91% in *Drosophila*. That comparison was not pre-specified and is exploratory. In the model, then, ketamine's timing faults abolish quorum reliably only when they act across the network (test T2b), and placing them on feedback connections deepens the loss of agreement without causing it. We keep the feedback proposal in this weaker form, as a contributing factor whose test lies in recordings of directed connectivity (U. Lee et al., 2013), and Appendix D records both revisions.

F.7 Simulation results in detail (Section 6)

Appendices B and C report every test; this section summarises them. The simulations ask whether the models behave as the hypothesis says, not whether brains do (Section 4.7). The percolation model is run on random networks of 302 to 1,000,000 nodes with 20 seeds per size in the validation study (up to 100,000 nodes in the glial study, and at 10,000 nodes with 10 seeds in the sensitivity analysis); the phase-oscillator model on the *C. elegans* (20 seeds), *Drosophila* (6 seeds) and human (20 seeds) connectomes; and the Wilson-Cowan model, for the congestion and synchrony-surge tests, on the human connectome (10 seeds per setting). A count such as 18 of 20 is the number of seeds in which a test held. A percolation-model test counts as held only if it holds in every seed at every network size, a pre-specified test passes if it holds in at least 80% of seeds (C.7, C.10, C.11) or if the lower bound of its Wilson 95% interval lies above one half (C.12 to C.14; Appendix A.6), and the other phase-model and Wilson-Cowan tests are reported as counts, without a pass rule. Tests fixed before their run are distinguished from analyses found afterwards (post hoc), which were then replicated or are labelled as such, and from consistency checks, which follow from the model's equations and

show only that the code implements them. Figures 1 to 4 (Section 6) and 5 and 6 (Appendix C.8) plot the main results: the transition and inertia of the percolation model (Figures 1 and 2), coherence on the three connectomes (Figure 3), the broadcast of a stimulus (Figure 4), the block of propagation in plant tissue, which occurred in every run with 17 to 35% of junctions closed on average (Figure 5), and the congestion test (Figure 6).

Result	Status	Evidence
Agents that cause transmission failure and agents that cause timing jitter both abolish quorum	Transmission failure: built in at the doses chosen. Timing variability: on the human connectome, fails or passes depending on how a connection inferred from diffusion MRI is counted (Appendix C.13)	These tests could not fail with the maximum effects chosen (limitation 25), so they show that the mechanism operates, not that it is right. Percolation: every seed at every size (C1). Phase model, where quorum is read as global coherence at least half its awake value rather than as the giant component of P3 (Section 4.7): 20 of 20, 6 of 6 and 20 of 20 seeds for each agent class (T2a, T2b); also with measured conduction delays (D1), with inhibitory connections signed (SG1, SG2), with fixed timing offsets (S4) and on rewired connectomes with the same degrees (N1). With a natural-frequency spread of 0.25 instead of 0.5, the GABAergic-like agent abolished it in no seed of 5 (phase-model sensitivity, Appendix B.6). Read with DPC's own window and quorum (Appendix C.12, fresh seeds, fixed before the run), transmission failure abolishes quorum on all three connectomes under every condition, and the oscillator dynamics bring the loss forward relative to static percolation; timing variability alone abolishes it on *C. elegans* and *Drosophila* (at the concentration static percolation on its draws predicts) but on the human connectome in no seed of 20, under every condition, although global coherence falls below half its awake value at 0.69 times the EC50. With one timing draw per connection, or with both directions required, timing variability alone abolishes quorum there in 20 of 20 seeds, at 1.15 and 0.60 times the EC50; the human verdict depends on how a human connection is counted (Appendix C.13)
The quorum threshold of a connectome is predictable from its wiring	Approximately (measured threshold 2.0 to 5.5% above the generating-function estimate, 10 draws; within sampling error for *C. elegans* and human, not for *Drosophila*); with direction kept, the directed generating-function estimate lies at 0.91, 0.96 and 0.96 of the measured threshold (Appendix C.14)	The measured threshold lies 2.0%, 5.5% and 3.4% above the generating-function estimate on the *C. elegans*, *Drosophila* and human connectomes (Appendix B.5); the gap exceeds seed-to-seed variation only for *Drosophila* (2.0% and 3.4% are 0.7 and 1.8 standard errors). Tree-based estimates of this kind are known to be accurate on clustered networks as long as their mean shortest-path length is small (Melnik et al., 2011), so this reproduces a known property rather than a new one. With direction kept, quorum needs 1.8 and 3.4 times as many timing-coherent connections on the *C. elegans* and *Drosophila* wiring, and the directed generating-function estimate lies at 0.91, 0.96 and 0.96 of the measured directed threshold (Appendix C.14); the human connectome comes from diffusion imaging, which gives no direction (each of its 17,865 connections is recorded in both directions), so its directed value only splits each connection into two independent links and is not reported as a result. On degree-preserving rewired copies, DPC quorum is lost at the same concentration within one step in almost every seed (C.12: WN and GN tests), so the threshold depends on the degree distribution, not on the particular wiring
Reference feedback sharpens the transition	Consistency check; failed the every-seed rule by one seed	19 of 20 seeds at 302 nodes, every seed at every larger size (C4), and every seed at every size on fresh seeds (Appendix C.14); positive feedback through a threshold is expected to sharpen it
Hub damage lowers the loss concentration more than random damage	Consistency check, sensitive to calibration	Every seed at every network size with λ calibrated to 1.08 (C5), but 9 of 20 seeds at 10,000 nodes with the earlier target of 1.11 (Appendix D) and 19 of 20 at 302 nodes on fresh seeds (Appendix C.14); expected for networks with heavy-tailed degree distributions (Callaway et al., 2000; Albert et al., 2000)
Fluctuations rise before the collapse	Held against independent phases, a low bar (limitation 33)	The pre-specified surge at the collapse failed (T1: 7 of 20, 0 of 6, 5 of 20). A rise of at least tenfold on the conscious side of the threshold, found post hoc (T1b: 18 of 20, 6 of 6, 20 of 20), passed as a pre-specified test on seeds not used in the study that suggested it (T1c: 16 of 20, 6 of 6, 20 of 20). But the ratio is measured against the awake value, which is very small when awake coherence is close to 1: the susceptibility at loss is similar across conditions (median 0.41 to 0.78 on the human connectome), and the ratio fails where awake coherence is low, as with measured conduction delays (awake coherence 0.41; 0 of 20 seeds) or without reference feedback (3 of 20, 1 of 6). The result may therefore reflect the awake baseline rather than a rise before collapse; a comparison at matched coherence is in Appendix C.12. The corrected test, which compares the fluctuations with those of independent phases at the same mean coherence, passed on all three connectomes (C.12: T1d, 17 of 20, 6 of 6 and 16 of 20 seeds), as did slower fluctuations at the threshold (T1e, every seed); growth with network size held only between the human and fly connectomes (T1f)
Reduced arousal delays emergence more than it shifts induction	Consistency check	Every seed at every size (C6); it follows from the model's equation, since the reference error $\lambda(1-S)(1-A(c))$ is close to zero on the conscious branch

Result	Status	Evidence
Sleep loses quorum at the measured human share of down-states but not at the rat share	Mixed	Quorum is lost when about 15% of populations are silent at once (0.152 in the largest networks); in the sensitivity analysis the threshold fell between the rat and human shares in 6 of the 11 non-reference settings with an awake quorum (both quorum fractions, arousal level 0.3, λ for the ratios 1.07 and 1.11, baseline jitter 1 ms; three of the 14 settings that can affect sleep, W = 1 ms, σ_b = 10 ms and mean degree 4, have no awake quorum), and not at the other extreme of the integration window or mean degree, at arousal 0 or 0.7 or at the strongest feedback; the rat share (0.095) abolishes quorum in no seed (O3), which implies that rats keep quorum in NREM sleep (limitation 27), and the human estimates (0.25 and 0.271) in every seed (O4, O5); the model equates a share of time spent in down-states with a share of populations silent at once; deep anaesthesia has no conscious state to return to without arousal drive (E3)
Glial extension (P9)	Not tested	Damaged myelin lowers the loss concentration (M1) and the microglial loop shapes emergence (G1 to G3) in every seed, but all four are consistency checks that follow from the equations; the one test, whether silencing astrocytes restores quorum in sleep, failed in every seed (A1) but could not have passed, since the share of populations it leaves silent already loses quorum (O4)
Learned arrival times make constant delays harmless (P4b)	Failed	Pre-specified (Appendix C.10): with learning, coherence fell even at zero concentration and, without conduction delays, quorum was lost on average below 0.13 times the EC50 with every agent (L1, L3 and L4: no seed); with conduction delays, learning first raised awake coherence (0.41 to 0.62, every seed); L2 held in every seed only because quorum was lost earlier, not later
Neural inertia from reference feedback (P5)	Percolation model only	Reproduced in the percolation model once λ is calibrated, up to a ceiling below the largest measured ratio, on the calibration networks; on fresh seeds the ratio calibrated to 1.08 is 1.071 at 10,000 nodes and 1.066 at 1,000,000 (Appendix C.14); absent from the phase model (Section 5.4); whether neural inertia exists in humans is contested (Kuizenga et al., 2018). With DPC's own readout, the phase model shows no inertia at any feedback up to 1.5 times its locking threshold (C.12: I1, I2). The percolation model on the connectome graphs reproduces it on the *C. elegans* and human graphs once calibrated (PI1, PI2), within the measured range on held-out human seeds (PI4: 17 of 20) but not *C. elegans* seeds (13 of 20), and on the fly graph not at all
The sensory area keeps processing while broadcast collapses (T3)	Held in most seeds of two connectomes	17 of 20 *C. elegans* and 16 of 20 human seeds, but 0 of 6 *Drosophila* seeds (see below); it also holds on rewired connectomes (N2: 19 of 20 and 20 of 20), so, within the limits of the broadcast measure, it reflects the degree distribution rather than the particular wiring. The broadcast measure has a floor, and the human count cannot be read as propagation (limitation 31)
A slower reference rate protects more against timing faults (P8)	Failed	The slower rate raised the loss concentration (F1, every seed) but its distinctive prediction failed in every seed (F3; in the fly the two gains were equal). In the Wilson-Cowan model, burstiness rose after the synchrony surge mainly at the middle coupling in the exploratory run (B1: 23 of 30 seeds, against 4 and 3 of 30), less clearly in the confirmatory run (18 of 30, against 5 and 15 of 30), and the gaps between bursts (B2) could rarely be measured in 1 s windows
Congestion causes the excitation of induction	Failed	Hub activity rose by at least 10% in at most 1 of 10 seeds in every setting (W1 to W4), in an exploratory and a confirmatory run; replaced by survivor lock-in with collective fluctuations (Section 5.1)
Ketamine leaves an internal agreement cut off from the senses	Shown on *C. elegans* only, with its control	Sensory-pathway faults kept agreement but halved broadcast in only about half the seeds (K1: 12 of 20, 2 of 6, 10 of 20); local islands of coherence did not form (S1, S2: no seed); timing faults on feedback connections alone rarely abolished quorum (K2, pre-specified: 1 of 20, 0 of 6). With the floor of chance locking removed and a longer window, a test fixed in advance passed on *C. elegans* (K1c: 20 of 20 seeds) and could not be evaluated on the human connectome, where 2 of 20 seeds had an awake broadcast at least twice the floor (Appendix C.11). Read with DPC's own quorum, the disconnection holds in every *C. elegans* seed (C.12: K1d) and the same timing error on as many random connections does not produce it in 18 of 20 seeds (K1e), so it is specific to the sensory pathways; the human connectome again could not be evaluated

Results moved from the Section 6 table. The rows below were in the Section 6 table before 22 October 2026, with the same columns; rows whose counts give network sizes (C4 to C7, O1 and O2, the glial checks) report the percolation model on random networks, and the plant-tissue row a tissue model, so their counts sit in the first count column, which in those rows does not mean *C. elegans*.

Claim	Kind	Pre-specified?	Could it fail?	*C. elegans*	*Drosophila*	Human	Appendix
Oscillators bring the transmission-failure loss forward relative to static percolation (T0a-D)	Test	Yes	Yes	20/20; 1.52 against 2.06 EC50	6/6; 1.40 against 1.90	20/20; 1.82 against no loss	C.12
Oscillators bring the timing loss forward (T0b-D)	Test	Yes	Yes	9/20, fails	0/6, fails	0/20, fails	C.12
Coherence is lost before quorum under timing variability	Measurement	Yes	n/a	0.650 against 0.903 EC50	0.600 against 1.000	0.700 against 1.150	C.13

Claim	Kind	Pre-specified?	Could it fail?	*C. elegans*	*Drosophila*	Human	Appendix
Losses sit at a common share of counting connections (SP1, SP1-1)	Consistency check, reclassified after the run	Yes, as tests	No	20/20	6/6	20/20 (SP1-1)	C.12, C.13
The coherence loss is synchronisation-like (SP2)	Test	Yes	Yes	20/20	6/6	20/20	C.12
The quorum threshold follows from the degree distribution	Consistency check	No	Yes	Approximately: 2.0% above the estimate (10 draws), within sampling error	5.5% above, beyond it	3.4% above, within it	B.5
The quorum threshold with direction kept	Measurement	No	n/a	1.8 times as many connections	3.4 times	No direction measured	B.5
The directed threshold follows from the in- and out-degree distribution	Estimate, no verdict	Yes	n/a	Estimate 0.91 of the measured threshold	0.96	0.96	C.14
The wiring matters beyond the degree distribution (WN, GN)	Test	Yes	Yes	2/20, fails	Not run	0/20 to 3/20, fails	C.12
Fluctuations surge at the collapse (T1)	Test	Fixed before the reported run, after earlier runs (Appendix D)	Yes	7/20, fails	0/6	5/20	C.3
Fluctuations rise tenfold against awake before the threshold (T1b; T1c)	Post hoc; T1c a test	T1c, on seeds not used in the study that suggested it	Yes, but inflated by a near-zero awake value	T1b 18/20; T1c 16/20	6/6; 6/6	20/20; 20/20	C.3, C.7
The excess grows with connectome size (T1f; across connectomes, not a finite-size test)	Test	Yes	Yes	Not run	6/6 (against human)	13/20, fails (against *C. elegans*)	C.12
Feedback sharpens the transition (C4)	Consistency check	Fixed before the reported run, after earlier runs (Appendix D); rerun on fresh seeds	Barely	19/20 at 302 nodes, all at larger sizes; every seed at every size on fresh seeds			B.4, C.14
Hub damage lowers the loss concentration more than random damage (C5)	Consistency check	Fixed before the reported run, after earlier runs (Appendix D); rerun on fresh seeds	Yes	Every seed with λ calibrated to 1.08; 9 of 20 at 10,000 nodes with 1.11; on fresh seeds 19/20 at 302 nodes, failing the every-seed rule, and every seed at larger sizes			B.4, C.14, D
Reduced arousal delays emergence more than induction (C6); arousal action not required (C7)	Consistency checks	Fixed before the reported run, after earlier runs (Appendix D); rerun on fresh seeds	No	Every seed, also on fresh seeds			B.4, C.14
Neural inertia in the percolation model on the connectome graphs (PI1, PI2; held-out PI4)	Test	Yes	Yes	20/20; PI4 13/20, fails	No inertia	20/20; PI4 17/20	C.12

Claim	Kind	Pre-specified?	Could it fail?	*C. elegans*	*Drosophila*	Human	Appendix
Neural inertia, phase model (S3; I1, I2)	Test	I1, I2	Yes	S3 2/20; I1 0/10, I2 0/20	S3 0/6	S3 5/20; I1 0/10, I2 0/20	C.2, C.3, C.12
Strong feedback traps small networks unconscious at zero concentration (O1, O2)	Test, no verdict	Fixed before the reported run, after earlier runs (Appendix D)	Yes	4/20 at 302 nodes, 2/20 at 1,000 (O2: 4/20 at both), none larger; on fresh seeds 2/20 and 1/20 (O2: 6/20 and 3/20)			B.4, C.14
Glial checks (M1, G1 to G3); astrocytes restore quorum in sleep (A1)	Consistency checks; A1 construction check	Yes	No; A1 could not pass	Every seed; A1 0/20			B.7
Learned arrival times make constant delays harmless (L1 to L4)	Test	Yes	Yes	L1, L4 0/20; L2 20/20, only because quorum was lost earlier	Not run	L1, L3, L4 0/20; L2 20/20	C.10
Constant offsets are harmless without learning (S4; D1)	Post hoc comparison	No	Yes	Earlier loss than jitter in 20/20	Same step in 6/6	Earlier in 18/20; delays: 1.16 rather than 1.57 EC50	C.3, F.4
The stimulated area keeps processing while broadcast collapses (T3; N2 rewired)	Test	Yes	Yes, but the measure has a floor	17/20; N2 19/20	0/6 (convergence caveat)	16/20, not interpretable; N2 20/20	C.3, C.4
The same faults on feedback connections abolish quorum (K2)	Test	Yes	Yes	1/20, fails	0/6, fails	Not run	C.7
Local islands of coherence outlast global coherence (S1, S2)	Test	Yes	Yes	0/20	0/6	0/20	C.3
A slower reference protects more against timing faults (F3); it delays loss (F1)	Test	Yes	Yes	F3 0/20, fails; F1 20/20	0/6; 6/6	0/20; 20/20	C.3
Congestion raises hub activity before collapse (W1 to W4)	Test	Yes	Yes			At most 1 of 10 seeds per setting, in both runs	C.5, C.6
Synchrony surges before collapse (X1)	Post hoc; replication without a pass rule	No	Yes			29/30 at the middle coupling, 18/30 and 1/30 at the others; replication 8 to 10 of 10 at the middle, 2 to 5 of 10 otherwise	C.5, C.6
Propagation in plant tissue fails at a percolation-like block	Test	No	Yes	Every run, at 17 to 35% of junctions closed			C.3

The fly. In *Drosophila*, T3 held in no seed: at the reference settings the broadcast left after quorum was lost stayed at 0.75 to 1.21 of its awake value. The fly's awake broadcast is low (about 0.06) and did not converge (Appendix C.4). With twice the simulated time, the broadcast after loss fell to 0.30 and 0.43 of the awake value in the two seeds checked, which meets the post hoc criterion T3b but not T3, because the sensory neurons' entrainment also fell slightly (from 0.837 to 0.826 and 0.836). By the reading rule fixed before the convergence study, the fly result is reported with this caveat. Its awake broadcast may itself be mostly the floor of chance locking described in limitation 31; the fly was not included in that check.

Ketamine, a gap for DPC's models. We expected the hypothesis to explain why consciousness under ketamine is disconnected rather than lost, and it does so only in part. In the phase model, ketamine-like timing faults abolish quorum reliably only when they act across the whole network (T2b). Confined to the sensory pathways (K1) or to feedback connections (K2), they left quorum intact in almost every seed, and when they grew with connection length quorum was lost but no local islands of agreement, which dreaming would require, outlasted it (S2; Appendix C.3). Post hoc, such islands did form when the shared reference was removed: under the GABAergic-like agent, local coherence outlasted global coherence in 18 of 20 *C. elegans*, 5 of 6 *Drosophila* and 18 of 20 human runs without reference feedback, against none of the same runs with it, so in the model the reference suppresses them; with a sensory stimulus applied, islands also formed with the reference present (10 of 20 *C. elegans* and 6 of 6 *Drosophila*

runs). Confined to the sensory pathways, the same faults did cut the senses off from the surviving agreement once the broadcast measure was corrected, on *C. elegans* (K1c), and with quorum read as P3 states it this held in every seed while the same faults on as many random connections did not produce it (K1d, K1e; Appendix C.12). Read that way, timing faults across the whole network abolished quorum on the human connectome in every seed under two of three rules for counting a connection, one timing draw per connection (T2b-1) or both directions required, but in no seed under the rule used first, which counts each connection inferred from diffusion MRI as two independently timed connections, either of which suffices (T2b-D); the other two rules were fixed after that failure had been seen. Whether, in the model of a human brain, ketamine's assigned fault type causes unconsciousness by itself therefore depends on how a human connection is counted (Appendix C.13). The hypothesis therefore accounts for loss of consciousness under ketamine on two invertebrate connectomes and, under two of three counting rules, on the human one, and, in the model of one nervous system, for the disconnection of the senses, but not for the preserved complexity and dreams that distinguish it from propofol (Sarasso et al., 2015); Section 5.6 gives the details.

F.8 Objections, predictions and protocol in detail (Sections 7 to 9)

Response. DPC is explicitly mechanism-agnostic at the molecular level (P4). DPC proposes [H] that GABAergic agents act chiefly by increasing transmission failure $p(c)$, which corresponds to omission faults, and that ketamine acts chiefly by increasing timing jitter $\sigma_0(c)$, which corresponds to timing faults; the spurious activity ketamine also causes is not modelled. Ketamine does affect timing in cortex: in two macaques, subanaesthetic doses dilated the estimation of elapsed time and flattened anticipatory ramping activity in the lateral intraparietal area (LIP; Waters et al., 2026, a preprint). That is interval timing within one area, not the timing of signals between areas that DPC requires, so it is consistent with the proposal but does not test it. Both kinds of fault reduce $\pi(c)$ and can therefore abolish the giant component. Ketamine's preserved PCI and dream reports (Sarasso et al., 2015) indicate disconnected consciousness rather than unconsciousness. Of DPC's two accounts of it, sensory-pathway faults failed their first test (K1) but passed a test fixed in advance on *C. elegans* once a floor in the broadcast measure had been removed (K1c), and faults concentrated on feedback connections were not supported (K2; Section 5.6). We state this as an open problem rather than a solved case.

Response. The molecular and synaptic effects of anaesthetics are evolutionarily conserved: presynaptic syntaxin 1A in flies (Zalucki et al., 2015) and mitochondrial complex I in nematodes (Kayser et al., 1999) are examples. Conservation at the molecular level is expected under P4. At the systems level, the observed effect in both species is a loss of coordination rather than of activity: correlations between *C. elegans* head neurons collapse while each neuron remains active (Awal et al., 2020), and isoflurane fragments the integrated structure of fly brain activity (Leung et al., 2021). Appendix C tests the model on the connectomes of both species, with 20 seeds for *C. elegans* and, because each run takes 30 to 65 minutes, 6 for *Drosophila* in both the cross-species study and the pre-specified tests (Appendix C.7), so the *Drosophila* confidence intervals are wider. The percolation argument applies to any network of coupled units, but the threshold depends on topology, and the behavioural endpoints differ (immobility in nematodes is not evidence of unconsciousness). DPC therefore makes a **restricted** claim: in any nervous system whose coordinated behaviour depends on timing-coherent communication across units, anaesthetics should abolish that coordination at a percolation-like threshold. We do not claim that the human thalamocortical architecture is scale-invariant, and we draw no conclusion about consciousness in invertebrates.

Objection. Anaesthesia often increases large-scale synchrony rather than reducing it: propofol produces strong, spatially coordinated frontal alpha (Cimenser et al., 2011; Purdon et al., 2013), and the slow oscillation is shared across cortex. A hypothesis in which unconsciousness follows a loss of coherence appears to predict the opposite.

Response. DPC distinguishes the shared rhythm from the timing of the content carried relative to it. The rhythm is the phase reference of P5; integration requires that signals between populations arrive within the window around it (P2). The two can move in opposite directions. Recordings during propofol induction show this dissociation. As consciousness is lost, frontal cortex enters a hypersynchronous alpha state; primary somatosensory cortex still produces its early response to median nerve stimulation, but the later responses, which draw in temporal and parietal regions, disappear, and the late response is smaller on single trials in which ongoing frontal alpha has greater power and greater coherence over long distances (Supp et al., 2011). At the onset of propofol-induced unconsciousness, spiking is confined to brief windows of the slow oscillation that are not aligned between distant cortical sites (Lewis et al., 2012), and global coherence shows no coordinated delta activity (Cimenser et al., 2011). In two macaques, anaesthetic doses of ketamine and dexmedetomidine increased phase locking in prefrontal cortex within and across hemispheres, yet neighbouring prefrontal subregions of the same hemisphere became less aligned in phase, which the authors relate to large travelling waves, while homologous areas of the two hemispheres became more aligned (Bardon et al., 2025): more synchrony, organised differently from waking. In mouse cortex, different anaesthetics synchronise the spontaneous activity of layer 5 pyramidal neurons, and of no other cortical cell type, across large distances, and the change coincides with loss and recovery of consciousness (Bharioke et al., 2022). A coherent rhythm with fragmented content is also what P8 describes: a slower reference with activity batched into synchronised bursts, although P8's distinctive prediction failed in the model (test F3, Section 6). The measure that DPC requires is therefore the timing coherence of content between populations, such as stimulus-locked interareal phase locking, and not the coherence of ongoing rhythms (prediction 1). The models in this paper do not generate the rhythms themselves (Section 10, item 18).

PCI (Casali et al., 2013) is an established index, not an invention of this paper. Cortex is perturbed with transcranial magnetic stimulation (TMS). Significant evoked source activity is binarised into a matrix of L elements (sources × time samples), and the Lempel-Ziv complexity c_L of that matrix (Lempel and Ziv, 1976) is normalised by its source entropy:

$$\mathrm{PCI} = \frac{c_L \cdot \log_2 L}{L \cdot H(L)}, \qquad H(L) = -p_1 \log_2 p_1 - (1-p_1)\log_2(1-p_1)$$

where p_1 is the fraction of significant (value 1) elements. PCI is high in conscious states and low in unconsciousness induced by propofol or xenon (Casali et al., 2013; Sarasso et al., 2015). PCI is estimated from many repeated stimuli at a stable state (Casali et al., 2013), so it is measured at concentration plateaus, not along a ramp; the ramp measures (S, its variance and autocorrelation) and the plateau measure (PCI) come from separate epochs. DPC predicts that PCI falls with S across plateaus for GABAergic agents and remains relatively preserved under ketamine (Section 7.2). PCI is supporting evidence; it does not test DPC specifically.

The prediction table before the restructure, with its fuller measurement and falsifier details; the changes of Section 8 (the edge rule of prediction 1, the falsifier of prediction 3, the single-neuron clause of prediction 8 and the measurement of prediction 15) are applied.

#	Prediction	Measurement	Result that would falsify DPC
1	LOR coincides with the collapse of the giant component of the functional network built from stimulus-locked inter-areal phase locking, the timing of content rather than of ongoing rhythms, and the collapse is sharper than the change in local power; the coherence of ongoing alpha may rise at the same time (Section 7.8)	TMS-EEG or single-pulse electrical stimulation, or other repeated stimuli, during slow concentration ramps; for each stimulation-recording pair, $\hat{\pi}$ is the share of trials with a detected response whose single-trial latency lies within $\pm W/2$ of the pair's awake median latency, and an edge exists when $\hat{\pi}$ exceeds a cut-off chosen, separately for each subject, so that the awake network lies near its percolation threshold, with W fixed in advance in milliseconds (Section 8)	Gradual, linear decline of the giant component with no threshold behaviour at LOR, or a giant component above q^* at concentrations where unconsciousness is established by more than unresponsiveness (P4c)
2	Signatures of a critical transition appear just before LOR: rising variance and autocorrelation of the order parameter (critical slowing down, a generic early-warning signal of transitions; Scheffer et al., 2009)	Time series of $S(t)$ during induction	No increase in variance or autocorrelation before LOR under carefully controlled slow induction. Existing evidence leans against it: human electrocorticography (ECoG) dynamics became more stable as patients became anaesthetised with propofol (Alonso et al., 2014), and in whole-hemisphere ECoG in monkeys the number of modes at the edge of instability fell smoothly during loss of consciousness, under ketamine-medetomidine and under propofol (Solovey et al., 2015); only thalamocortical recordings in mice under ketamine-xylazine showed diverging fluctuations and a hysteresis loop (Hwang et al., 2012)
3	Inter-areal lag *variability*, not mean lag, predicts LOR across agents; for GABAergic agents the variability is expected to arise through the reference feedback of P5, which adds timing error as the giant component shrinks, so response failures should rise before or with latency variability, whereas for ketamine variability should lead	Distribution of lag across trials for frontal-parietal connections	Mean lag or local power predicts LOR as well as or better than lag variability across propofol, sevoflurane and ketamine
4	Induction-phase excitation scales with sensory and recurrent load	Compare beta and gamma surges at induction with and without sensory deprivation, matched for concentration	No difference in surge magnitude between high- and low-load conditions. The model's version of this account already failed (Section 5.1), and sensory deprivation may also change arousal, which must be controlled. Scalp beta and gamma at light anaesthesia are contaminated by muscle activity, which itself changes with arousal, so the comparison needs neuromuscular block with an isolated forearm, or muscle components removed and a muscle index reported, and preferably source-space or intracranial gamma
5	Neural inertia ($c_{\mathrm{off}} - c_{\mathrm{on}}$) correlates with the strength of thalamocortical reference coupling (λ)	Thalamocortical phase-locking in animal models, with optogenetic or pharmacological modulation of thalamic drive	Manipulating thalamocortical coupling changes LOR or emergence thresholds without changing hysteresis. In the percolation model λ is calibrated, not measured, and the oscillator model shows no robust hysteresis (Section 4.3)
6	Targeted disruption of hub connections lowers the concentration required for LOR more than random disruption of the same number of connections	Animal models with focal inactivation of hub versus non-hub regions	Equal threshold shifts for hub and non-hub disruption. Many network accounts make the same prediction, so it does not single out DPC; in the model it holds in every run at every network size tested at the reference parameters (Appendix B, test C5), except one of 20 seeds at 302 nodes on fresh seeds (Appendix C.14), but not at several settings of the sensitivity analysis

#	Prediction	Measurement	Result that would falsify DPC
7	Ketamine reduces coherence with external stimuli more than coherence among internally generated cortical activity	Stimulus-locked versus spontaneous inter-areal phase-locking under ketamine	Equal or greater loss of spontaneous internal coherence, while PCI remains high. The model's version of this proposal failed as first specified (test K1) and passed, with a corrected measure, a test fixed in advance on *C. elegans* (K1c, Section 5.6); the prediction is kept as an empirical test
8	Trial-to-trial variability in the latency of responses *between* areas rises at concentrations just below loss of responsiveness, before mean latency changes appreciably, while single-neuron timing stays precise (P4)	Intracranial recordings of responses propagating between areas at stepped effect-site concentrations, in patients with electrodes implanted for clinical reasons (for example epilepsy monitoring; Krom et al., 2020); a full ramp up and down with repeated stimulation may not be obtainable in one patient, so data would be pooled across patients with differing electrode coverage; the single-neuron clause needs microelectrodes for single-unit timing, which clinical ECoG alone cannot provide	Inter-areal latency variability unchanged until responsiveness is lost, or rising only after mean latency. Evoked-response variability in rat auditory cortex is known to depend on anaesthetic depth, being highest at medium depth (Kisley and Gerstein, 1999), but it has not been measured between areas relative to loss of responsiveness. Under propofol, intracranial responses to single pulses become more variable after loss of consciousness without a change in first-peak latency (Zelmann et al., 2023), and the median latency of the earliest response component lengthened from 22.0 ms awake to 26.4 ms under propofol anaesthesia during surgery in another study (van Blooijs et al., 2024); neither measured timing variability between areas across graded concentrations
9	At the onset of propofol frontal alpha, phase locking between distant regions is better preserved in the alpha band than in broadband activity, and alpha onset follows a rise in timing variability (P8)	High-density EEG or intracranial recordings during slow induction	No preservation of alpha-band long-range phase locking relative to broadband, or alpha onset unrelated to timing variability. The model's version of P8's distinctive prediction failed (test F3, Section 6); the prediction is kept as an empirical test
10	The stimulus intensity needed to wake a sleeper increases with the share and the spatial extent of down-states, measured directly, beyond what overall slow-wave power predicts (P7)	Intracranial or high-density EEG with graded arousing stimuli across N1-N3, measuring down-states and their extent	No relation between down-state share or extent and arousal threshold once slow-wave power is accounted for. The relation with slow-wave power itself is already reported: high delta power raises the arousal threshold in rats (Neckelmann and Ursin, 1993), and in human N2 sleep arousal is more likely during periods of weaker slow-wave activity and stronger inter-areal phase coherence (Bastuji et al., 2021)
11	Damaged myelin lowers the anaesthetic concentration needed for loss of responsiveness (P9)	Loss-of-righting-reflex dose-response in demyelinated mice (for example after cuprizone) against controls; as a secondary, observational test in multiple sclerosis, white-matter integrity on diffusion imaging against propofol effect-site concentration at LOR under a protocolised target-controlled induction, adjusting for age, disability and medication that changes anaesthetic requirement	No difference, or reduced sensitivity. Existing evidence does not settle it: induction time at a fixed concentration was unchanged in one cuprizone study, which found slower emergence (Wang et al., 2024), and longer in older cuprizone mice in another, with small groups (Tot et al., 2024)
12	Microglial depletion narrows neural inertia, $c_{\text{off}} - c_{\text{on}}$ (P9, P5)	Slow up-and-down concentration ramps at steady state in microglia-depleted and control mice	Inertia unchanged after depletion. In one study genetic depletion of microglia did not hasten emergence (Cao et al., 2026)
13	Extending the finding that loss of an astrocytic calcium pathway lowers NREM delta power and increases microarousals (Bojarskaite et al., 2020): suppressing cortical astrocytic signalling lowers the arousal threshold in proportion to the fall in the share of down-states (P9, P7)	Chemogenetic or genetic suppression of astrocytic calcium, with sleep recordings and graded arousing stimuli	Fewer down-states with no change in the arousal threshold. The model's version of this effect (test A1) could not have passed, so it did not test the prediction
14	At matched depth of slow-wave activity, responsiveness returns faster and with less hysteresis on arousal from NREM sleep than on emergence from anaesthesia (P7)	Graded arousing stimuli in N3 sleep compared with stepped reduction of propofol in the same volunteers at matched slow-wave activity, measuring the time and stimulus intensity at which responsiveness returns	Equal or slower return of responsiveness from sleep at matched depth. A faster return from sleep is also expected because no drug has to clear, so the distinctive part is the smaller hysteresis at matched depth

#	Prediction	Measurement	Result that would falsify DPC
15	Between the concentrations of recovery and loss of responsiveness, a brief arousing stimulus (for example central thalamic stimulation) restores responsiveness that outlasts the stimulus; above the concentration of loss, responsiveness returns only while the stimulus lasts (P5, P7)	Central thalamic or cholinergic stimulation at steady-state concentrations inside and outside the induction-emergence window in the same animal, comparing how long responsiveness persists after the stimulus ends	Responsiveness that never outlasts the stimulus at concentrations where induction and emergence differ, or that outlasts it equally at every concentration

The protocol of Section 9.2 in its earlier, fuller form:

1. Estimate pairwise phase synchrony in frequency bands defined in advance. A measure robust to volume conduction, such as the weighted phase lag index, discards coupling at zero and near-zero lag, which is the coupling P2 counts (directed phase transfer entropy on source-reconstructed high-density EEG has already shown changes in directed connectivity at propofol-induced loss of consciousness; Chen et al., 2022); it is therefore paired with a lag-sensitive measure on intracranial or source-reconstructed data, where volume conduction is tractable, and the giant component from the weighted phase lag index alone is reported as a lower bound.
2. Construct a functional network per time window and connection, using surrogate-based significance thresholds.
3. Compute the order parameter $S(t)$ (fraction of nodes in the giant component), its variance and lag-1 autocorrelation, and the distribution of inter-areal lag variability.
4. Fit the model in Section 4 to S as a function of effect-site concentration, separately for induction and emergence, and estimate π_c, q^* and λ. These may not be separately identifiable from S alone, so the fit will report their joint uncertainty (a profile likelihood or posterior), not point values only.
5. Compare agents (propofol, sevoflurane, ketamine, dexmedetomidine as a non-GABAergic sedative) and compare with natural sleep, which differs neurophysiologically from anaesthesia (Akeju and Brown, 2017). Unresponsiveness under dexmedetomidine is usually rousable, and most interviews after it report experiences, mainly dreaming (Radek et al., 2018); DPC therefore predicts [H] a giant component at or near quorum under dexmedetomidine at loss of responsiveness, with sensory broadcast reduced, which separates it from propofol at matched responsiveness.
6. Control opioids and surgical stimulation: opioids such as remifentanil lower the hypnotic dose and change the electroencephalogram, and noxious stimulation raises arousal by itself. Use either a fixed opioid effect-site target or opioid-free volunteers, analyse stimulus-free epochs, and compare agents only at matched opioid (Kuizenga et al., 2018, found that remifentanil changed whether induction and recovery concentrations differed).
7. Pre-specify (ideally in a public registry) the hypotheses, frequency bands and thresholds before analysis. Appendix E does this for the predictions that existing open recordings can test (predictions 1, 2, 3 and 8, in part), with the datasets, measures and pass rules fixed before any test was run.

F.9 Limitations in full (Section 10)

1. **Borrowed words.** Consensus and quorum are used in their everyday sense only (Section 1.4) and do not imply that the brain implements an algorithm. The predictions in Section 8 are stated in neurophysiological terms so that they can be tested without the words.
2. **Measured, calibrated and free parameters.** W and the baseline jitter are assumed: they are values measured within neurons or local circuits and applied between cortical areas, where they have not been measured (Section 4.7). λ is calibrated to one measured inertia ratio (Appendix B): 1.08, for isoflurane in mice (Friedman et al., 2010), whose authors showed that the hysteresis is not fully explained by pharmacokinetics. This target was adopted after results at an earlier target, the human propofol ratio 1.11 (Iwakiri et al., 2005), had been seen; at 1.11 consistency check C5 held in only 9 of 20 seeds at 10,000 nodes, and at 1.08 in every seed (Appendix D). The choice is therefore not independent of the results. The human propofol estimates of 1.07 and 1.11 are not usable as calibration targets: their authors found no consistent hysteresis (Warnaby et al., 2017) or describe the two concentrations as similar (Iwakiri et al., 2005), and a prospective study designed to test human neural inertia found none for propofol, though it found it for sevoflurane (Kuizenga et al., 2018). The other reported ratios from 1.07 to 1.63 give other values of λ, which Appendix B reports and with which the sensitivity analysis (Appendix B.6) repeats the tests. The largest measured ratio, 2.85 for halothane in mice (Friedman et al., 2010), could not be reproduced: as λ grows, the mean ratio reaches only about 2.0 to 2.4, rising with the arousal level, before some seeds no longer recover at zero concentration (Appendix B.3). The model's neural inertia is therefore bounded, and an agent with inertia as large as halothane's lies outside it. On networks not used for calibration the calibrated λ does not reproduce its target: the ratio is 1.071 ± 0.014 at 10,000 nodes and 1.066 ± 0.007 at 1,000,000, just below the lowest published ratio (Appendix C.14). The quorum q^*, the oscillator coupling and the maximal effects of each agent class are free parameters; the sensitivity analysis (Appendix B.6) reports how the results depend on them.
3. **Simplified topology.** Random-graph percolation ignores spatial embedding, modularity, directionality and synaptic dynamics. It also treats each connection as either coherent or not, although a fibre bundle between regions contains many synapses, so that synaptic failure weakens a bundle rather than removing it.

4. **Measurement confounds.** EEG phase synchrony is affected by volume conduction, reference choice and changes in signal-to-noise ratio. These must be controlled before connectivity changes are interpreted.
5. **Ketamine.** DPC's account of disconnected consciousness under ketamine is provisional: its first version failed its first test (K1) and passed, with a corrected measure, a test fixed in advance on *C. elegans* only (K1c, and again with its control, K1e; the human connectome could not be evaluated), the revised version failed its pre-specified test (K2) and is kept only as a contributing factor, and neither explains why complexity is preserved under ketamine (Section 7.2). Read with DPC's own quorum, the timing faults assigned to ketamine abolish quorum on the human connectome under two of three rules for counting a connection, but not under the rule used first, which counts each connection inferred from diffusion MRI as two independently timed connections (Appendices C.12 and C.13); under that rule, either ketamine also causes transmission failure there, or its fault type is not timing variability alone, or the human model lacks what makes timing faults decisive.
6. **Criticality.** Claims that cortical networks operate near criticality, first made from neuronal avalanches in rat cortical slices and cultures (Beggs and Plenz, 2003), remain an open question; a functional magnetic resonance imaging (fMRI) analysis of propofol-induced unconsciousness reported a departure from critical dynamics (Tagliazucchi et al., 2016), and cortical recordings (electrocorticography and magnetoencephalography) in humans and macaques reported near-critical slow cortical dynamics during consciousness (Toker et al., 2022). DPC does not require self-organised criticality; it requires only a percolation-like threshold in the coherent subgraph.
7. **Experience.** DPC addresses the conditions for experience and report, not why such integration is accompanied by experience.
8. **Arousal is modelled only as a share of the reference.** Without an arousal term, the reference simulator can remain unconscious even at zero concentration after strong reference feedback or loss of hub nodes, which patients do not. P7 adds an arousal drive that sustains part of the reference and is suppressed by the agent. With it, most runs recover, but not all: some stay trapped unconscious at zero concentration (test O1: 4 of 20 seeds at 302 nodes and 2 of 20 at 1,000, none in larger networks), and more do after a hub lesion (test O2: 4 of 20 seeds at 302 and at 1,000 nodes, none in larger networks; Appendix B). The arousal level itself is a free parameter, because the reported inertia ratios from 1.07 to 1.63 are reproduced at every level tested (the halothane ratio of 2.85 at none, limitation 2), and the arousal systems are represented by a single number rather than by their own dynamics (Kelz et al., 2008).
9. **Neural inertia depends on the model class.** The percolation model reproduces the measured loss-to-recovery ratio when λ is calibrated, but in the phase-oscillator model on the human connectome the reference feedback produced no hysteresis distinguishable from run-to-run variation at any strength tried (Appendix C.2). Hysteresis is therefore not a robust consequence of P5 across model classes, the percolation model cannot reach the largest measured ratio, and part of its agreement with the data is a calibration (Section 5.4); other mechanisms, including adaptive feedback (Kim et al., 2018) and noise-driven switching between metastable states (Proekt and Hudson, 2018), remain candidates.
10. **Local processing is narrower than local module integrity.** In the simulations, spontaneous coherence within anatomical modules generally fell together with global coherence; it did not persist inside every module. What persisted was stimulus-driven entrainment of the stimulated sensory area while propagation to the rest of the network failed. Section 7.4 is stated in that narrower form, which is also what intracranial recordings show (Krom et al., 2020).
11. **Paradoxical excitation.** The congestion account failed in the model (tests W1 to W4). In the oscillator model, fluctuations of global coherence peak at or after the loss of quorum rather than clearly before it, so the pre-specified test T1 failed; they do rise at least tenfold on the conscious side of the threshold, a post hoc finding (T1b) that replicated on seeds not used in the study that suggested it (T1c). The revised account of Section 5.1, survivor lock-in with collective fluctuations, therefore rests on a replicated but generic threshold behaviour and on a synchrony surge reproduced only at one coupling, and interneuron antisynchrony (McCarthy et al., 2008) remains the more developed explanation.
12. **Natural sleep.** The model represents NREM sleep only by withdrawal of arousal and by down-states that silence either whole populations or single connections. The two accounts give different results (Appendix B), and which dominates in the brain is not known. The shares of time spent in down-states come from rat recordings and from a human estimate that combines two studies; the share of populations silent at any one moment, which the population account needs, has not been measured directly. REM sleep is not modelled, and the account of sleep as a pause that frees resources for maintenance (Section 5.6) is a hypothesis that the simulations do not test.
13. **Constraining each agent independently.** If the failure probability $p(c)$ and the jitter $\sigma_0(c)$ of each agent could be chosen freely, DPC could fit any outcome. The hypothesis is falsifiable only if these are measured independently for each agent, for example as synaptic failure rates and spike-timing variability in slice or in vivo recordings at known concentrations, before network-level predictions are compared with data.
14. **Fixed delays and jitter.** In the models constant offsets and measured conduction delays are not harmless (S4, D1), and a learned expectation of arrival phase failed (L1 to L4); how constant delays would be made harmless is open (Section 3.4).
15. **One fly, one sex, few seeds.** The *Drosophila* results rest on a single adult female brain (Dorkenwald et al., 2024) and, because each run takes 30 to 65 minutes, on 6 seeds per condition in both the cross-species study and the pre-specified tests, on a coarser concentration grid (Appendix C.1), so their confidence intervals are wide. The shorter settling of the *Drosophila* runs also weakens the collapse of broadcast, so test T3, which held in no *Drosophila* seed, is reported with a caveat (Appendix C.4 and Section 6). A complete connectome of the male central nervous system, including the ventral nerve cord, has since been published (Berg et al., 2026); repeating the cross-species study on it, once its terms of use are confirmed, would test whether the results generalise across individuals and sexes.
16. **Glial regulation is represented only by parameters.** The glial loops of P9 have their own dynamics, but the model holds each fixed during a sweep, and their strengths (γ, g, s_a) are free parameters; λ was not recalibrated with the microglial

loop. The evidence comes from mice. Astrocytic effects on anaesthesia differ in sign between regions (Lin et al., 2023; Liu et al., 2023; Zhao et al., 2024); several earlier astrocyte studies used a mouse line whose transgene is also expressed in neurons (Fujita et al., 2014); and astrocytic calcium signalling did not modulate hippocampal plasticity in one study (Agulhon et al., 2010). The microglial depletion findings (Cao et al., 2023; He et al., 2023) used the drug PLX5622, which also speeds the clearance of anaesthetics, and genetic depletion did not reproduce the faster emergence (Cao et al., 2026); the remaining support is genetic loss of the P2Y12 receptor (Cao et al., 2023), and no study has measured neural inertia with microglia manipulated. The ageing evidence is only suggestive: the propofol concentration at which half of volunteers lose consciousness falls with age, from 2.35 μg/ml at 25 years to 1.25 μg/ml at 75 (Schnider et al., 1999), but loss of neurons and receptor changes are equally possible causes.

17. **Mammalian timing in every species.** Both models use the same mammalian timing values for all three connectomes: a 40 Hz reference, a 3 ms integration window and 3 ms baseline jitter. Invertebrate nervous systems do not share this timescale. An identified *C. elegans* sensory neuron recorded in situ lacks classical Na^+ action potentials, and the other neurons recorded share its membrane properties (Goodman et al., 1998), although one class of sensory neuron, AWA, fires all-or-none calcium action potentials (Liu et al., 2018). The cross-species results therefore test the network part of the hypothesis, whether each wiring diagram has a sharp quorum threshold and where it lies, which depends only on topology and on timing error relative to the window. They do not show that worms or flies integrate on a millisecond timescale. Accordingly, the critical jitter in Appendix B should be read as a multiple of the integration window (for *C. elegans*, about seven windows), not as a value in milliseconds. A direct test in *C. elegans* would use whole-brain calcium imaging under slowly stepped isoflurane, building the functional network from correlated activity over seconds and measuring whether its giant component collapses at a threshold and recovers at a lower concentration; Awal et al. (2020) showed that the correlations collapse, but not whether they do so at a threshold or with inertia.
18. **The models do not generate the rhythms they assume.** Neither model produces alpha, the slow oscillation or burst suppression from its own dynamics: the percolation model has no rhythm, and the phase model imposes a single reference frequency. The phase model's global order parameter measures the coherence of the reference rhythm and of content together, so it cannot show the dissociation between a hypersynchronous rhythm and fragmented content described in Section 7.8. Separating the two requires measuring stimulus-locked coherence apart from ongoing coherence in the model, as prediction 1 requires in data.
19. **One phase reference where the brain has several rhythms.** P5 assumes a single shared reference, but perception appears to be sampled by several rhythms that depend on modality, task and region (VanRullen, 2016). Several references would make quorum a condition on more than one timing domain at once, and how they interact under anaesthesia is not modelled here.
20. **No thalamus in the human connectome.** The 998-region structural network (Hagmann et al., 2008) is cortical, so the thalamocortical phase reference of P5 enters the models as a parameter, not through thalamic anatomy.
21. **Many tests, some added later.** Many tests were run across organisms, network sizes and parameter settings without correction for multiple comparisons. Several analyses were added after earlier results (T1b, X1, X2 and the re-specified A1); they are labelled post hoc or re-specified, and the model-revision history in Appendix D records each change.
22. **Consistency and construction checks are not support.** Checks such as C2, C3, C6, C7, M1, G1 to G3, SP1 and SP1-1 follow from the model's equations (C4 and C5 are expected from percolation theory but can fail in small networks), and C1, T2a, T2b, T2a-D and T2a-1 could not fail at the chosen settings (A1 could not pass). They confirm that the code implements the model and are not counted as evidence for the hypothesis.
23. **Concentration is relative.** In the models EC50 is the concentration of half-maximal circuit effect, not of LOR; quorum is lost at 0.88 to 1.8 EC50 depending on the network, so no single mapping to clinical concentrations follows. Predicted propofol effect-site concentrations at LOR in the Michigan dataset ranged from 0.8 to 3.0 μg/ml (Appendix E.2); published values differ with the pharmacokinetic model and ke0.
24. **No quantitative comparison with rival accounts.** The paper does not fit DPC and the accounts of Section 2.3 to the same data; prediction 3 and the protocol of Section 9 are where they would be compared.
25. **Quorum loss is guaranteed by the chosen maxima.** With transmission failure of up to 95% and added timing error of up to 40 ms, longer than a whole 25 ms cycle at 40 Hz, quorum loss in tests C1, T2a and T2b could not fail at the reference parameters. The one exception found is a narrower spread of natural frequencies in the phase model (0.25 instead of 0.5), with which the GABAergic-like agent abolished quorum in no seed of 5 (Appendix B.6), so in the phase model whether quorum is lost depends on the oscillator parameters as well as on the agent. Those results show that the mechanism operates as stated, not that it is supported; what they add is where the loss happens and that both routes reach it.
26. **The model mostly needs more transmission failure than is measured.** On the connectomes, quorum is lost only when 85 to 90% of transmissions fail with the direction of connections ignored, or 61% and 72% with it kept on the *Drosophila* and *C. elegans* wiring, the only connectomes with measured direction (Appendix B.5), whereas isoflurane at 0.7 mM, about twice its MAC, inhibited the exocytosis triggered by single action potentials by 62% in cultured rat hippocampal neurons, measured at 4 mM extracellular calcium, well above the physiological level (Baumgart et al., 2015). The inhibition at the lower concentrations at which responsiveness is lost was not reported there, so on that ground alone the gap between what the model needs and what has been measured would be larger than this comparison shows. Read with DPC's own quorum in the oscillator model, the GABAergic-like agent abolishes quorum at 74 to 81% transmission failure with one timing draw per connection, at 75 to 87% with C.12's rule and at 43 to 75% with both directions required (the maximal failure times the fractional effect of the Hill dose-response at the mean loss concentration, Appendices C.12 and C.13). Only two values fall below the measured 62%: the both-directions rule on the human connectome (43%) and the directed *Drosophila* wiring in the percolation model (61%). Read literally, the model mostly reaches unconsciousness only beyond measured anaesthetic effects,

unless timing error contributes much more than assumed or the threshold is lower in real networks. The comparison is not like-for-like: the measurement is the release probability at single synapses, whereas the model's failure is that of a whole inter-areal connection made of many synapses, which fails less often than any one of them; and the high calcium of the measurement and the high concentration at which it was made bias it in opposite directions.

27. **Rats should stay conscious in NREM sleep, in the model.** The rat share of down-states never abolishes quorum (test O3), so the model implies that rats remain conscious during NREM sleep unless their down-states are larger or more synchronised than the share of time they occupy. The comparison also treats a share of time as a share of populations silent at once.
28. **Pass rules, realisations and multiplicity.** Percolation checks count as holding only in every seed, the pre-specified tests at least at 80% of seeds (C.7, C.10, C.11) or with a Wilson 95% lower bound above one half (C.12 to C.14; Appendix A.6), and the cross-species results are reported as counts. Seeds are realisations of one wiring diagram, not independent brains, so the counts describe the model's own variability, not generality across individuals, and none is corrected for multiple comparisons (item 21).
29. **Results that depend on the calibration choice.** Switching the reference ratio from 1.11 to 1.08 (Appendix D) changed some outcomes: with λ calibrated to 1.11, the hub-lesion test C5 held in 16, 17 and 9 of 20 seeds at 302, 1,000 and 10,000 nodes (archived report `validation_full_20261018T123515Z`, run before the code corrections of 19 October), against every seed at the reference value. λ and the arousal level are not separately identified, since every arousal level tested reproduces every reported ratio up to 1.63, and λ was not recalibrated when the Hill coefficient was varied. Results close to such edges are fragile, and the sensitivity analysis (Appendix B.6) shows where. The calibration also used the same ten random networks at 10,000 nodes as ten of the twenty validation networks at that size, so at that size the validation is not independent of the calibration; the tests of Appendix C.12 use calibration seeds disjoint from their test seeds. Repeated on seeds disjoint from both (Appendix C.14), every claim held at every size except C5 in one of 20 seeds at 302 nodes, so the every-seed verdict is again that a claim failed, and the loss-to-recovery ratio was 1.071 ± 0.014 at 10,000 nodes, below the 1.08 to which λ was calibrated, and 1.066 ± 0.007 at 1,000,000.
30. **P6 is not tested.** In the Wilson-Cowan model, each region's inhibition is balanced while it receives long-range excitation, so a region cut off from that input has a single, low-activity steady state and no second state to alternate with. In 10 realisations at each of the three global couplings, this held for all 998 regions, with excitatory activity at most 0.12 on a scale of 0 to 1; the steady state was stable in all but at most 2 regions, which by linear stability analysis would oscillate at about 10 Hz instead (post hoc diagnostic, `simulator/results/diagnostics/`). Accordingly, at the highest concentration, hub activity fell on average to 1 to 19% of its awake value, depending on the coupling (a post hoc reading of the runs in Appendices C.5 and C.6). Testing P6 would need local circuits that sustain activity on their own, which adaptation then destabilises into up and down states, as in the whole-brain model of Sacha et al. (2025). That model also bears on the clause that the default dynamics are agent-specific: in it, GABAergic and NMDA-antagonist actions both produced slow waves, each with increased adaptation, and they differed in how closely functional connectivity followed anatomical connectivity rather than in the form of the dynamics.
31. **The broadcast measure has a floor.** Broadcast is the locking of the non-sensory populations to the stimulus over two stimulus cycles, and a coherent population rotating at its own frequency shows some locking to any rhythm over so short a window. A post hoc diagnostic repeated K1 with a phantom stimulus of zero strength (`simulator/results/diagnostics/`). On *C. elegans*, the broadcast left at the highest concentration was at about the level of locking to the phantom (median 0.093 against 0.090; with a window four times longer, 0.039 against 0.026, above it in 15 of 20 seeds but by less than half of the awake excess), while awake broadcast clearly exceeded it (0.212 against 0.093); on the human connectome, even awake broadcast barely exceeded it (0.040 against 0.029). The ratio tests built on broadcast (K1, K2b, T3 and N2) therefore compare quantities that both contain this floor. Their verdicts are reported as fixed, but they cannot be read as measures of propagation: the human T3 count is not interpretable. On *C. elegans*, with the phantom's locking subtracted from both sides of the ratio, K1's criterion holds in 16 of 20 seeds, and in 20 of 20 with the longer window; this reading was post hoc, and a test with a floor-corrected broadcast and a window of eight cycles, fixed before its run on fresh seeds, then passed on *C. elegans* (K1c: 20 of 20 seeds) and could not be evaluated on the human connectome (Appendix C.11); the same correction for T3 (T3c) was not run.
32. **The oscillator model did not implement P2 and P3 as stated.** Until the readout of Appendix C.12, quorum on the connectomes was read as global phase coherence at least half its awake value, and timing error weakened coupling smoothly rather than excluding late signals (Section 4.7). Every connectome result before C.12 is therefore a result about agreement among coupled oscillators, not a direct test of the window and the giant component. The readout of C.12 now provides that test; it changed one central result: timing variability alone did not abolish quorum on the human connectome. Appendix C.13 shows this depends on how a human connection is counted.
33. **The fluctuation result may reflect the awake baseline.** The tenfold rise of fluctuations before collapse (T1b, T1c) is a ratio to the awake value, which is very small when awake coherence is close to 1; the susceptibility at loss is similar across conditions, and the ratio fails where awake coherence is low, as with measured conduction delays (Section 6). In the matched-coherence comparison of Appendix C.12 the excess survived on all three connectomes (T1d), so the model's threshold does show collective, slowing fluctuations; recordings in monkeys, however, found cortical dynamics becoming more stable, not less, during loss of consciousness (Solovey et al., 2015), and the test on open recordings (Appendix E, H2b, labelled OD2b in Section 6) was inconclusive. The model's fluctuation measures are computed on global coherence R, not on S: the variance and autocorrelation of S were not computed. The bar of T1d, $\chi/\chi_{\mathrm{ind}} \geq 2$, corresponds to a mean pairwise correlation of fluctuations of only $1/(N-1)$; T1e samples once per jitter correlation time, over which the Ornstein-Uhlenbeck jitter keeps an

autocorrelation of e^{-1}, so part of its lag-one autocorrelation is the jitter's own. Neither measure shows a peak at the threshold.

34. **NREM sleep and dreaming conflict in the model.** With the human estimate of down-states, quorum is lost in every seed (O4), yet dreams are reported from NREM sleep (Siclari et al., 2017), and in the model no local islands of agreement outlasted the loss of quorum (S2). For both to hold, a giant component above q^{*} would have to survive in a subset of regions, for example posterior cortex, while down-states silence others; the model, which silences populations at random, cannot produce this.